ADDRESSES

AND

PROCEEDINGS

FOURTH

National Conservation Congress

INDIANAPOLIS

OCTOBER 1-4, INCLUSIVE

1912

ADDRESSES

AND

PROCEEDINGS

FOURTH

National Conservation Congress

INDIANAPOLIS

OCTOBER 1-4, INCLUSIVE

1912

The Project Gutenberg eBook of Proceedings [of the] fourth National Conservation Congress [at] Indianapolis, October 1-4, 1912

Title: Proceedings [of the] fourth National Conservation Congress [at] Indianapolis, October 1-4, 1912

Creator: United States. National Conservation Congress

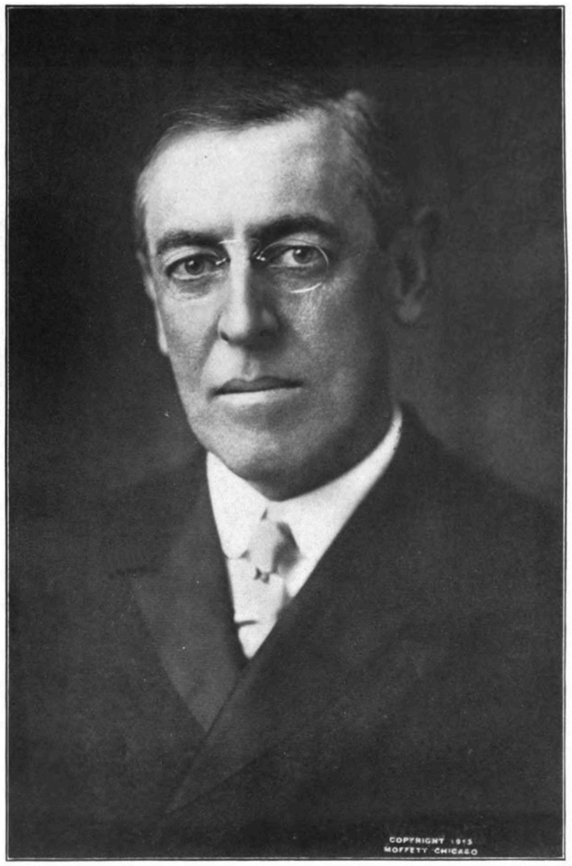
Release date: January 4, 2023 [eBook #69706]
Most recently updated: October 19, 2024

Language: English

Original publication: United States: National Conservation Congress, 1911

Other information and formats: www.gutenberg.org/ebooks/69706

Credits: Bob Taylor, Bryan Ness and the Online Distributed Proofreading Team at https://www.pgdp.net (This book was produced from scanned images of public domain material from the Google Books project.)



WOODROW WILSON
PRESIDENT OF THE UNITED STATES

PROCEEDINGS

Fourth
National Conservation Congress

Indianapolis

OCTOBER 1–4, INCLUSIVE, 1912

“Let us conserve the foundations of our prosperity”

(Declaration of the Governors, 1908)

INDIANAPOLIS
NATIONAL CONSERVATION CONGRESS
1912

WM. B. BURFORD PRESS
INDIANAPOLIS,—IND.

CONTENTS.

OFFICERS AND COMMITTEES, 1912.

President,

JOHN B. WHITE, Kansas City, Mo.

Executive Secretary,

THOMAS R. SHIPP, Washington, D. C.

Treasurer,

D. AUSTIN LATCHAW, Kansas City, Mo.

Recording Secretary,

JAMES C. GIPE, Indianapolis, Ind.

Executive Committee.

E. LEE WORSHAM, Atlanta, Ga., *Chairman*.

J. LEWIS THOMPSON, Houston, Texas.
W. A. FLEMING JONES, Las Cruces, N.M.
WALTER H. PAGE, New York.
GEORGE C. PARDEE, Oakland, Cal.
DR. H. E. BARNARD, Indianapolis, Ind.
MRS. PHILIP N. MOORE, St. Louis, Mo.
BERNARD N. BAKER, Baltimore, Md.
HENRY C. WALLACE, Des Moines, Iowa.
GIFFORD PINCHOT, Washington, D. C.

Local Board of Managers, Indianapolis.

RICHARD LIEBER, *Chairman*.

JOSEPH C. SCHAF, *Vice-Chairman*. JAMES W. LILLY, *Treasurer*.

L. H. LEWIS, *Secretary*.

FREDERIC M. AYRES.
GEORGE L. DENNY.
EDGAR H. EVANS.
CARL G. FISHER.
C. G. HANCH.
O. D. HASKETT.
ALBERT E. METZGER.
WILLIAM J. MOONEY.
W. H. O'BRIEN.

Vice-Presidents.

Arkansas—E. N. PLANK, Decatur.
California—FRANCIS CUTTLE, Riverside.
Colorado—I. S. T. GREGG, Golden.
Connecticut—PROF. J. W. TOUMEY, Hartford.
District of Columbia—DR. HARVEY W. WILEY, Washington.
Florida—T. J. CAMPBELL, Palm Beach.
Nebraska—PROF. E. A. BURNETT, Lincoln.
New Jersey—E. A. STEVENS, Hoboken.
New York—DR. W. T. HORNADAY, New York City.
Ohio—J. C. RODGERS, Mechanicsburg.
Oklahoma—T. C. HARRICE, Wagoner.

Georgia—L. R. AKIN.

Illinois—BALLARD DUNN, Chicago.
Iowa—PROF. P. G. HOLDEN, Ames.
Louisiana—HENRY E. HARDTNER, Urania.
Massachusetts—PROF. F. W. RANE, Boston.
Missouri—HERMAN VON SCHRENK, St. Louis.

South Carolina—PROF. M. W. TWITCHELL, Columbia.
South Dakota—GOV. R. S. VESSEY, Pierre.
Texas—W. GOODRICH JONES, Temple.
Washington—A. L. FLEWELLING, Spokane.
Wisconsin—HERBERT QUICK, Madison.

STANDING COMMITTEES, 1912.

Forests—H. S. Graves, Washington, D. C., Chairman; E. T. Allen, Portland, Ore.; Major E. G. Griggs, Tacoma, Wash.; William Irvine, Chippewa Falls, Wis.; George K. Smith, St. Louis.

Minerals—Dr. Joseph A. Holmes, Washington, D. C., Chairman; Dr. Charles R. Van Hise, Madison, Wis.; Dr. I. C. White, Morgantown, W. Va.; C. W. Brunton, Denver, Col.; John Mitchell, New York City.

Lands and Agriculture—Prof. L. H. Bailey, Cornell University, Chairman; Prof. George E. Condra, Nebraska; Prof. J. L. Snyder, Lansing, Mich.; F. D. Coburn, Kansas; Charles S. Barrett, Union City, Ga.

Education—Dr. C. E. Bessey, Lincoln, Neb., Chairman; Dr. David Starr Jordan, Leland Stanford University, Oakland, Cal.; Dr. Edward E. Alderman, University of Virginia, Charlottesville; Dr. E. C. Craighead, Tulane University, New Orleans, La.; Prof. E. T. Fairchild, Topeka, Kas.

Vital Resources—Dr. William H. Welch, Johns Hopkins University, Baltimore, Md., Chairman; Prof. Irving Fisher, Yale University, New Haven, Conn.; Dr. J. N. Hurty, Indianapolis, Ind.; Hon. A. B. Farquhar, York, Pa.; Dr. Oscar Dowling, Shreveport, La.

Homes—Mrs. Matthew T. Scott, Washington, Chairman; Mrs. Harriet Wallace-Ashby, Des Moines, Iowa; Mrs. J. E. Rhodes, St. Paul, Minn.; Mrs. Sarah S. Platt-Decker,[1] Denver, Col.; Mrs. Amos F. Draper, Washington, D. C.

Child Life—Hon. Ben B. Lindsay, Denver, Col., Chairman; Dr. Samuel M. Lindsay, New York City; Judge Henry L. McCune, Kansas City, Mo.; Mrs. Carl Vrooman, Bloomington, Ill.; Dr. Anna Louise Strong, Seattle, Wash.

Food—Dr. Harvey W. Wiley, Washington, D. C., Chairman; F. G. Urner, New York; Prof. F. Spencer Baldwin, Boston, Mass.; J. F. Nickerson, Chicago, Ill.; Lucius P. Brown, Nashville, Tenn.; E. H. Jenkins, New Haven, Conn.; M. A. Scovelle, Lexington, Ky.; Prof. Geo. A. Loveland, Lincoln, Neb.

Civics—Ralph Easley, New York, Chairman; Albert Hall Whitfield, Jackson, Miss.; B. A. Fowler, Phœnix, Ariz.; H. M. Beardsley, Kansas City, Mo.; Francis J. Heney, San Francisco, Cal.

General (including Domesticated Animals and Wild Life)—Dr. W. T. Hornaday, New York, Chairman; Dr. L. O. Howard, Washington, D. C.; Mrs. Minnie Maddern Fiske, New York City; Dr. John Muir, Martinez, Cal.; D. Austin Latchaw, Kansas City, Mo.; Prof. Geo. A. Loveland, Lincoln, Neb.

Waters—Hon. J. N. Teal, Portland, Ore., Chairman; Hon. Joseph E. Ransdell, Lake Providence, La.; Walter S. Dickey, Kansas City, Mo.; Hon. Herbert Knox Smith, Washington, D. C.; W. K. Kavanaugh, St. Louis, Mo.; Dr. W. J. McGee, Washington, D. C.; Prof. Geo. F. Swain, Harvard University.

National Parks (including Mammoth Cave, Ky., and Adjacent Lands)—Dr. W. J. McGee,[1] Washington, D. C.; Dr. Henry F. Drinker, South Bethlehem, Pa.; Hon. William P. Borland, Kansas City, Mo.; Hon. Gifford Pinchot, Washington, D. C.; M. H. Crump, Bowling Green, Ky.

FOOTNOTES:

[1] Deceased.

OFFICERS AND COMMITTEES, 1913.

President,

Charles Lathrop Pack, Lakewood, N. J.

Vice-President,

Mrs. Philip N. Moore, St. Louis, Mo.

Executive Secretary,

Thomas B. Shipp, Indianapolis, Ind.

Treasurer,

D. A. Latchaw, Kansas City, Mo.

Recording Secretary,

James C. Gipe, Indianapolis, Ind.

Executive Committee,

E. Lee Worsham, Atlanta, Ga., *Chairman*.

Walter H. Page, New York City.
J. B. White, Kansas City, Missouri.
B. N. Baker, Baltimore, Maryland.
Dr. Henry S. Drinker, S. Bethlehem, Pa.
George E. Condra, Lincoln, Neb.
Joseph N. Teal, Portland, Oregon.
Dr. Henry Wallace, Des Moines, Iowa.
Dr. George C. Pardee, Oakland, Cal.
Thomas Nelson Page, Washington, D. C.
Gifford Pinchot, Washington, D. C.

Mrs. Emmons Crocker, Fitchburg, Mass.

[2]*Standing Committees.*

Forestry—Henry S. Graves, Chairman, Forest Service, Washington, D. C.; E. T. Allen, Yeon Portland, Ore.; J. B. White, Long Building, Kansas City, Mo.; W. R. Brown, Berlin, New Hampshire; E. A. Sterling, Secretary, Real Estate Building, Philadelphia, Pa.

FOOTNOTES:

[2] At the time the Proceedings went to press the other standing committees had not been appointed.

CONSTITUTION

OF THE

NATIONAL CONSERVATION CONGRESS

As Amended by the Fourth Congress.

ARTICLE 1—NAME.

This organization shall be known as the National Conservation Congress.

ARTICLE 2—OBJECT.

The object of the National Conservation Congress shall be: (1) to provide a forum for discussion of the resources of the United States as the foundation for the prosperity of the people, (2) to furnish definite information concerning the resources and their utilization, and (3) to afford an agency through which the people of the country may frame policies and principles affecting the wise and practical development, conservation and utilization of the resources to be put into effect by their representatives in State and Federal Governments.

ARTICLE 3—MEETINGS.

Section 1. Regular annual meetings shall be held at such time and place as may be determined by the Executive Committee.

Section 2. Special meetings of the Congress, or its officers, committees or boards, may be held subject to the call of the President of the Congress or the Chairman of the Executive Committee.

Section 3. After a call of the Executive Committee by the Chairman, and after all members of the Committee have been notified of the meeting in sufficient time to be present, three members shall constitute a quorum for the transaction of business.

ARTICLE 4—OFFICERS.

Section 1. The officers of the Congress shall consist of a President, to be elected by the Congress; a Vice-President to be elected by the Congress; a Vice-President from each State, to be chosen by the respective State delegations; one from the National Conservation Association and one from the National Association of Conservation Commissioners; an Executive Secretary, a Recording Secretary, and a Treasurer, all to be elected by the Congress.

Section 2. The duties of these officers may at any time be prescribed by formal action of the Congress or Executive Committee. In the absence of such action their duties shall be those implied by their designations and established by custom. In addition, it shall be the duty of the Vice-Presidents to receive from the State Conservation Commissions, and other organizations concerned in Conservation, suggestions and recommendations and report them to the Executive Committee of the Congress.

Section 3. The officers shall serve for one year, or until their successors are elected and qualify.

Article 5—Committees and Boards.

Section 1. An Executive Committee of seven, in addition to which the President of the National Conservation Association, the President of the National Association of State Conservation Commissioners, and all ex-Presidents of the Congress shall be members, ex officio, shall be appointed by the President to act for the ensuing year; its membership shall be drawn from different States, and not more than one of the appointed members shall be from any one State. The Executive Committee shall act for the Congress and shall be empowered to initiate action and meet emergencies. It shall report to each regular annual session.

Section 2. A Board of Managers shall be created in each city in which the next ensuing session of the Congress is to be held, preferably by leading organizations of citizens. The Board of Managers shall have power to raise and expend funds, to incur obligations of its own responsibility, to appoint subordinate boards and committees, all with the approval of the Executive Committee of the Congress. It shall report to the Executive Committee at least two days before the opening of the ensuing session, and at such other times as the Congress or the Executive Committee may direct.

Section 3. An Advisory Board, consisting of one person from each national organization having a conservation committee, shall be created to serve during that Congress and during the interval before the next succeeding Congress. The board shall report to and co-operate with the Executive Committee.

Section 4. The President shall appoint a Finance Committee of five, three from the members of the Executive Committee and two from the Advisory Board, whose duty it shall be to plan ways and means of increasing the revenue of the Congress, and to prepare a budget of expenditures. The Chairman shall be a member of the Executive Committee.

Section 5. The Executive Committee shall appoint, in consultation with the Vice-President from the State, a State Secretary whose duty shall be to work with the State organizations for the special interests of the Congress. Such Secretary shall report progress to the Executive Committee.

Section 6. A Committee on Credentials shall be appointed, consisting of five (5) members, by the President of the Congress not later than on the second day of each session of the Congress. It shall determine all questions raised by delegates as to representation, and shall report to the Congress from time to time as required by the President of the Congress.

Section 7. A Committee on Resolutions shall be created for each annual meeting of the Congress. A Chairman shall be appointed by the President. One member of the committee shall be selected by each State represented in the Congress. The committee shall report to the Congress not later than the morning of the last day of each annual meeting.

Section 8. Permanent committees, consisting of five members each, on each of the following five divisions of Conservation: Forests, waters, lands, minerals and vital resources, shall be appointed by the President of the Congress. The Committee on Vital Resources is to consist of six subordinate committees as follows: Food, homes, child life, education, civics, and general (including wild life, domesticated animals, and cultivated plants). These committees shall, during the intervals between the annual meetings of the Congress, inquire into these respective subjects and prepare reports to be submitted on the request of the Executive Committee, and render such other assistance to the Congress as the Executive Committee may direct.

Section 9. By direction of the Congress, standing and special committees may be appointed by the President.

Section 10. The President shall be a member, ex officio, of every committee of the Congress.

Article 6—Arrangements for Sessions.

Section 1. The program for the session of each annual meeting of the Congress, including a list of speakers, shall be arranged by the Executive Committee. The entire program, including allotments of time to speakers and hours for daily sessions and all other arrangements concerning the program, shall be made by the Executive Committee.

Section 2. Unless otherwise ordered, the rules adopted for the guidance of the preceding Congress shall continue in force.

Article 7—Membership.

Section 1. The personnel of the National Conservation Congress shall be as follows:

Officers and Delegates.

Officers of the National Conservation Congress.

Fifteen delegates appointed by the Governor of each State and Territory.

Five delegates appointed by the mayor of each city with a population of 25,000 or more.

Two delegates appointed by the mayor of each city with a population of less than 25,000.

Two delegates appointed by each board of county commissioners.

Five delegates appointed by each national organization concerned in the work of Conservation.

Five delegates appointed by each State or interstate organization concerned in the work of Conservation.

Three delegates appointed by each chamber of commerce, board of trade, commercial club, or other local organization concerned in the work of Conservation.

Two delegates appointed by each State, or other university, or college, and by each agricultural college, or experiment station.

Honorary Members.

The President of the United States.

The Vice-President of the United States.

The Speaker of the House of Representatives.

The Cabinet.

The United States Senate and House of Representatives.

The Supreme Court of the United States.

The representatives of foreign countries.

The Governors of the States and Territories.

The Lieutenant-Governors of the States and Territories.

The Speakers of State Houses of Representatives.

The State officers.

The mayors of cities.

The county commissioners.

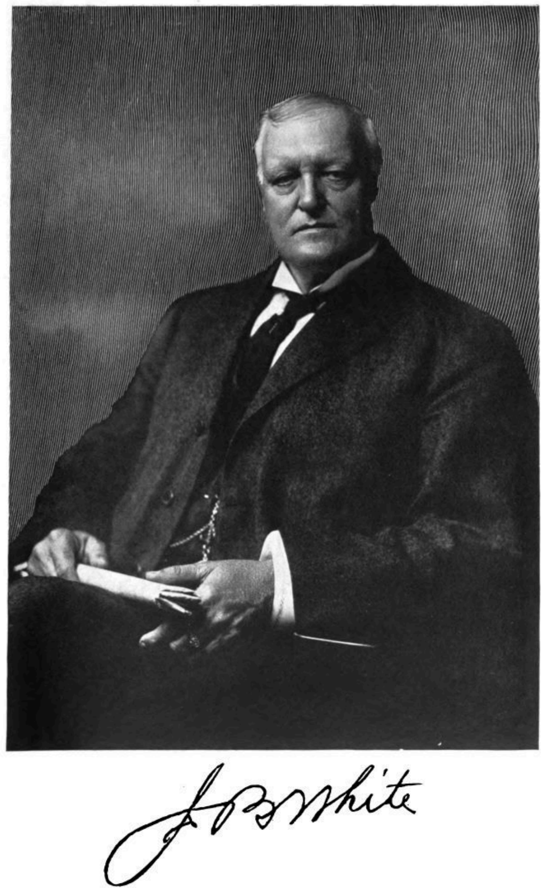
The presidents of State and other universities and colleges.

The officers and members of the National Conservation Association.

The officers and members of the National Conservation Commission.

The officers and members of the State Conservation Commissions and associations.

Section 2. Membership in the National Conservation Congress shall be as follows:



J. B. White (signature)
OF KANSAS CITY, MO.,
PRESIDENT, FOURTH NATIONAL CONSERVATION CONGRESS

Individual membership: One dollar a year, entitling the member to a copy of the Proceedings and an invitation to the next year's Congress, without further appointment from any organization.

Individual permanent, or life membership: Twenty-five dollars, entitling the member to a certificate of membership and a copy of the Proceedings and invitations to all succeeding annual Congresses.

Individual supporting membership: One hundred dollars, or more, entitling the member to a certificate of membership, a copy of the Proceedings, and an invitation to all succeeding Congresses.

Organization membership: Twenty-five dollars, entitling its delegates to the Proceedings, and an invitation to the organization to appoint delegates to the next Congress.

Organization supporting membership: One hundred dollars, or more, entitling the organization to appoint one delegate from each State, each of whom shall receive a copy of the Proceedings.

Article 8—Delegations and State Officers.

Section 1. The several delegates from each State in attendance at any Congress shall assemble at the earliest practicable time and organize by choosing a Chairman and a Secretary. These delegates, when approved by the Committee on Credentials, shall constitute the delegation from that State.

Article 9—Voting.

Section 1. Each member of the Congress shall be entitled to one vote on all actions taken *viva voce*.

Section 2. A division or call of States may be demanded on any action, by a State delegation. On division, each delegate shall be entitled to one vote; provided (1) that no State shall have more than twenty votes; and provided (2) that when a State is represented by less than ten delegates, said delegates may cast ten votes for each State.

Section 3. The term “State” as used herein is to be construed to mean either State, Territory, or insular possession.

Article 10—Amendments.

This Constitution may be amended by a two-thirds vote of the Congress during any regular session, provided notice of the proposed amendment has been given from the Chair not less than one day or more than two days preceding; or by unanimous vote without such notice.

RESOLUTIONS.

FOURTH NATIONAL CONSERVATION CONGRESS.

The Fourth National Conservation Congress, made up of delegates from all sections and from thirty-five States of the Union, met in the City of Indianapolis, do hereby make the following declarations:

Recognizing the natural resources of the country as the prime basis of property and opportunity, we reaffirm the declaration of the preceding Congress, that the rights of the people in these resources are natural, inherent, and inalienable; and we insist that these resources shall be developed, used and conserved in ways consistent both with the current and future welfare of our people.

We put chief emphasis on vital resources and the health of the people; and since health and brains are the first and most important factors of efficient life, we urge the adoption of all rational and scientific methods which will lead to their building-up.

To be well born is the primal requirement, and the first step to make sure that children shall be well born is to stop the multiplication of those bearing hereditary defects of body and mind. We believe that science is capable of solving the problem satisfactorily and that improvement is possible under existing conditions. We earnestly urge its consideration by the public.

We believe that every State should have wisely ordered health laws, with officers empowered to enforce them, and also that a National Department of Health should be created, comporting with the dignity and importance of the cause. This department should work effectively for the promotion of the physical and hence the moral and intellectual health of the people.

The accurate registration of births and deaths, which has been called the ‘Bookkeeping of Humanity,’ is a fundamental necessity for a study and knowledge of disease, and for all public health work. Therefore, we affirm our belief in the importance of vital statistics registration, and recommend that all States now without proper vital statistics adopt as early as possible the model bill for the registration of vital statistics indorsed by the United States Bureau of the Census, and by many prominent professional and scientific bodies.

We urge the strengthening of laws safeguarding the health and the lives of workers in industrial establishments; and we commend to the employers of labor all practicable safety devices and proved preventive measures against illness and injury and physical inefficiency; and we urge upon the other States the investigation of accidents by elevators and the enactment of laws similar to those on the statute books of Pennsylvania and Rhode Island.

We commend the activity of all individuals and organizations and governmental agencies to put an end to such work by children and women as impairs the health of the race. Childhood is our greatest resource, and its right to protection in growing to a normal maturity is inalienable. We deplore the ignorant use of medicines; and we call upon all humane and educational agencies to teach the waste and danger of any drug-habit.

We earnestly advocate the employment by communities and manufacturing concerns of such methods of sewage disposal as will render their waste products harmless to health and utilize them in the restoration of soil fertility; and we urge the enactment by States of laws prohibiting stream pollution and by the Federal Government of such legislation as will prevent the pollution of interstate and coastal waters.

Uniform State legislation regulating the refrigeration of perishable food stuffs is advisable, therefore this Congress recommends that its Food Committee be requested to study the questions involved in the production, collection, sanitary preparation, transportation, preservation and marketing of perishable foods and to report its findings to the succeeding Congress as a basis for uniform legislation.

In view of the enormous losses annually sustained by the agricultural interests of the United States on account of the ravages of injurious insects, which might be kept more under control by an increase of insect-eating birds, we urge the passage of Federal laws for the protection of all migratory birds; and the passage of State laws for the prohibition of spring shooting and of the sale of game.

We reaffirm the great importance of our fishery resources, which are threatened with serious diminution. We urge upon Congress and the States to provide more liberally for the propagation and preservation of food fishes.

LANDS.

We keenly recognize the need of the people of the country for more complete and accurate knowledge of their land and its conditions than is now available, in order to promote their economic, social and intellectual well-being and to conserve scattered individual energy;

We recognize that such data should be collected by a general series of State and National surveys arranged in the order in which they will be most accurate and effective and that many of these are already in progress;

This Congress earnestly points out the following kinds of data of which the people have need and the approximate order in which it should be collected, namely:

1. A thorough geographical survey of public boundaries and cultural features.

2. Of the form or topography of the earth's surface.

3. Of the geology, including the structure and economic deposits of the earth's crust.

4. Of the kinds and distribution of soils in their relation to agricultural operations.

5. Of the climate in its local variations and relation to crops and industry.

6. Of the surface and underground water supply of the country in its local and regional relation, including flood and storage problems.

7. Of various biological, crop and forestry conditions and relations.

8. And of many other surveys of a more specialized character and local application which may be adequately carried forward on the basis outlined above.

We urge the several States and the Federal Government to examine their existing agencies to determine whether they are completely and effectively fulfilling these functions.

Further, we reaffirm the action of the last Conservation Congress in approving the withdrawal of the public lands pending classification, and the separation of surface rights from mineral, forest and water rights, including water-power sites, and we recommend legislation for the classification and leasing for grazing purposes all unreserved lands suitable chiefly for this purpose, subject to the rights of homesteaders and settlers, on the acquisition thereof under the land laws of the United States; and we hold that arid and non-irrigable public grazing lands should be administered by the Government in the interest of small stockmen and home-seekers until they have passed into the possession of actual settlers.

FORESTS.

Believing that the necessity of preserving our forests and forest industries is so generally realized that it calls only for constructive support along specific lines—

We commend the work of the Federal Forest Service, and urge our constituent bodies and all citizens to insist upon more adequate appropriations for this work and to combat any attempt to break down the integrity of the national forest system by reductions in area, or transfer to State authority.

Since Federal co-operation under the Weeks law is stimulating better forest protection by the States, and since the appropriation for such co-operative work is nearly exhausted, we urge appropriation by Congress for its continuance.

We recommend that the Federal troops be made systematically available for controlling forest fires.

Deploring the lack of uniform State activity in forest work, we emphatically urge the crystallization of effort in the lagging States toward securing the creation of forest departments with definite and ample appropriations, in no case of less than ten thousand dollars per annum, to enable the organization of forest fire work, publicity propaganda, surveys of forest resources and general investigations upon which to base the earliest possible development of perfected and liberally financed forest policies.

We recommend in all States more liberal appropriation for forest fire prevention, especially for patrol to obviate expenditure for fighting neglected fires, and the expenditure of such effort in the closest possible co-operation with Federal and private protective agencies; and also urge such special legislation and appropriation as may be necessary to stamp out insect and fungus attacks which threaten to spread to other States. We cite for emulation the expenditure by Pennsylvania of $275,000 to combat the chestnut blight, and the large appropriation by Massachusetts to control insect depredation, and urge greater Congressional appropriation for similar work by the Bureau of Entomology.

Holding that conservative forest management and reforestation by private owners are very generally discouraged or prevented by our methods of forest taxation, we recommend State legislation to secure the most moderate taxation of forest land consistent with justice and the taxation of the forest crop upon such land only when the crop is harvested and returns revenue wherewith to pay the tax.

We appreciate the increasing support by lumbermen of forestry reforms and suggest particularly to forest owners the study and emulation of the many co-operative patrol associations which are doing extensive and efficient forest fire work and also securing closer relations between private, State and Federal forest agencies. Believing that lumbermen and the public have a common object in perpetuating the use of forests, we indorse every means of bringing them together in mutual aid and confidence to this end.

MINERALS.

We reaffirm the opinion of the last Conservation Congress that mineral deposits underlying public lands should be transferred to private ownership only by long-time leases with revaluation at stated periods, such leases to be in such amounts and subject to such regulations as to prevent monopoly and needless waste; and that in case of doubt as to availability of such mineral deposits, or while they are waiting exploration, surface rights to the land should be transferred by lease only under such conditions as to promote development and protect the public interest. Natural and manufactured fertilizing materials should be limited and regulated by law.

Since present conditions in the mining industry result in heavy and unnecessary loss of life and great waste of natural gas, coal and other mineral resources, we call to public attention the need of specific and uniform laws for the betterment of these conditions—laws as rigid and comprehensive as we enact for the protection of life and for the right use of property in any other fundamental industry.

WATER POWER.

We reaffirm the previously expressed belief of the Conservation Congress than all parts of every drainage basin are related and inter-dependent, and that each stream should be regarded and treated as a unit from its source to its mouth.

Recognizing the vast economic benefits to the people of water power derived largely from interstate and navigable rivers, we favor public control of their water power development; and we demand that the use of their water rights be permitted only for limited periods, with just compensation in the interests of the people.

COUNTRY LIFE.

We applaud the betterment of conditions affecting country life, such as good roads, and organizations for co-operative buying and selling; and we urge the study of rural credit systems whereby the farmer may more easily borrow capital at a reasonable rate of interest.

We applaud the work of making rural schools fit rural needs.

DR. W. J. McGEE.

We here place on record our sense of the deep loss by the country through the untimely death of Dr. W. J. McGee, a member of a Committee of this Congress, a scientific man of broad attainment, and of the widest human sympathy, whose helpfulness in these Congresses and many similar meetings will be sadly missed.

THE EXHIBIT.

We mention with appreciation the work of the Committee on Exhibits, Mrs. Philip N. Moore, Chairman, which made the instructive health exhibit under the management of Dr. Winthrop Talbot.

We record our grateful appreciation of the hospitality and helpfulness of the State Government of Indiana, and of the City Government of Indianapolis; and of the Local Board of Managers, Mr. Richard Lieber, Chairman; of the Reception Committee, Mr. Albert E. Metzger, Chairman; of the Commercial and Industrial organizations which, through the Commercial Club, made the Congress here possible; of the State Board of Agriculture, and of the Claypool Hotel, for their helpful courtesies and generous co-operation; and we thank the newspapers of Indianapolis for their unusually generous and accurate reports.

We wish to assure the retiring President, Captain White, of the heartiest appreciation of the Congress and of the country for his generous and efficient administration of the complicated business of the Congress; and Mr. Thomas R. Shipp, the Executive Secretary, for his zealous labor and good judgment and skilful management; and Mr. James C. Gipe, the Recording Secretary, for his energy and efficiency; and Colonel John I. Martin, the Sergeant-at-arms, must add one more vote of thanks to his ever-lengthening collection.

FOURTH NATIONAL CONSERVATION CONGRESS.

OPENING SESSION.

The Congress convened in the Murat Theater, Indianapolis, Indiana, on the morning of October 1, 1912, President J. B. White in the chair.

President White—The Fourth National Conservation Congress will now come to order, and the audience will please rise while the Rev. Dr. F. S. C. Wicks invokes Divine blessing.

Invocation.

Infinite and Eternal One, we would open our Congress with an acknowledgment of Thee as the Giver of every good and perfect gift. Thou hast placed us in a rich and fertile land, teeming with the things needful for Thy children, and Thou hast laid upon us the great responsibility of conserving these resources so that these blessings will extend to our children's children and to all generations forevermore. To Thee be all the praise and the glory. Amen.

ADDRESSES OF WELCOME.

President White—On behalf of the State of Indiana, your fellow citizen, the Honorable Charles Warren Fairbanks, will address the Congress in words of welcome. (Applause.)

Mr. Fairbanks—Mr. President, Ladies and Gentlemen: Indiana has frequently been honored by the presence of conventions of national importance. Our countrymen, engaged in various and vast pursuits and in the consideration of a large variety of questions, religious, social, fraternal, economic and political in their character, have assembled here from time to time to take counsel together with respect to the subjects engaging their particular attention, and to the advancement of our common welfare.

Our State has a hospitality for all who are engaged in promoting the moral, material and political well-being of our rapidly multiplying millions. I will not be misunderstood, I know, when I say that we have never more heartily welcomed to our midst any body of men than we now welcome the Fourth National Conservation Congress. (Applause.)

We recognize in this great assembly one of the most beneficent agencies for good which has taken on the form of systematic organization, national in its scope. It is not sectional, but is as comprehensive in its purpose as the ample limits of the Republic. It takes thought, not of the few, but embraces within its generous purpose one hundred millions of people of all conditions and without suggestion or partiality for white or black, native or alien born. How vast and how vital the field of its activities!

How full of promise such an assembly as this is! It is, Mr. Chairman, a complete answer to the pessimist. No thought of commercial gain has brought you here; a spirit of altruism, love of country and of mankind has been the impelling motive which has caused you, at your own expense, to leave the comforts of your homes and firesides and your daily vocations to come here and deliberate upon great themes of larger interest to the great community of which you are a part than to yourselves.

You hold no commission from the government, yet your service is of profound importance to it. You are not public servants in a narrow sense, but in a broad sense you freely serve the public in the best possible way.

The lesson of men voluntarily devoting themselves to the betterment of their fellows without the thought of sordid gain is a fine one and must impress itself in a very vivid and beneficial way upon the minds of others and tend to elevate the entire mass. What tends to impress us with our interdependence and to stimulate a feeling of homogeneity among us as this movement does is of incalculable benefit. It is a splendid thing for people to fellowship together in this manner, to take counsel of each other with reference to questions concerning the common good. It shows that we are interested more in what concerns the great body of the community than in what concerns ourselves.

General Harrison, gifted statesman and our fellow citizen, once very happily expressed the fact of the strength of confederated numbers in a good cause. He told of an engagement during the great Civil War when he was colonel of an Indiana regiment that was fighting in the midst of a woods and thicket. The enemy was pressing hard in front and fighting every inch of ground with a desperation that was unsurpassed. The Indiana regiment was feeling the shock of war in an extreme degree, and was almost on the point of discouragement. They felt they were fighting the battle alone. But in the course of time, as they emerged into a savannah, they saw a New York regiment, with its battle flags flying to the breeze; and over there another regiment from Kansas, and a shout of victory went up all along the line, for they found they were not a mere detachment, but part of a great army fighting for a common cause.

So it is a fine thing to feel that we are part of a great army fighting for a common cause—for home and country, rather than detached units fighting for ourselves. (Applause.)

Conservation is comparatively new in the vocabulary of our modern domestic economy, but it is a great word. It has come to be one of the greatest words of the human language from a practical standpoint. It is a continent-wide word in America. Conservation in some aspect of the subject touches every community, every city, every State and every individual. In other words, in a vital degree, it touches the welfare of one hundred millions of American citizens. Its importance is just beginning to be appreciated. Great today, but greater tomorrow in the progress of affairs. (Applause.)

A good Providence endowed us so abundantly with the prime necessities of our being that we have not fully realized the fact that there was either a possibility or danger of dissipating them. We were wont to boast of our inexhaustible resources. Nature has been prodigal, and we have been prodigal in the use and abuse of what she had so generously placed at our hands.

The forests—how vast and how majestic! We were obliged to fell them for the plow and the harvest, and for homes and cities. We came to look upon them as in our way—obstructions to our progress, as in a certain sense they were, but in a large way they were not. And we carried the work of demolition to the danger point before we realized our mistake. What nature had been centuries creating for us we frequently recklessly destroyed in a day.

The soil, the primary source of human life and strength, was rich beyond compare. In the laboratory of nature the chemical elements had been so nicely compounded that, to use a familiar simile, the farmer had only to tickle the land with a hoe and it laughed with the harvest. In time, Mother Earth began to resent neglect and abuse, and the crop yield diminished; but that mattered little to the unthinking, for there were still vast areas of virgin soil and the food supply was adequate to our needs. In the course of time, however, there were no unoccupied lands to be pre-empted, no fresh soil for the asking.

MILLIONS COME TO OUR SHORES.

Millions of men and women flocked to our shores every ten years from every land beyond the seas, seeking home and opportunity; millions every decade were added to our population at home by natural increase. Students of statistics came to realize that in the face of an increasing demand for food supply at home, regardless of the millions in the Old World dependent upon our granaries, soil exhaustion was a subject of very vital importance, a crime, if you please, not by the statute, but by moral law; and this may be said with respect to the reckless or ignorant dissipation of all our natural resources.

We are in a very real sense trustees of the fields and forests, mines and other sources of wealth, not to use and abuse at our will, but rather to use for our own reasonable necessities and then to transmit them unimpaired, so far as possible, and if possible increased in life-sustaining power, to our children. (Applause.)

By no other method can our civilization be perpetually maintained upon the highest level and the Republic kept in the forefront of the nations of the world. The man who owns and tills the soil, who owns and fells the forest, who owns and mines the coal, has no moral right to abuse his ownership; no one has a moral right to waste patrimony which must support not only the owner but the man who is not the owner, and whose continued comfort and existence must depend upon the wisdom with which the owner of the soil and forest and mine uses them.

The importance of Conservation derives emphasis from the rapidity with which our population grows. Our cities will not only multiply in number, but their inhabitants will increase, population will become congested everywhere, and the demand upon our natural resources will be greatly increased. We have added nearly ninety millions to our population in one hundred years. One hundred years ago we were small in numbers compared with the older countries. We have outstripped all but the older empires and republics of Continental Europe. Take Russia, with her 172,000,000; take India, with 325,000,000, and China, where they are building a republic upon the ruins of an empire, with her 400,000,000, and the United States stands fourth in magnitude of population among the nations of the world, having outstripped all but these, and with the present ratio of increase, in one hundred and fifty or two hundred years we will stand not the last of these great populous countries. And what does this signify? It signifies that the great subject of Conservation that you are taking hold of with such intelligent, patriotic interest, will be the overmastering question then as it is today. (Applause.)

Who can put a practical limitation upon a definition of Conservation? Conservation of our natural resources does not go far enough. The public health falls within the subject of Conservation in the fullest and best sense, and that is susceptible of many subdivisions. Conservation of the minds and morals, Conservation of our political institutions—all of these and many more subjects of but little less importance engage the attention of such men as are assembled here.

I understand, Mr. Chairman, that the human side of Conservation is to receive particular emphasis in this Congress. I am glad it is so. We have been so long concerned with the physical resources that we have failed to give proper credit to really a larger aspect of Conservation. As important as is the conservation of our natural resources, far more important is the question of conserving the health, conserving the intellect, conserving the morality of the one hundred millions of people we have. (Applause.) I have known men who were more solicitous regarding the health of a fine horse or dog than the health of the family. I have sometimes seen (but not in any of the States from which any of you come) ladies that had a more affectionate solicitude for a fine cat or a fine poodle than for the members of her household. (Laughter.) We are getting beyond that. We are coming to appreciate that that greatest assets in the United States today are men and women, and we must know how to conserve them.

There is manifest and gratifying awakening upon this subject throughout the country. We have not begun to appreciate the possibilities in this field. Men of science, the microscope, the laboratory and carefully gathered and well-digested statistics have opened up a new world to our vision. Physicians and surgeons have been exploring the mysteries of the physical man and familiarizing themselves with the perils of his environment and learning how to arrest the work of his destroyer.

They have learned how to locate his worst enemies by the use of the searching eye of the microscope, enemies who destroy more thousands than those enemies who come with fleets and armies and flaunting banners. It was not the Chagres river and Culebra cut which defeated the French Company in the construction of the Panama Canal, but the mosquito.

An American physician opened up the way to the completion of this work of world-wide moment by destroying the insect which had successfully defeated the French. The white plague, which takes such tremendous toll annually, is now under siege from every quarter, and science will in due time win a new victory in removing this scourge. Better sanitation in cities, villages, schoolhouses, workshops, homes, on farms and in cities, guarding our water supply against pollution, insuring pure food and pure drugs and their better preparation, are a few of the imperative requirements of the day. And when I speak of pure food and drugs, Dr. Wiley comes to my mind. (Applause.) He has to do with an aspect of practical Conservation that will entitle him and his associates to perpetual remembrance in the United States. (Applause.)

These are all practical questions, the importance of which cannot be over-emphasized. They concern the health and happiness of many millions of people and the destiny of the Republic itself.

INDIANA NOT INDIFFERENT.

Indiana has not been indifferent to this great movement. It has taken up the work of Conservation with full appreciation of its magnitude and its direct bearing upon the present and future of the State. Our interests are so diversified that our conservationists in all branches of the movement find full opportunity for the exercise of their activities.

We have an agricultural college which is doing much to advance agriculture, horticulture, stock raising and the like along advanced lines. Farmers are being interested in the necessity of cultivation of the soil and the importance of seed selection, drainage and the like. We have farmers' short courses instituted by the college which are proving of immense value. We are conserving the health of the livestock upon the farms. Sanitation has played an important part in this branch of work, as it has upon the human side.

We have a board of forestry supported by the State, and a Forestry Association organized by the people; also a commission to protect the food supply of our lakes and streams. These are only a few of the evidences of our progress in Conservation.

We are conserving with particular care the health of our school children with admirable results. We have learned, somewhat slowly perhaps, that sound bodies and sound minds should and can go together, and that to educate the mind and allow the body to become diseased is false economy on the part of the State and is nothing short of a crime, committed through either our ignorance or indifference.

We have sought to guard against and cure occupational diseases which impair and disqualify so many wage earners. We have more and more sought to throw around them such safeguards as well protect them against injury and death, and then to provide an adequate measure of compensation in case of accident as one of the legitimate burdens upon industry of the community which ultimately rests upon the public.

HAVE REDUCED ACCIDENTS.

During the last fifteen years we have made much advance in the conservation of the health of our people. By rigid factory inspection we have reduced accidents to our workmen from machinery and by improved sanitation we have protected their health. We have also rigidly inspected our mines with like results.

In fifteen years diphtheria has decreased sixty per cent., consumption has decreased in this same period six to eight per cent.; deaths from typhoid fever have fallen in the last two years from almost two thousand to 936 in 1911. Education, better living, improved sanitation, and an efficient State

Board of Health, with its excellent organization of health officers in every locality, the co-operation of the press in the education of the people and support of our health officers, have accomplished a great work in increasing in a very considerable degree the health, vigor and happiness of our people.

The net result of it all is told in the vital statistics of the State. In the last fifteen years the duration of life has been increased from 38.7 years to 44.6 years.

We are advocating the creation of a State Conservation Board with supervisory power over all subjects of Conservation now committed to separate and independent boards or commissions, so as to more effectively co-ordinate their efforts in a scientific manner, avoiding duplication and intensifying the work. It is suggested that a building be erected by the State for the proper accommodation of the entire Conservation service.

We regard this as a matter of great importance, and there is no doubt whatever that the State will liberally respond to the prevailing sentiment in favor of broadening the work of Conservation. It never pursues any parsimonious policy in supporting whatever concerns the education, health, moral safety and welfare of our people, so far as this may be appropriately accomplished under the law.

It is not inappropriate in this presence to observe that the Conservation of our political fabric must not be left out of consideration. This is a matter we must always hold uppermost in our minds, lest we allow harm to come to our priceless heritage.

Partisan utterance would, of course, contravene good taste, and I shall not offend against it; but I may suggest with propriety that we should hold fast to the fundamental principles of republican government, which have been our guaranty of liberty and human rights and of orderly progress for a century and a quarter.

The political wisdom of our forefathers has been abundantly vindicated in our experience. Older countries in continental Europe and in the Orient are turning toward us more and more and fashioning their political institutions after ours.

We need not be quick to surrender the present well-tried guaranties we have of justice and the rights of men for theories which neither upon good reason nor upon experience are commended to our best judgement.

The program which lies before you is full of the promise of entertainment and instruction. Men of wide experience, students of our economic and social needs, will lay before you the rich fruit gathered by them in the fields of their activities. Specialists in many branches of the great work of Conservation will make you their debtors. I shall not, of course, attempt to anticipate the subjects upon which they will enlighten you.

Custom, my friends, alone has led me to make the observations in which I have indulged in extending you welcome on behalf of the State of Indiana. It is quite unnecessary to occupy your time in discharge of this pleasant duty, which but for his enforced absence would have been performed by the distinguished Governor of the State.

You would understand me, I know, if I merely said "Welcome." You would know that it was no perfunctory utterance, but that it came from the bottom of the Hoosier heart. In a sense we do not look upon you as our guests; we prefer to regard you as members of our household. (Applause.)

President White—The thanks of the delegates, the thanks of the visitors, and the thanks of the people of the United States are due and will be given to the Hon. Charles W. Fairbanks for this most intelligent address, this statement of the principles that lie at the heart of every true conservationist. (Applause.) He has taken a forward step, he has led in the great movement in his own State, and he is now president of the Indiana Forestry Association.

I want to say that it is very fortunate for the people of the country that this address, and others that will follow, will be published and sent broadcast over this great land. We are going to teach the principles of conservation in every home.

It is now fitting that the next speaker should be also a conservationist—a conservationist of a different type, but no less a true conservationist, for at his hands, through his work, has come to the City of Indianapolis a reduction in fire loss from $600,000 to $300,000 annually. He is President of the Merchants' and Manufacturers' Insurance Bureau, and has practiced conservation in a most practical manner by reducing the fire loss and saving money to the people. We who have investigated that subject in Germany and other countries know how necessary it is that it should be brought home to us here in our cities and our homes. I now have the pleasure of introducing to you the Chairman of the Local Board of Managers, Mr. Richard Lieber. (Applause.)

Mr. Lieber—Mr. President, Ladies and Gentlemen: It is a very great pleasure and a distinguished honor to welcome you to our city upon this auspicious occasion. The City of Indianapolis deeply appreciates your coming and knows that through participation in your assemblages and deliberations it will materially profit in those matters which are of such vast and comprehensive benefit to its citizens. From here, through your able and learned speakers, potential knowledge will be disseminated throughout the length and breadth of our beloved country, which, in its application, will increase the happiness, contentment and usefulness of our people.

You have come here to consider most serious problems regarding the conservation of national wealth, more particularly that of vital resources, and above all, the conservation of human life.

For that reason, coupled with our welcome, is our expression of thanks for your coming, for "your worth is warrant of your welcome."

The thought of conservation is comparatively new. It marks a new era in the development of the country, and nowhere are its lessons more intensely needed than in a country like ours, vast in its expanse, relatively sparsely populated and apparently inexhaustible in its natural riches.

But are these riches inexhaustible? Can we go on in the manner of our fathers and forefathers, who frequently had to destroy in self defense?

Not since the days of the migration of nations, not even since the legendary days of the fall of Troy has the world witnessed anything like this stupendous conquest of a virgin continent. It is an intensified Iliad of modern days. No comparison with former ages can suffice. What are even the wondrous tales of Moses' messengers of the great land where "floweth milk and honey" compared with the gigantic proportions and abounding riches of this modern promised land?

That the pioneer, coming to this land was destructive before he could be constructive is a matter of historical truth. It could not have been otherwise.

He fought civilization’s battle, that civilization may enjoy peace and prosperity. But some of these destructive habits of the settler have taken root in our being and destruction has continued where construction was needed. What have the American people not wasted! Land and water, fish and game, coal, natural gas and too many other riches. Above all, how many useful and dear lives are drawn into the surging maelstrom of our national waste through indifference, carelessness and greed!

We find ourselves confronted here with the anamorphosis of civilization.

Human sacrifice belongs to a dark and unenlightened day, but the human sacrifice in mills and mines, in railroads and sweatshops in our time is a dark blot upon our civilization. (Applause.)

In this mad chase after things material at any cost, we must pause, for a nation will become unbalanced in its natural progress if its spiritual and intellectual advance be retarded.

Conservation wishes to bring about a more harmonious blending of these national needs. It teaches a wholesome regard for created values, it preaches the sanctity of a child’s life and the economic value of our boys’ and girls’ health, and aside from general consideration where is an application of conservation ideals and principles more needed than in our cities. We must learn that a good man’s or woman’s example in the community is more beneficial and of greater force than a mere ordinance. Virtue, righteousness and high principle spring from the seen of teaching that has fallen in mind and heart; they are inculcated but cannot be legislated. (Applause.)

CHARLES LATHROP PACK

PRESIDENT, FIFTH NATIONAL CONSERVATION CONGRESS

Would it not in this connection be braver for us fathers and mothers to speak openly to our boys and girls concerning the dangers that beset them in their course of life end thus turn the energies of their lives into the board avenues of light, strength and usefulness than to let them be drawn into the abysmal chasm of a veritable hell of human waste. Would it not be better to save, to lessen the inflow, than to clog the mouth of this human sewer by police orders after prudery, hypocrisy and cowardice have filled it? (Applause.) We are everlastingly treating symptoms instead of diseases, attacking effects instead of causes, and we persistently thereby aggravate the malady.

Let us have more light of thought, more air of true freedom and a deeper and more sympathetic understanding of our own needs and those of our fellow man that we may be enabled to show the folly of vice, the contentment of virtue; that we may alleviate pain and want, and that the warmth of human sympathy may send hope to the hopeless, courage to the faltering and faith to the despondent.

With these fervent wishes the City of Indianapolis welcomes the Fourth National Conservation Congress. (Applause.)

President White—These words of welcome, coming from a different point of view, are felt deeply by us all. We feel the spur of duty still greater.

It is very fitting that another side of conservation should be heard from. The business men, the local business organizations of a city have done a good work for conservation. Human efficiency is one of the greatest forces that move the world, and systematic organization is one of the greatest powers towards efficient conservation of life and of all material progress. A business man knows that his success depends upon perfect organization, and that perfect organization is just as necessary to the conservation of every natural resource.

I have the pleasure of introducing to you Mr. Winfield Miller, of Indianapolis, who speaks on behalf of the local business organizations. (Applause.)

Mr. Miller—Mr. President, Ladies and Gentlemen: When I was honored by the commercial organizations of Indianapolis with the invitation to extend for them a few words of greeting and welcome to this National Conservation Congress, I looked into the biggest book, the Dictionary, for a definition of the word "Conservation." I found the word concisely defined to mean "the art of preserving from decay, loss or injury." While the definition is not extended, it is comprehensive and can be readily amplified to cover every phase of the question.

I then turned to the greatest book, the Bible, and read that early edict which still holds good, "In the sweat of thy face shalt thou eat bread." Over this ancient decree and its cause, there have been volumes of theological commiseration, but in the light of subsequent history, it is now generally agreed that man has been a greater force in the garden of the world that if he had remained in the Garden of Eden.

The thought occurs, however, that resting under the edict of life-long toil man would, from an early period, have practiced conservation in all things.

But he soon discovered that "the earth and the fulness thereof" were his, and, as ever, has been injuriously careless of results.

Again, he was not left without hope. The same great authority, in language and grandeur of thought unsurpassed, gives a promise of perpetual inspiration, in this, that "While the earth remaineth, seed time and harvest, and cold and heat, and summer and winter, and day and night shall not cease." This promise, according to accepted chronology, has the confirmation of forty centuries of time and gives man the assurance of a continued field in which to do his work. The earth, the air, the waters are his environment; they are immutable, unchangeable. The animal, vegetable and mineral kingdoms furnish him food, clothing, shelter, life. Their best use should be his first and highest consideration.

Nature has been prodigal in her gifts to man. Her kingdoms have been his to rightfully exploit. But too long and too often has selfish and neglectful exploitation been his purpose and practice.

There is abundance for all if nature's forces are properly conserved and her products fairly distributed. But some men, in their greed and haste, have grabbed a thousand-fold more than their necessities or happiness required. They eat their bread in the sweat of the other man's face. On the other hand, the many have been ignorantly neglectful of the opportunities of their environments—so that life presses hard, too hard. Avarice, ignorance, waste, have linked arms to the detriment of civilization.

We must strive for the necessities of food, clothing and shelter. These sustain animal life, which is worth while; but animal life, endowed with the highest moral and mental strength, is the goal to be reached, for the summit of man's ambitions should shine with human comfort and happiness. Conservation is the road to that summit and this National Congress has convened to further blaze the path and light the road. (Applause.)

Inventions of the last century, mostly within the half century, have injected into the field of travel and communication means that excite profound admiration; chemical analysis of the air and soil have shown that the food supply of the world, if nature's forces are properly conserved, is without limit; while the mighty strides made in the better understanding of the physical needs of man himself insure the race at large improved health and longer life.

May I briefly indulge in a few common illustrations? The telegraph, the telephone, the automobile, steam and electric power save time and shorten distance. In that part of commerce relating to traffic we have caught the spirit of conservation. The railroad builder no longer takes the route of the least resistance in construction, but applies the geometric proposition that the straightest line is the shortest distance between two given points, works to that end, meets the difficulties of engineering, reduces gradients, and practically builds his road along the line of least resistance, conserves time, saves energy, increases efficiency and lessens rates.

The school books tell us of the “Seven Wonders” of the ancient world—the Pyramids, the Hanging Gardens of Babylon, and so on; all to gratify the vanity of kings and queens; not one for the advancement of civilization.

In a little more than two years the dream of four centuries will be realized—the Panama Canal will be completed. The distance from the Occident to the Orient will be shortened seven thousand miles—the truly modern wonder in advancing civilization and practical conservation. (Applause.)

While the physical aspects of this mighty work, as they relate to the traffic and commerce of the world, stand out in bold relief, but little less, if any, in achievement, is the practical demonstration that scientific sanitary methods can clean the plague spots of the world and make them healthy and habitable for man.

Who can compute the saving of time and energy this mighty work will bestow on the generations to come. Long after the passions of this generation have ceased, history will record the names of the strong men who have brought to full consummation this great waterway, as true benefactors of mankind.

At this time, our public press is ecstatic over the great harvests of 1912 that promise such abounding prosperity. Some writers are so extravagant as to say that the bountiful yields from our soil make an epoch in history. To speak of one crop only, the corn, or Indian maize crop, spreads over 108,000,000 acres, and is estimated to be 3,000,000,000 bushels. How enormous! At fifty cents a bushel, its money value would be $1,500,000,000, or $16.00 to every man, woman and child in the United States. Measured in bread, there would be enough to give to each of our 93,000,000 of people a five-cent loaf for more than 320 days, or nearly one full year. As gratifying as this is, the average yield is only twenty-seven bushels per acre; while it is shown that, by proper selection of seed, cultivation and fertilization of the soil, easily twice the yield could be produced, which would double the benefits enumerated.

How often do we pass a barren field, the soil too impoverished to grow wire grass, nettles, or thistles. The every-day farmer will tell you that a crop or two of clover will restore the necessary plant life to the soil of that field, and again make it blossom like the rose. He knows from practical observation and experience the cure, if he cannot scientifically trace the cause of the transformation.

Truly truth is stranger than fiction. Back of the restoration of a thousand barren fields restored to productiveness in the simply way named, lies one of the world’s greatest romances in patient scientific investigation that will continue to bring untold benefits to mankind. You know the story of Professor Nobbe, of Forest Academy, Germany. He also knew that clover and other legumens of the plant family would restore fertility to the soil. But why? After long and exhaustive study, labor and experiment, he found that the clover family were nature’s chemist of the soil; that by an invisible, intangible cord of attraction they drew from the inexhaustible reservoir of the air nitrogen so necessary to plant and animal life.

We are told that “nitrogen is what makes the muscles and brain of man; that it is the essential element of all elements in the growth of animals and plants, and, significantly enough, it is also the chief constituent of the gunpowder and other explosives with which the wars of the world are waged. The single discharge of a 13-inch gun liberates enough nitrogen to produce scores of bushels of wheat.”

Some day, through this agency, man may turn his attention entirely from war to the production of food, and in that hour true conservation of life will have reached its triumph.

We are further told that four-fifths of the air we breath is nitrogen, and that four-fifths of the atmosphere around us is nitrogen, so that if mankind dies of nitrogen starvation, it will die with food everywhere in and about it.

So that, while the human race may be but from three to six months behind abject starvation, the fact begins to appear that through science “mankind has just begun to sound the world’s capacity for food production and that it is practically limitless.”

The proper conservation of the soil by the application of the research of scientific discovery means increased yields of all plant crops, with but little greater expenditure of energy. This would enable the producer of food and clothing to sell more pounds, bushels and yards at less cost, and still reap as great reward for his labor as at present. This would forestall the Malthusian doctrine that population increases faster than the means of subsistence and, still better, would help to solve the high cost of living that presses so sorely upon the millions throughout the world today. Man is a productive machine; so the more machines of the highest type the world possesses the better for the world.

This conservation movement that is so strongly taking hold of the minds of thinking men and women, is so big, so broad and so comprehensive that it covers every phase of human thought and activity in what is best and highest for the individual as well as organized society. It is education in the broadest sense.

The Golden Rule is not only a statement but a living principle. To teach that a just distribution of nature’s gifts to each individual who is willing to earn and conserve his share is a recognition of that principle.

The City of Indianapolis esteems it a high honor to have this Congress with us. To all of its members, and especially to the distinguished men from other lands who have come to give us their best thought upon the various questions affecting this great movement, we extend our most cordial welcome and greeting, and our deep appreciation of your presence.

Our commercial organizations also cordially join in holding the door of welcome and hospitality open to you, and bespeak for your deliberations their kindest sympathy and deepest interest.

President White—This is a proud day for the officers of this Congress, for its delegates from the different States, and for the friends of Conservation everywhere, to be welcomed so hospitably, not only for ourselves who are strangers within your gates, but generously because of your sympathy in the great cause for which we stand. The citizens of your great city are noted for their public spirit, for their broad culture, and as being always found in the van with the army of those of progressive ideas. It is very fitting that the State of Indiana should have this Congress within its borders because of the immense interest shown and all the valuable help given by its citizens in the conservation of all natural resources, especially of human life and soil fertility.

To become the best, to do the best for all in a community, we must each develop the best within us, and must find our greatest reward in something far beyond the mere accumulation of dollars. Our community of interests recognizes a reciprocity of duty each to and for the other. Our title to the regard of our fellow men must come from our devotion to them and our love of humanity and its highest ideals, and not from selfish zeal in accumulating monetary wealth, which only represents the toil, the waste, and the necessities of human lives. This has been and is the age of commercialism. The measure of success has been gauged by the amount of money accumulated. In the language of Goldsmith, our country was in danger of descending to a condition "where wealth accumulates and men decay." But I believe a turning point has been reached; and that we are entering upon a new era, a more glorious conception of higher duties for mankind; so that it shall not be asked: "What hath he taken from others in the competitive struggle for existence," but rather: "What hath he given to others of himself for their advancement and development?" He who lives only for himself and does not plant for those that are to come after him, lives in vain. I believe the time is near at hand when a man shall be regarded with pity and as very poor indeed, who has nothing but money selfishly gained for selfish use.

The Conservation Congress of the United States has a great field to occupy. Its labors are for the betterment of its citizens in every way. Its work is to seek for the best methods to do the greatest good to all for this and for future generations. And in this there is no partisan politics; but it is such good national politics, that each party will strive only in seeking for the best methods for the common good.

Human life, with its possible attainments, is far beyond valuation in money. We should reverse the tables; and instead of human life being estimated in dollars, the dollar should be valued only for what it can do for greater humanity. Dr. Holmes, Director of the United States Bureau of Mines, in illustrating waste of material resources, says that in producing one half billion tons of coal, we waste or leave underground one quarter of a billion tons. And then only eleven per cent of the energy in coal is utilized; nearly ninety per cent is lost through inefficiency of boilers, engines and dynamos. How great a per cent. are we wasting of human life and human efficiency? We will have abundance of all the necessaries of life, and even of life itself, if we wisely save, wisely develop and protect, and wisely use. Human life is our greatest asset, and its waste is a permanent loss. The wealth of the nation is in its men, thrifty, honest, capable and patriotic men —in their moral and physical health, the foundation of their highest efficiency. The milestones of a nation's progress are recorded in the history of every generation. In India, the average duration of life is twenty-five years; in Sweden more than fifty years; in the State of Massachusetts (the State where most careful records have been taken) it is over forty-five years. Wherever sanitary and highest medical science has been applied, it has been found possible to increase the span of life. In Europe it is said to have doubled in three and a half centuries. The report from Massachusetts shows an increase of fourteen years in the past century. So this humanitarian cause is surely a most economic, worthy and profitable one. In figuring from the standpoint of capital, Prof. Mayo Smith estimates that men and women between fifteen and forty-five years of age are worth an average of one thousand dollars. But figuring from a human standpoint, they are worth all that there is, money being only one of the tools to work with in effecting exchange of commodities, and the products of brawn and brain. We want to increase the ratio of the value of man to the soil, of man to all and any of his products, of man to money, and to put man first all the time. (Applause.)

We will increase the fertility and productiveness of the soil and we will enlarge the scope and increase the efficiency of the man. We waste in production as well as in consumption. In agriculture we will say that we will make the soil produce so many bushels per acre per man. The man will be first in his wise application of labor and methods and of means to an end. The "limits of subsistence" under what political economists used to call their "law of diminishing returns" has no fear for the conservationist. The developing of human intelligence is enlarging the production of the soil. Irrigation, where possible, and where impossible the science of what is called dry farming brings increasing results. Old farms in Europe produce more than they did 300 years ago, and this will prove true with us, and there will be no starvation for the human race because of increasing population.

And so will it be with all other industries, occupations and professions. He will be greatest who accomplishes most for man. For the brotherhood of man was the world made and the fullness thereof. Such freedom as may benefit any individual and does not in any way work to the injury of others is natural justice to all. Competition shall be robbed of the “red tooth and the bloody claw,” and co-operation and development for the good of all shall be the supreme object of all our efforts.

We will protect our watersheds by growing forests, and learn to control our floods, prevent soil erosion, and store the water, and convert its power into electricity, and from electricity produce light, heat and power in undiminishing quantity forever. In nearly every State there is daily flowing to waste power enough, if arrested and utilized on its way to the sea, to turn every wheel of industry and to move the traffic of commerce, and furnish light and heat for every city. It is said that the wheel does not turn with water that is past, but other wheels farther down the stream do, and the power is used again and again and finally pumped back by the sun to the mountains and plains to forever repeat the process of service to mankind. New discoveries are being made, and the use for by-products is being multiplied so that they are often found to be of greater use than the product from which they are derived. We must protect our forests by preventing forest fires. Government and State appropriations must be made sufficient for this purpose. In the report of the Conservation Commission to the President, it is stated that fifty million acres are burned over annually, and since 1870 there has been lost each year an average of fifty lives and fifty million dollars’ worth of timber. The lumbermen’s interests are to prevent fires and to stop waste; and they are anxious to co-operate with the State and with associations for this purpose, and are already doing so in many places. The true, saving features of forestry are becoming better understood, and better applied; and we will save our forests, and will grow trees, wherever necessary and profitable, the same as any other crop; and there will be no timber famine in the near or distant future. Our foresters are studying the experience of France, Austria, Italy, and Switzerland, coupled with our own experience, and we are making successful progress. In Kansas five years ago, according to President Waters of the State Agricultural

College, there was only one school that taught agriculture. Now nearly five hundred high schools and more than six thousand rural schools are teaching the principles of agriculture, forestry and domestic science.

The Commissioner of Internal Revenue reports that for the official year 1912 ending June 30, the people of the United States drank more whisky and rum and smoked more cigarettes than ever before in its history. The smoking of over 11,221,000,000 cigarettes exceeded the record of 1911 by nearly two billions. Does this make for or against human efficiency? In this huge traffic is it the man or the dollar that stands first in importance? Popular education will be the source of protection, that all may have a fair chance for a useful life. There are other great factors of vice and crime leading to national decay. Also there is the enormous waste of human life by our railroads, mills, mines and factories amounting to tens of thousands annually, and those permanently injured and made a burden to themselves and to society to tens of thousands more. In no civilized country in the world is this loss anywhere near as great in proportion to work accomplished as in the United States. The greatest part of this immense loss can be prevented. (Applause.)

Here is thought and work for those in the department of vital statistics and those in charge of our health departments, who are laboring for the conservation of human life. Surely there is a great moral and economic need for this national organization. May this Congress, which now begins the work of its program, prove to be another step in advance of its predecessors in the labor of love and of progressive activities. The work in this vineyard is for both men and women; for him with one talent as well as for him with ten talents. Conservation should be taught in our schools and preached in our churches. It is a call of and for all the people.

In the language of the official call of this Congress, the objects of this Congress are “to provide for discussion of the resources of the United States as the foundation for the prosperity of the people; to furnish definite information concerning the resources and their development, use and preservation, and to afford an agency through which the people of the country may frame policies and principles affecting the conservation and utilization of their resources and to be put into effect by their representatives in State and Federal governments.” (Applause.)

President White—The preliminary organization has now been completed. It was expected that the President of the United States would be present to honor this occasion, at the opening of this Congress, or it was at least hoped that it would be possible for him to do so, but before he knew that he would send a personal representative he wrote a letter of greeting to the Congress, which I will now read:

MESSAGE FROM THE PRESIDENT.

Beverly, Mass., September 7, 1912.

Hon. J. B. White, President National Conservation Congress, Bemus Point, N. Y.:

My Dear Mr. White: Inasmuch as I have had to deny myself the pleasure of being present at the opening of the National Conservation Congress on October 1st, I want to take this means of conveying to the officers and delegates my very cordial greetings and good wishes for a most enthusiastic and instructive session.

You who know of my very real interest in the conservation of our national resources need no assurance of my hope that your meeting in every way may be a success, and I only want to say that that interest has not diminished in the slightest.

May your deliberations be productive of great good in promoting the cause of Conservation and in enlisting public interest in the solution of the problems which must be met in giving the people of the present day the benefit of the nation’s resources, while at the same time insuring to posterity its full heritage.

Sincerely yours,

WILLIAM H. TAFT.

It was afterwards found possible for the President to be represented personally, and he has sent the Honorable Henry L. Stimson, Secretary of War, to represent him here at this Congress. I now take pleasure in introducing to you Secretary Stimson. (Applause.)

ADDRESS BY THE SECRETARY OF WAR.

(Representing the President of the United States.)

Mr. President, Ladies and Gentlemen of the National Conservation Congress: I unite very sincerely in the congratulations which the former speakers have tendered to you on your assembling here in such an important and such a noble cause.

I am very happy to be here as the representative of President Taft—happy, both because of the interest which I know he feels in the great movement for Conservation, and also because of my own personal association with and enthusiasm for that movement. Four days ago, when I was busily engaged in inspecting one of the army posts of northern Wyoming, in the far away Rocky Mountains, I received an urgent telegraphic request from the President, that I should come here today and attend your meeting on his behalf. And the 1,600 miles or more which separate Fort Yellowstone from Indianapolis, may serve at the same time as a measure of the President's interest in your meeting, and a measure of the depth of my own unpreparedness to speak to you today. I know, therefore, that you will understand and pardon me if I talk to you rather informally, merely as one friend to others interested in a great common cause, and confine the brief remarks which I shall make to one of the phases of Conservation with which I have become familiar through the work of the War Department.

Parenthetically, I might say that inasmuch as the Department of War is not usually considered as a particularly appropriate agency for the conservation of life, I will have to hark back to the material side of Conservation in some of its aspects which have been presented at your former meetings.

Of course, the main work which the Federal Government performs in regard to Conservation is done through the Department of the Interior. Incidentally, the truest of all indications of the interest with which the President regards the conservation of the natural resources of this country lies in the character and attainments of the man whom he has placed at the head of that Department in order to conserve them—Walter L. Fisher. (Applause.) You will all of you remember how his thorough investigation and clear-cut decision of the famous Cunningham claims, has settled and disposed of, in the interest of the people, one of the most bitter controversies of our cause. You are also undoubtedly familiar with the careful investigation which he made last year into the very complicated and serious problems of conservation which confront our Nation in Alaska; and with the luminous address with which he reported his conclusions and pointed out a solution of these questions. Though the work of his Department in investigating and conserving in the public interest, the water power sites which remain on our public lands, and our remaining beds of coal and phosphate, has not attracted so much attention as his work in these former more controverted matters; yet there is, I think, a very general and well founded feeling throughout the country that in all these matters the interests of the people of the United States are being thoroughly protected by the Secretary of the Interior in accordance with the most intelligent and thorough views of Conservation. (Applause.)

I allude to his method of thorough investigation because I think it is characteristic of the attitude of the President himself toward this whole subject. In order that progress should be real, it must be based upon carefully ascertained facts. In dealing with the problems of Conservation, we are dealing with problems which are new to our Nation. As the honorable speaker who first addressed you pointed out, we have only recently passed from an era of exploitation into an era of Conservation. We are surrounded by thousands of our fellow countrymen who have been brought up to honestly believe that the only method of developing the country is to turn its resources as rapidly as possible over into private hands. In putting into effect, therefore, the new policy to which the Nation has now come, there must be care taken lest false steps, or the injustice which may come from hasty action, may not produce a reaction which will delay or imperil the reform. As a former District Attorney, representing the people in the enforcement of the law, I have long had it impressed upon me how essential it was that no hasty, or harsh, or intolerant action, taken in the heat and controversy of a jury trial, should thereafter imperil the entire work and by producing injustice, and a subsequent reversal in the higher courts, bring some great reform into disrepute or temporary delay. Patience, thoroughness, and courage, mark the only pathway towards permanent progress and reform. (Applause.)

Now, the subject which I am going to call to your attention briefly this morning is one of those few matters where my own Department, instead of the Department of the Interior, touches upon the problems of national Conservation. It is also a subject the history of which, I think, exemplifies clearly the importance of the methods to which I have just alluded. I wish to point out to you the attitude of the administration as to the Federal regulation of water power in our navigable rivers.

It is needless to remind such a body as yours of the importance of that sphere of Conservation. All our other present sources of power—such as coal, wood, oil, and the like—are limited, and will be eventually exhausted. Water power alone is permanent. And just as we are coming to learn more and more the value of that permanence, we are simultaneously, through the development of electricity, learning to transmit its energy to greater and greater distances. No other subject occupied more keenly the attention of the past session of Congress, or was more vigorously debated upon the floors of that body.

For many years our national policy, or rather lack of national policy, towards our waterways and our water power, has presented a singular inconsistency. On the other hand, we have been spending hundreds of millions of the taxpayers' money on the improvement of the navigation of our great inland waterways. On the other, we have been granting away permits for the construction of dams on these same rivers and waterways, which will create water power of incalculable and increasing value; and we have been doing this without exacting *for* the taxpayers any return or compensation whatever.

I believe it was not until the administration of President Roosevelt that any effort was made to obtain for the public any compensation for the water power which was thus granted away. Mr. Roosevelt demanded in his veto of the James River bill, and in several other messages, that no permits for dams in navigable rivers should be granted without a reservation of proper compensation to the public for the grant thus made. His action was courageous and right. But there were not as yet in the hands of the public sufficient carefully ascertained facts upon which the constitutional power of the Federal Government to take such action could be confidently based. And there was therefore great ground for misapprehension in the public mind of any action attempting to take such a position. A bitter controversy at once arose with those advocates of States' rights, who contended that the Federal Government had no rights whatever in connection with water power, that under the Constitution its powers were limited solely to navigation, and that water power was an entirely separate and distinct sphere, falling wholly within the jurisdiction of the several States. Such advocates contended that for the Federal Government to exact compensation for a water power right, simply because it was in a position to withhold the permit altogether if it wanted to, was little better than legalized blackmail; and the progress of the reform was stubbornly and for a long time successfully contested.

Even as late as 1906, the General Dam Act, passed by Congress and approved by the President, conveys to the Executive no clear right to exact compensation for these grants. It has remained for Mr. Taft's administration, following the method of patient investigation and research which I have above mentioned, to collect the facts necessary to solve the problem, and to show from these facts that the jurisdiction of the Federal Government over navigation must necessarily include jurisdiction over water power as an incident of the navigation.

Most of the rivers of this country are long and comparatively shallow. In order to make them commercially navigable, there has become prevalent among engineers a method of improvement known as the "slack water" method or the method of "canalization." The method consists in building throughout the length of these rivers, a series of dams and locks, by which the river is converted into a succession of deep pools, adequate for commerce of a far more important character than could use the river in its unimproved condition. In fact, many rivers which are not capable in their natural state of being used at all commercially, can by this method be made useful and available for important commerce.

Now, most of the dams thus constructed in a "slack water" improvement, particularly in the rapid portions of the streams, will create water power of commercial value. It is also manifest that if the commercial value of the water power thus created can be realized by the Government and turned into further river improvement, the improvement of navigation on our rivers can be greatly expedited, and the expense to the general taxpayer very much lessened. And, on the contrary, unless this is done, the complete improvement of the river will be just so much delayed and postponed. The water power developed is thus shown to be intimately connected with navigation. It is a by-product of the improvement which can be turned into further improvement. And from the standpoint of constitutional law, it makes no difference whether the dam in question is to be erected by the Federal Government or by a private corporation. If it is a dam which is to assist the navigation of the river as well as to create water power, the power of the Government will be complete. What the Federal Government can constitutionally do itself it can do through an agent.

The corps of army engineers to whom are referred all proposed bills in Congress granting permits for dams for water power have been accordingly, under Mr. Taft's administration, directed to investigate and answer specifically four questions in every report. They are directed to ascertain in regard to every such bill:

First, is the river on which the dam is to be created a navigable stream subject to being improved, either now or in the future of the country, at the expense of the general taxpayer?

In the second place, they are asked whether the dam which is sought to be constructed will form an essential part of any such improvement.

Thirdly, whether the dam will create water power of commercial value.

Fourthly, whether the value of that water power will tend to increase with the growth and development of the Nation.

You can see for yourself the pertinence of such questions. Once answered in the affirmative, there is a case presented upon which the jurisdiction of the Government's power can rest.

Trial has now shown that the answers to these questions are nearly always in the affirmative. And as a result of the information thus obtained we are in a position now, unlike our position six years ago, where we can take a step forward, and hold permanently the ground thus gained.

There is now laid before Congress a sure foundation upon which we can rest our national right to exact the fair value of these grants. Investigation has regularly brought out the fact that each one of these dams is essentially connected with navigation, and that a failure to preserve the right to regulate them and to exact compensation for the power created is throwing away a valuable national asset.

The issue has been sharply drawn by President Taft, and his position clearly stated in his message, submitted on the 24th of last August, vetoing the bill which proposed to grant authority to build a dam in the Coosa River. The Coosa River is in Alabama. The bill in question sought to authorize the Alabama Power Company to build a dam suitable for the development of navigation in that river, and at the same time to create water power for the exclusive benefit of the corporation. It contained no provision permitting the Federal Government to exact any compensation for the rights of water power thus granted. The bill was strongly urged by powerful leaders of both houses of Congress. It was also vigorously opposed by the leaders of the conservation movement of Congress. But it ultimately passed. The President vetoed it in a message which asserts in unqualified language the duty of the Federal Government to reserve to itself the right to exact proper compensation. (Applause.) The President says on this point:

> "I think this is a fatal defect in the bill, and that it is just as improvident to grant this permit *without* such a reservation as it would be to throw away any other asset of the Government. To make such a reservation is not depriving the States of anything that belongs to them. On the contrary, in the report of the Secretary of War it is recommended that all compensation for similar privileges should be applied strictly to the improvement of navigation in the respective streams—a strictly Federal function. The Federal Government by availing itself of this right may in time greatly reduce the swollen expenditures for river improvements which now fall wholly upon the general taxpayer. I deem it highly important that the nation should adopt a consistent and harmonious policy of treatment of these water power projects which will preserve for this purpose their value to the Government whose right it is to grant the permit."

There are few subjects of equal importance with the proper improvement of our great river systems. We are behind many of the nations of Europe in our appreciation of this importance. The development of our rivers is not only vitally important for the commerce that they will thus carry, but even more for the regulative effect which they should and can have upon the freight rates of the railroads with which they compete. If Mr. Taft's position is sustained, it means that all the potential value of these streams can be turned toward the improvement of their navigation. As he says, it offers one of the possible solutions for our swollen river and harbor appropriations. On the other hand, it also means that the hand of the nation is to be kept on this great national asset of our water power; and that this great subject of water power regulation will be handled comprehensively, consistently, and with due regard for the wants of the Nation as a whole.

If, however, the view of the opponents of the President prevails, it means that this necessary improvement of our rivers will be greatly postponed, and that all the expense of such improvement will have to be borne by the general taxpayers of the Nation. And it further means that the closely related subject of our water powers on these navigable rivers, instead of being treated nationally and broadly, will be subject to the piecemeal policies of forty-eight different States. Seldom is there presented an opportunity to apply the principles of conservation simultaneously to two such important subjects as river transportation and water power regulation. (Applause.)

President WHITE—I am sure we all appreciate the address that has just been delivered by our distinguished representative of the President. It has left upon our minds the significance of the importance of protecting those natural resources that are permanent and which should not be given away to private individuals, or corporations.

We will now hear some announcements.

Mr. THOMAS R. SHIPP (Executive Secretary)—The section of which Dr. Wiley is chairman, the section on "Food", will meet this afternoon at four instead of tomorrow morning. The meeting will be held in the Palm Room, Claypool Hotel, and will be open to the public. The fact that Dr. Wiley is at the head of this section and will preside and speak will make it of great interest to delegates. In addition to Dr. Wiley, there are other gentlemen of national reputation on this question who will speak. An invitation is extended to all delegates to attend this meeting this afternoon at 4:00 o'clock.

President WHITE—If there are no further announcements we will adjourn until this afternoon at 2:00 o'clock.

SECOND SESSION.

The Congress was called to order by President White at 2:00 o'clock p. m.

President WHITE—Gentlemen, we are a little late in getting together this afternoon, owing to the late adjournment of the morning session.

We have a program for four days, a most entertaining one. Those that do not get here will miss something, while those of us who are here are going to gain something.

The audience will please rise while the Rev. Dr. A. B. Storms invokes the Divine blessing.

INVOCATION—*Our Heavenly Father, we wait for a moment, asking for the blessings of Thy grace upon us. We need Divine guidance in all our counsels; may we be guided by Heaven. We return Thee thanks for Thy great kindness, for the bountiful harvest, for the resources with which Thou hast stored the earth. We thank Thee for the revelation of Thy love, for the redemption that speaks of the worth of Thy children. We thank Thee for all the impulses Thou hast set in motion for bringing good out of evil, for bringing men to their best. We pray for the guidance of the divine spirit that in all these councils which have for their purpose the good of our kind, we may have such guidance and be sustained by such grace that permanent good shall come.*

May Thy blessing rest upon this Congress, upon all it represents, upon our people and Nation. May this be a people whose God is the Lord, we ask in the Redeemer's name. Amen.

President WHITE—The first thing on this afternoon's program is a report from Dr. George E. Condra on "What the States are Doing." He is President of the National Association of Conservation Commissioners. We are very much interested to know how far the spirit of Conservation is being taken up and applied in the different States. We will now hear Dr. Condra.

Dr. Condra—Mr. President and Delegates: This report, prepared at the request of the Executive Committee of the Congress, is based on data received from several Governors, and the conservation organizations of various States. It can not be given in detail, for that would require too much of your time. Neither do we deem it advisable to treat the subject State by State, for it would call for needless repetition. Consequently the data are reviewed subject by subject corresponding to the leading departments of Conservation, and the States are mentioned only in connection with the progress they have attained in each department. It is assumed that: 1. You are in full sympathy with State Conservation and its co-operation with Federal agencies. 2. You do not expect to hear overdrawn statements. 3. You wish a review of such conservation facts as are really worth while in development. 4. You know how natural resources control industrial development. 5. You agree that the leading resources in the United States are mineral fuels, iron, water, soil, plant and animal life, the varying importance in the distribution of which determines to a considerable extent the locations of industrial and commercial centers, and that these resources are not distributed according to state lines, but that development is influenced to some extent by State laws.

COAL.

The importance of coal in our country is much greater than most people suppose. It is well distributed. The amount of power derived from it is many times that of all our man power working every hour of the day. The annual production of our coal leads that of Great Britain by a wide margin. The ranking States in output are Pennsylvania, West Virginia, Illinois, Ohio, Indiana, Kentucky, Alabama, Colorado, Iowa, Wyoming, Tennessee and Maryland. Wyoming is thought to contain even larger natural stores of coal than Pennsylvania. Mr. Edward W. Parker, head of the coal division of the United States Geological Survey, reports over two trillion tons of unmined coal west of the 100th meridian, lying principally in the Great Plains and Rocky Mountain provinces, and in smaller districts farther west.

MRS. PHILIP N. MOORE

OF ST. LOUIS, MO.,
VICE-PRESIDENT, FIFTH NATIONAL CONSERVATION CONGRESS

It is evident that there is much more coal for future consumption than most conservationists have claimed. This is a pleasing fact, for it indicates that industry should not be seriously hampered by lack of this source of power for many years to come. The argument that there is enough coal and to spare is used, however, to further selfish ends. It causes bad management of coal lands at many places. Notwithstanding the fact that the United States is so favorably endowed with coal it is coming to be known that some of the better bituminous and anthracite grades most favorably located are doomed to early exhaustion. The rapid increase in the use of these is causing some of the eastern States to show deep interest in conservation.

During the year the conservation of coal was directed mainly towards larger recovery from the mines, to the study and prevention of mine accidents, especially those caused by explosions, to improving the methods of use whereby more power is derived, and to the saving of by-products in coke making. The National Bureau of Mines lead in most of these investigations. Several States, mostly in the eastern province, studied the same problems, as for example, Pennsylvania, West Virginia, Tennessee, Virginia and Alabama. Illinois was equally active in the interior province. Practically all coal mines are now inspected by delegated authority.

The bee-hive coke oven produced relatively less coke during the year than the by-product oven in which is recovered coal tar and other useful products of considerable value. Investigations definitely proved that the most economic way to use certain soft coals is in the manufacture of producer gas. The culm heaps in the Scranton and Wilkesbarre districts were drawn upon more than formerly for the production of the smaller sizes of washed coal. This is an important utilization of what formerly was waste.

It would seem that every one in this Congress should be deeply interested in the conservation of coal whether his State produces it or not, for the permanence of this resource has a power relation, one that affects the industrial and social development of the whole country.

PETROLEUM AND NATURAL GAS.

These are uncertain resources as to their occurrence and permanence of development. The amount of production, however, is very large, coming from several States, and having increased from about 63,000,000 barrels in 1900 to over 200,000,000 barrels in 1911. New pools were developed in each province, though the annual production fell off at places. The largest developments were in California and Oklahoma, yet Illinois, West Virginia, Ohio, Texas, Pennsylvania, Louisiana and Indiana were important producers, as they have been for several years. No new conservation movements were inaugurated except that California took more definite steps to prevent waste at the wells. Adequate tankage, together with a high degree of attainment in refining, are the two leading factors in the conservation of petroleum. This industry is a splendid example of conservation for special interests, yet the public is supplied with many useful commodities, such as kerosene, gasoline, waxes, paraffine, oils, etc. Kerosene has a close commercial relation to the gas engine and the automobile industry. The price of gasoline, for reasons not fully understood by the speaker, made a marked advance. Just how this may affect the future building of gas engines is not known.

The production of natural gas is even more erratic than petroleum. It is readily used in the manufacture of brick, tile, glass and cement. A lack of permanence gives to the gas-using industries a migratory character, a movement to and from gas regions. For several years such plants have operated up to their full market capacity in Kansas and Oklahoma. The financial depressions of 1907 and of the past year checked some of factory building in and near the gas regions. For about two years certain companies have been diligent in selling equipment for making gasoline from natural gas.

The conservation movement is partly responsible for the decrease in waste of natural gas. Formerly the unused wells of Oklahoma and Louisiana, especially, were allowed to cast their millions of feet of fuel into the air without even a remonstrance from the States. Flambeaux burned night and day in the streets of small towns and many persons between Indiana and Texas were then heard to say that gas is cheaper than matches. The States stand indicted. This wrong to nature and to present and future industry cannot be repaired. The deed is done, and our only hope is that we may escape without having to suffer for such an offense.

IRON ORE.

This is the basis of iron and steel manufacture. It supplies the materials used in harnessing the power of fuel and water and has importance in mining, transportation, smelting and milling. The industries connected with iron and steel making in the United States are conducted in a much larger way than in Great Britain and Germany next in rank. The increasing use of steel by railroads, in highway construction, ship-building, the making of engines and farm machinery, and for large building is causing many persons to wonder how long this progress can continue unhampered. Is there no limit to our high grade ore and to the development of the gigantic enterprises dependent upon coal, iron and steel? What appears to be the correct answer to this question has been made by good authority. It is that the supply of high grade ore, like that now used, is not permanent—that it will not last many years. If this is true, the time will be when it will become necessary to mine less desirable ore, grading lower and lower as production continues. This, without doubt, will have an unfavorable effect upon our whole industrial organization.

The history of iron in the United States is most interesting. It shows that one by one many of the small districts were abandoned for the richer fields of the Lake Superior region, the Birmingham and Guernsey districts. The States that lost out in this change now realize that production may again return to their borders when the richer and larger deposits are exhausted. In consequence of this several States are beginning investigations looking to the future utilizing of low grade ores. At the smelters more than usual thought is given to the quality of output, making it more durable or otherwise better suited to the use for which it is intended. Experiments are under way for the purpose of testing out hydro-electric smelting in parts of California and other western States where the ore is distant from coal.

Much of the iron and steel conservation is directed by corporate interests in whom the ownership of ore and the development based thereon are definitely established.

WATER RESOURCES.

Dr. W. J. McGee, whose death we mourn, once said that “water is the prime necessity of life.” He also discussed its importance for drinking, in navigation, for power, and in the production from the soil of such materials as food and clothing.

The drinking water of the country and small towns is obtained principally from underground through wells and springs. A few States are trying to improve their domestic water supply by making sanitary surveys, noting the relation of the wells to drainage from lots, privies and other dangerous sources. Typhoid epidemics, due to sewage entering the water system, occurred in several towns. More than usual activity was manifest in making careful studies of streams in their relation to floods, drainage, power, sewage, water supplies and navigation. The Lakes and Rivers Commission of Illinois has gathered and published more data than other States in this line. The subject, “Navigation of Inland Waterways,” with special reference to the Mississippi and its “Lakes to the Gulf Route,” received new impetus principally because of its relation to the Panama Route. The Gulf States are now supported by Illinois especially in a campaign for larger attainments in this development.

Irrigation had a good year, especially so in the Rocky Mountain and Great Plains regions. The irrigation development is an important contributor to the larger industrial life of the whole country.

LAND AND SOIL.

The United States has vast areas of land of many kinds. The soil of this land is our greatest physical resource. Its fertility feeds the crops and is therefore of fundamental importance in agriculture and industrial development based thereon. Nevertheless, it is true that many disregard this great fact in their farm management. They conserve their own selfish interests and not the state. Just how to develop the State's view point in land management is not known. The southern States, in co-operation with the United States Department of Agriculture, are making progress in the solution of this problem. In many places there, the farmers are showing rapid improvement in crop rotation and methods of cultivation.

In Texas and Florida, much of the wet alluvial land is being improved by drainage. The Levee and Drainage Board of Texas surveyed over 300,000 acres last year and constructed 100 miles of levees. Land valued at twenty dollars an acre became worth seventy-five dollars to one hundred dollars at a cost of thirty dollars per acre. Deep floods of the Mississippi River did great damage in Louisiana, Mississippi, Arkansas and Kentucky, causing the Delta region to put forth a plea for National aid in draining the wet lands of the South. It does seem that their plea for support should not go unheeded when such a vast, fertile area lies unreclaimed.

Nearly every State is studying soil erosion, the tenant system and land taxation. Dr. E. N. Lowe, State Geologist of Mississippi, reports that his survey endeavored to secure the enactment of a law that would tend to check the great losses in the northern part of the State caused by soil erosion. The bill was opposed by a prominent senator on the ground that it would interfere with the personal rights of land owners. The bill did not pass, but Dr. Lowe is to conduct a campaign of education before the next Legislature is convened. The difference in viewpoint here shown, is the difference between the meaning of "legal" and "right." Does any one have the right to ruin the land?

Co-operative soil surveys were carried on during the year in the various States with complete success. Every State now sees the need of reliable study and mapping of its soils, to serve as a basis for farm management, taxation, and real estate. At a recent meeting of the National Tax Association, held in Des Moines, Iowa, the relation of land surveys and taxation was discussed with considerable detail. It was the conclusion that land value maps should be prepared by soil surveys to serve as a physical basis of taxation.

FOREST RESOURCES.

Though originally endowed with vast areas of forest on public domain, some having great value, our Federal Government was slow to develop effective measures for its protection, utilization and future growth. One generation stripped the forest from the agricultural lands of the central west; and their posterity turned the trick with interest in the west. No wonder many persons took advantage of such an occasion as was presented in the Rocky Mountain and Pacific regions to help themselves where the public treasury was free for the asking, not having been carefully surveyed and evaluated. The large timber owners are not alone to blame for this history, which in considerable part is not what it might have been. It is time, then, to close the chapter and to turn our attention to present day events in so far as they are related to forest conservation. Now the State foresters and Federal forces closely co-operate with the large lumber producers along several lines. The Weeks Law, recently enacted, furthers co-operative effort in the prevention of forest fires. One of the first States to take advantage of this law was Wisconsin. Then came applications from New Hampshire, Montana and most other forest States. New York appears to lead in perfecting State patrol. Most States in the Appalachian province have perfected their patrol systems. Oregon appears to lead in the Pacific region.

Colorado of the Rocky Mountain province is fighting the Forestry Commission, the Conservation Commission and Federal agencies, under the guise of State Rights. Here the National Government has large reserves and is meeting the expense of fire protection. Certain State men are diligently spreading the doctrine of State Rights, claiming that the Federal Government should cede its domain to the State. Such a sentiment is echoed, but not so forcefully, in a few other western States. The opponents of this doctrine claim that the States do not have the means to patrol the forest, and that the State Rights people are making the campaign for selfish reasons—to secure ownership of the forest.

During the year, many cities and States added to the area of their parks and forest reserves. The Maryland Legislature voted $50,000 for this purpose. The Appalachian bill passed the last Congress, providing funds for use in establishing reserves in the Appalachian province. A start in this development has been made at several places. It is reported, however, that land speculators are interfering with the project by securing options on land that is wanted for the reserves.

The work in general tree planting and forestation progressed about as usual. Promoters handle eucalyptus propositions in California with varying degrees of success. Many States, especially in the middle west, are planting catalpa for the production of posts. One of the largest problems in several States, as in Oregon, Washington, Wisconsin and Michigan, is that of utilizing the cut-over land. Some of this is suitable for farming, but much of it is classed as forest land. The problem then is one of reforestation, which cannot be done economically on most of the land because of high tax. The tax problem is closely related to and by many thought to be the controlling feature in the reforestation of land in private ownership. The Wisconsin and the Oregon Conservation Commissions are studying the problem. Louisiana has passed laws intended to promote timber planting on large holdings.

A few States published helpful literature on economic species of trees suitable for forestation, shade and decorative purposes. A little volume by the New Jersey Forest Commission, title "Planting and Care of Shade Trees," is a model that other States may well follow.

Following in line with the recommendations of this Congress, and in harmony with the policies of state foresters and the Federal Bureau, considerable progress was made during the year in forest surveys and forest studies. Fully half of the States are doing this work under the direction of their geological surveys, forest bureaus, or Conservation Commissions. Maryland and Rhode Island have completed such surveys.

Several large lumber producers report improvements in the way of saving practically all of the timber. When one wants to cite an example of extreme waste in lumbering, he usually refers to the Pacific region, perhaps not realizing that the method of utilization may be determined by commercial limitations. Be that as it may, it is pleasing to know that some companies in the West, as for example, the Smith Lumber & Manufacturing Company, are installing by-product plants. The company above named is building a fiber plant to utilize the waste mill products by the sulphate process, and to extract the turpentine from the red fir.

VITAL RESOURCES.

More than usual progress has been made in recent years in learning that living things, whether forest, forage, cereal crops, game, fish, farm animals or man, are natural resources subject to development.

Perhaps the greatest result of the Conservation movement is found in its helpfulness in improving the life and lot of people. Such a stimulus is needed, for it certainly is time society should conserve its men and women not only in working efficiency but in fitness to be fathers and mothers as well.

Most States have departments to promote good seed, fish and game resources, and the breed and health of animals. Some of the publications issued by these departments are most attractive and valuable as, for example, the reports on birds by the North Caroline Geologist-Natural History Survey.

More than usual State activity is now put forth in improving the stock, health, life and working efficiency of people. To further this end there is inspection of water, milk, food, drugs, and factories. Several States are making preliminary sanitary surveys; others conduct investigations under the head of "conservation of people." It has been learned that the public health can be markedly improved by observing a few simple safeguards that prevent sickness and disease. This calls for education, and perhaps for organized inspection of both the home and the school. State medical colleges begin to realize their duty in preventive medicine, and in some cases show a willingness to co-operate with health organizations in extension work in the conservation of public health. A number of the Southern States have taken important steps to rid their sections of typhoid, tuberculosis and the hook worm disease. Mississippi reports marked progress in this line. The Louisiana Health Train is known to all. The exhibits at this Congress indicate the great progress attained by Dr. Hurty and others in their fields. In closing the discussion in this department it should be noted that practically all parts of the country show a deep interest in the work of Dr. Wiley and the fight he has made for pure food. It is further evident that there is a strong demand for a Federal health department to work in co-operation with the state departments.

CONSERVATION ORGANIZATION.

Several State departments are related to Conservation work, as for example, the Geological Survey, Soil Survey, Natural History Survey, Forest Commission, Public Service Commission, Pure Food Department, Health Department, and Experiment Stations. So, since most of these have been in existence for several years, we should know that conservation work is not a new thing. The various forces were united into a definite movement, however, in 1908, following the Governors' conference at Washington. Immediately after the adjournment of that meeting the Governors appointed State conservation commissions to serve their respective States. Unfortunately, many commissioners were selected mainly because of their political affiliations. In some cases the selections were made wholly on the basis of ability to serve. Such Commissioners were chosen from among public spirited citizens, and the State and university departments closely connected with industrial development.

Practically all commissioners chosen because of political affiliation did very little work. Most of them were not reappointed after changes in State administrations. The non-political commissions did better work as a rule, and soon received financial support and statutory authority from the State for a wider range of activity. The commissioners with this authority are now appointed by the Governor, or they become commissioners by virtue of their connection with certain university and State departments named in the State laws. The tendency is to make the commissions entirely non-political and to give them full charge of certain natural resource surveys and the State supervision of development, at least to some extent. A resolution passed by the Trans-Mississippi Commercial Congress of this year is of interest in this connection. It reads thus: "We favor the selection of Conservation Commissioners from among those who are actively engaged in State surveys, in the investigation of conservation problems, or in the development of public welfare." It further urges that the work of such commissioners be done along non-political lines and in co-operation with Federal Conservation efforts. Most States have conservation commissions.

The best organized work is in New York, Rhode Island, Oregon, Nebraska, Wisconsin, Kansas, Utah, West Virginia and New Mexico. The New York Commission has three commissioners, a secretary, assistant secretary, deputy commissioners and several engineers, all well paid. The Commission has full authority to investigate and supervise the development of water, forest, fish and game resources. Rhode Island stands next to New York in organization, duties, and results attained. Its commission has full charge of the natural history survey, supervises the development of natural resources, and conducts an educational campaign. Nebraska's commission is non-political, composed principally of heads of departments in the University, who also direct the various State surveys. The duties of the commission are largely in supervision and education. A Conservation Survey unites the efforts of the University and State departments in systematic surveys of the water, soil and forest and in making careful field studies of the leading economic problems. Nebraska holds a Conservation Congress each year with a large attendance. This Congress has great value in unifying State development. It is under the guidance of the Commission, Conservation Survey and public spirited citizens and is an open forum for the discussion of development problems. The duties of other State Commissions are similar to those of the States above described. Utah is directing its effort mainly in the line of making non-political maps.

The Conservation Commissioners together with other persons directing State development have an organization called the National Association of Conservation Commissioners. It meets each year as a department of this Congress. The object of the association is co-operation, in which each State is able to learn of the progress attained in other States.

That the conservation activities in the various States are benefited by the different sessions of the National Conservation Congress is very evident. The influence also of the National Conservation Association is helpful.

In concluding this division of my report, I wish to emphasize the facts that the State conservation commission is coming into a broad field of work, that it must stand for the best interests of the State as a whole, that an important part of its activity is to unite the efforts of departments now existing in a co-operative work that has practical value to the State. Such commission must be composed of broad-minded men, preferably those who have a thorough acquaintance with the departments represented. The commissioners should be free from political entanglements, and refrain from using their position for selfish ends. They should stand for the greatest good of their States first, last and all the time.

Does your State have a commission of this kind?

SURVEY BASIS OF DEVELOPMENT.

The survey idea is now popular. In fact it may be abused in some cases, especially where the work is done with a lack of scientific spirit and undue rapidity. Such effort has no place in the conservation survey which seeks to determine useful facts, those really worth while in development.

In harmony with the spirit of the year which calls for the fundamentals, we have the following resolution by the Trans-Mississippi Congress, passed in its last session:

> “Recognizing the natural resources as the physical basis of development, we urge the States of the Trans-Mississippi region to make surveys of their leading resources under competent direction, and to publish reliable reports upon the same. We favor such reorganization of the State conservation commissions as will qualify them to make inventories of natural resources, to study industrial problems, and promote the proper development of the respective States.”

This demands that Conservation be placed on a survey basis. Just that thing is the order in many States under the leadership of conservation activity and through the co-operation of State and Federal agencies. During the past year, progress was made in co-operating the work of the different surveys, making them of greater value to the people and State. It is now understood and agreed that the following lines of information should be determined and made available for use in the development of each State as soon as possible consistent with good work and reliable results and in about the order herein named. The points considered in the complete survey are:

A. Topography. By topography is meant the surface features of the land. Topographic maps have many uses in development. Maryland, New York, Connecticut, Rhode Island, Oklahoma, and a few other States are now mapped.

B. Structure. By this we mean the underground make-up of a region, the kind and arrangement of the materials of the land. Structure is directly related to mineral resources, topography, water supply, and soils.

C. Surface Water and Drainage. This refers to the amount of run-off, to such as streams, lakes, marshes, and has importance in irrigation, navigation, fish culture, city water supplies, etc. Illinois is leading in this survey.

D. Ground Water. This is water in the land. It is the source of crop water and the largest source for domestic and town supplies. The amount of water held by the soil is even more important than the quantity of rainfall. The depth to the water table, and the quantity, and quality of well water are of great importance in agricultural regions.

E. Climate. The elements—temperature, sunshine, wind, humidity, and rainfall—are recognized as having importance and should be known for every part of our country. The latest movement is for facts in local climate, even that of the farm or certain parts of it, and of the soil.

F. Soils. The relation of soil to industry is generally known. The soil survey classifies and describes the soils as to origin and properties, and maps them accurately so that the farmer may know definitely the kinds and their distribution on his farm. Farm management demands intelligent comprehension of soil characteristics.

G. Native Life. The native plants and animals of a country represent the natural selection of the fittest for the conditions encountered. The life of a region reflects the topography, soil, and climate under which it lives. In new territory the native plant life reveals to the keen student much concerning the soil and climatic conditions. In older communities undisturbed patches of vegetation tell the same story. By studying such life the qualities needed in cultivated crops may be fairly well determined and the losses incident to haphazard experimenting avoided. Native life then needs to be considered in a rural survey because: (1) It gives a summary history of soil and climatic influences; (2) it may lead to economic production of certain native types of plants and animals; (3) it presents concretely the problem of utilization of waste lands; (4) it will give emphasis to the need of utilizing our lakes and streams as a source of food supply.

H. Social and Industrial Conditions. If a move into new territory is contemplated, the questions of vital interest are not only of the natural and industrial conditions but also in regard to social conditions. By this is meant the classes of people as to race and culture, and the opportunities offered for advance in social and intellectual lines. These characteristics of people are closely associated with their occupations. The pursuits of the people are largely dominated by the physical basis of industry. Hence the social survey must recognize this influence if it is to correctly interpret conditions as they exist. Data of most vital interest in the social rural survey pertain to the following lines: (a) History of settlement. (b) Condition of agriculture. (c) Education. (d) Religion. (e) Recreation. (f) Sanitation. The industrial conditions of a region are practically determined by its physical features. The development is further related to the biological and social life. Hence the industrial survey must be based on these fundamentals if it is to be comprehensive.

In closing this review of the fundamentals in surveys it should be understood that: 1. The physical and biological surveys should come first, since they are necessary for accurate work in other investigations. 2. The special surveys of industries, rural and urban life should be made from the common basis of physical and biological conditions and extended into their respective fields.

It should be recognized that the broad controls affecting industry are structure, topography, drainage, climate, soils and native life, but that they do not have equal importance in any and every locality. Any one of them may be the controlling feature with the rest of minor importance.

It is not a pleasing fact to know that most States have not yet accurately mapped their lands, waters and forests. The departments responsible for this work should receive adequate financial support and the people in turn should demand results.

What progress has your State made in these lines?

RELATION OF EDUCATION TO CONSERVATION.

The State universities of the Middle West especially are meeting their obligation to the people by training students for real work—for efficient service. Such institutions, by their instruction, surveys and extension departments further the development of practically every line of industrial activity in the State. From these centers are directed geological, soil, water, sanitary, social, farm management and other surveys. Consequently the professors and advanced students get a good work-out and the citizens are caused to look to the institution for assistance in practically every development problem that arises. The State universities that are giving the largest service in this line appear to be Wisconsin, Minnesota, Cornell Agricultural College, Illinois, and Nebraska. It is my great privilege to be connected with one of these.

Unmistakably, the present tendency is to associate the public service State departments and conservation activities more with higher education, taking them from the field of politics. This noticeable feature in the rearrangement of conservation activities of the past year is worthy of consideration by all States.

CONSERVATION OF BUSINESS.

The different lines of business are conserved in many ways. This applies to practical developments in improving the process involved in handling commodities all the way from manufacture to sale; to trade, in the direction of economy in buying, transportation and sales; to farming in improving methods of cultivation, the better care of stock, and in less buying on time; to more economic use of school and church buildings; to the building and maintenance of good roads and clean streets, and to the improvement of public service generally. So there is room for practical conservation in many lines. It prevents waste, increases efficiency, and thereby decreases the cost of commodities. A very general movement for good business is the feature of the year. It is promoting real business by demanding that it be done on the square and free from fraud. This is working a public good.

At another time, I will discuss the subject, “Land Frauds or Get-Rich-Quick Schemes,” with special reference to their effect upon real business.

CONCLUSIONS.

As a summary conclusion, you will permit me to enumerate the things that stand out in the progress of the year.

1. The prominence of Conservation on many State and National programs.

2. The tendency to place State development on a survey and fact basis.

3. Development of co-operation between State and Federal agencies.

4. Demand by the public for reliable land classifications, soil, sanitary and agricultural surveys.

5. Interest in soil fertility as a basis for agricultural development.

6. The affiliation of Conservation organizations with educational departments and removal from politics.

7. Discussion of Lakes-to-the-Gulf Route and success attained in presenting the cause of drainage in the Mississippi delta region.

8. Modernizing of State universities, making them of greater value to the State.

9. The determined demand for vocational training in the public schools.

10. A demand for less extravagance in public service.

11. Taxation of cut-over lands.

12. Perfection of forest control.

13. The very general recognition that people are the most important natural resource subject to development.

14. Increased regard for sanitation throughout the country.

15. Massachusetts minimum wage law for women.

16. A determined and widespread movement on the part of social workers to eliminate the social evil.

17. Widespread movement against fraud and the assistance given to the movement by the Postoffice and National Reclamation departments.

18. More than usual discussion of co-operative enterprises and methods of distribution.

19. Rapid progress in the building and maintenance of good roads.

20. A growing tendency for the citizens in every part of the country to outgrow provincialism; to come into respect and appreciation for the people and institutions of every State; to recognize the fact that the home State is but a part of the Union and larger world in which people live not to themselves alone but in helpful relationship with all others.

President White—This was a very interesting address, which we allowed to extend beyond the time, because it is a summary of Conservation work during the past year in all the States. Heretofore, we have had a report from a representative of each State, but it was thought advisable this year to have these reports condensed into one paper, a work which Dr. Condra has done most admirably.

The next address, which is of the greatest interest, is on the subject of "Human Life, Our Greatest Resource," and the name of the gentleman who is to deliver it will be a sufficient guaranty that it will be replete with interest, and will be useful to every one of us who listens. I now introduce Dr. William A. Evans, of Chicago.

(Dr. Evans failed to return his manuscript for insertion in the Proceedings.)

President White—We must hasten on, for we have some other addresses that will be very interesting to the children. There are a great many present that have come no doubt to see the wild life pictures. So we shall have to hasten in order to reach them.

Dr. Bessey, who was to have been next on our program, will be here at 3:30 o'clock.

We shall now call on Mr. E. T. Allen, of Portland, Oregon, whose subject is "Conservation Redefined."

Mr. Allen—On a hot afternoon, a bare-footed boy, on his way home from school, in western Washington, eager as any school boy for the swimming hole, or whatever waiting attraction had kept his eye on the clock since about 2:00 o'clock, stopped, hesitated, then clambered down a steep, brushy slope to the stream at its foot, filled his hat with water, climbed up the hill again laboriously so as not to spill his burden, and put it on a camp-fire some voting citizen had left burning by the roadside. It still smoked, so he went back twice, three times. About then, the man who told me this story came along and asked the boy why he made it his business to put out that fire.

"Why, it told in a little book I got at school," was the reply, "why every one should try to stop forest fires. It told what grown-up people can do by being careful and passing laws and such, but it said a boy may do as much as anybody by putting out some little fire with water or dirt before it gets big."

Now, the action of that school boy, and of the teacher who handed him the booklet, and of the State authorities who instructed her to do so, and of the man who wrote the booklet and enlisted the State's co-operation in its distribution to a hundred thousand children, and of the timber owners through whose protective association that man was hired and the cost of printing and distributing that booklet was paid, was Conservation. It was forest Conservation, definitely conceived, definitely executed, and with an exceedingly definite result.

About a month ago I was talking to an extremely intelligent man, a scientific man whose life is devoted to bettering humanity. He said, "Allen, do you believe in Conservation?"

Rather astonished, I replied, "It's my trade, isn't it?"

"Oh, I don't mean forest protection, like putting out fires and making trees grow, but forest Conservation—Pinchotism, tying up everything for future generations."

Now that man's conception was the result of Conservation activity, certainly. Without our agitation there would have been no counter agitation. No doubt he has read of these congresses every fall and of countless other forms of our work. But, apparently, only one interpretation, and that a mistaken one, had ever reached him in a form definite enough to make an impression. How else can you account for getting effort and sacrifice from the irresponsible barefoot school boy, but no realization by a citizen of the highest type that Conservation wants his help in some way that he can give it?

To what extent these remarks apply to your work along other Conservation lines, I am not competent to say. In forestry, there has been, I will not say too much debate, but certainly too little other use of our Conservation machinery in presenting clear-cut principles of forest economics in the specific local forms and with the specific local needs that are necessary to engage and direct accomplishment. This is true of what we do at these meetings and more true of what we do when we go home.

What our forests need most is more patrolmen, more trails and telephones for them to use, more funds and organization to marshal fire-fighting crews when required, better fire laws and courts that will enforce them, public appreciation that forest fire departments are as necessary as city fire departments, more consideration for life and property by the fool that is careless with match and spark, realization by more lumbermen that it pays in more ways than one to do their part, State officials who will handle State lands intelligently, tax laws that will permit good private management, consumers who will take closely-utilized products, and a few other things that demand specific study and specific action. Very few will follow automatically after any amount of agitation under the general term of forest Conservation. Do you suppose this would have sent the boy down the hill after water? No more will it write a good forest code and drive it through the devious channels of legislation. No more will it organize a hundred busy lumbermen and install a trained co-operative patrol. No more will it supply the necessary systematic campaigning to teach the people of your State and mine in just what ways their homes and pocketbooks are touched by every injury to forests or forest industry and exactly what they, as individuals, must do to prevent such injury.

Without decrying their sentimental aspects, these are business problems. They call for all the exact facts, all the systematic planning, all the decisive action, all the appeal to human motives, selfish and otherwise, that are essential to any business. We have a commodity to offer. By whatever name we call it, fire prevention, reforestation, or more vague yet, forest Conservation, we are really offering prosperity insurance. It must be paid for by the community in currency of individual and collective effort, by individual care with the forest and by public policies enforced at public expense. To make the community pay for this commodity requires the same methods that make it buy life insurance; the same devising of a sound, attractive policy that the buyer can see and understand, the same skilful advertising, the same personal persuasion by its agents. I believe that if this were a congress of life insurance agents they would be talking mostly of just these things, particularly of improved methods to close with procrastinating “prospects,” with a view to putting these methods into the most definite kind of practice the day after they got home. We do not need argument on the merit of Conservation any more than they would on the merit of life insurance. We are converted, or we would not be here. But we need a whole lot of instruction in salesmanship, and I believe we fail to make this the feature of these congresses that it might be.

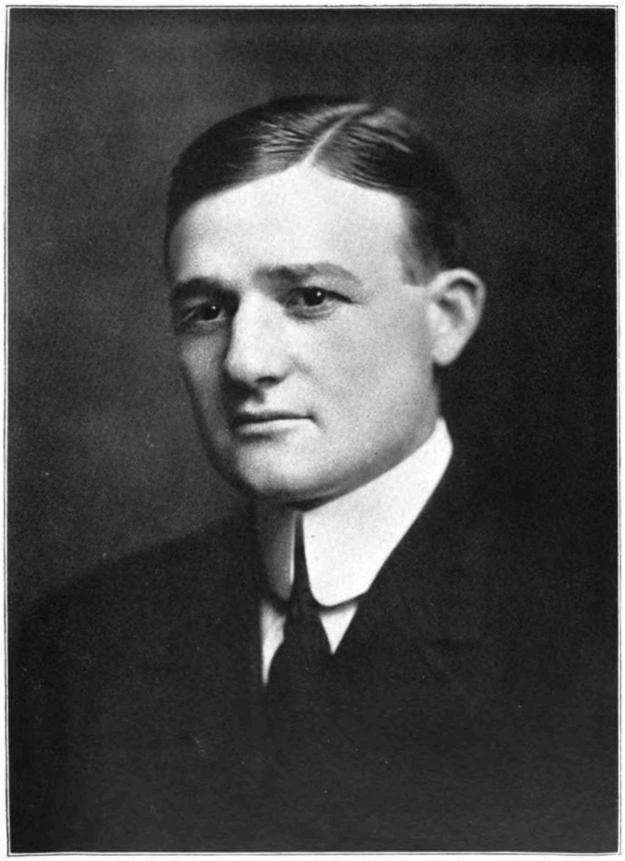
Let us look ahead, we agents of prosperity insurance, to see what is to be done after *we* get home.

The Government needs little but our moral support. The Federal Forest Service is our highest authority in technique, the national forests are our most conspicuous examples of practice. But the task of the Forest Service is stupendous, not only in protecting these vast forest areas and the lives within them, but also in replanting denuded areas and managing great timber sales, so new growth will follow. Congress does not appropriate anything like enough for this work. The forest rangers out West are working for you and me, not for Congress. We want more of them, and better facilities for their work, and it is up to us to say so at the right time, to the right men, and so emphatically that there will be no misunderstanding. Petty politics and "retrenchment" would not be practiced so much more vigorously when dealing with the lives and resources of the people than when dealing with the "pork barrel" if we Conservationists were half as free with telegrams as we are with resolutions. Yes, this means you. So long as you stand for having the appropriations for preserving the Nation's forests from three to twenty times less per acre than the lumberman is spending on his contiguous holdings, or for any congressional attack upon the integrity of the national forest system, your Conservation preachments are going to the wrong address or are not properly spelled in words that look like votes.

There is even greater need of definitions that apply to the situation of our States. Many have done nothing. Others have ill-balanced laws passed by some one agency without due consideration of the needs of others or of the greater need of bringing all into harmonious co-operation. In few is there a far-seeing comprehensive policy financed and executed. Here, of all places, forest Conservation must narrow itself to specific issues. Scattered ideas do not pass good laws or prevent the passage of bad ones. Propaganda work must be as forcible and carefully directed as blows with an ax, to cut out one by one the local foundation of every obstacle. In presenting our remedy we must prove our knowledge of the principles and technical frame-work which will insure freedom from politics, just distribution of cost, effective organization, strict and enforceable fire laws, systems of patrol and fire-fighting, facilities for educating lumber men and public management of

State-owned lands, fair taxation, and, above all, co-operation with and stimulation of endeavor by private owners. Without such knowledge, and skilful publicity and campaigning, your very success in general agitation may result in legislation worse than none.

All this involves considerable knowledge of the problems of the private owner. After all, he controls most of our forest area. His use of it, our use of it, and the effect of our relations on our joint use of it, largely determine our forest destinies. And there is entirely too much forgetting that forests are useless unless used; that not forests, but forest industry, is what we really seek to perpetuate. Except from their protection of stream-flow and game, the community has little to gain from forest preservation unless it also preserves, on a profitable and permanent footing, the industry that makes forests usable and worth preserving, that employs labor, affords market for crops and services, pays taxes, and manufacturers and distribute an indispensable commodity. Forest wealth is community wealth, but not without forest industry to coin it.



E. LEE WORSHAM

OF ATLANTA, GA.,
CHAIRMAN EXECUTIVE COMMITTEE, FOURTH NATIONAL CONSERVATION CONGRESS

The lumberman as a class, because he is honest and useful as a class, should be accorded the same encouragement as a captain of desirable industry that is accorded the leader in agriculture or irrigation who develops possibilities of utilizing our resources and supplying our needs. And as a class, because whatever may be true of the past he now sees his livelihood dependent upon forest preservation, he is a stauncher supporter of forest reform than any other class. He will utilize the present crop closely, and grow a new one, whenever these are business possibilities. The most efficient and liberally supported fire organizations in America are the lumbermen's co-operative patrols inaugurated in the Pacific Northwest and now spreading eastward. Most of our best State forest legislation has been promoted by lumbermen. Where this is not true, we can make it true quickest, as Judge Lindsay has found with his boys, not by censure and compulsion that make them sullen or antagonistic, but by learning their troubles and working with them hand in hand toward the ends which in the very nature of things must in the long run be of mutual advantage. And this means that we must talk a common language; that here, too, forest Conservation must be expressed in practical terms of fire prevention, just taxation, and business encouragement.

What I have said of propaganda for State and private action applies to our appeal to the ordinary citizen, with this difference that because his number is greater, and his interests are more varied, we must add to the list of our specific personal arguments and to the list of our publicity mediums for carrying these arguments. Every vocation, every trait of character, every selfish and unselfish motive, has its best avenue of approach.

Immediate tangible results come most surely from immediate injury. Even good laws are of small use unless the public of today is sufficiently warned to insist on their enforcement. Do not think me lacking in ideals when I say that our greatest need is vigor and skill in appealing to human selfishness. The altruist comes to us unsought. But to reach the hand with the torch, the vote withheld, the word unspoken; we must find the man, make him listen, and show the cost of forest destruction to his particular home and pocketbook. We will not have forest Conservation till we have done this, and we will not do it until we master and apply the technical knowledge of mediums and psychological appeal that go into any successful advertising campaign.

The definitions of Conservation I have outlined are those used by the Western Forestry and Conservation Association. Its field is the five Pacific forest States—the Nation’s woodlot—containing over half the country’s standing timber and capable, by reason of rapid growth, of growing an adequate supply forever. In this field we practice what we preach. Our constituent local patrol associations spend from $300,000 to three quarters of a million a year, all paid by lumbermen, but protecting your resources and mine.

Our booklet reached that school boy and three hundred thousand more. Through every modern avenue of publicity—newspapers, circulars, posters, railroad folders, telephone directories and a dozen others—we carry the lessons of forest economics to every citizen in terms he can best understand

and apply. Although you had not made that scientific man style himself a conservationist, we had secured his help in passing a model fire law. We wrote that law. Under it State, Government and lumbermen work hand in hand to protect practically every forest acre, sharing the cost, and the lumbermen in that one State contribute $150,000 a year.

But, best of all, we provide a common meeting ground for all four agencies in our entire territory, each having the hearty support and confidence of the others, and we talk only of our joint business of actual, practical, constructive work. We talk not needs, but methods, and find means to apply the methods.

We believe in this National Congress of Conservationists. We think it will enter a permanent future of still higher usefulness when it develops a more sectional organization, giving the real workers in every branch opportunity to get the very most out of meeting their own colleagues, and this not only in the technique of application but also in the lagging art of promoting the prosperity insurance of Conservation in terms and policies the public can understand and cannot evade.

President WHITE—It will now be necessary to drop a curtain in order to arrange a screen for the illustrated lectures that are going to follow, so everyone will retain their seats. We shall not be detained long. While the curtain is dropped, Secretary Shipp will make some announcements.

The announcements were made by the Secretary.

President WHITE—Dr. C. E. Bessey, of the University of Nebraska, having now arrived, will read his valuable report for the Standing Committee on Education:

Dr. BESSEY—Your committee recognizing that in the field of education we must for a time provide for a propaganda of suggestion and information, to be followed ultimately, when the public mind has been adequately wakened, with plans for a campaign of aggressive activity, now presents the following as a preliminary report. And while we feel confident that even at this stage something may be done more than the inauguration of a campaign of agitation, it is certain, nevertheless, that it is agitation more than anything else that we can best promote at the present time. And we must not belittle the importance of this stage of our work, for in every great movement there is first the period of agitation during which the "seers of visions and the dreamers of dreams" talk, and urge, and plead, with increasing vehemence and increasing confidence.

It is our privilege now to promote such a work of agitation. Accordingly our suggestions are all made with reference to this preliminary phase of our work.

There are three principal lines along which this preliminary work may be developed—namely, in the communities, in the schools, and in our law-making bodies.

I. WORK IN THE COMMUNITY.

Here we have to change the feeling of apathy, and carelessness, and irresponsibility, to one of active, conscientious responsibility. In this task we have to deal with the men and women and children who constitute the community. We must influence all of them. We must reach them in such a way that there will grow up in the community a better feeling with regard to the world we live in, and a clearer appreciation of our relation to it in every way. They must be led to see that the world is to be used, not destroyed. Just as the child has to be taught that his toy is to be enjoyed, and played with, but not wantonly destroyed, so we must bring the men and women in the community to see that preservation, and not destruction, is the higher duty. That citizen is the better one who leaves to the next generation a better world than he found; whose use of Nature's soil, and water, and plants, and animals, leaves Nature still the rich storehouse in which others after him may find these unimpaired, and in abundance.

How shall such a high sense of responsibility be developed in the community? How may we awaken this larger and deeper altruism? How can we bring the men and women of this generation to see that they are stewards of their Master's estate?

Your committee commends three agencies as rendering effective service:

(a) *Public Lectures.* For these we may rely upon public spirited men who are primarily interested in Conservation, as well as many whose affiliations to different branches of natural science have prepared them to appreciate the purposes of this propaganda. To these we may add the great number of ministers of the gospel who nearly to a man may be depended upon to favor the movement, and to speak for it as occasion offers. Last of all we may confidently enumerate the teachers in the public schools and the higher educational institutions, and from them we may certainly secure many regularly prepared addresses and many more less formal short helpful talks. The influence of all of these presentations can scarcely be measured beforehand, but we confidently predict that in a few years we shall find that there has been a decided change in the general attitude of the community from one of ignorant indifference to a more or less intelligent interest.

(b) *Articles in the Public Press.* We believe in the power of the public press as a molder of the opinions of the community, and feel that we must enlist the interest and co-operation of the newspapers throughout the country. To do this generally will require carefully considered, nation-wide plans; but a great deal may be done in every locality by the printing of the addresses referred to above. Where this is not possible abstracts may always be published, as well as summaries of shorter talks and discussions. Now and then a short, pointed article should be prepared and printed in the local paper. Here we feel the need of admonishing writers to be brief. No communication should attempt to be exhaustive. Better far to say a little at a time, and to come back to the subject again and again, than to say it all at once. Short, suggestive articles are generally read, while long ones usually become so dry that few read them.

(c) *Books and Pamphlets.* For certain classes of people the appeal through the more permanent form of publication is far more effective, and therefore there is in our work a need of the book writer, and the writer of pamphlets. Here, quite naturally the writer must possess to a marked degree the ability to present the matter in such a sustained way that his book or pamphlet will be read throughout. Probably the most effective writing of this class is that which appears in our illustrated magazines where by the aid of half-tone reproductions of striking photographs the interest of the reader is held much more certainly. Such articles collected into small books or pamphlets would go far towards stimulating a proper state of mind in regard to the conservation of our natural resources.

It occurs to us also to suggest that now and then our state experiment stations might quite legitimately devote a bulletin to Conservation.

II. WORK IN THE SCHOOLS.

While the community as a whole is receiving such suggestions as are possible through the agencies mentioned—lectures, addresses, newspaper articles, books and pamphlets—there is a vastly more effective means at our disposal in the public schools, dealing as they do with no less than twenty millions of children. We suggest that teachers everywhere be urged to include in all the studies that pertain to nature something in regard to the preservation of natural objects. This need not be much in amount, and it should be brought in with care and wisdom. We are reminded that once a very good cause was much discredited in the schools by the rash unwisdom of its advocates who insisted upon such an overdose of advice and admonishment that acute nausea resulted. So we would suggest that in the following studies care should be taken on the one hand to suggest conservation while on the other hand still greater care should be taken not to overdo the matter.

(a) *Nature Study.* Along with an appreciation of Nature there should be inculcated the feeling that others after us should have the opportunity of enjoying the same beauties that we have.

(b) *Geography.* As now generally presented this deals more with the earth and what it contains, than with its political divisions. Thus the soil, the forest cover, the streams, the water supply, all fall within this rejuvenated science, and here most readily can be inculcated the principles of conservation, as applied to the soil, the forests, the streams, and the underground waters.

(c) *Botany.* When the pupil's attention is more specifically drawn to the plant covering of the earth, in the study of botany, it is not at all difficult to impress upon him the desirability of preserving the vegetation of the present day for the generations that are to come after us. No lover of plants can contemplate with pleasure the thought that for the botanists of the twenty-first century certain curious orchids, some rare trees, and possibly some Golden Rods, may be as completely extinct as are the Paleozoic Calamites and Lepidodendrids. The latter perished from the face of the earth, and we know of them now only by the fragments that have been preserved in the fossils which we dig up from the old rocks. Extinction has been the fate of many a plant, and extinction of plants now living is by no means improbable. The botanical teacher should preach the doctrine of preservation, the preservation of the plants of the present for the people who come after us.

(d) *Zoölogy.* So, too, the teacher of zoölogy should improve his opportunity to help create a feeling favorable to the conservation of the present animal life. Especially do we need a propaganda of conservation in relation to the birds of the country. And here we remark that there are methods of presenting this part of zoölogy which emphasize rather the living bird in the tree than the dead bird in the cabinet. And these methods are happily displacing those that suggested if not required the death of every bird studied. We are well aware of the fact that it is not so much the killing of birds for study that threatens the extinction of some species, as the wanton killing for the sake of killing, and as in the case of birds of fine plumage, the killing for the money value of the dead birds. Yet we realize that the place to begin is to educate the children of the schools not to kill birds for any purpose. When they have regard for the life of a bird they may be trusted not to kill one needlessly.

(e) *Geology.* In this the pupil comes to see the foundations of the earth, fortunately little of which man may injure or deface. And yet how thankful we are that on the hills of New England there have been preserved in their original ruggedness the great masses of granite that have withstood the elements for millions of years. And who is not gratified that the great wall of the Palisades on the Hudson River has been saved for all time? These cliffs were valuable for crushing into gravel for road-making, and for the quarrying of building stone, but certain men of finer sensibilities felt that the Palisades had a far higher value for their grandeur and beauty. And so the Palisades were saved.

We need more of this fine sense of the value of rocks, and lakes, and waterfalls, and cliffs, and mountains, and of the need of their preservation.

(f) *Conservation Clubs.* Aside from much that may be done in school classes to foster a spirit of conservation something further may be accomplished by taking advantage of the club forming instinct of children. Conservation clubs, Conservation leagues, Conservation guilds, pacts, societies, or what-not, may be suggested by the wise teacher, who can discreetly keep himself in the background while the youngsters do the work. If a nauseating namby-pambyism can be avoided such clubs may be joined by even the most vigorous of boys, the very class in whom it is desirable to develop the spirit of conservation.

III. WORK THROUGH LEGISLATION.

What has been already outlined is probably enough for the present, but the American people are not satisfied unless something is done in the way of enacting our ideas into laws. In the present condition of society we act as though we thought it quite impossible to do anything on a large scale without having the sanction of a direct law in regard to it. We are only very slowly learning that some of the best of human activities have been developed independently of legislation, and no doubt the time will come when we shall not be so anxious to have our plans formulated into laws found in our statute books. But for the present we may suggest the following legislation as helpful. We purposely avoid suggesting the passage of laws dealing with details. They must come later, when the conservation sense of the public has been adequately aroused. Here we may consider state and national laws.

(a) *State Laws.* These may well include those intended to preserve rare birds, and in some places certain rare plants which are in danger of extermination. To these may also be added provisions for the preservation of important natural features, as forests, waterfalls or massive rocks that lend interest or beauty to the general landscape.

(b) *National Laws.* These may deal with larger problems, as the preservation of certain widely distributed birds. Naturally, too, it is the National Government that must take the initiative in regard to the conservation of the great forests, waterways, waterfalls, and the features in the national parks and reserves.

Carefully drawn laws, both State and National, covering the foregoing will no doubt aid the cause of Conservation. Too much must not be attempted. More good will result from a constant vigilance with regard to the passage of bad laws which give away the heritage of the community, than from attempts now to formulate a general conservation code.

Respectfully submitted,

Charles E. Bessey (Chairman),

David Starr Jordan,

EDWIN A. ALDERMAN,

E. T. FAIRCHILD,

EDWIN B. CRAIGHEAD,

Committee.

President WHITE—We have all been very much interested in this valuable contribution to Conservation, coming from such distinguished contributors as were on this committee, and I desire, for the officers of the Conservation Congress, to thank the committee for its admirable report. I feel that every delegate here would like to join in an expression of thanks for such an interesting and such a helpful paper, which will go forth to all sections of the country. All those who desire to so express thanks please rise to their feet. (The entire audience rose to its feet.)

This is a very grateful and pleasant expression of thanks. I thank you.

We will now be entertained by an illustrated address by Dr. T. Gilbert Pearson, of New York City, Secretary of the National Association of Audubon Societies. The subject is "Bird Slaughter and the Cost of Living."

(Dr. Pearson's address, which, unfortunately, was not recorded by the official reporter, was heard with keenest interest by a large audience and was interrupted by frequent applause. The speaker prefaced his illustrated lecture with a vivid statement of wild life conditions, which was heard with closest attention.)

President WHITE—I am sure you have been entertained by the very excellent address we have just heard. And there is another interesting address to follow. I want every one of you to know we are having a very interesting Congress and a very large attendance. This afternoon there have been three section meetings going on: one, I understand, with about one thousand people in attendance. All belong to the Conservation Congress.

We will now listen to a discussion of "Federal Protection of Migratory Birds," by Dr. W. T. Hornaday, Director of the New York Zoölogical Park.

Dr. HORNADAY—Mr. President, Ladies and Gentlemen: The subject presented to this Congress by the Committee on Conservation of Wild Life is one of the most practical subjects that you could possibly imagine. It touches the market basket and the dinner pail, and I know of nothing that can come much closer home to a family than that. Within the last three months, in the City of New York, we have had riots in our streets on account of the high cost of certain articles of food.

Whenever I have an opportunity to stand before an audience and speak in behalf of wild life, "I would that my tongue could utter the thoughts that arise in me."

We have reached the period now when it is absolutely necessary for us to adjust our ideas according to new conditions. I am trying to place before you conditions as they exist throughout the United States today, and I think when that has been done the facts will suggest to you the logical conclusion. The trouble is that our system of protecting wild life is nine-tenths absolutely wrong. We are confronted today by a slaughter of wild life throughout the whole United States, throughout the whole continent of North America, and throughout the world, that is absolutely appalling.

Now, in the City of New York there are several national organizations which make it their business to keep in touch with the conditions of wild life throughout the world. Unless a person takes pains to keep in touch with those conditions, as those national organizations do, you lose sight of the things that are actually going on and which ought to be of common knowledge. But our lives are so busy, there is so much to do, the days are so short, and we are so pressed for time that we grasp only the things that come close to us.

Now, take the slaughter of bird life, it is not like the cutting down of a forest. When a forest is cut down the stumps are left to be constant reminders of the destruction for days, for weeks and for years. When your bird life is destroyed, it simply fades from view. It fails to return in the spring and you go about your day's business and you see the beauties of the forest and field, but you forget to what extent the birds have disappeared. It is a difficult thing to obtain an accurate estimate of the decrease in the general volume of wild bird life throughout a given year, but it is possible to obtain such estimates. Now, there is in the United States a tremendous force at work destroying wild life. The force that is preserving wild life is not nearly so large and not nearly so active. I will show you presently a picture especially designed to bring this home to you. Dr. Pearson has set before you many beautiful pictures showing bird life in protected areas. That points an important moral which I do not wish to forget. It means that if we are diligent, if we reform our system and our laws we can to a very large extent bring back the vanished bird life. There is hope for the future. Today we are confronted by the prospect of a country gameless and birdless everywhere except in the protected areas. We all know how important the game preserves and the protected bird areas are. We cannot have too many of them; they cannot be too large. But there is a vast volume of bird life that cannot be protected in the preserves, the migratory phase of bird life, which we cannot control except for short periods of the year.

I believe that the subject we are now bringing before you is one in which it is possible for the members of this Conservation Congress to achieve a practical result of the greatest magnitude and in the shortest possible time and with the least effort of any subject that will be presented to this Congress. I know that is a large order, but I think that before I conclude you will agree with me that my proposition is not exaggerated.

When I was assured that I could have the honor and the privilege of speaking to this Congress on the subject of wild life, the first thought that occurred to me was to endeavor to place before you some ocular proof of the slaughter of wild life that is now going on at so terrific a rate. I gathered from my side table a collection of pictures that had dropped into my hands from various portions of the United States and outside, and those pictures I wish you to see now. They will tell a story of their own with very few words from me, and after that we will come to the logical conclusion.

Dr. Hornaday here gave an illustrated lecture which was thoroughly enjoyed.

President White—The Congress will now stand adjourned until 8:15 o'clock this evening, when Dr. Harvey W. Wiley will speak, at Tomlinson Hall.

A large reception was given by the officers of the Congress and the Local Board of Managers to the speakers, delegates and visitors, at 7:30 o'clock, Claypool Hotel.

THIRD SESSION.

The Congress was called to order by President White, in Tomlinson Hall, at 8:30 o'clock p. m.

President White—We are a little late opening this meeting, because we are trying to do so much in different places, and we do not all get in one place at the same time. But I am glad to see such an enthusiastic meeting here tonight. The audience will rise while the Rev. Dr. Allan B. Philputt, of the Central Christian Church of this city, invokes the Divine blessing.

INVOCATION.

Lord, our God, we ask that Thy blessing may rest upon us in what we believe is work well-pleasing to Thee and for the upbuilding and welfare of our common humanity. We pray Thee, bless Thy servants who have gathered here to instruct and lead us on with the mighty host of those who are willing to follow in the good ways that shall be pointed out for the preservation, not only of our material resources, but for our moral, intellectual and spiritual well-being. We pray that strength may be given those who lead, and guidance and light, and the heartiest co-operation on the part of all our citizenship. May we be interested in these things which will add to our happiness, and wealth, and peace and plenty, and by which we may also come to a better knowledge of Thee and Thy laws. May Thy blessing rest upon all the sessions of this great Congress, especially upon those who have sacrificed time and means to come here and give themselves unreservedly to this great cause. May Thy favor rest upon those present, may Thy blessing be upon those who are strangers within our city, and may hospitality be unbounded, may sympathy and cordiality flow from heart to heart until we feel the strong ties that bind us, not only in one State, but with every State in our great Republic. This we ask through Jesus Christ, our Lord. Amen.

President White—I have a communication to read to this audience from an old, well-known and well-loved conservationist, one of the great leaders in conservation work. I do not think there is any politics in this. I will read it.

“Omaha, Neb., September 30, 1912.

Capt. J. B. White, National Conservation Congress, Indianapolis:

Please tell the Congress I am keenly sorry to be away. I should be with you, except that I believe I can do the cause of Conservation more good where I am. We are working to make this continent a better home for a better race. It is a great task. I wish you the best of meetings and complete success.

GIFFORD PINCHOT.”

The speaker of this evening is well known to us all. He has impressed himself and his subject upon the people of this great country in the past few years, and he needs no introduction from me. I have long wanted to know how old people managed to grow old and keep looking young. I do not mean to infer that the speaker of this evening is getting old, as I understand he has a boy only about a year old (applause); but I have found out his age, by persistent and tactful undertaking, and, being in pursuit of some way of living to a good old age myself, I inquired as to his habits. I will not give them to you now, except to say that he told me, briefly and epigrammatically, that he doesn’t smoke, he doesn’t drink (applause), he doesn’t chew, and he says he doesn’t swear (applause)—only occasionally. (Laughter.)

I now take great pleasure in introducing to you Dr. Harvey W. Wiley, who will speak on the subject “The Conservation of Man.” (Applause.)

Dr. Wiley—The National Conservation Congress has at its previous meetings discussed in a most illuminating and helpful way the great problems of Conservation as applied to the soil, to the forests, to the mines, and to the running streams.

I do not suppose it is proper, with an audience of this kind, to refer to earlier papers, but I do believe I am the first person who ever made a public address in this country upon the subject of Conservation, and I am certain, as far as I know, that I am the last one that is making such an address. But as long ago as 1893 and being a very old man, as you have heard, I can remember that far back—I made an address on the conservation of the soil, so I am really the father of the conservation movement in this country as well as of a very fine boy. (Applause.) I miss my dear friend, Gifford

Pinchot, whom I love as a brother, but who has fallen into the patent medicine habit and is giving us "absent treatment." I am not at all sure that he is doing a better work out there than he would be here. In the words of the Scotch poet, "I hae ma doots." But still we were glad to hear from him and know he has not lost interest because of the strenuous political life he is now compelled to lead.

With this great work, from its inception, I have been in deepest sympathy and have collaborated in such a manner as I could to further it. The work accomplished has produced benefits which are difficult to measure by any standard which can be properly appreciated. The American people have come to believe in the application of a single standard of value and this is a scientific principle with which, as a rule, I would have no quarrel, but unfortunately the single standard which Americans have been taught to value is that which pertains to the almighty dollar. The Conservation Congress, however, has not been blind to the fact that the standards of ethics, health, morality and happiness are of even far greater value than that of money. Nevertheless, in order to present the subject in a manner easily grasped by the American people, attempts have been made to measure the value of health and life by a money standard. As a justification of this, we have the procedures of the courts, based upon statutory enactments, which fix a money value upon life, although in many cases, after mature deliberation, it has been found that the life for which compensation has been asked, was of small value. In like manner, in the treatises which have been written on the public health and its value as a national asset, it has been attempted to portray in dollars the most precious of all human possessions, namely, life. And, in point of fact, it is not wholly unscientific, though undoubtedly unsentimental, to thus value human existence. All useful members of a community render services of some kind, for which payment is made in the coin of the realm. Following one of the established customs of great financial operations, it has been customary to capitalize the human life on its earning capacity, either active or prospective. The infant and the child, measured upon an actual earning capacity, would have practically no value, but this would be an unscientific method of determining worth, because of the fact that the infant and the child represent the necessary preparatory stages of earning capacity. Based upon this fact they both have a real monetary value.

I shall not take up the time of this address with any effort to ascertain the actual values which may safely be assigned to the infant, the child, and the grown-up person. This has been carefully and sufficiently accomplished by other investigators. Abraham Lincoln said that in so far as efficiency is concerned the human race may be divided into three classes, namely, one, those who work effectively; two, those who work to no purpose, and three, those who do not work at all. Judging by rigid standards which have been set up by students of efficiency, class one is probably the least numerous of the three. Class two is composed of well-meaning people who do work, are willing to work, and anxious to work, but who do not know how, and therefore waste their energies. Class three is made up of the idle rich, the idle poor, and that considerable portion of our population incapacitated by disease or otherwise exempt from taking part in any useful employment.

FUNDAMENTAL PRINCIPLES OF THE CONSERVATION OF MAN.

Primarily, in the study of the conservation of human efficiency, that is of man, man himself and knowledge of what he is, and what he has been, within the years in which man has been studied, in a scientific way, is of the utmost importance. Unfortunately, we have not access to a universal system of demography, inasmuch as only a few countries have adopted scientific demography in its entirety. The world descriptions of human life, health, and efficiency are, therefore, exceedingly fragmentary. We are too apt to base our ideas upon personal acquaintance and knowledge of the efficiency of man, than upon a scientific study thereof, and yet, in order to have a proper view of the subject of the conservation of man, the actual state of his health and his capacity for useful labor must engage our attention.

The Division of Vital Statistics of the Census Bureau has done much to furnish the student of humanity with fundamental data, and first of all let us consider what is the expectation of life in the various countries according to the latest authorities which can be secured. The Division of Vital Statistics has prepared the following table, which is to be accepted as the most authoritative which is accessible. No claim is made, of course, for entire accuracy, but it is sufficient to show what the condition was in this country twelve years ago. It is reasonable to suppose that conditions have improved somewhat in the twelve years which have passed since the compilation of the data submitted.

EXPECTATION OF LIFE IN VARIOUS COUNTRIES ACCORDING TO LATEST LIFE TABLES.

(The “expectation of life” is sometimes known as the “mean after-life time,” “average after-life time,” “mean duration of life,” and “average duration of life.” Data are from the international tables in *Statistik des deutschen Reichs*, Bd. 299, *Siechetafeln*; the French *Statistique internationale*; the English Registrar-General’s Report; Supplement, 1891–1900, and Census Bulletin No. 15, Twelfth Census, Tables for the United States, or rather for that part of it having fairly complete registration of deaths, will be published in connection with the Reports for 1910, now in preparation.)

EXPECTATION OF LIFE IN YEARS.

		Males.			Females.		
Country or State.	Years.	At Birth.	One Year.	Ten Years.	At Birth.	One Year.	Ten Years.
England and Wales	1891–1900	44.13	52.22	49.63	47.77	54.53	51.97
Healthy Districts	1891–1900	52.87	59.13	54.16	55.71	60.53	54.46
France	1901	45.31	53.10	49.25	48.69	55.34	51.53
Italy	1899–1902	42.83	50.67	51.25	43.17	50.08	51.00
Austria	1900–1901	37.77	49.17	48.22	39.87	49.31	48.54
Belgium	1891–1900	45.39	53.51	50.32	48.84	55.88	52.78
The Netherlands	1890–1899	46.2	54.8	51.7	49.0	56.2	53.0
Sweden	1891–1900	50.94	56.25	52.79	53.63	58.04	54.61
Massachusetts	1893–1897	44.09	52.18	49.33	46.61	53.58	50.70
German Empire	1891–1900	40.56	51.85	49.66	43.97	53.78	51.71
New South Wales	1891	49.60	——	50.89	52.90	——	53.39
India	1901	23.63	——	34.73	23.96	——	33.86
District of Columbia (white)	1900	41.64	49.30	46.37	45.77	52.89	49.90
Massachusetts (white)	1900	44.29	53.13	50.15	47.80	54.96	51.70
New Jersey (white)	1900	44.06	52.05	49.27	48.27	54.45	51.59

One of the most remarkable facts presented by the above table is in the marked increase in the expectation of life after the age of one year. In other words, the terrible infant mortality, which prevails in all countries, is so great that the expectation of life at birth is a number of years less than at the age of one year. In England and Wales, the infant mortality decreases the expectation of life at birth, in round numbers by eight years; in France and Italy about the same; in Austria, by eleven years; in Sweden, by six years; in the German Empire, by eleven years; in Massachusetts, by nine years. In the report, of the Bureau of the Census on Mortality Statistics, printed in 1909, and referring to the calendar year 1908, data are collected from seventeen States, the District of Columbia, and seventy-four registration cities, comprising a total of 51.8 per cent. of the total estimated population of the country. The total number of deaths registered in this area in 1908 is 691,574, corresponding to a death rate of 15.4 per 1,000 of population, which is said to indicate a remarkably favorable condition of the public health.

In the mortality statistics for 1910, two years later, the registration area, which included in 1910 an estimated midyear population of 58.3 per cent. of the total population of continental United States, the deaths reported were 805,412, representing a death rate of 15 per 1,000 population. The death rate for 1909 was only 14.4 per 1,000. While these variations are marked, the work has not been carried on for a sufficient length of time to do more than to warrant an expression of opinion that the death rate in this country is generally receding. It varies as shown, on both sides, having decreased very considerably from 1907 to 1909, but increased to a very marked degree in 1910 over 1909. The registration area covers the following States in toto, and some of the principal cities in the other States: California, Colorado, Connecticut, Indiana, Maine, Maryland, Massachusetts, Michigan, Minnesota, Montana, New Hampshire, New Jersey, New York, North Carolina (municipalities of 1,000 population and over in 1900), Ohio, Pennsylvania, Rhode Island, Utah, Vermont, Washington, and Wisconsin.

The extension of the system of registration to a larger area and number of population and the improvement in the efficiency of securing data are all to be considered in comparisons of very small periods of time. For one hundred million of population a death rate of 15 per 1,000 indicates a total of 1,500,000 deaths per annum. This figure may be accepted as being sufficiently accurate for all practical purposes at the present time as representing the death rate of today in the United States.

Comparing the United States with other countries and giving the expectation of life at birth as the basis of comparison, we may safely assume that the average expectation of life for the United States is in round numbers 44 years. Comparing this with the other countries we find that Sweden, Holland and New South Wales have a lower death rate than the United States. England, France, Belgium and Holland have almost the same death rate. The German Empire and Austria have a higher death rate. India is the banner country for shortness of life, the expectation of life in India being a little over half that in the United States.

WHAT ARE THE DISEASES WHICH ARE MOST ACTIVE IN CAUSING THE DEATH OF OUR PEOPLE?

In the registration area of 1910, 154,373 infants under one year of age died, in round numbers one-fifth of all the deaths. Assuming the total deaths to be 1,500,000, the number of children dying in the United States every year under the age of one year is 300,000. A striking illustration of the danger of the hot months for children under 2 years of age is shown by the fact that the number of deaths from diarrhœa and enteritis for July and August was 12,535 and 12,565 respectively, while in February the deaths from the same causes were 1,373. From these data it is evident that during the hot two months nearly 1,000 infants under the age of one year die every day in the United States.

The report of the Division of Vital Statistics shows that beginning with the second month of life diarrhœa is the most serious cause of infant mortality. While infantile diarrhœa and its allied disease, enteritis, is the most frequent cause of death among infants, the greatest destroyer of the human race, without respect to age, is tuberculosis, which caused 10.7 per cent. of the deaths from all causes in 1910. Next in importance in destructiveness is found organic disease of the heart, causing 9.5 per cent. of all the deaths. For all ages diarrhœa and enteritis come third in fatality with 7.8 per cent. Close after this comes pneumonia with 6.7 per cent. Kidney disease causes a mortality of 6.6 per cent.

The number of deaths from tuberculosis during the year 1910 was 160.3 per 100,000, or for 100,000,000 people 160,300. The death rate from tuberculosis from 1900 to 1909, inclusive, was 183 per 100,000. Apparently the death rate for tuberculosis is decreasing.

The number of deaths from cancer in 1910 was 76.2 per 100,000, or a total of 76,200; the highest death rate ever recorded from cancer. Evidently the deaths from career are increasing in proportion to the population.

I wish sometimes that every house in this country could be burned to the ground, if the people could escape. Why? Because tuberculosis and cancer are house diseases, and if every house were burned, we would not have them any more—at least until we built new houses. But we can purify our houses, we can live out doors, we can sleep out doors most of the year, and by the teaching and practicing of the principles of hygiene and sanitation we need not burn our houses at all. But people do not know, and worse than that, they do not care. They take no interest in such things. If you were discussing the tariff tonight, the house would not hold the people; if you were discussing trusts, there would be no standing room; but when you discuss this tariff on human life—they are not interested.

Organic disease of the heart: The number of deaths in 1910 was 141.5 per 100,000, which is a very large increase over that of the preceding year of 129.7 per 100,000. The total number of deaths from heart disease was 141,500.

Pneumonia: The death rate from pneumonia for 1910 was 147.7 per 100,000, making a total of 147,700 deaths from this disease. The death rate from this disease increased considerably over that of the preceding year.

Kidney disease: The total number of deaths from kidney disease in 1910 was 99 per 100,000, making a total of 99,000 for an estimated population of 100,000,000. This includes all forms of kidney trouble, nephritis and Bright’s disease.

Typhoid fever: The death rate from typhoid fever was 23.5 per 100,000, a total of 23,500 for the estimated population of 100,000,000.

You older men like me who were in the war know that war is hell—not because you are shot—that is glory; but because you die of disease; and if you will read the military history of the Civil War, so-called (I do not know why, for it was not so very “civil”) you will see that while one man died of wounds, four died of disease, because we did not understand the principles of serum prophylaxis. We are not going to have in the next war four men die of fever where one is killed in battle.

One of the curious features in connection with typhoid fever is that some of the most sparsely settled States show the highest rates of fatality, for instance the number of people dying in Colorado of typhoid fever is 41.9; in Montana, 39.9, and Utah, 37 per 100,000. Only one of the thickly populated States equals this—Maryland, 40.7 per 100,000. Some of the lowest death rates for typhoid fever were found in New Hampshire, 10.7; Massachusetts, 12.4; Rhode Island, 13.6; Vermont, 14; New Jersey, 14.5, and Connecticut, 14.7. Of cities of 100,000 population or over in 1910, Omaha, Nebraska, showed the highest rate, namely, 86.7; Minneapolis, Minn., 58.7; Kansas

City, Mo., 54.4; Atlanta, Ga., 50.1; Birmingham, Ala., 49.5; Nashville, Tenn., 48.9; Milwaukee, Wis., 45.7; Spokane, Wash., 45.4, and Baltimore, Md., 42. The lowest rates shown for some of the large cities were those of Bridgeport, Conn., 4.9; Paterson, N. J., 7.1; Cincinnati, O., 8.8, and Cambridge, Mass., 9.5 per 100,000. These cities seemingly are as well protected against typhoid fever as some of the cities of Europe, where death rates are as follows: London, 4; Edinburgh, 2; Dublin, 10; Paris, 7; Brussels, 19; Amsterdam, 7; Copenhagen, 3; Stockholm, 4; Christiania, 2; Berlin, 4, and Vienna, 4 per 100,000. Thus, evidently in such cities as Cincinnati, Berlin and London, death from typhoid fever is no longer a terror.

Measles, which is supposed to be almost a harmless disease, causes a large number of deaths, the death rate for 1910 being 12.3 per 100,000 population, or a total of 12,300 for the estimated population. In some cities the number of deaths by measles was almost as high as that by typhoid fever, notably in Pittsburgh, Pa., 33.1; Providence, R. I., 31.9; Kansas City, Mo., 28.4; Lowell, Mass., 28.1; Albany, N. Y., 23.9; Columbus, O., 23.6; Buffalo, N. Y., 22.1, and Richmond, Va., 21.1 per 100,000. Scarlet fever is not so deadly a disease as measles, since the fatalities per 100,000 for 1910 was 11.6. Death rates from this disease were high in the following cities of 100,000 population or over: Buffalo, N. Y., 53.6; Lowell, Mass., 41.2; St. Paul, Minn., 30.2; St. Louis, Mo., 27.1; Kansas City, Mo., 23.2; Milwaukee, Wis., 22.3; Pittsburgh, Pa., 22.2; Rochester, N. Y., 21.4, and New York, N. Y., 20 per 100,000.

Whooping cough produced as many deaths as measles and scarlet fever, the death rate for 1910 being 11.4 per 100,000 population. Diphtheria and croup produced a death rate of 21.4 per 100,000 population, or a total of 21,400 for the estimated population.

Influenza, or “la grippe,” caused a death rate of 14.4 per 100,000 population for 1910. This disease is less prevalent than for the preceding ten years. The above data are sufficient to show the principal causes of death, old age, unfortunately, being so small a factor as to be almost negligible in the compilation.

It might be interesting to extend these vital statistics to a greater length, but a sufficient number of data have been given to establish some of the fundamental principles which should guide physicians and the sanitarians of the future in their work.

THE MEANS OF AVOIDING AVOIDABLE DEATH.

The question which is now presented for discussion at this Congress is, How can avoidable death be successfully avoided? I have not included in the discussion of this question the deaths by accident, which are lamentably all too frequent in this country. The motor car, the aeroplane, the railway, and the steamboat, still continue their deadly work in increasing violence as our population grows denser. It is easy to understand how the State could do much toward preventing these unfortunate accidents. No doubt concerted action on the part of the States will soon be perfected to prevent so many of the horrible catastrophes, whose descriptions form the principal reading matter, after murder and suicide, in the morning journals. And this leads us to say that murder as a means of ending human life is more prevalent in this country than in any other country of the world, and in consideration of the features which relate to the conservation of man the prevention of murder should receive particular attention.

A study of the above data reviewed in connection with the known etiology of disease, shows clearly where the work of the conservation of man, especially by the prevention of disease, should begin and on what line it should be prosecuted. To this end it is sufficient to call attention to the fact that diseases are naturally divided into two classes: those which are communicated and those which are produced by the conditions of the personal environment. Physicians are pretty well agreed at the present time that disease is rarely inherited, therefore, most of the causes which produce death are those which come from without, or those which are developed from within by improper habits of life. But one may inherit deficient vitality and thus fall an easy victim to an infectious disease. The point for us to consider most particularly in this connection, is to what extent we can prevent these diseases, that is, those which are contracted from without.

EDUCATION OF FUNDAMENTAL IMPORTANCE.

It would be well to classify the efforts which we are making for the prevention of disease in some systematic order. I will begin, therefore, with the one which is the most important of all, and that is education.

In order to secure proper protection for the citizen, he must be made to understand that he needs it. Further than this, it must be made plain that the protection of the individual from communicable disease is not by any means wholly within his own power. Unless the State acts, the individual in many cases is powerless; hence education beginning in the family, continued in the public school, and illustrated in practical adult life, is the most important feature of prophylaxis. Into the details of education I cannot go, but one thing I do with to insist upon, namely, that the child should be taught early, frequently and constantly, that most of the disease he has to fear are like enemies in the dark. I need not refer again by detail to the statistics of mortality, but simply would say that if the diseases which produce some of the most deadly inroads into humanity, such as tuberculosis, measles, whooping cough, scarlet fever, diphtheria, croup and typhoid fever, are solely communicated to the individual from without, they are the diseases which the State must help the individual to avoid. On the other hand, organic diseases of the heart, nephritis and Bright's disease, are apparently more of a personal character, due either to inherited weak qualities or to errors of diet or faults of metabolism. These are diseases which we should be taught to avoid by strict attention to personal hygiene. They are not, so far as known, communicable, and therefore the State can do little, aside from educational work, towards their prevention. Another disease which may be partly communicated and partly the result of improper nutrition, is enteritis, and especially infantile diarrhœa, diseases which by proper education might be almost wholly avoided.

DISEASES OF UNKNOWN GENESIS.

There remain two great causes of human death, namely, cancer and pneumonia, which are still practically beyond control, because of our ignorance of their etiology or our powerlessness to prevent their progress. These diseases are considered communicable, that is, they are induced by specific infection, but the methods and the exact nature of the infecting germs are still subjects of investigation. It is true that we are told of the organism which produces pneumonia, and it is said to be constantly in the mouth of even healthy people, and we read almost monthly of the discovery of the real cause of cancer, but in spite of all this, these diseases remain as a rule unknown in character and are gigantic and terrible enemies which we have to fight in the dark. To one point attention should be called in regard to the increase in such diseases as those of the kidneys and the heart, that are essentially diseases of old age, just as tuberculosis and typhoid fever are diseases of early life. In proportion as we save people from tuberculosis and typhoid fever, just in that proportion will we save men and women who subsequently become victims of old age diseases. Therefore the increase in the number of deaths due to these causes may be an index to the increasing longevity of the people, instead of the opposite.

It is of course a question, which unfortunately we are unable to decide for ourselves, as to whether we should be saved from tuberculosis and typhoid fever for the express purpose of being killed by cancer, kidney lesions and diseases of the heart. Upon the whole I think, however, that terrible as these diseases are, especially cancer, most people would rather die of cancer at 70 than to succumb to tuberculosis at 30. But in the great problem of the conservation of human life we must not lose sight of the fact that many experienced and competent investigators are devoting their whole time to revealing the secret of these dread diseases, which still baffle the skill of the physician. We may hope in the near future that at least pneumonia and cancer may be put upon the same footing as typhoid and tuberculosis, that their actual genesis will be disclosed, and thus the road made clear toward their prevention. It is along these lines that education must go, because we cannot develop a public sentiment for the protection of life and health except by the desire of the people to live and be well, and the education of the youth and the adult is the best method of securing that result. When the people are educated, then we can successfully introduce the other methods of saving human life.

PREVENTION OF COMMUNICABLE DISEASES.

It is a self-evident fact, granting a disease to be of communicable origin by a specific germ, that the disease may be prevented if its victim be protected from infection. In other words, such diseases as tuberculosis, typhoid fever and others of the same character, which are undoubtedly communicated from individual to individual, could be wholly exterminated if the opportunities for communication were destroyed. We may assume, therefore, that all specific diseases due to a specific organism are capable of elimination by the simple exclusion of the organism.

Based on this are the great factors of prevention, namely, quarantine and segregation, which are practically one and the same. It stands to reason that an infected center should be removed or so isolated as to be no longer dangerous. For the same reason the infected center should not be allowed to enter a new community. Based upon this principle our systems of quarantine and segregation should be greatly strengthened. It is not a question of the wishes of the individual in this case; if it were, no ship would be detained and quarantined, and few people would go to a smallpox hospital or tuberculosis sanitarium. The principle of the welfare of the race as superior to the interests of the individual is dominant in these particulars. Tennyson, who foresaw many of the great truths of science, has beautifully presented this principle in his well-known stanza:

> “Are God and nature then at strife,
> That nature sends such fearful dreams?
> So careful of the type she seems,
> So careless of the single life.”

In the protection of the public health it will become as much the duty of each State and Nation to provide sanitary detention camps for infectious diseases and rigidly enforce residence therein, as it is to watch the border and establish strict quarantine.

IMMUNITY.

It is evident, however, that it will take a long course of education and almost revolution in the sentiment of the people, to establish a system of segregation and quarantine as rigid and as perfect as that which is outlined. What then is the next best resort? I answer immunization. If we cannot keep the infectious organism from contact with the human body, we should endeavor to make the body immune from its ravages. There are two methods which might be adopted; the one which could be most generally practiced is that of good nutrition, proper housing, fresh air, pure water and pure foods. The child that sleeps in the open and eats an abundance of pure, wholesome foods and takes a proper amount of exercise, will escape most of the diseases of infancy and grow into manhood with a body immune to almost every infectious germ. I need not go into detail in regard to the actual mechanism of immunity to prove the fact that a well-nourished body, sustained by blood of high nutritious power and bearing its untold millions of organisms, armed cap-a-pie to destroy intruders, is a sufficient illustration of immunity. The physiologists will describe to you the nature of the phagocytose opsins, and the hormones by means of which this immunity is secured.

For the above reason the campaign for pure and wholesome food lies at the very foundation of the protection of the public health. It is a mistaken idea that a food is not to be condemned unless it produces diseases. A food is to be condemned which is in any way so debased as to undermine nutrition and impoverish the blood, and thus open the door of the body to the invitation of every germ that may be coming along the road. Thus the addition to foods of bodies which in themselves are not poisonous or harmful, but which debase the product and make it less palatable or less nutritious, is a crime of the same magnitude as that of adding to the foods poisonous and deleterious ingredients or of suffering it to fall into advanced stages of decomposition.

What a sorry spectacle, in the light of these facts, was presented at the Fifteenth International Congress of Hygiene and Demography at Washington last week, when Professor Long, member of the Remsen Board, which has validated the use of some of these poisons, attempted to justify the addition of an active drug to the food supply of the nation! Such an act was so foreign to the purposes of the Congress as to constitute an unpardonable anachronism. Dr. Long was one of the most enthusiastic protagonists of benzoate of soda in the Federal Court in Indianapolis when those who secured the appointment of the Referee Board in defiance of law sought to force the people of Indiana to eat their adulterated products. The people ask for bread and Dr. Long and his assistants give them a stone in the form of the moribund benzoate.

Of a similar pernicious and mercenary character was the paper presented by Professor Sedgewick, of Boston, in which he urged the use of infected oysters and diseased meats as human foods. Professor Sedgewick was one of the principal witnesses in the celebrated egg case in New Jersey, where he testified that eggs so decomposed as to produce death when injected into guinea pigs were wholly fit for human food if sufficiently disguised in taste and smell by baking! Oysters, according to Sedgewick, should be classified into good, to be eaten raw by the rich, and bad, to be cooked and eaten by the poor. Meats of diseased animals should also be eaten by the poor, unless so badly diseased as to be physically seen to be unsound.

This is the doctrine of modern hygiene according to its prophets Long and Sedgewick. I cannot subscribe to these doctrines. There is plenty of clean food for both rich and poor. To excuse processes of growing food animals, and manufacturing foods which permit and condone unsanitary methods and introduce active drugs into the finished products, stimulates and encourages reprehensible practices, which all interested in the public health should condemn. Happily the Federal courts, both in New Jersey and Indiana, were unconvinced by such specious arguments, and condemned the very processes which were praised and defended before the world's congress of sanitarians.

The workers for the conservation of man do not yet fully realize the great importance of the food supply of the country as a means of producing immunity of disease. The well-nourished body is clad in armor and bears an impenetrable shield which enables one to march into the midst of dangers and for the most part escape unscathed. All power and ethical spirit, therefore, to the men who are chosen to administer the food laws, in order that they may realize the importance of their office to the health of the people, and the life and efficiency of our citizens. Let them learn to put a heart and a soul into science.

IMMUNITY OF HEREDITY.

We are all familiar with the common phrase, the foundation principle of eugenics, "He inherited a good constitution." It is undoubtedly true that we come into the world with widely different vitalities. The true principles of scientific immunity to disease therefore lie imbedded in the human life principles of long past aeons. The ideals of eugenics, as formulated by Francis Galton and elaborated by his nephew, the son of the immortal Darwin, are but irridescent dreams. If man is to be bred scientifically, there must be many selected mothers and a very few high grade fathers. The human race is not yet ready to face the problem in the true light of science and contemplate a race of males of which 75 per cent. are eunuchs. This is kako- instead of eu-genics. As long as the heart is whole, men and women with only one lung will fall in love. For untold centuries to come we must be resigned to human race composed principally of scrubs. But there is one principle of eugenics which can be and ought to be put into practice. It has been done partially in some States, especially in Indiana. It should be generally adopted. The degenerate, the vicious and the imbecile should not be allowed to propagate. These are classes of society that have no right to multiply. Before proceeding further in restricting parenthood let us see that individuals of both sexes, criminally vicious or imbecile, are segregated or rendered impotent. And even here only the typically bad cases are to be treated. It would be too nice a question for the jury if there was a doubt of any kind, even inconsiderable. Among those of average intelligence, education should do the rest. Teach those who are physically diseased the duty of celibacy. Persuade and not force them.

INDUCED IMMUNITY.

Another method of securing immunity in the human organism is by the development of some morbid condition of a nature similar to or identical with the disease to be combatted, so as to produce in the system anti-bodies, specifically adapted to fight the particular disease which has generated them. The principle of immunization by this method rests upon the successful experiments, or rather observations, respecting a given virus. Jenner's observations in regard to smallpox were purely empirical, and it remained for Pasteur to develop a scientific basis of induced immunization. Serum-therapy is by no means half so important as serum-prophylaxis, and here again comes the importance of education, because there is still a very large and respectable body of our citizens who resent any interference on the part of the State with their rights as regards medical relations. It looks almost like tyranny to force a citizen to subject himself to inoculation of any kind when his own belief in the efficacy of the process is hostile and where he resists enforced immunization. But here again the right of the people asserts itself and thus justifies compulsory vaccination. While education can do much to remove this prejudice, we must expect to always have with us those who conscientiously resent inoculation, and condemn all efforts to prevent disease.

Since, because of lack of care and proper supervision, grave disorders and disease and sometimes death result from the practice of inoculation, the State owes a special duty to its citizens in seeing that all forms of inoculation materials, no matter what their nature may be, are of the purest and best. Of course, the thought presents itself that induced immunization is only a confession of inability to protect the health by isolation of the invading virus. It is something like the pasteurization of milk, which is a mute tribute to insanitary conditions, uncleanly cows, and long keeping; but here it seems that there is no choice left. The impossibility of complete isolation, at least for many years to come, is apparent, and hence the desirability of general immunization becomes obvious. The successful inoculation which has lately been accomplished against typhoid fever is another promise of what the future may bring in the way of immunization by induction. Meanwhile it is the part of wisdom for those who seek the public welfare by the conservation of life to urge both prophylaxis and immunization, in the hope that the infecting centers will become so few and so remote that good nutrition, and all that it implies in a sanitary way, will eventually become a sufficient protection against communicable diseases.

THE SUPERVISION OF DRUGS.

Hand in hand with the supervision of our food supply, we should not forget the control of drugs. I am far from believing that drugs are an efficient remedy for all human ills; in fact, I am convinced that they are not. They are at best only adjuncts, except in those cases where specifics have been discovered, as in the case of quinine and malaria, and the arsenic compounds, which have proven so useful in combating syphilis. But without discussing the efficiency of drugs, I think we will all admit that as long as they are articles of commerce they should be pure and of constant strength. To this end we should support, with all our enthusiasm and ability, the efforts which are made to perfect the pharmacopœia, and to standardize and purify the drugs of commerce.

THE CONSTANT THREAT OF PROPRIETARY MEDICINES.

In this connection I cannot refrain from alluding to one of the greatest dangers of drugs, and that is, their indiscriminate use by the laity. The fakers that pretend to find sovereign remedies for every disease, through the medium of the newspaper and the periodical, of the postal card and the circular, inflame the minds of the people and induct them into indiscriminate drugging. One can generally, by taking up a paper in any locality and scanning its columns even carelessly, see the wonderful vogue of these fakes and crimes. Such falsely praised substances as Peruna, Kilmer's Swamp Root, Duffy's Malt Whiskey, and the whole brood of wretched specifics, serve to illustrate the great danger to which we are subjected. But the worst of it all is that through the carelessness of physicians, and sometimes through their criminal pretentions, habits are formed for certain drugs, such as cocaine, opium and its products, chloral and alcohol, which enslave their victims, weaken their vitality, and invite disease. I think I do not exaggerate it when I say that the drug habit, no matter how induced, is a menace to the American people. No matter how slight the ailment or how easily controlled, the first advice and the first act is to "take something," no matter what, or whoever may recommend it, for every imaginable ailment. The effect of this continual drugging upon the human body is more easily imagined than described. The nerves and stomachs of our people are gradually succumbing to the bombardment of pills, pellets and powders. For the sake of gain every possible influence is brought to bear upon the American people to increase the consumption of drugs. The danger is so imminent and so acute that it is hoped that through the means of education a public sentiment may yet be awakened in this country which will protect our people against all these nefarious concoctions. I would not for a moment in any way curtail the right of citizens to consult accredited physicians, no matter to what so-called school they might belong; but it is the duty of the State, as an additional safeguard, to the health and life of our people, to see to it that no one sets himself up as a physician unless he has qualified himself in the fundamental principles of anatomy, hygiene and physiology, to understand the human body and its operations. We are too prone to tolerate physicians who tell you that the blood which supplies the brain passes into the cranium altogether through the canal of the spinal cord. Charlatanry, quackery, and ignorance in the practice of medicine should be rigidly suppressed. The people of the nation who have freedom of choice should not be left helpless victims of avarice and ignorance.

DANGERS OF STIMULANTS.

In addition to drugs, as commonly considered, the people of our country are also subjected to imminent dangers in the use of stimuli, which have no food value and which induce activities that are beyond the power of the system to sustain. I refer especially to such beverages as tea, coffee, and alcoholic drinks and the manufactured articles containing their active principles, such as coca cola and all the great army of "olas," and to tobacco, as an illustration of additional dangers to which we are likely to succumb. In spite of the fragrance of the coffee, and the aroma of the tea, and the flavor of the rum, and the dreams of the pipe, I am inclined to the belief that it was a sad day for humanity when these things were first brought to the attention of man. In so far as intellectual development is concerned, I find the nations of antiquity, and especially the powerful nations of Greece and Rome, developed to be leaders in architecture, masters of painting and sculpture, and geniuses in poetry and expression, without the aid of any of the stimuli which the artist, the poet and the writer are supposed to depend upon today.

It would indeed be a happy day for the community if all of these stimuli, as appetizing as some of them are, could be relegated to the scrap heap, and the art of their use forever lost. (Applause.) Meanwhile, we all understand that this Utopian condition is at present impossible, and hence we must content ourselves with education and with legal control to prevent the abuse of these bodies and to eliminate the injury which they have done. Temperance may always be practiced, even where prohibition fails. It is therefore the duty of every one concerned with the public health to urge the extremest moderation in the use of tea, coffee, tobacco, and alcoholic beverages, in the hope that the injuries which have already been wrought may be avoided in the future, and temperate indulgence take the place of unbridled consumption until the day of final elimination arrives.

SUMPTUARY LAWS.

In the interest, therefore, of the public health and the lengthening of life and increasing the efficiency of man, we must bring ourselves to the point of acknowledging that the State should control things which in themselves are injurious and unnecessary must be established. In other words, the individual’s rights, so dear to every lover of freedom, the cardinal principle of democracy, must give way to the public good. No one has any right to practice any habit, or induce others to do so, which in itself is likely to prove injurious to humanity. I would leave to the individual the largest freedom in everything that is good, and restrict his activities to the lowest minimum in everything that is bad. I would not make of man a machine, nor would I desire that he should live in an environment which in any way would tend to affect his evolution and progress injuriously, and so I preach what seems to me the only solution of all these evils—education, temperance, legal restriction of abuse, and leave the rest to the manly part of humanity.

If I can in my life just put one nail in the coffin of quackery and false medicine, I will not have lived in vain; if by my voice I can get one man or woman interested in a healthy way of living, my work will not be in vain; if I can save one infant from premature death, my life will be well spent.

I believe when you conserve a man physically you conserve him mentally and morally, and then sin and sorrow and suffering will pass. There are only two learned professions in the world that are necessary—one is agriculture and the other is teaching. If you feed men right and teach them right, there will be no law breaking, and hence we will need no lawyers; there will be no sickness, so we will need no physicians; and when you have a country that is so happy as to have no law breakers or sick people, you will not need anybody's help to get you into heaven, so we can do away with the ministers. (Laughter.)

THE PROLONGATION OF LIFE.

What is in sight in the way of prolonging human life? I have briefly laid down what seems to be the fundamental principles of the conservation of man and the prevention of disease. If this plan can be carried out, is there any hope to be offered to man of greater freedom from disease and a longer life? I answer unhesitatingly in the affirmative. Why should we be content with an average life of 44 years? There is historical evidence to show that man's greatest activities are developed with experience and that the age between 60 and 70 is more productive for one who has lived in accordance with nature. It is shown from statistics that we die sixteen years before we reach the maximum usefulness of man. I would like to see more old age. I would like to see more men and women with gray hair and more wrinkled faces than I can see today. To all this, objection may be made that a place must be made for the young man and young woman; that the old man and woman keep the young from development and usefulness. But to this I reply, that there is infinite opportunity for good work offered to all. If we can secure a race free from disease, endowed with all those qualities of mind and body which make for human efficiency, we need not ask that every one become eminent and wealthy, but each can perform the duties which come to him in a way to develop a uniform excellence of the human race. We have room in the country for millions of people. We welcome the infant and the child, but let us keep the man and woman. There is room for all.

This is my message to you tonight—the conservation of man—not only his health, but his life, the most precious possession man has. (Applause.)

Col. John I. Martin—I move that the fullest acknowledgment and thanks of this gathering be and are hereby tendered to Dr. Wiley for his very interesting and splendid address.

The motion was seconded by many delegates and carried unanimously.

After announcement by the Secretary, the Congress adjourned until 9:30 o'clock Wednesday morning.

FOURTH SESSION.

The Congress convened in the Murat Theater, Indianapolis, on the morning of October 2, 1912, and was called to order by President White.

President White—I want to take this occasion to state that there is a child's welfare exhibit richly worth seeing. It is an education in itself and is installed at the Capitol building. Every one should embrace the opportunity of seeing it.

The audience will now arise while the Rev. Dr. Harry G. Hill, of Irvington, invokes Divine guidance.

INVOCATION.

Father of us all and maker of all that is good, source of all light and life and love, early upon this morning, the second day of this Congress, we bring our respects to Thee and bow before Thee as the One worthy of worship, and invoke Thy blessing and benediction upon all who meet with us. We thank Thee that Thou hast so bountifully blessed us, and would have Thee, through these ministrations, at this time impress upon our minds that we are stewards of a great wealth, and we ask Thee to help us that we may so minister that there may be an equal distribution to all Thy people of the great goods with which Thou hast endowed us. May we hold above everything else the wealth of human life, and may we look to our work as that of making a better world, a better place for men.

May Thy blessing rest upon the deliberations of this hour, on all those who are participating in this Congress, and may it go on and do much good in the years that are to come, that Thy knowledge shall be in the hearts and minds of men, and they shall serve Thee and make this is a great opportunity to increase Thy rule and kingdom through Christ, our Lord and King. Amen.

President White—In the study of Conservation in this Congress we are getting around to the fundamental basis of all vital conservation. We are getting to the point where Conservation should have first begun—the study of human life as a national asset. It was Pope who said “The proper study of mankind is man.”

I take please in introducing to you, as the first speaker of this morning session, one who has had a great deal to do in the actual figures, the actual statistics, the actual knowledge of why human life is a national asset and why it should be conserved. I take pleasure in introducing to you Mr. E. E. Rittenhouse, of New York City, Conservation Commissioner of the Equitable Life Assurance Society of the United States, whose subject is “Human Life as a National Asset.” (Applause.)

Mr. Rittenhouse—The National Conservation Congress has been engaged in the noble task of guarding posterity against the waste of our natural resources by the present generation. It has had a most far-reaching influence, for its purposes are in tune with public sentiment, and with the spirit of the age. It has now given another and still more commanding reason for its existence by joining earnestly in the campaign for the conservation of our “human assets.” This is a field of usefulness that will endure for all time. However important the protection of our natural material resources may be, our greatest obligation to posterity is to preserve the health, virility and morality of our race.

The first and most important item in humanity’s Bill of Rights is *the right to live*.

The primary purpose and function of organized society is to guard the lives of its members from needless destruction. Liberty, education, wealth and other earthly blessings are important—but we must be alive to enjoy them.

The nation with the keenest sense of justice and the highest standard of intelligence and morals—virtues which some of us modestly claim for our people—is the one which should place the highest value upon human life and surround it with the greatest protection.

How would our civilization rank by this method of measurement? What have we already accomplished in preventing life waste? What is our present loss? How can it be reduced?

We may well rejoice over the achievements of the patient heroes of the laboratory and of the unselfish and devoted men of medicine who have provided, disseminated and applied the knowledge of prevention so far as it has gone. To them, to the press, the clergy, and the other good men and women who have helped spread the gospel of disease prevention belong the chief credit for the reduction of the death rate by nearly 25 per cent. in the past thirty years.

To these benefactors of our race is also due the honor of initiating and developing the widespread interest which now prevails throughout our country in the conservation of health and life. They have demonstrated that morbidity and mortality can be reduced—that human life can be prolonged by spreading and applying our present knowledge of the science of disease prevention. At the close of last year we had to the credit of these life savers over 400,000 lives that would have been lost that year if the death rate of 1880 had still prevailed.

If the present thirst for knowledge of health and life conservation continues to increase, it is not only possible, it is reasonably certain that during the next thirty years the present death rate of 15 per 1,000 population in the registered area will be reduced to 10.

While we have every reason to felicitate ourselves upon this wonderful result of the spread of life-saving intelligence, we must not overlook these facts:

1. That this great life-saving movement is still in its infancy.

2. That it has been directed almost wholly against preventable contagious diseases, and that the waste of life from these maladies has only been reduced—the loss is still excessive.

3. That while we have reduced the mortality from these diseases common to infancy and early adult life, the degeneration diseases of middle life and old age, against which we have waged no war, have been steadily increasing.

4. That we have increased the average length of human life only by increasing the proportion of people living in the younger age periods, while the average duration of life of those who pass into middle life and old age has been constantly shortened.

In other words, we are still furiously burning the candle at both ends—slower at one but faster at the other.

It is important that this point should be clearly understood. It is natural to conclude at first glance that if we are saving these lives of the younger age period that naturally there are more older people to die, but that does not follow. In the first place, we are dealing with a death rate, the death rate for 1,000 population not in the bulk, and while it is true that the passing of these lives over into the older age period does affect that rate, it only affects it slightly. It has been asserted also that the lives saved from these communicable diseases have been weakened and that they die early after passing into middle life. It is also true that that does not explain the extraordinary increase in the death rate in the older age period. In England and Wales they have the same reduction in the death rate of communicable diseases common to the earlier age period, but not any increase above forty.

With all its blessings modern civilization has introduced hazards, habits and conditions of life which may not only invite, but which have increased in many ways, physical, mental and moral degeneracy.

What excuse have we Americans to offer for the excessive waste of human efficiency and human life from which the Nation is now suffering?

Surely we can not plead ignorance nor poverty, for we have both the knowledge and the money wherewith to stop this annual sacrifice.

How can we explain our growing contempt for the value and sacredness of human life? There is no other civilized country where this greatest of all assets—the most precious gift of the Almighty,—is held so cheaply as in this glorious land of ours.

And why do we continue to view with indifference the constantly multiplying evidences of the mental and physical degeneracy of our race?

We may agree that in the long run the trend of humanity is ever upward, and that this is but a temporary reaction, but can we afford to rest wholly upon the hope that race deterioration will automatically cease when our people have had time to adjust themselves to modern conditions? Wise men doubt it. This problem will not solve itself; this adverse tendency will be checked only when our people are made to see conditions as they actually exist, and are aroused to the need of correcting them.

This is our task. Let us briefly survey it.

In order to measure the effectiveness of the Nation's life Conservation work, and the magnitude of the task remaining undone, we must now compare our efforts not with those of the past, nor with those of other communities or countries, but with our present loss from preventable and postponable sickness and mortality.

What are the principal items of life waste?

What evidence have we of degenerate tendencies? Here are some of them—the estimates are from competent sources and are based upon official records.

AN INDICTMENT.

Our birth rate is steadily declining, and at the same time the span of life is steadily shortening.

Twenty-seven per cent. of our annual deaths are of babies under age 5; 200,000 of them die from preventable disease; about 150,000 of these are under age 1.

To offset this waste of life, large families are demanded. Would it not be well to stop this needless destruction of infants before asking for an increase in the supply?

Of the 20,000,000 school children in this country not less than 75 per cent. need attention for physical defects which are prejudicial to health.

Insanity and idiocy are increasing.

Diseases of vice, the most insidious enemy of this and future generations, are spreading rapidly according to medical men. So far we have lacked the moral courage to openly recognize and fight this scourge.

The alcohol and drug habits are constantly adding new victims to the degenerate list and to the death roll.

Suicides are increasing and now reach the enormous total of about 15,000 annually.

Lynchings and burnings-at-the-stake continue, and are common only to our country.

Attempts upon human life by individuals and mobs under trifling provocation, or none at all, are obviously increasing.

Over 9,000 murders are committed every year, and it is estimated that but an average of 116 murderers are executed for these crimes. We have the appalling estimated homicide record of over 100 per million population as against 7 in Canada, 9 in Great Britain and 15 in Italy.

In the United States the death rate above age 40 has increased steadily for years (about 27 per cent. since 1880), while it has remained virtually stationary in England and Wales.

The important organs of the body are wearing out too soon—the diseases of old age are reaching down into the younger age periods.

The death rate from the degenerate diseases of the heart, blood vessels and kidneys, including apoplexy, has increased over 100 per cent. since 1880. These diseases claim over 350,000 lives annually.

The doctors tell us that fully 60 per cent. of these deaths are preventable or postponable if the disease is discovered in time.

Periodical health examinations would detect these chronic diseases in time to check or cure them, but aside from the efforts of the Equitable Life Assurance Society and another smaller company, no public campaign to educate our people to this vital need is being carried on.

All of our money, all of our energy, seem to be directed against diseases that can be communicated. Is not a life lost from Bright's disease as valuable as one lost by typhoid fever?

The annual loss from pneumonia aggregates 135,000 lives, a large portion of which is due to weakened bodily resistance resulting from these degenerative affections.

Cancer, a baffling disease to which our people in their present physical condition are highly susceptible, claims 75,000 lives annually and is increasing very fast. Deaths from external cancer alone have increased 52 per cent. in ten years.

Pellagra, a deadly plague new to this country, is increasing rapidly in some of our Southern States, and it excites but slight public concern.

Over 150,000 Americans are destroyed annually by tuberculosis. We know how to prevent it, but our taxpayers object to the expense and leave the battle almost wholly to charity.

Nearly a million afflicted people are spreading the poison of tuberculosis to the well, with virtually no official restraint or supervision because of the expense.

Over 25,000 Americans are still sacrificed annually to the preventable filth disease—typhoid fever. About 300,000 suffer from it and are more or less impaired by it.

Other germ diseases are wasting more lives than typhoid and tuberculosis combined. We are warring against them, but compared to the lives still being lost our efforts are feeble and only partially effective.

Over 90,000 Americans are killed annually by accidents and various forms of violence. Our efforts to prevent the steady increase of this waste have failed.

The annual economic loss due to preventable disease and death is conservatively estimated at $1,500,000,000, and our life loss at about $250,000,000.

To prevent fire waste our cities spend through the public service approximately $1.65 per capita, and to prevent life waste, 33 cents per capita.

It is estimated that 1,500,000 of our people are constantly suffering from preventable disease, and that during the next ten years American lives equaling the population of the Pacific Coast and Rocky Mountain States (about 6,000,000) will be needlessly destroyed if the present estimated mortality from preventable and postponable disease continues.

These are the conditions we are asking our people to correct. Is there anything unreasonable in the request?

The money loss is stupendous, but if this does not impress our people, surely they should be stirred to action when they reflect upon the immeasurable sum of sorrow, suffering, poverty, immorality, crime and the hereditary degeneracy which results from this wholesale wrecking and destruction of human life from preventable cause.

RACE SUICIDE.

We are not only reducing the fertility of our race and also shortening the span of life, but we are permitting at least 650,000 lives to be destroyed annually which we could save by the application of simple and well-known precautions.

This is the real race suicide problem.

If we would save these lives, they together with their natural offspring would solve the problem of maintaining an adequate surplus of births over deaths. What we need is not necessarily larger families, but *more* families. A larger number of small families is surely preferable to a smaller number of large families.

THE DOLLAR AND THE DEATH RATE.

The primary duty of conserving our human assets resting with the State, it is obvious that the State must lead in the national movement. It is, therefore, the first duty of every individual and of every unofficial organization interested in this efficiency and life-saving campaign to rally to the support of the public health service.

We must not only teach the individual how to guard his life against preventable disease and accident, we must educate our communities to the need of an effective public health service to enforce sanitary regulations and otherwise guard the health and lives of their members.

But it takes money to carry on a great educational movement, and it takes money to conduct the public health service.

The war against preventable disease and death is therefore in the final analysis, a struggle between the dollar and the death rate.

So far the dollar is ahead. The body politic seems still to prefer a high death rate to a slight and temporary increase in the tax rate.

“How much,” says the American taxpayer, “will it cost to reduce this annoying death rate to the lowest possible limit?”

“About $1.50 per capita at first, much less later on,” answers the health officer, “and you will gain immeasurably by the increase in the wealth and happiness of the community.”

“Very well,” says the taxpayer, “here is 25 cents; we will save two bits’ worth of these lives. The rest will have to die. We have much more important places for our money; we must improve the streets and roads, beautify our cities with much needed parks and public structures. We must improve our harbors and rivers, build canals, and encourage commerce generally. Besides, we are absolutely obliged to use about two and a half billion dollars this year for automobiles, jewelry, candy, alcoholic drinks, tobacco, diamonds and other similar urgent needs of life. What is the loss of a few hundred thousand lives compared to these vital necessities?”

And so the health officer plods along with his two-bit appropriation and naturally runs a two-bit health service. His own fitness and efficiency may be 100 per cent., but the effectiveness of his department only 15 per cent. because of the 25-cent limit.

TRIFLING WITH A SOLEMN DUTY.

National Government.—Of all the money provided by the people for the expenses of the National Government only about 1.3 per cent. is used for the conservation of health and life.

Our national health corps has an international reputation for efficiency and achievement. Although the service is under-manned and its personnel underpaid, the patriotism and high sense of duty of these able and energetic men have spurred them to the performance of the very highest service to their country and to humanity. They have not only jeopardized their lives, but numbers of them have sacrificed health and life in the performance of duty.

Through their discoveries in the science of prevention, they have been the means of saving thousands of lives, not only for one year but for all years to come. They have won the admiration of the American people and deserve their most hearty support.

And yet, when it is proposed to co-ordinate the various public health activities of the Government in order to increase the efficiency and usefulness of this splendid body of men, the interest of our countrymen in this service seems to end with admiration. For notwithstanding our confidence and appreciation we have permitted a small but active body of people who are more concerned in treating disease than in preventing it to block the consummation of this thoroughly sensible and business-like consolidation of the various bureaus under one responsible head.

We have many educational agencies at work throughout the country which are directly or indirectly arousing public interest in health conservation, but this experience emphasizes the need for a permanent central organization to stimulate interested people to back up their judgment with action, and no organization is better fitted to render this invaluable service than this National Conservation Congress.

At the last session of this Congress Dr. Harvey Wiley told you something about the dangers of impure food, drink and drugs, and what was being done to guard the public against them. Your individual interest was excited. How long did it continue? Were any of you inspired to give actual support and assistance in the enforcement of the pure food laws or to any other official public health activity? To be interested and to agree is not enough—again, we must act, individually as well as collectively, and stimulate others to act.

States.—The same lack of practical support of the public life-saving service exists in most of the States. The appropriations for the public health work of our State departments can only be characterized as trifling. The exception is Pennsylvania, which is paving the way for a fully adequate health service, as was explained to you at the third session of this Congress in the able paper of Mr. A. B. Farquhar.

The appropriation for the Pennsylvania State Health Department is about 48 cents per capita. Arkansas makes none at all, the State of New York spends about 1.7 cents; Massachusetts, 4.2; Florida, 10; Indiana, 1.8; Kansas, 2.7; Virginia, 1.9, and so on.

Municipalities.—We have many cities with active and efficient health officers, but there is not a city in this country with an adequately equipped and financed health department. Not one of them has sufficient financial support to successfully perform its task, which must be measured by the preventable sick and death list in each community. And we must not confine this list to contagious affections. It must include an educational campaign against all preventable diseases.

The duty of the State to teach our people, through the health departments, how to avoid preventable disease of all kinds that they may live healthful and productive lives, is just as imperative as is the duty of teaching them, through our schools, how to avoid illiteracy and how to live intelligent and useful lives.

While health appropriations have increased over former years, all of our cities place the value of property far above that of human life in applying measures to prevent waste. Here are a few examples:

In 1911, fifty of our important American cities, with an annual preventable death list of 117,724 people (which means an economic loss of at least $200,000,000) spent through their public service to prevent life waste, an average of 30 cents per capita, and through their fire departments to prevent fire waste, $1.63 per capita.

Here are a few examples: Providence, R. I., spent for health conservation, 11 cents; for fire prevention, $1.99 per capita; Portland, Ore., health, 13 cents; fire, $1.91; Minneapolis, health, 14 cents; fire, $1.67; Louisville, health, 12 cents; fire, $1.36.

In 1910, 184 American cities could spare but two per cent. of their total public appropriations for the public health service. The average for all expenses was $16.54 per capita. Of this but 33 cents was for the public health. Seventy-one of these 184 cities spent less than 15 cents per capita for the public health, and among these are such cities as Quincy, Ill., 2 cents; Lansing, Mich., 5 cents; Rockford, Ill., 6 cents; Scranton, Pa., 7 cents; Bridgeport, Conn., 9 cents; Portland, Ore., 10 cents; Harrisburg, Pa., 12 cents; Jersey City, N. J., 13 cents; Springfield, Ill., 14 cents.

There are many of our largest cities that are well below the average of 33 cents per capita. Among them: Toledo, 15 cents; St. Paul, 17 cents; Minneapolis, 18 cents; Indianapolis, 20 cents; Kansas City, Mo., 20 cents; Milwaukee, 20 cents; Cincinnati, 21 cents; Chicago, 22 cents; St. Louis, Mo., 26 cents; Buffalo, 27 cents; San Francisco, 28 cents.

The natural result of this sort of economy is that the health laws we have are not properly enforced.

How can we benefit from the pure food laws, for example, while we refuse to provide the means of enforcing them?

The great city of New York has an ably administered health department, but it has only thirty inspectors to supervise over 27,000 food dispensing establishments. The request of the health officer for an inspection force of 209 men has been steadily ignored for years.

How do you suppose the meat ordinances of Philadelphia are enforced where the people allow the health department but seven inspectors to watch over 8,000 meat shops and slaughter houses?

How can the eight pure food inspectors in Kansas be expected to enforce the pure food laws in the drug and grocery stores, the meat shops, bakeries, etc., in 800 towns? These inspections must be made frequently to be of any value.

These are not exceptions, they are examples.

Could anything be more absurd from a business point of view than this record of “economy” in providing for the public life-saving service?

HOW SOME COMMUNITIES SAVE MONEY.

Some prosperous American communities hold human life so cheaply that they maintain no public health service at all. Others—and there are many of them—have a mere skeleton service. The citizens imagine that if they appoint a health board consisting of doctors, all will be well with them. The suggestion that the board be provided with money to carry on its functions would be regarded as wanton extravagance.

There are scores of cities and towns which select a doctor to head the health department and expect him to earn his living by practicing his profession among the very people over whom he is supposed to exercise police authority in enforcing sanitary and other health regulations.

There are cities of from 5,000 to 100,000 population that hire a doctor on the “part time” plan as chief health officer, and pay him a trifling salary. Whether he is a competent sanitarian or in any way skilled in the prevention of disease is a matter of little concern to them. The fact that they are saving a few dollars in his salary fills them with joy and indifference as to the consequences to the community.

I know of a thriving, wealthy young city in the South of 130,000 population with a substantial preventable death rate which saves as much as $800 annually in this way.

I know of a prosperous New England city of 40,000 population with but three people in its health department—two of them are “part time” employes. It is a six cents per capita department *and 50 per cent. of the annual deaths in that city are of children under five years of age*.

In theory, we must all stand ready to serve the State when called upon, even at personal loss. But does it not seem the height of absurdity to expect a competent professional man to leave his practice to take charge of these under-manned and under-financed health departments at the small salaries which our States and cities offer them? If he does his duty, he is sure to make enemies during his term of service, and if he is an able man he will certainly lose money by leaving his practice.

Surely we offer our health officers every inducement to follow the line of least resistance.

A SAMPLE GROUP OF CITIES.

An investigation was recently made of forty-four Illinois cities averaging in population about 16,000; fifteen of them had over 20,000 population, and three had over 50,000.

> The average salaries paid the chief health officer amounted to the magnificent sum of $300 annually.
>
> Twelve of these cities paid nothing for health protection—and this included three cities of 22,000 population and one of 30,000 population.
>
> One city of 26,000 population employed a layman as a health officer.
>
> In one of 22,000 the police matron served as “health officer” when she was not otherwise engaged.

> Twenty-nine of these cities made no pretense of supervising their milk supply.
>
> Only nine of them had isolation hospitals for contagious diseases.
>
> Thirty-one of them kept no mortuary records whatever.

These conditions exist in a prosperous agricultural and manufacturing State—and they can doubtless be found to exist in almost any State in the Union.

AGENCIES THAT CAN HELP.

These are unpleasant facts, but they give us an idea of the way we are performing the primary function of government—the guarding of human life against avoidable destruction.

We have now briefly considered the extent of the waste of the most vital asset of the nation, and how we are conserving it, or rather how we are not conserving it.

Now let us rejoice over the fact that we not only know how to reduce this waste, but that thanks to those who have aroused the life conservation sentiment in this country, a general improvement in the public health service is taking place in many States and cities. The experiment has been successful. We now know what we can do. We have the wealth and knowledge, and the machinery is organized throughout the country to rapidly correct our appalling record of life waste. Our work is to induce our people to use it.

Every business and social organization should do its full share of this work.

The life insurance institutions of this country have a constituency of 25,000,000 policyholders. These policyholders are directly interested in the promotion of longevity, not only from the humanitarian but from the financial viewpoint; for the lower the mortality among policyholders, the greater will be the saving and the larger the dividends to policyholders, which means a reduction in the cost of their life insurance. It is estimated that about $50,000,000 is lost annually by postponable mortality among the insured.

The Equitable Life Assurance Society, with which I have the honor to be connected, is endeavoring to do its part not only in conserving the lives of its policyholders, but in stimulating community action. The Metropolitan Life Insurance Company of New York is also rendering a valuable public service in Conservation along somewhat different lines. Two or three of the small companies, and perhaps the same number of fraternal insurance societies, have also given it attention. Let us hope that the time is near at hand when the other two hundred-odd life insurance companies, and the fraternal societies as well, will also increase their usefulness to their policyholders and the public by joining in this great work.

SOME SUGGESTIONS.

This Congress will be asked to do and to advocate many things, for there are a multitude of independent activities connected directly or indirectly with this general subject. Among others I sincerely trust the following suggestions will be duly considered:

1. To encourage business institutions, civic, social and religious organizations which have influence over any considerable number of people to join in at least some of the many phases of the life conservation campaign.

2. To encourage the education of the individual to adopt healthful habits of life—to avoid the intemperate life, which means excess in eating, drinking, working, playing —and unhealthful indulgence in indolence as well.

3. To encourage communities to establish and maintain ample public health organizations consistent with the magnitude of the work in hand.

4. To advocate the organization of local health leagues as a stimulus to public interest and to give aid and support to the public health service.

5. To encourage the slowly growing sentiment for a rigid supervision (and isolation if necessary) of tubercular victims, which is the only way in which this devastating plague can be stamped out.

6. To advocate the employment of civic nurses in the health service, who may also act as health inspectors and aid in educational work.

7. To advocate the issuance and distribution by the States or municipalities of an official prevention manual to teach the public how to avoid preventable disease.

8. To urge every individual to go to his or her doctor for periodical health inspections to detect disease in time to arrest or cure it.

9. To urge employers of labor to give their employes these examinations free as a part of their efficiency and welfare program.

10. To encourage philanthropy, now so generously contributing for the care of the sick, to also enter the field of disease prevention which it has so far quite generally neglected.

Human life is our paramount asset. Its conservation should be your paramount issue.

President White—The audience is certainly indebted for this great and interesting paper. It is hard to get over stubborn facts and figures, especially where figures are facts.

A great doctor once shocked the people of the country by saying that everybody should be chloroformed when they arrived at the age of sixty. From this paper it would seem we ought at least to reach the age of sixty, the age of being chloroformed, and, better still, we had better so conduct the board of health and so support liberally the board of health in our city that people may just begin to live when they get to be sixty.

I now have the pleasure of introducing Prof. Irving Fisher, of Yale University, who has given this subject many years of study and research and will now speak to you upon the "Public Health Movement." (Applause.)

Prof. Fisher—The Conservation movement is a movement to prevent waste. When the Conservation Commission was appointed, four years ago, emphasis was placed on the wastes of our natural resources, but by the time the Commission made its report, it had come to the conclusion that by far the most serious as well as the most preventable wastes are the wastes of human life.

A generation ago it was a common impression that the average human lifetime was fixed as by a decree of fate. When I was in college one of our reverend instructors showed us a mortality table and said with great impressiveness: "There is no law more hard and fast than the law of mortality." I believed it, and even yet many people are under this delusion. Pasteur did much to introduce a more optimistic view. He stated his belief in these immortal words, "It is within the power of man to rid himself of every parasitic disease." He staked this opinion on his own wonderful laboratory revelations as to germ life. Today we can confirm his words by absolute statistics. And now his successor, Metchnikoff, has surpassed even Pasteur in optimism. Metchnikoff is devoting himself to the question of the prolongation of human life and already gives us a vision of the time when centenarians will be regarded merely as in the prime of life and when the normal span of a century and a quarter will be a frequent occurrence.

The growing consciousness that human life is not a fixed allotment, which we must accept as our doom, but a variable, which is within our power to control, has recently led to extraordinary exertions all over the world to save human life. This impulse has gained strength also from the great and almost universal decline in the birth rate. Old countries like France, and new countries like Australia, are confronted with the specter of depopulation. Consequently, as human life becomes scarce, it becomes precious—like any other commodity! These two facts, the consciousness that much mortality is preventable, or at any rate postponable and the fact that increasingly fewer babies are being born in the world, are together operating to produce a great health movement throughout the world. Nothing will stop it until the whole world is convinced of the paramount importance of this problem of human Conservation.

This world-wide movement for the conservation of human life has expressed itself in many ways—in medical research; in societies for preventing tuberculosis, infant mortality, social diseases, alcoholism, and vice; in the growth of sanatoria, dispensaries, hospitals and other institutions; in an immense output of hygienic literature, not only technical books and journals, but also popular articles in the magazines and daily news papers; in the constant agitation and legislation for purer foods, milk supply, meat supply and water supply; in the movement to limit the labor of women and children and to improve factory sanitation; in the establishment of social insurance in Germany, England, Denmark and other countries; in the improvement of departments of health; in the spread of gymnastics, physical training and school hygiene; in the revival of the Olympic games and the effort to revive the old Greek ideals of physical perfection and beauty, and last, and most important, in the sudden development of the science of eugenics.

In the summer of 1911 was held in Dresden a unique world’s fair, devoted exclusively to health—the International Hygiene Exhibition. In this were shown the fruits of the whole movement in all lands—except, alas, our own; for to our shame it must be said that we, as yet, are among the backward nations in this movement for the conservation of human life. Our Congress was asked to appropriate $60,000 to erect a building and supply an exhibit to show what we have done for our part in this movement, but Congress thought it could not afford so large an expenditure for so small (!)

an object, and the result was that from the millions of people who visited this exhibition one constantly heard the question asked: “Where is the United States?”

And those few Americans who did go to visit the exhibition found that other nations had far outstripped us in this movement for national sanitation and health. Some of the achievements already attained by other nations should be recorded among the wonders of the world. One is the striking decline of the death rate in the city of London. Within two decades, London’s death rate has virtually been cut in two and is now only thirteen per thousand, or less than that of most cities one-fiftieth its size.

Probably, however, the greatest achievement of any country is that of Sweden, where the duration of life is the longest, the mortality the least and the improvements the most general. There alone can it be said that the chances of life have been improved for all ages of life. Infancy, middle age and old age today show a lower mortality in Sweden than in times past, while in other countries, including the United States, although we can boast of some reduction in infant mortality, the mortality after middle age is growing worse and the innate vitality of the people is, in all probability, deteriorating. The reason why Sweden of all countries has succeeded in improving the vitality of middle age and old age, while other nations have failed, is, I believe, to be found in the fact that Sweden, of all nations, has seen the problem of human hygiene as a whole instead of partially. In most other lands, and particularly in the United States, public health has been regarded almost exclusively as a matter of protection against germs; but protection against germs, while effective in defending us from plague and other epidemics of acute diseases, is almost powerless to prevent the chronic diseases of middle and late life. These maladies—Bright’s disease, heart disease, nervous breakdowns—are due primarily to unhygienic personal habits. Medical inspection and instruction in schools, as well as Swedish gymnastics, have aided greatly in the muscular development of the citizens of Sweden. Swedish hard bread has preserved their teeth. The Gothenburg system is gradually weaning them from alcohol. There has even been a strong movement against the use of tobacco. Other countries are tardily following in the path which Sweden has trod so successfully.

The significant fact is that Sweden has not hesitated to attack the problems of personal habits. I believe we must have a revolution in the habits of living in the community if we are going really to realize the promise of Metchnikoff and others as to the prolongation of human life. Health officers in this country have not regarded it as a part of their duty either to live personally a clean, hygienic life, or to teach others to do so, or even to investigate what those conditions of well-being are which make for personal vitality.

I can remember, thirteen years ago, talking with a doctor in Colorado as to the habits of living of his patients. I said to him, "You tell me that tuberculosis is a house disease, and that the reason it exists is because people do not open their windows. Why, then, do you not tell your patients they must open their windows, or sleep out of doors?" He said, "I wouldn't dare to do that; I would lose my practice. They would think I was a crank and meddling in their personal affairs." Today that battle has been largely won. Today, not only in Colorado and California, and in the places where there is perpetual sunshine, sleeping out of doors is common and not confined to invalids, but indulged in by the community generally. Even in New England and throughout the country you will find sleeping balconies going up all over. The change has even affected in some degree the architecture of the country, and while as yet only a minority of the people sleep out of doors, yet I believe it is true that the majority of the people in the United States have far more air in their sleeping and living rooms today than ten years ago. The fact which the doctor in Colorado did not dare tell his patients thirteen years ago, has in some way been told to the people of the United States.

But there are many other things that need to be told, after we are sure that they are true. When we have, through our National, State or municipal officers made thorough investigation and have been able to discover the actual truth as regards eating and drinking, hours of work, recreation and play—all those facts that go into what may be called personal habits, then we may gradually overturn existing unhygienic habits of living. John Burns attributes a large part of the great reduction in London mortality to the improved personal habits of working men, particularly in regard to alcohol. In this country, Dr. Evans, both as health officer of Chicago and later as health editor of a Chicago newspaper, has shown how public instruction in personal habits can be made effective, and it will be largely through affecting personal habits that the life insurance companies will improve the longevity of their policy holders.

Scientific men today have reached substantial agreement that alcohol is a poison. When everybody understands this, the days of alcohol as a beverage will be numbered. Sweden in the thirties was called drunken Sweden, but today the antialcohol movement there has converted Sweden into one of the soberest of countries.

But the use of tobacco, tea and coffee ought also to be investigated, so that we may know how far they are deleterious, and to spread this knowledge among the people.

Fashions are in their essence changeable and the time will come when the world will not be built on fashion but on reason. Japan has made more rapid progress in civilization than any other nation, because the late Mikado resolved and publicly stated that the institutions of Japan must not be tied by tradition but must be based on reason. When we have replaced tradition by reason, we shall have gotten a solid basis for civilization, and this must apply to ancient customs and habits of every kind. I am firmly convinced that we are looking at only one-half of this public health movement as long as we confine ourselves to the acute or infectious diseases. We shall not get more than half the results obtainable until we realize that there must be a revolution in the personal habits of the people.

Yet the United States, in spite of its shortcomings, has some special triumphs to record. We have, through hygiene under Colonel Gorgas, made it possible to dig the Panama Canal. We have virtually abolished yellow fever on our shores and in Cuba. We have nearly eliminated hook worm disease in Porto Rico and are gradually doing the same in the Southern States. We have found a cure for spinal meningitis. We have, in New York, made an object lesson in the last year of reducing the summer death rate of infants in a striking manner. We have, by individual milk stations in Boston and other cities and in individual sanatoria, dispensaries and other institutions, demonstrated that the death rate from specific diseases can often be cut in two.

Yet we have depended altogether too much on private initiative. In New York the summer death rate of infants was reduced chiefly through the work of the milk committee and individuals like Nathan Straus. The elimination of hook worm disease and the discovery of the cure of spinal meningitis came through the gifts of Mr. Rockefeller. It is well that individuals should apply themselves to these problems and without such personal interest they could never be solved. Nevertheless, progress will be many times as rapid when the problems for the nation are managed in a national way. There are three great agencies to which we must look for the saving of human life in the future and it has been the object of the Committee of One Hundred on National Health, of which I am President, to help stir these three agencies into activity in this country. They are the public press, the insurance companies and the Government.

To a limited extent, all of these agencies have increased their health activities in recent years. A few years ago, popular articles on public health were seldom seen because the public and the press thought the subject of disease uninteresting and repulsive. Today, on the other hand, one can scarcely pick up a popular magazine without finding not only one but several articles dealing with questions of public health; and it has been found possible not only to make these articles interesting, but, by emphasizing the positive, or health side, instead of the negative, or disease side, to render them attractive and beautiful. And yet, as Dr. Wiley has said, the newspapers in spite of all the good they are doing with their right hands are, with their left hands, in their advertising columns trying to undo that good by advertising the fraudulent part of the "healing" profession who are trying to line their own pockets at the expense of the lives of the public.

The second great agency from which I believe we may expect wonderful results in the future is life insurance. As our committee pointed out to the Association of Life Insurance Presidents several years ago, life insurance companies can save money by preventing deaths just as fire insurance companies have saved money by preventing fires, and steam boiler insurance companies have saved money by preventing explosions. Since this suggestion was made, a number of progressive life insurance companies have tried the experiment. The Metropolitan and the Equitable have established departments of human conservation and a number of other and smaller companies have undertaken similar enterprises. The Postal Life Insurance Company has recently published the statistical results of their experience, worked out in a most careful manner, and have demonstrated absolutely that it pays life insurance companies to save human life. This being the case, we may expect life insurance companies in the future to become active in life conservation. Already there are probably fifteen million policy holders in the United States insured in companies which are trying to do something for their health—through medical examinations, instruction in hygiene, utilization of visiting nurses, participation in civic health movements and otherwise. To save human life merely to save money is sordid enough, but it is well to harness commercial motives, when possible, in the service of humanity.

The third, and most important, agency is the government. State and National health offices are becoming yearly stronger and more efficient; and yet much remains to be done, particularly by the National Government. We need a National Department of Health or a Department of Labor which shall include in its operations the conservation of human life. We have already passed the phosphorus match bill to prevent one of the worst industrial diseases—phossy jaw. We have passed effective legislation in regard to interstate commerce in prostitution. We have established a Children's Bureau and a Bureau of Mines to prevent industrial accidents in mining. We have enacted suitable legislation in regard to cocaine and habit-forming drugs. We have a Pure Food Law and laws for the inspection of meats. Yet, as Dr. Wiley, Mrs. Crane and others who have watched the operation of these laws at close range well know, they need to be executed with a stronger hand.

The truth is that as yet we have only made a feeble beginning in public health work, especially in this country. We need first of all to do what Sweden has done for a hundred and fifty years—namely, to keep proper vital statistics. Vital statistics are the bookkeeping of health, and we cannot economize health any more successfully than we can economize money unless we keep books. At present only a little over half of the population of the United States has statistics of its deaths, while the statistics of the births are as yet nowhere sufficiently accurate to be called real statistics.

Our National Statistician, Dr. Wilbur, illustrates by a story how much better we keep our commercial books than our books of vital statistics. In a Western State a girl was entitled to a fortune when she became twenty-one. Reaching, as she supposed, her twenty-first birthday, she laid claim to the fortune. Much to her surprise, her father said, “But you are only nineteen;” and then the two tried to look up the records. They had no family Bible, they had no public record office to go to, and they were at sea as to how to discover exactly the date when she was born. However it suddenly occurred to her father, who was a farmer, that the very day his daughter was born a calf was born on his farm and the birth of the calf had been recorded. In that way he established the date of the birth of his daughter.

In view of the great slack of our vital statistics, therefore, we cannot measure even the death rate, much less the number of preventable deaths in the United States. All that we can do is to study carefully the registration area and on this basis to work out certain minimum figures.

Four years ago, as a member of President Roosevelt's Conservation Commission, I endeavored to do this and to report on the condition of our "National Vitality." I found, after getting together all the statistics available and taking account of the degree of preventability of different diseases as estimated by experts that, out of some 1,500,000 deaths annually in the United States, at least 630,000 are preventable. Of these preventable deaths, the greater number are from seven causes. These seven causes include three great diseases of infancy, then typhoid fever, which usually makes its attack in the twenties, then tuberculosis, accidents in industry, and pneumonia which come in the thirties.

Now 630,000 unnecessary deaths per year mean over 1,700 unnecessary deaths per day or more than the lives lost on the Titanic disaster. The nation cannot continue indifferent to hygiene as it gradually dawns on the public that for lack of hygiene we suffer a Titanic disaster every day of the year. The popular imagination was deeply stirred by the image of 1,600 helpless human beings suddenly engulfed in mid-ocean. That was a vivid dramatic picture which the blindest of men could see and understand. It led to immediate official action on both sides of the Atlantic to safeguard human life at sea. Yet on land we lose three hundred and sixty-five times as many lives as this every year and never stop to add it up. They are scattered and diffused throughout the land—a Wilbur Wright lost from typhoid, a handful of miners in an explosion, some railway employes in an accident, some victims of lead poisoning, a little army of infants, here a few and there a few. Yet these deaths are just as real and mean an infinitely more serious loss than were the deaths from the Titanic disaster. Moreover, they could be as easily prevented.

And concomitant with this unnecessarily great death rate, there is, of course, a colossal aggregate of needless sickness. We have no real statistics, but by analogy with English statistics we may assume that, on the average, for every death per annum there are two persons sick during the year. This makes about three million people constantly lying on sick beds in the United States, of which, on the most conservative estimate, at least half do not need to have been there.

If, now, on the basis of these figures, we try to compute how much human life is needlessly shortened in the United States, we find that it is shortened at least fifteen years. Again, if we translate these preventable losses into commercial terms, we find that, even by the most conservative reckoning, this country is losing over $1,500,000,000 worth of wealth producing power every year.

What does this mean? To us individually, it means that we are losing a large part of our rightful life not only by death itself which cuts off many years we might have lived, but also from diseases and disabilities which are not fatal but cripple the power to work and mar the joy of living. I believe I am far within the facts when I venture the opinion that the average man or woman in the United States is not doing half of the work nor having half of the joy of work of which the human being is capable.

With all this room for improvement before our eyes, it is not surprising that the zeal of the health movement is growing fast. Each success serves as justification for further effort.

One of the most encouraging symptoms of progress is the great attention which is being paid to public health in the present political campaign. All three of the party platforms included planks in behalf of public health. The Democratic and Progressive platforms were particularly explicit and emphatic and all the candidates have emphasized health in speeches and in their record in public life. The Democratic campaign managers are carrying out plans to make progressive health legislation prominent in the campaign.

These and other indications augur well for better legislation, more energetic enforcement of the law and, above all, a more appreciative public sentiment as to the transcendent importance of the conservation of human life. It is now reported that the Hon. Dr. Roche, Secretary of State in Canada, is in strong sympathy with the proposal there for the establishment of a Federal Department of Health and the Republic of China is reported to have already established such a department.

From all these indications of actual activity as well as from the logic of the situation we are justified in predicting that an age of human conservation is at hand. Men and women are waking to their responsibility to the race. Eugenics will be a watchword of the future. To squander our natural resources is ignoble indeed, but far worse is it to squander our vital resources. The most sacred obligation of each generation is to bequeath its life capital unimpaired to the generation which comes after. Scourges like typhoid and tuberculosis must be swept off the face of the earth. Habit-forming drugs, including alcohol (and even tobacco, especially for young boys) must be recognized in their true light as means of depleting the vitality of nations. Prostitution and the white slave traffic must be condemned anew as robbers of the race. Industries which kill and maim, poison or infect their workers, which deform and stunt little children, which incapacitate women for normal motherhood, which through overlong hours of toil close each successive day's work with progressive exhaustion, must be controlled. Machinery was made for man, not man for machinery. Immigration which drains European public institutions of their criminal, insane, feeble-minded and other defectives and delinquents and sets these creatures loose in America to breed with and contaminate our population, must be regulated. Marriage laws and customs must be adjusted so as to discourage or forbid the procreation by the unfit. All these and other hygienic and eugenic reforms will be realized as fast as public sentiment becomes educated to the solemn responsibilities and higher valuations of human life.

The noblest task, therefore, which I can conceive for any man is to aid in erecting true ideals of perfect manhood and womanhood. Our ideals, though improving, are not yet worthy to be compared with those of Japan or Sweden and the ideals even of these countries have not yet reached the level of those of ancient Greece still imaged for us in imperishable marble. With superior knowledge our health ideals should excel those of any other age. These ideals should not stop with the mere negation of disease, degeneracy, delinquency and dependency. They should be positive and progressive. They should include muscular development, a sound mind in a sound body, integrity of moral fiber, a sense of the splendor of the perfect human body as a temple of the human soul, a sense of the enjoyment of all life's proper functions. As William James said, simply to breathe or move our muscles should be a delight. The thoroughly healthy person is full of joy and optimism. He rejoiceth like a strong man to run a race. Said Emerson, "Give me health and a day and I will make the pomp of emperors ridiculous." Our health ideals should be nothing short of an abiding sense of the sweetness and beauty, the nobility and holiness, of human life.

President White—We have had wonderful addresses this morning from the distinguished speakers upon this question of conservation of human life.

I wish now to announce the Committee on Credentials: Mr. E. T. Allen of Portland, Ore., Mr. Volney T. Foster of Chicago, and Col. W. A. Fleming Jones of Las Cruces, N. M.

I wish also to announce the Chairman of the Committee on Resolutions: Mr. Walter H. Page of New York. The different State organizations will report to him a member for that committee from each particular State. It will be well to report to Mr. Page either tonight or early in the morning.

We all need to be put under authority. We find people are not taking good care of their health, of themselves or of the community. We will now hear from Dr. L. E. Cofer of Washington, D. C., Assistant Surgeon-General of the United States Public Health Service, who will address the Congress on the subject of "Authority In Health Control."

Dr. Cofer—Mr. President, Ladies and Gentlemen: I do not think any invitation has ever been received with more satisfaction than the one for a representative of our service to appear before you. It was received with the greatest gratification.

The Public Health Service is absorbed in the work of health conservation and Surgeon-General Blue evinced the greatest interest in your invitation for him to send a representative to explain the scope of the work being performed and discuss the question of authority in connection therewith.

This topic is now receiving the consideration of many authorities on public health matters, and on this account one may approach the subject in a hopeful attitude. I say "hopeful" because public health as an institution is rapidly growing, and its practical value is becoming more and more manifest, and sanitary science is not now nearly so far in advance of its practical application as it was even a few years ago. The possibilities of sanitation in the general advancement are being made a part of all high ideals of government, so that it is not to be wondered at that the general government should be called upon to do its share. The difficulty lies in determining just what the government should do in aid of public health and just what should be left to the States and municipalities.

History furnishes no precedents for this Nation to follow. It is almost useless to seek a model for our guidance in some foreign country. A nation with our conditions of boundary and magnitude, with millions of immigrants coming to our shores from all parts of the earth, has its own salvation to work out in the public health as well as in many other problems. In other words, we must rely upon ourselves, whether we proceed in haste or by feeling our way step by step. There is a marked divergence of sentiment growing in regard to national health control. One is that the government should do far more than it is now doing towards the protection of the public health, another that too much is expected of the National Government, and that there is a tendency on the part of State governments to call upon the Federal Government for service which should be performed by the States themselves, but which service is asked for largely in the interest of economy. These widely differing ideas in regard to the apportionment of public health responsibility lead us to a consideration of the provisions of the Constitution of the United States relative to this matter. These provisions are contained in Section VIII, paragraphs I and 3:

> Section VIII. The Congress shall have power—
>
> Par. 1. To lay and collect taxes, duties, imposts and excises, to pay the debts and provide for the common defense and general welfare of the United States; but all duties, imposts and excises shall be uniform throughout the United States.
>
> Par. 3. To regulate commerce with foreign nations and among the several States, and with the Indian tribes.

It has been held by some that the powers of the National Government, relating to public health, are restricted to paragraph 3, which gives the right to Congress to regulate commerce, and, in regulating commerce, to so regulate it as to prevent its being a carrier of disease. Others have held that under the general welfare clause, in paragraph 1, Congress has the right to legislate for the public health.

Should the latter interpretation be the correct one, Congress could establish the national health control over States and municipalities with regard to municipal and domestic sanitation, with all details as to house drainage, plumbing, sewerage, and disposal of garbage, water supply, ventilation, school houses and public buildings ventilation, examination of milk supply, food and drugs, disposal of the dead, disinfection of dwellings, etc. Would it be desirable for the National Government to have such authority? Would it be tolerated by the people? It is a fact that the American people have already decided this question when the old National Board of Health was abolished.

The National Board of Health was created by an act of Congress, approved March 3, 1879. Another act was approved June 2, 1879, clothing the board with certain quarantine powers, but this last act was limited to a period of four years, at the expiration of which time Congress declined to renew it. The National Board of Health, therefore, had an active existence from 1879 to 1883. The act establishing the board remained upon the statute books until February 15, 1893, when it was formally repealed by Congress.

To state the case concisely, the National Board of Health was not in accord with the spirit of American government, and the people rejected it. Now, what do the American people want? I will not attempt to answer this question, but will suggest that they want a general sanitary administration which is capable of steady development, and yet may be subject to such modifications as the changing conditions of our country may necessitate, a sanitary policy which can be made to expand until it will answer the public needs not only for the present but even for decades to come.

Its direct aim should be the ultimate intelligence and education of the average citizen in matters relating to his personal health, and the health of his commonwealth. No better plan for sanitary government appears at the present time than one modeled upon the structure of the general government itself. Broadly stated, this sanitary policy expects of each State a sanitary autonomy whose influence should be appreciated by every individual in every hamlet, however small, in its domain. It contemplates a State pride in the development of sanitation, a self-reliance and an unwillingness to surrender functions or call for aid from the general government excepting after the clearest convictions of propriety or necessity. This policy expects from the general government that since it controls commerce, both maritime and interstate, it will prevent commerce from conveying disease; that it will respect the sanitary institutions of the States; that it will have such organizations and establishments as properly belong to its sphere of action to supplement where States fail, and to enable it to wield its peculiar power when urgency demands.

As an apt illustration of this conception of authority in health control, let us consider the present activities of our Federal Public Health Service. These are as follows:

1. The prevention of the introduction of infectious and contagious diseases.
2. The sanitary regulation of foreign commerce.
3. The observance of international sanitary treaties.
4. The prevention of the spread of infectious and contagious diseases from one State to another through co-operation with State and municipal health authorities.
5. The collection and dissemination of sanitary information.
6. The conduct of scientific research in matters pertaining to the public health.
7. The enforcement of sanitation in Federal territory and in connection with Federal administrative affairs.

THE PREVENTION OF THE INTRODUCTION OF INFECTIOUS AND CONTAGIOUS DISEASES.

The chief national quarantine law is that approved February 15, 1893, amended and extended by acts of Congress approved August 12, 1894, March 2, 1901, and June 19, 1906.

Under these acts the maritime quarantine administration has become national, many state stations having been voluntarily surrendered to the Government, others supplanted by the General Government, because of failure to comply with government regulations, and others superseded by direct authority of law.

The diseases excluded from the country by the national quarantine establishment are cholera, yellow fever, smallpox, typhus fever, leprosy and plague.

Some quarantine stations are inspection stations only, but many are large institutions, comprised of hospitals, quarters, barracks for detention of crews and passengers, wharves and disinfecting machinery, and boarding vessels, all requiring good administrative ability on the part of the commanding officer, who must also be expert in the detection of disease.

When a ship from a foreign port arrives off a port of the United States, it is met by a quarantine officer for inspection under the national regulations. Fifty medical officers of the service are engaged in this work at forty-seven separate stations, extending along the Pacific, the Gulf and Atlantic coasts from Alaska to Portland, Me. Without the quarantine certificates given these officers and the bill of health obtained at the foreign port, the ship would not be allowed entry by the collector of customs and without his permit it would be unlawful for the ship to unload its cargo.

At a few ports, not more than three or four in number, this inspection is made by a State quarantine officer, a relic of the system which prevailed prior to 1893, when quarantine was considered a State rather than a National function. They are obliged, however, to enforce the National regulations, and are subject to inspection by the Federal officers, and if they fail or refuse to comply with the United States regulations the President is authorized to detail an officer of the Government for that purpose.

In addition to the diseases remanded by quarantine, others are excluded under laws relating to immigration, and for this purpose at the principal ports of entry there are also stationed seventy medical officers, who, during the past year, for example, examined more than 1,280,000 immigrants, certifying more than 30,000 of them on account of physical and mental defects. The immigration laws exclude persons afflicted with any loathsome or any dangerous contagious disease, or having mental or physical defects which may affect their ability to earn a living.

Humanity requires the treatment in hospital of immigrants arriving sick with ordinary as well as prohibitive diseases, and the large hospitals connected with the stations are under the professional conduct of service officers.

Although the immigration stations are under the control of commissioners attached to the Department of Commerce and Labor, nevertheless the medical officers are subject in their professional work to supervision by the Public Health Service, and their instructions as to the medical inspection of aliens are prepared by the Surgeon-General and approved by the Secretary of the Treasury.

THE SANITATION OF FOREIGN COMMERCE.

At certain foreign ports and at certain times, depending upon the presence of the various quarantinable diseases, either in the foreign ports of departure or in the country contiguous thereto, officers of the Public Health Service are detailed by the President to serve in the offices of the American consuls, to assist them in enforcing the quarantine regulations for foreign ports. These officers keep themselves informed of the prevalence of contagious disease in these cities and the surrounding country. They sign a bill of health which certifies that all the regulations required to be enforced at foreign ports on vessels leaving for the United States have been complied with.

This involves a knowledge of the point of origin of the freight and passengers, disinfection of material from an infected locality, the personal inspection of passengers, particularly steerage passengers, and their detention if necessary. The power of enforcement of these regulations lies in the above mentioned act of Congress approved February 15, 1893, which imposes a penalty of $5,000 upon any vessel from a foreign port seeking to enter a port of the United States without this consular bill of health. The consul can legally refuse a bill of health if the regulations are not complied with.

In this connection it may be said that officers of the Public Health Service are stationed constantly at such ports as Hongkong, Shanghai and Amboy, in China; Yokohama and Kobe in Japan; Salina Cruz, Manzanillo and Puerto Mexico in Mexico; Guayaquil, Ecuador; La Guaira, Venezuela, and Havana, Cuba. During the summer of 1911, on account of cholera conditions prevailing in Italy, Russia and France, there were officers of this service detailed in the offices of the American consul at Naples, Genoa, Palermo, Messina and Catania, in Italy, at Libau in Russia, and at Marseilles, France. In addition to this, officers were ordered to several other foreign ports of departure, there to confer with the American consular officers as to the enforcement of the regulations for foreign ports, and for the purpose of insuring uniformity of procedure.

The State Department has done much to assist in the quarantine and sanitary work in foreign ports, through the interest it has aroused in the said work on the part of its consular corps.

THE OBSERVANCE OF INTERNATIONAL SANITARY TREATIES.

These treaties or conventions establishing them have been ratified by the Senate of the United States, as well as by the other governments.

The International Sanitary Bureau of American Republics at Washington was founded by the International Conference of American States held in the City of Mexico in 1901. That conference also called for international sanitary conventions, which are now held every two years. Two have been held in Washington. The object of the conventions is to freely discuss all matters relating to the public health and particularly those which affect the American Republics, and the purpose of the international Sanitary Bureau is to encourage the execution of the resolutions or agreements decided upon by the conventions. The convention held in Washington in 1905 drew up a treaty with regard to the quarantine treatment of cholera, plague and yellow fever, which was signed ad referendum by the official delegates, and has been confirmed by practically all of the American Republics. At the meeting in Mexico in December, 1907, action was taken which has brought the International Sanitary Bureau at Washington into relations with the International Office of Public Hygiene at Paris.

The International Office of Public Hygiene at Paris was formally inaugurated December 9, 1907. It is the outgrowth of international sanitary conferences at Rome, Venice and Paris, with regard to the bubonic plague. The following governments are represented: Algeria, Argentina, Australia, Belgium, Bolivia, Brazil, British India, Bulgaria, Egypt, Canada, France, Great Britain, Holland, Italy, Mexico, Peru, Persia, Portugal, Roumania, Russia, Servia, Sweden, Spain, Switzerland, Tunis, Turkey and the United States.

Each of these governments has agreed to pay its pro rata of the expenses necessary to maintain the international office. The principal object of the office is to collect and bring to the knowledge of the participating States facts and documents of a general character relating to public health, especially as concerns infectious diseases—notably cholera, plague and yellow fever—as well as the measures taken to combat these diseases.

PREVENTION OF THE SPREAD OF INFECTIOUS AND CONTAGIOUS DISEASES.

These operations are conducted under two laws. One is the national quarantine act of 1893, already referred to, which contains practically the same provisions for interstate as for maritime quarantine. The other is the annual law passed by Congress appropriating an “epidemic fund” which contains a provision that it may be used in aid of State and local boards of health in the enforcement of their quarantine regulations, as well as those of the national service—to be used, however, only against certain specified epidemic diseases, viz., cholera, yellow fever, smallpox, typhus fever and bubonic plague.

Now, with these two laws in hand, and when the appearance of any of the above-named diseases in any State so require, the officers of the Public Health Service are at once upon the scene with the double object of seeing that the Treasury Interstate Quarantine Regulations are enforced by the State or local authorities and to offer aid, as authorized by law.

When aid is extended, the Government’s funds must be expended by its own officers, and the latter are therefore placed in charge and have the co-operation and assistance of the State or local authorities. They have, therefore, the support of the State and local laws and regulations, as well as those of the Federal Government. This is fortunate, since experience has shown the importance, in a Republic like ours, of local sympathy and support.

THE COLLECTION AND DISSEMINATION OF SANITARY INFORMATION.

The Public Health Bureau, through its Division of Sanitary Reports and Statistics, compiles and publishes each week a pamphlet called the Public Health Reports. It contains a statistical report from all cities in the United States of more than 10,000 inhabitants, and some others, giving the morbidity and mortality in each city with regard to twelve diseases and the total mortality from all diseases. It contains also a statement of the weekly mortality in some 120 foreign cities from thirteen communicable diseases. It gives special information concerning quarantinable diseases and sanitary measures in the United States and foreign countries. The foreign information is received through the United States consuls and service officers abroad.

Collective investigations are being made of the prevalence of pellagra, infantile paralysis and leprosy.

A compilation has been prepared of state laws bearing upon reporting diseases, with a view to increasing the collection of morbidity statistics and bringing about improved methods and greater uniformity in their collection.

CONDUCT OF SCIENTIFIC RESEARCH.

In the District of Columbia, in a commodious building, the Public Health Service has its hygienic laboratory, a research laboratory exclusively for public health investigations. It is conducted in four divisions, viz., bacteriology and pathology, chemistry, zoölogy and pharmacology. This organization brings under the same roof, and in intimate association, scientific workers in each of these several branches, interesting facts developed in one line of investigation being made freely known to the investigators in other lines of research.

Officers are detailed to receive instruction in this laboratory, thus enhancing the scientific attainments of the corps and giving opportunity for selection of those best qualified for permanent detail in research work. In this manner specialists have been and are being developed on various subjects, such as typhoid fever, pellagra, hookworm disease, infantile paralysis, scientific disinfection, etc.

Public Health Service officers may be found in the States investigating other diseases than those named in the epidemic law, viz., typhoid fever, infantile paralysis, cerebro-spinal meningitis, hookworm disease, malaria, pellagra, dengue fever, milk sickness, etc. These investigations are usually made at the request of State health authorities. The bureau at Washington, on receiving a request from a city or locality for expert aid, invariably refers the request to the State Board of Health before compliance.

The laws permitting these investigations are, first, the interstate section of the quarantine law of 1893; and second, the act of Congress approved March 3, 1901, providing a building for the hygienic laboratory for investigations of contagious and infectious diseases and matters relating to the public health. As the investigations require laboratory examinations, they come within this last named law and the appropriation which supports it.

In various States of the Union, there are thirteen establishments engaged in the production of vaccines, antitoxins and serums, which play so important a part in modern therapy. The variation in the potency and the occasional impurity of these products caused Congress to pass an act July 1, 1902, requiring a license for their manufacture for sale in interstate traffic.

ENFORCEMENT OF SANITATION IN FEDERAL TERRITORY.

In the Philippine Islands, where the government is by commission and a legislature, much work of value to the public health is performed in the bureau of science under the insular government. There are, however, in the several ports of the Philippines medical officers of the Public Health Service under appointment from the Treasury Department in Washington, engaged in the transactions of both incoming and outgoing quarantine. Two of these officers, in addition to their supervision of the national quarantine, are also director and assistant director, respectively, of the public health of all the Philippines.

In Hawaii you will also find medical officers conducting the national quarantine. They are also assisting the territorial health board in preventing the recurrence of plague by the extermination of rats and continuous bacteriological examination of those captured. One of these officers is the official sanitary adviser of the Governor of Hawaii, and is carrying on a campaign for the eradication of disease-bearing mosquitoes.

Here also may be observed the leprosy investigation station, also controlled by our officers, both on the island of Molokai, where hospital and other accommodations have been erected under the law of March 3, 1905, appropriating $100,000 for this purpose, and at the receiving station at Honolulu, where cases are seen in the earlier stages.

In Porto Rico public health officers are enforcing the United States quarantine regulations under the acts of Congress relating to Porto Rico and national quarantine. The campaign which has practically eradicated plague from San Juan is being conducted by the Federal Public Health Bureau.

In the Canal Zone you will find two commissioned officers enforcing quarantine regulations at Ancon on the Pacific and Colon on the Atlantic.

These officers are loaned to the Isthmian Canal Commission. This is an important adjunct to the work of the canal, because it would be useless to clean the zone if fresh importations of disease were permitted.

I will now devote a few words to the Health Bureau organization in Washington by means of which all the functions or activities above described are administered under one head.

THE ORGANIZATION OF THE BUREAU OF PUBLIC HEALTH SERVICE.

The law which changed the name of the Marine Hospital Service and made it a Public Health Service was approved July 1, 1902. This law fixed the status of the officers, enlarged the hygienic laboratory and gave it an advisory board, provided for the conferences with the State and Territorial Boards of Health, provided for compilation and publication of statistics, and directed that the President should prescribe rules for the conduct of the service and the uniforms of its officers and employes.

It also provided for a Public Health and Marine Hospital Bureau at Washington.

By an act of the Congress approved August 14, 1912, the name of the Public Health and Marine Hospital Service was changed to Public Health Service. The public health functions and duties of the service were extended and certain changes were made in the salaries of the officers.

The Public Health Service is under the supervision of the Secretary of the Treasury, and is in charge of the Surgeon-General, who has six Assistant Surgeons-General in charge of the Bureau Divisions. These divisions are as follows:

1. Foreign and Insular Quarantine and Immigration.

2. Domestic (Interstate) Quarantine.

3. Personnel and Accounts.

4. Marine Hospitals and Relief.

5. Scientific Research and Sanitation.

6. Sanitary Reports and Statistics.

The above lengthy description of our present public health activities has been necessary not only in order to demonstrate their character, and scope, but also as an illustration of the variety of legal authority existing for the enactment on the part of the general Government of public health work.

This paper will not admit of the incorporation into it of the national laws relating to public health which are now operative, but a careful inspection of these laws will demonstrate that they will admit of such interpretation as would make possible an almost unlimited amplification of our present public health activities, the limit being only one of appropriations and officers.

Careful analysis of the present health laws and activities will also show that the Government is seeking to control nothing which any other public health organization would wish to control. The foundation of the national public health service is in the quarantine law of February 15, 1893, referred to above. The quarantine service is today almost entirely national, notwithstanding a local sentiment for State or municipal control, which exists in two or three cities only, and which it is believed is destined to a short tenure for the following reasons:

It must be admitted that maritime quarantine should be a national affair. It is a concomitant to commerce, over which under the Constitution the national government has absolute control, and it naturally belongs to that department of the government regulating commerce in other respects. In other words, it seems especially appropriate that quarantine should be one of the functions of the Treasury Department which registers, licenses and enrolls all merchant vessels of the United States, inspects the hulls, boilers and machinery of such vessels, determines the number of passengers which said vessels may carry and provides for the housing and rations of the crews.

Besides this, it carefully examines all pilots upon American vessels and determines upon granting them licenses. It enforces the navigation laws and aids vessels in distress by an efficient revenue cutter service. It also provides for the care of the sick of our merchant marine. Then why should it relegate to a State authority, or health officer of some small port, the one remaining act of surveillance over vessels, namely, the determination as to whether they may be admitted to entry from a sanitary standpoint? Why should it be left to a local appointee, responsible only to a mayor or governor, the power to determine whether all the people and the merchandise on vessels destined for all ports of the United States, shall be permitted to enter without detention; and why should it give this local officer power to detain such vessels; and further than that, why should such local officers desire that power?

In the same way, the other activities of the Public Health Service conflict in no way with the functions and prerogatives of the State and local boards of health. Therefore, the term “national health control” is a misnomer. The term “national health co-operation” would be much more descriptive of the conditions actually existing. The interstate health activities above described must of necessity be governmental functions. The duties and responsibilities connected with them could not be discharged by States with any degree of uniformity. Therefore, interstate commerce laws are considered as appropriate national enactments, and their operation encroaches upon no State or municipal rights.

It may be said with a feeling of conviction that the health control in the United States today is just exactly in accordance with the desires of the people. The people know that their State and municipal boards are being aided by the health activities of the national government rather than being encroached upon. In addition to this the Federal Public Health Service and the State and municipal boards are acting in harmony to the following ends: They are controlling commerce, in order that commerce may not be clogged, and where necessary they are laying the net of healthful restraint for purposes of good.

The government is receiving the good-will and co-operation of the State and local health authorities in its work of catching and throwing back the diseased persons who seek entrance to our shores in the great Waves of immigration. They stand together to check the merchant or the manufacturer when he is ready or willing to risk the lives of the people by furnishing improper or impure food or drug products. They stand together to frustrate the lawyer who seeks by illegal technicality in the behalf of an individual, or steamship company perhaps, to force a way around a sanitary barrier erected for the protection of the people at large. Again, the municipal, State and government health authorities are standing together to stimulate the knowledge of our legislators in public health needs and are combining their knowledge to insure reasonable appropriations for the carrying out of general public health projects.

The mission of the three classes of sanitarians above mentioned may go still further. It may go to the extent of prodding the conscience of the tardy doctor, and even to the sweeping aside of the sentimental obstructions which the unenlightened are able to put in the path of the conservation of life. There is ample law for present and probably for future needs, and the control of national health remains, after all, today where it has remained in the past, and where it always will remain, that is, with the American people, not solely with the government, nor with the State or municipal health agencies. Each of the great nations of the world has gone about the direction of its public health work in its own way, and always with the realization that the ideal is not necessarily the practical, and what is best today may be supplanted by better tomorrow.

To summarize the situation, we have today State boards of health in control of State sanitation, operating under proper and ample State law. We have municipal health organizations operating under their own legal authority, and finally we have the United States Public Health Service, operating under several laws, as stated before, more far-reaching in their scope than is indicated by the activities pursued under their authority. The people, apparently, are satisfied so far as the Public Health Service is concerned. When the people want anything more they will demand it, and if available appropriations will not admit of compliance with such requests they will be forthcoming. Therefore, I am at a loss to suggest what additional health legislation is necessary or desirable to be engrafted upon that already existing in this country, and I am unable to see the necessity for any different plan of organization so long as the people, in whose behalf the organization is being maintained, are satisfied.

In closing, I wish to say that I have endeavored simply to place various facts before this Congress, and while I do not pretend to have exhausted this branch of the subject, I fear that I can not say the same with regard to your patience.

President White here requested Dr. Henry Wallace, of Des Moines, Iowa, to take the Chair.

Chairman Wallace—We are now ready to hear the report of the Committee on Lands and Agriculture. The first speaker will be Dr. George E. Condra, of Lincoln, Neb., whose subject is "Land Frauds, or Get-Rich-Quick Schemes."

Dr. Condra—Mr. President and Delegates: Some of you may recall the fact that the speaker has briefly outlined this subject at each of the preceding Congresses, under the head, "Conservation of Business." The discussion offered at this time is based on reliable information secured from many States. It is largely the result of field work. The data are presented according to the viewpoint of Conservation and should be so considered.

Do you fully realize that the principles of Conservation are permeating every department of human industry, improving the processes, increasing efficiency, and promoting common honesty, that the idea of equity is increasing in force? That it is being extended to business not for the purpose of holding it in check, but primarily for protection against fraud? This movement for square dealing certainly is in order for business is sore with graft and tracked by fraud at every turn. Plain it is that many transactions in the realm of commerce fall outside the sphere of true business. They grade from those that are doubtful on through to those that are plainly fraudulent and therefore criminal. The term "business," however, has a splendid meaning which should be conserved. It symbolizes honesty, stability, honor and reliability. Sharp practice, double dealing and doubtful promotion are but parasites and should be so regarded. They have no legitimate place in business and are being eliminated.

Several persons have spoken in this Congress on pure food, eugenics, etc. Their messages will tend to make people healthier and better fit to be fathers and mothers. All this is good. Dr. Wiley and others have emphasized the importance of pure food and health laws, but how many go back of this matter of health and food to the land, or source of our food and raiment and show the great need for pure land laws? (Applause.) The State trains its sanitary engineers, lawyers and physicians for their life work. It examines the lawyers and doctors before permitting them to practice, but how about land agents? They are good and bad. Many of them have no special qualifications for their work and should not be permitted to do about as they please without restriction, promoting this and that deal which may or may not have merit. Grant me your closest attention and I will point out certain classes of fraud that operate in connection with the development of mineral lands, irrigation, fruit lands, eucalyptus culture, drainage, dry land farming and the small tract propositions.

Promotion of Mineral Land.—The amount of money sent from the country and town and city to doubtful mine promoters is enormous. The return for this outlay is small, in some places less than one cent for each dollar. Yet the public does not fully realize that nearly all reasonably sure propositions are not available for wanton promotion, that a mere prospect is not a mine, and that fraudulent promoters are hurting the mining business.[3]

Oil and Gas Promotion.—The excitement caused by a developing oil field is intense. Agriculture gives way to a spirit of speculation and overvaluation and everything looks good to an investing public. Fabulous returns appear to be in sight for all who invest in time. This gives opportunity for professional promoters to do their work, sometimes on a large scale. They claim a sure thing even when wildcatting. So they send unwarranted prospectuses broadcast and the money harvest is on. It is difficult to place the criminality of such procedure. We only know that it works out badly as a rule. You should know that it is bad business to accept the unqualified statements of most oil and gas promotion concerns as a basis for investment. These persons and concerns interfere with legitimate development and should be brought under control.

Irrigation Schemes.—The Federal Government spends vast sums in developing the irrigation resources of several dry land States. Such reclamation is of economic importance. Furthermore, many reliable individuals and private companies do as well and even better in developing some projects. As a result of successful irrigation thousands of happy homes are made where once was only dry land. Notwithstanding this fact there are fraudulent irrigation promoters. Scheming individuals sell illegitimate propositions which can not succeed because of lack of water, unsuitable land or heavy graft. Such promotion has gone on to such an extent as to call for severe criticism by many practical irrigationists of the West, and the Reclamation Department of the Federal Government is increasing its diligence in checkmating the work of persons who attempt to promote bad projects.

Fruit Land Promotion.—Have you visited the great fruit districts of Oregon, Washington and other Northwestern States? Do you know what care is there given to the cultivation and marketing of apples especially? The fruit is so perfect in form and color. It is accurately graded for the Eastern and foreign markets. These splendid successes are widely known and are taken advantage of by scheming persons who promote the sale of any and all kinds of land in and near fruit districts. One of the leading fruit men of Washington says that thousands and thousands of dollars are going into the hands of concerns that are sure to fail and that the fruit business is being hurt by such operations. The trouble of it is that the average investor does not know that the fruit business is highly specialized, and that many matters concerning soil, exposure, climate, markets, etc., not known to him, are the features that determine success and failure. Furthermore, the fraudulent promoter does not know, neither does he care.

Doubtful promotion of this kind is not confined to the Northwest alone. It has hurt the South and may do damage to New York and other States in which are lands well suited for fruit raising, if the proper authorities do not conserve the larger interests of the industry and State against promoters.

Eucalyptus Promotion.—For many years the forests of the United States have been in process of depletion. Some have seen in this, and with good reason, an approaching timber famine. The alarm has been sounded, and the demand has gone forth for better methods in timber utilization, for fire protection, and tree planting. This is the right thing without doubt, but it affords a loop-hole for promoters. It is understood, also, that some trees grow faster and are more all-purpose than others. The eucalyptus are of this kind. They are of many kinds. Such trees can not be grown on any and every type of soil and are limited somewhat by climate. It so happens that California, because of its soil and climate, is the leading State in culture of eucalyptus. It has several successful groves and larger plantings, yet the situation is promoted for all it is worth, and perhaps more. The public (in the Central and Western States) is worked by carefully-planned selling schemes. The fact is that there is too much graft in some of them. The process has gone on to such an extent as to cause the friends of eucalyptus planting to sound a warning against such procedure. This should cause investors to make a more careful inquiry of reliable persons, not controlled by the promoters, before parting with money. The trees must have suitable soil, climate, and care.

Drainage Schemes.—One of the largest lines of development in the United States is in the field of drainage, whereby swamp and flood lands are improved. The amount of land that either has or can be reclaimed by drainage is said to be about 75,000,000 acres. The Federal Government, various States, companies, and individuals, are doing this work. Much of such development is well founded, yet there are bad deals, which might be called deliberate steals in some cases. Examples of these exist in a few States and much money has been squandered on projects that can never succeed. Teachers, ministers, farmers, merchants and others are victimized. In the language of one of Florida's representatives at the National Irrigation Congress of this year, "Persons selling certain wet lands of Florida are practicing fraud and should be prosecuted as criminals. They are hurting the good name of Florida and swindling people in the North." This person severely criticized certain cities of the North as being promotion centers. Further comment is not necessary.

Dry Land Deals.—Much dry land promotion is fraudulent, caused in part by misinformation on the part of agents, but due to some extent to deliberate misrepresentation. For instance, there are places in Texas, Oklahoma, Kansas, Colorado, western Nebraska, Wyoming, and other States subject to such promotion. The fact is that a part of the land in the dry area of each State named is well suited for dry farming, but that unscrupulous agents sell anything and everything to unsuspecting persons as being good, awaiting the plow and successful development. So it is that geographic position has been overworked. The following points are sometimes overdrawn in securing sales:

a. The idea that nearly all agricultural land is under cultivation.

b. The notion that dry farming methods are successful on almost any kind of dry land.

c. That the climate, referring to the rainfall especially, is becoming more favorable for agriculture in dry regions as the years go by. This notion, used in deceiving thousands of people, is greatly in error.

d. Advantage is taken of such fluctuations in rainfall as occur from year to year and at more or less regular periods, ten to twelve years apart. During the wet years the country is boomed; at dry times the people move out and industry wanes. These ups and downs are recurring features on certain areas not permanently suited for farming. The process works havoc with the misguided settlers, hurts a State that encourages it, and brings no lasting beneficial results to land men who manage the operation.

Apparently, Nature is no respecter of persons, especially so on the dry, sandy lands. It is coming to be known that there is no permanent change for the better in rainfall, frost belts or any thing of the kind. Some lands are better suited for grazing than for ordinary farming and should be so managed.

The speaker is pleased to be the servant of a State that stands strongly against misrepresentation of land values. Such a policy works out the greatest good in the long run. It breeds a healthy demand for a fact basis of development and minimizes the tendency to “stand up” for the home State by unwarranted “boostings.”

Land Schemes in General.—There are many other land projects. The public has invested largely in small tract propositions in Florida, Texas, and other States. Much of this promoted land has considerable value, but some of it is over-estimated, and many investors are quite apt therefore to lose all or nearly all of their money. Certain kinds of land look more inviting during one season of the year than at another. For example, there are places in Texas and Mexico to which the promoters take their victims in the dry season and to other lands during the wet season. This year the speaker heard a Texas representative declare, in a national meeting, that many of the small tract propositions, together with certain other land schemes of his State, are filled with fraud. He criticized northern people for promoting Texas. This should serve at least as a warning to unthoughtful investors. The good agricultural propositions of Texas and elsewhere are handled by responsible land agents.

The movement for the reclamation of the so-called abandoned lands of some of the older States is quite apt to be hurt by unreliable promoters.

Misrepresentation and Overvaluation.—Not only do some promoters misrepresent propositions for the purpose of receiving gain therefrom, but they often advance the sale price unduly. Many examples of this kind have come to my attention. Two weeks ago I received a prospectus from Oklahoma, advertising lead and zinc land for sale at $6.00 a block, twenty feet square, making 1,089 blocks in the tract of ten acres. This would be $6,534 for the land. I happen to know the region and own land close to the small tract. The fact is that one can purchase such a place at $10.00 or less an acre, or at not to exceed $100 for ten acres. So the difference between $100 and $6,534 is too much of an advance for those who invest. What do you think of such a deal? The persons handling it use the general statement of a geologist which recites the fact that the geological formation that contains zinc and lead in the Joplin District, some thirty miles distant, extends through the promoted land. This statement has no specific importance, but is sufficient for persons who accept the "get-rich-quick" bait. It is my judgment that Oklahoma should not permit such a clean-up. (Applause.)

The public craze for land makes it easy for promoters to do their work. Many farmers, dominated by a spirit of consideration for their children, accept the "spiel" and assurance of the "dopster," sell in agricultural regions and move onto nearly worthless land, believing that it will become about like the old home place in time, and that each child will then have a farm and home. May we not say that he who deceives a family in this way is a mean man? (Applause.) Can you think of a worse service to a community? Certain railroads are not free from blame in that they promote this traffic. The farmer who accepts the bad "dope" is also to blame. It has taken a long time for the people to learn that mere belief, opinion, and sentiment are not strong enough forces to overcome the influence of land not suited for agriculture.

If our land seekers could realize how important and far-reaching is this matter of choosing favorable places for home building, they would be less easily led astray. They would consider soil, climate, water supplies and other necessary conditions of success, as they actually exist, and be governed less by the old arguments and slogans so often used for land development in general. They would pay less attention to deceptive literature written for the special purpose of securing emigrants and sales. They would inquire into the methods whereby this phase of the land business is carried on, and avoid being carried off of their feet, especially when on “home-seekers’” excursions and worked by a well-organized plan.

Formerly, the newer States encouraged the work of grafting land men. Time has shown, however, that this was bad business and really a drawback to permanent development. The present trend is to conserve the interests of those who go onto and manage the land, making it easier for them to succeed. They are assisted by the publicity of useful facts and the censure of fraud. Furthermore, it is coming to be recognized that State emigrant agents, agricultural experiment stations, soil surveys and Conservation Commissions should not lend their support to any interest other than that which brings the best results to the people of the State. They should stand for the policies that insure permanent development and do so as their plain duty. Do you know how public men are urged and tempted to further the interests of promotion concerns and that there are plenty of opportunities to sell one’s influence? That it requires diligence and courage to rightly serve the State? Happily, our public-spirited citizens who have at heart the best and largest interests of their States, stand strongly against misrepresentation whether unintentional or not. They claim that doubtful promotion serves only in closing deals, and in directing settlers to the land, but that in the long run the process works a positive harm to the misguided people and to the State as well, if the land is not suited for habitation. Fortunately, most States are coming to this viewpoint. They have learned that it pays to tell the truth when transplanting a population and directing the permanent development of a State.

Where do you delegates stand on this proposition, and what is to be the attitude of your States?

Promoters’ Methods.—Do you know the signs of fraud? They are exposed in the method used in securing money from the community. The plan is about as follows: A selling scheme is perfected. It is constructed in a way that leaves no flaws, apparently. Each agent learns the scheme; he becomes skilled in applying it to the different types of individuals. Too often it is of little concern whether the project has merit or not. The chief object is to get money. Extravagant claims are made in which returns of 100 per cent. or more a year are said to be a sure thing. The influence of nationality, church, and fraternal orders are brought to bear in securing sales. The support of persons with good standing in the community is secured. Those who assist the promoter are given a reduction for their influence. The dope is given them often and systematically. So they soon realize the greatness of the project. This is promotion psychology. The land is offered at high enough price to permit reduction for quick sale, which bait works in many cases. Persons filled with greed for money are easy victims. The above kind of thing, though less common than formerly, is practiced in most States, and the wonder of it is that it can continue and why it is permitted to continue. It is fraudulent and should be stopped entirely if we are to conserve the interests of good people.

Effects of Land Fraud on Local Business.—Many families lose enough through fraudulent entanglements to give a college education to the son, a piano to the girls, and general improvements for the home or farmstead. The drain is away from home and school. Perhaps the greatest loss is the people who are lured to places where in many cases they are less well off than in the old home. Persons who lose in bad deals become suspicious of real business done by reliable men in the community. They refuse to invest in local developments in which the returns are sure, though smaller than those promised by promoters. Many are put out of business entirely by land frauds.

Do you agree with me in that it is not good business to farm the land, cash in its fertility and then scatter the proceeds among grafters? Let us quit chasing the ends of the rainbow, and turn our attention more towards the right use of the fruits of our labor in education and home building. (Applause.)

Regulation.—There are many laws for the conservation of business. The Federal Government prosecutes persons who make fraudulent use of the mails. There is opportunity under the law to recover on account of misrepresentation; but these laws are not sufficient. Public sentiment is now ripe for the enactment of special laws to conserve business against land frauds. Nebraska has made a special study of the subject, reduced fraudulent procedure by the force of publicity and public opinion, and will pass special conservation laws in its next Legislature. Kansas has gained distinction by the enactment of the well-known “Blue Sky Law.” This is good so far as it goes. It provides for registration, reports, supervision and penalties. Many States, as, for example, Wisconsin, Wyoming, and Texas, are to undertake legislation of this kind at the next sessions of their Legislatures.

Provision should be made in the special act against land frauds for field examination and report upon properties offered for sale. This field work might be done by the State Soil Survey, or the State Conservation Commission.

An essential feature of the act will be the registration of realty agents and the furnishing of proof that they are competent and reliable. This will reduce the number of land agents and insure the responsibility of those permitted to do business. The Western realty men are now framing a law of this kind to meet the needs of the various States.

Apparently there is no opposition to the proposed legislation for it is to conserve business and eliminate fraud. It is sure to receive the support of all unless we except those who make gain through doubtful promotion. If opposition appears before the various Legislatures it will have the embarrassing position of being on the side of fraud.

Summary.—Let me close this report with the following statements:

1. This discussion, though favorable to reliable land agents is against doubtful promoters.

2. Realty agents should have a practical knowledge of land classification, soil types and the land business.

3. Reliable and competent real estate agents have an important place in the State. They are against promoters and promotion values.

4. No one should deal with an agent who is not favorably known and is not good at the bank.

5. See the land you purchase. Also get a reliable report upon it from a competent, disinterested party. Base your transaction on facts—not on opinions. Get a good title and not a mere promise to deliver.

6. Keep out of the “get-rich-quick” schemes. Quit chasing the ends of the rainbow. If your fever gets too high, consult a banker.

7. As a rule, it is best to avoid the “home seekers’” excursions and “boom” literature, unless you are sure of your footing.

8. Consult disinterested old-time residents whose places show that they are actual, successful tillers of the soil in the locality where you are to buy. They will give you the farm value, and not the promotion value.

Ladies and gentlemen, are you ready to support in this important movement? (Applause.)

Chairman Wallace—I am sorry we haven't half an hour longer to give Dr. Condra to skin those skunks.

We will hear from Mr. Charles S. Barrett, President of the Farmers' Union, and finally from Mrs. Lund, of California. I want these speakers to show their appreciation, their gallantry, by giving her the last five minutes, and I am going to call them down unless they do.

Dr. Condra—It has been suggested that we close this discussion in one minute. I am very sorry that neither Mr. Barrett, or Dr. Bateman can be heard.

My friends, when a State puts upon its statute book an adequate law, no fake concerns will seek to do business in that State. That is true. Now, we ask that your committee be continued to the end that we may report the conditions of the soil and the development of the soil. I thank you and give ten minutes additional time to the lady.

Chairman Wallace—It is my great pleasure to introduce Mrs. Haviland H. Lund, of California, whose subject is the "Conservation of Land and the Man."

Mrs. Lund—It is a great pleasure to follow Dr. Condra, because his speech is such a good precedent for what I have to say.

If the masses of the American people knew what one man could accomplish for himself, physically and financially, upon from one to five acres of land, this knowledge would revolutionize the life of the Nation. The congestion in our cities is more than a country-wide menace. It is an unnecessary outrage. There is land, good, health-giving land, enough for all the people.

The conservation of the man has been too long overlooked. The commercial policy of the Nation could scarcely be called far-sighted—so wasteful have we been of all natural resources.

We have despoiled our forests, impoverished our soil, given away the public domain. Our labor conditions in many respects shame us in comparison with other nations. Looking about today, it would seem that our thought has been “Get all we can, no matter how, and waste it as we will, for after us, the deluge!” But a new commercial and political spirit is being born; a renaissance of righteousness is setting in, and the commercial leaders of the country are taking stock, as it were, of the actual situation.

Big business men are realizing that a healthy man is worth more in dollars and cents than a half sick one; it is recognizing that sanitation is a good investment. It is beginning to wake up to the fact that the children are more valuable producing machines when they are well protected, housed, fed and educated. The cry of the philanthropist to give because it was right and necessary that these conditions be ameliorated, has met with only sporadic response, but this new call to do the right thing because it pays in dollars to do it, is meeting a greater answer from the people.

Little Farms Magazine found it impossible to evade the responsibility imposed upon it by its readers. We roused them to a desire to go out upon the land—to try the new condition. They came to us for information. We could not go into the land business. We decided to form “Forward-to-the-Land Leagues” in all principal cities.

Moneyed men are not asked to contribute alms but only to invest their money at a nominal rate of interest, which the workingman with his own home and garden, with health and a living assured, is willing and able to pay. This has been proved where the experiment has been tried in the manufacturing cities in England, and in such communities as San Ysidro, Southern California, in our own country.

The work of the Little Farms Magazine in the founding of these Forward-to-the-Land Leagues has been unique and necessary. And its purposes two fold.

In the first place, it was of the utmost importance in meeting the grave problems confronting the nation, particularly that of the bringing our ratio of agricultural production where it safely balances the ratio of population, to have a medium by which knowledge of the intensive methods of agriculture could be brought to the individual.

The widespread interest in the forward-to-the-land movement, which has been taken up alike by press and magazine, has created a hunger for specific information which occasional columns of general news can not satisfy. Little Farms Magazine tells, specifically, how a small acreage will yield and has yielded, industrial independence. It quotes stories of those who have made good after leaving the old work of bookkeeping and clerking and taken a “little farm.”

The problem which the farm presents today is not the same as that of yesterday. The loneliness and isolation no longer obtains. The message that the Little Farms Magazine takes to the world today is that *scientific agriculture makes the acreage necessary for individual maintenance so small that social life can be developed on the farm in the most ideal manner*. The magazine advocates the upbuilding of the social center, with its library, its clubhouse and gymnasium, its moving pictures and mechanical music.

As I came through the country from the Pacific Coast and saw the empty acres of farm land waiting, and then entered the big eastern cities, and looked into the hopeless, pallid faces of its people, I could think that the earth, if it had a voice, would cry aloud with the cry of Him of long ago, who said: “How often would I have gathered thee as a hen gathereth her chickens, but ye would not.”

Chairman WALLACE—There are fifteen minutes left. If Mr. Barrett, President of the Farmers’ Union, is here we would be glad to give it to him.

Mr. BARRETT—Mr. President, Ladies and Gentlemen: Speaking for approximately three million American farmers, I can say with absolute accuracy that the primary article in the creed of Conservation should be the conservation of the man on the land.

In volume and variety of resources, the United States is the mightiest nation in the world. It is true that the British Empire may, through its dependencies, have a greater territorial reach, but from the standpoint of a continuous stretch of land and the body of acres cultivated and susceptible to cultivation, America admittedly leads the world.

The effect of this handicap is indicated not only in the present breadth of our domestic and international commerce, but to a greater extent in the promise of its more wonderful commercial conquests yet to come. The Nation is barely on the threshold of its destiny. That fact should not mislead us as to the difficulties in the way of making the destiny real, and not merely a boastful prophecy.

In the process of transmuting our possibilities into assets—what is the dominant factor? The American farmer. I challenge any of my distinguished audience to mention a single phase of commerce, one feature of trade, the smallest detail of actual subsistence that does not eventually trace back to the man plodding out there on the acres.

Napoleon said an army traveled on its belly. He could have said, with equal truth, that civilization travels on its belly. And the farmer is the factor that fills the Great American Stomach, and that keeps full every dinner pail, regarding which we have heard so much during political campaigns. More than that, he also clothes the armies of development. Nor must we forget that with the South's cotton as the lever, he keeps the international trade balance on the American side of the ledger. You tell me the manufacturer plays a large part in our current and our probable development. This is true. You tell me also, that what might be called trade-strategy, pure and simple—the proverbial "Yankee shrewdness"—is going to win for America the bulk of the world's business.

I do not dispute these assertions. But I answer: That back of trade-strategy and of dollar-diplomacy is—the American farmer. Without him, all would be in vain; without him, all of those resources we agree ought to be conserved would melt into impalpable air.

Let us admit, then, that the farmer is the keystone in the arch not only of national advance, but of sheer national existence. His problems, then, are the Nation's problems and his welfare, the Nation's welfare. No nation is stronger than its farmers. If the farmer is poorly nourished, if the Government is negligent of his rights, indifferent to his mental development and moral soundness, the way will be surely blocked to our national march forward.

It is to the vital interest of America to cultivate intensively not only the farm, but—what is more important—to cultivate intensively the farmer. What use to conserve our resources, unless we conserve the man behind the resources? The stability of national progress and of government itself is dependent upon conserving the farmer.

All of you within hearing of my voice may say: “We concede these facts. Are we not, right now, trying to aid the farmer, to conserve him, to intensively cultivate his possibilities and safeguard his rights?” And I answer: “Probably you are. But you can not help—you can not conserve—you can not cultivate the farmer unless you mix and mingle with him in the first person—not for twenty-four hours, but more likely for twenty-four months or twenty-four years.” I give full credit to the splendid intentions of the men who have tried and who now are trying to aid the farmer. But you can not adequately grasp his problem by using field-glasses from the convention hall or interviewing him over a long distance telephone, so to speak.

The scientists who are searching for secrets, the missionaries who are looking for converts, use neither of these methods. They go straight to the scene of battle. And so must all persons do, my friends, if they expect intelligently to conserve, to cultivate the American element which is the pivot of all other elements in this country. Study him at first hand, then your sympathies will be practical, not theoretic; your suggestions based on conditions, not on conjecture. Fight with him, side by side, in the ranks, day by day. That is the only way you can learn of the foes—not the least of which is his own weakness—which he has to combat, and what his victory means to the weal or woe of this common country of ours.

At this point President White reassumed the Chair.

President White—The ex-President of this Congress, familiarly called “Uncle Henry,” and, in dignified circles, Dr. Henry Wallace, but who doesn’t like the name and prefers “Uncle Henry,” will speak tonight, as will Judge Ben B. Lindsey, of Denver, Colo., the children’s friend.

The morning session is now at an end. We hope you will get back here at 2 o’clock, because we have a very full program.

FOOTNOTES:

[3] These statements are based on many specific examples of fraudulent promotion.

FIFTH SESSION.

The Congress reconvened in the Murat Theater, at 2:00 o'clock p. m., and was called to order by President White.

President White—On account of Professor Fairchild's being called away, having to leave on an early train, we will listen to his address first this afternoon. Professor Fairchild is foremost in the ranks of modern education, in teaching the conservation of human life, the conservation of the soil, and everything that goes to make up thorough manhood among the boys of the land. I now introduce to you Prof. E. T. Fairchild, of Topeka, Kan., President of the National Educational Association, whose subject is "The Duty of the Teacher."

Professor Fairchild—Mr. President, Ladies and Gentlemen of the Congress: With your permission I want to change my subject as printed. It is not the subject of my remarks this afternoon. I should like to call it "A Plea for More Equal Educational Opportunities."

In the few minutes allowed me, I can only hope to sketch briefly some of the conditions that confront us today. I shall have some things to say that represent definitely a great lack of progress, but that I may not be labeled as a pessimist, I wish at the beginning to state as my conviction that the present is the best moment educationally that the world has ever seen. Had I the time, I should like to describe to you the marvelous progress that has taken place in certain types of our educational activity.

The growth of our universities and colleges is little short of marvelous. In a single decade in these United States the increase in enrollment has been fully 98 per cent. This increase in enrollment has also been manifested in Europe, in England, where there has been a genuine increase in the number of provincial universities. The increase in enrollment in the past ten years is most marked in Germany. In Germany, where there have been no new institutions erected the increased enrollment in a single decade represents 60 per cent. Such is the history of the increased enrollment, which, with increased efficiency in the way of larger and more efficient faculties, has taken place in this country and in Europe. It is a world-wide movement, my friends, and so far as I can see, is a recognition that the best field of opportunity to the ambitious and capable youth is through the college.

Then we come to the story of the success of our high schools. Here again the growth has been phenomenal. Those schools in number and in enrollment have gone forward by leaps and by bounds. In a single State, in my own, if I may be pardoned for this allusion, let me tell you what has happened in five years. The increase in the number of high schools in five years is one hundred per cent. and the increase in the number of teachers one hundred and twenty per cent. This is simply typical of the condition all over these United States. Again we have a concrete instance of the conviction upon the part of our people that the boy or girl of today who is to have something like an equal chance tomorrow, should have the opportunities provided by our high schools. I wish I had time to expand before you this growth and its meaning to our nation, but it is not my purpose to discuss that at length. I want to say, however, that as the result of this marvelous activity and growth we have had in our high schools and our graded schools in the cities, we have reached a maximum term. We are having a constantly enriched curriculum and generous expenditures are being made everywhere. Modern buildings, the latest word in lighting and heating and ventilation are found everywhere. Thorough organization characterizes this type of our educational work.

A vital point in the development of this growth, this high organization, is the expert supervision that has charge of these schools everywhere.

It is upon these higher institutions, the universities, the colleges and the high schools that the emphasis of educational thought and interest has been bestowed. Here notable investigations are constantly in progress with a view to still greater efficiency. Here the active and moral influence of the best and wisest of our country finds expression. Every one is aware of these educational activities and all are proud of them. We are spending money most generously for the city grade schools, for the high schools, for the colleges and the universities.

But, good friends, now I come to the essential thing I wish to present to your attention. Of the twenty-five million boys and girls in the United

States today of school age a majority receive their foundation training in the rural schools, or in an environment that is characteristically rural. And what of these types of schools? Do we find the same advancement? Do we find the same public concern? Do we find the same skill and organization and supervision? Do we get such results as are being secured by city schools? They do not measure up in the character of teaching, friends, in the kind of courses of study, in the length of the term, in the results obtained and this is perfectly obvious to the student of rural schools. There has been no such progress. What we have too often is the untrained teacher in the rural schools; terms ranging from three months up to six and seven; buildings that lack in every modern application of light, heat, ventilation and sanitation and all attractiveness; ground that too often is the sore spot of the community; houses that too often represent or suggest the pest house rather than a place of learning; inadequate support financially and inefficient supervision.

I want to speak briefly, then, to you in the few minutes that I have left as to four vital defects, as I view them, that it seems to me are the defects that we, as a nation, ought to undertake to remedy.

First, we have too many untrained, inexperienced teachers in these schools. In saying this, my good friends, I am not unmindful of the fact that in thousands of communities in this country of ours are to be found rural conditions that are most pleasant, that there are thousands of teachers in rural schools consecrated to their work, performing a daily service for those boys of inestimable value; but listen, in one of the largest cities of this Union the superintendent of public instruction in a report said of 10,500 rural school teachers 9,400 are themselves but eighth grade graduates. I want to avoid being too explicit in pointing to the places, but I personally know of a State, a State that stands well above the average, I think, educationally in the United States, in which of the 8,000 rural school teachers 4,400 in 1910 had only such training as is found in the country school and not beyond the eighth grade. In too many places this condition prevails. I believe I am well within the truth when I say that of the teachers of the fourteen million boys and girls in the rural districts of America today who are being taught, more than fifty per cent. of those teachers have themselves an academic training that does not extend beyond the grades.

But, to return to the subject of the high schools, that I spoke of a moment ago. See how conditions have changed as to the training and kind and character of the teachers that must be placed therein. It is rare that you see a teacher of a high school who is not a graduate of a high school and in many cases of a normal school. In the rural school, the first defect is that we have too many poorly trained teachers.

The next thing I wish to speak of as a great defect in our present system is our manner of raising and distributing tax. You are aware that the prevailing unit of school organization in America is the district. In my own State we have 13,400 teachers; to boss, guide and direct those 13,400 teachers is an army of 30,000 school officers. By the way, he is the most numerous officer in this country. Within a radius of three miles you will run across a school officer in most of the States of the Union, a condition that makes for lack of uniformity, lack of singleness of purpose, the most wasteful, the most extravagant system that could be devised. But I want to speak a word in regard to taxation. The trouble is, good friends, that our system of distribution of taxes is utterly unfair and utterly prejudicial to the best interests of the child. On the one side of the road is a district having a splendid valuation with a low tax that may maintain eight months of school with a splendid teacher, a good building, and on the other side of the road, the maximum reached by law or gone beyond it, they are only able to supply the most inferior facilities for the boys and girls. The day must come when we shall have a prevailing system at least with the county as a unit for the taxes to be raised and distributed, so that the boy or girl who lives in some poor part of that county or State shall have the same opportunity as far as money will bring it to have a good teacher. The fact is that poor communities are the ones that ought to have the best teachers in all this land (applause), and that the contrary is too often true I am sure you will all agree.

Let me say as to the courses of study now a word or two. I have, good friends, said to you that there are twenty-five million boys and girls of school age in America, fourteen million of these in rural schools. Now, listen, of these fourteen million less than twenty-five per cent. are so much as completing the work of the grades in this, the morning of the twentieth century. If this does not spell tragedy then I have no means of interpreting these facts. Less than twenty-five per cent. To assign the reasons for this is difficult, but because of the kind and character of these schools, because they are lacking, because they are not making the progress, because they have not the attractiveness that our city systems have, is a reason why these boys and girls do not stay. But there is another and further reason. The course of study too often lacks vitality; somehow and someway we have not grasped the thought that the school has a larger and wider duty than consuming all its time and energy in text book knowledge. Somehow and someway we have failed to see there, as we are coming to see in our more highly organized system, that to interest that boy and girl, to send them out capable, self-sustaining citizens, we must do more than consume our time and energies on the text book knowledge. I should like to see a reasonably but not rigidly classified course of study with adequate attention to fundamentals, to large opportunity for hand work and with every possible connection between the experience of the school and the actualities of life. We must vitalize these schools. Another important thing in connection with our rural schools is this: the great majority of these boys and girls are denied high school privileges. Here in the city of Indianapolis with the splendid system of high schools that they have small wonder is it that the boy and girl in the grades if possible persevere in the work, looking forward always to the opportunity to get this liberal education afforded in the high school. Often this is not true in the country. As I said a moment ago, the great majority of these boys and girls are denied such opportunities, denied for geographical reasons, for financial reasons. If every township there could be created a rural high school, in its course of study emphasizing the things that are most needed in the lives of the boys and girls in that township, preparing them by a well developed and organized course of study for the great and important and practical business of life—if such an institution could be put within the reach of those fourteen million boys and girls, don't you agree with me that many more than twenty-five per cent. would finish the work of the grades in the hope that they, too, might enter these schools and enjoy their advantages. And so I say, there is another great defect that some way ought to be overcome.

Now, just a word or two further. The last defect that I will mention is the question of supervision. In my judgment the commanding reason for the development and growth of our city schools is the skilled supervision supplied by the city superintendent. If we could have like supervision in these schools in the country the development would be marvelous and it would be rapid and vital. We have county superintendents. They have them here in Indiana. We have them in our State, but in no single instance so far as I am aware, is this supervision adequate. First of all, to remedy the question of the supervision of our schools, the question of the superintendency should be taken absolutely out of politics. (Applause.) It is a crime against the children of this nation to select either a city superintendent or a county superintendent upon any other basis than educational qualifications. (Applause.) The children of Indiana, the children of every other State of this Union will never come into their own, good friends, until the supervising element is selected because of their being experts in the job they are looking for.

Now, just one other thing on that. It is perfectly preposterous to expect a superintendent in a county such as there are in my State, for illustration, to visit one hundred and fifty or more schools, going over roads in all times of year, in all conditions, to make his visits worth while. He may get there once a year. We ought to imitate Oregon in this respect. In Oregon they have subordinate superintendents, one for every twenty schools. There they can accomplish something.

My time is more than taken. You have been patient, as has your President. I thank you most sincerely. I only regret that I can only touch the edges of this problem.

In conclusion, you representatives of this National Conservation Congress, here is the problem. The great thing we need to do, first of all, is to make public everywhere the actual condition of the rural schools. Publicity is the first step; organization is the second; organization of national scope and of State scope. Give me twenty common people in any State in this Union and I will guarantee to see that the rural schools make more real genuine advance in the next five years than under ordinary circumstances they would do in ten years.

The country is the Nation’s great recruiting ground. Here we look for the best men and women of tomorrow who are to take leadership, who are to represent in their actions and in their lives the good red blood that characterizes the Anglo-Saxon race. Are we doing our duty when but a paltry three million five hundred thousand out of a total of fourteen million are not so much as accomplishing the work of the grades? (Applause.)

President White—Louisiana has been first and foremost in several phases of Conservation. Louisiana stands first in making forestry possible by wise and beneficial laws that encourage forestry, and I think Louisiana stands among the first in its State Board of Health, doing something worth while in every parish. I have the pleasure of introducing to you Dr. Oscar Dowling, of New Orleans, Louisiana, President of the Louisiana State Board of Health, who will speak on "Hygiene in Relation to Public Health."

Dr. Dowling—Mr. Chairman, Members of the National Conservation Congress, Ladies and Gentlemen: We are very glad to have this opportunity to appear before this great Congress. In the beginning I want to say that we owe much of our enthusiasm to the good work of the Indiana State Board under Dr. Hurty, and to your pure food department, under Dr. Barnard; also to Dr. Evans, of Chicago, and Dr. Wiley, of Washington. We have endeavored to imitate them in some ways, but nevertheless, in some ways we have fallen short.

Hygiene, the science of preservation and promotion of health, in some form, has been recognized by every nation since the dawn of civilization.

Among the people of antiquity, conquest and domination were directly dependent on physical vigor, hence their laws regulating this feature of national life. Among the Greeks, the health idea was embodied in the cult of Hygeia which arose hundreds of years before the Christian era, consequent probably to a devastating plague. In the early period of Rome, when courage and patriotism were cardinal virtues, physical development was provided for and emphasized. Social and political fluidity in the middle ages precluded the evolution of organized thought or systems in sanitary science.

Individuals set aside conventional thought and method and strove with Nature that they might learn her secrets; their work was not in vain, but with few exceptions their discoveries were unimportant.

The experimental method popularized in Baconian philosophy gave an impetus to the study of the physical sciences, but many decades passed before notable deeds were recorded. It was the nineteenth century, scientific in spirit and achievement, that made vital the long result of time and opened a perspective before undreamed of. The awakened health conscience of today is the crystallized result.

In scientific annals, the discoveries of the bacteriologist rank among the first. Perhaps, in the evolution of knowledge no truths are more potential. Within a generation the influence is marked, not only in relation to the individual and community, but in effect on the civilized world. The sanitarian with this knowledge was enabled to demonstrate control of environment. The success of the experiment has opened a new world just as surely as did the discovery of October, 1492.

The changed viewpoint of the relative value of hygiene in its application to life is due not wholly to the discoveries in medical science. It is one phase of the general awakening to the defects of the present social order: a manifestation of the modern attitude toward "waste." Efficiency implies economy, not alone of expenditure, but of material resource and vital force.

Conservation and preservation of the material wealth of the country is dominant in the intellectual activity of all enlightened people. But it becomes increasingly apparent that the Nation which conserves its mines, forests, soil and sources of power is poor indeed if its men lack virility and mental initiative. This thought is back of the public health movement. The impulse is in part commercial, in part scientific. It grows out of recognition of the futility of remedial and philanthropic measures and the conviction of the potentialities of science for human betterment. In import the movement is ethical and spiritual; it is beyond question the greatest of modern times.

This meeting is significant of the changed attitude toward the Nation's greatest natural resource—its people. The Congress is national, its purpose conservation, its main topic—to quote from the invitation—the conservation of vital resources. There is significance also in the topics selected for discussion in the health section. They relate to the larger aims of sanitary science. In the popular mind health work has reference only to superficial conditions, control of epidemics, cleaning of streets and similar activities, but the hygienist knows that sanitary regeneration means an attack on many existing institutions, customs, practices and methods that lie deep in the roots of the social structure.

Housing, child labor, industrial occupations, labor insurance, vital statistics, food supply, community methods and conditions are the subjects chosen for discussion. Their primary importance is apparent.

The period of twenty minutes allotted for the opening of this division makes imperative only brief suggestive statements of the essentials in their relation to public health and individual well-being.

Mr. Lawrence Veiller, in the Annals of the American Academy, says: “We have paid dear for our slums.... No one has ever attempted to estimate the cost to the Nation of our bad housing conditions, because it is an impossible task.... Who can say of the vast army of the unemployed how large a portion of the industrially inefficient are so because of lowered physical vitality caused by disadvantageous living conditions? Of the burden which the State is called on to bear in the support of almshouses for the dependent, hospitals for the sick, asylums for the insane, prisons and reformatories for the criminal, what portion can fairly be attributed to adverse early environment?” Describing surroundings, the author continues: “The sordidness of it all, the degrading baseness of it, unfortunately is withheld from the eyes of most of us. What it can mean to the people who have to live in the midst of it we can but faintly conceive. Let us frankly admit that these conditions result in imposing upon the great mass of our working people habits of life that are more compatible with the life of animals than with that of human beings.”

Moreover, not alone in the slums do these conditions exist. In almost every city of the Union, a few blocks from the main thoroughfares, there are congested districts unspeakably bad.

With the knowledge we now have of the relation to health and sickness of air, sunlight and propagating agencies of disease incident to dirt, it is nothing short of criminal to tolerate such conditions. If physical suffering only were the result, indifference would be unpardonable, but overcrowded homes, insanitary in every respect, make for low standards of decency and morality. Vice, with its correlatives, disease and pauperism result. Often crime and insanity make the chain complete. The conditions of life in the middle ages as recorded in history seem to us barbarous in the extreme; relatively, ours really are. Then, there was no certainty as to the effect of insanitary environment; the people did not know; we do, yet with inexplicable indifference communities not only let the worst obtain, but they permit a perpetuation of the system. Authorities stand aghast at the expense involved in the tearing away of a whole section of a city, but the cost of such a measure easily, often probably, may become a mere item in comparison with the economic loss from an epidemic of a virulent type.

It is a hopeful sign that a few enlightened municipalities have set an example in remodeling districts, not only in the erection of comfortable homes, but further in the establishment of healthful and beautiful environments. The housing problem is one of the most difficult and complex of our day. It can be solved only by enlightened legislation supported by public opinion.

About a century and a quarter ago the factory system began to develop with intensity in England. Later, in this country, it grew by leaps and bounds. Child labor with its attendant evils was a logical result. For nine years there has been systematic effort to control the unhygienic features of the system. Some good has been accomplished, but because of the nature of the problem progress is slow. The injury to the child is plainly apparent. Long hours in poorly ventilated rooms, with constant use of the same set of muscles, stunts and dwarfs the body; equally, the mind. Toil of this nature uses up the young life; it leaves the State the burden of caring for an individual hopelessly inefficient if not worse. But of more importance is the consequential physical deterioration. If these youthful toilers grow to maturity their bodies are devitalized; if they marry their children are almost invariably low in vitality. Hygiene in its application does not imply the remedy of existing conditions alone for the individual or the present; it looks to the future. Therefore, protection of the child is a principle of paramount importance.

Child labor laws are now more humane than a few years ago; conditions in many factories have been vastly improved. But as yet we are far from an ideal stage in the regulation and supervision of this feature of industrial life.

Every argument concerning the employment of children in factories may be applied to women engaged in similar occupations. In the mills and shops where women stand all day, where they endure for hours not only unhygienic environment, but in addition mental anxiety, where the whip “employed by the week only” is held over them, the nervous strain as well as physical exertion saps the very foundations of vitality. Investigations made by Dr. R. Morton of New York, show the health of industrial women is proving a serious thing in the United States, and unless conditions are bettered that there will be a general breakdown of the working women of the country. Nor is this the sum total of the consequences. In the children of these women low vitality is perpetuated. Records quoted by Dr. George Reid, Health Officer of Stafford, England, give the mortality of children under one year of age as greater among those of mothers who work in factories than among home mothers. Statistics compiled by him show the death rate one hundred and forty-five per one thousand births for infants of home mothers and two hundred and nine per one thousand births for infants of mothers who work in factories. The injury to the State is apparent.

On the question of prevention of occupational diseases, I cannot do better than quote the measures suggested by Dr. H. Linenthal, of Boston. They are: collection of accurate data about working conditions; data relative to the effect of occupation on mortality; proper medical instruction; reporting to health authorities specific industrial diseases; examination of all industrial workers; exclusion of minors and women from certain industries; sanitary laws for factories; regulation of dangerous trades by health authorities, and the carrying of an educational campaign of hygiene among employers and employes. The comprehensiveness of these measures indicates the extension of the problem. No movement of recent times is more humane and economic than the one termed industrial insurance.

The purpose is the capitalization of the workingman's energy at the time of his greatest productivity; the basic principle that every far-sighted social policy is founded more on energy reserve than money reserve. The aim is to secure for the nation the greatest possible reserve of bodily and mental force and power and physical and moral health.

The problem has been attacked in various ways by different countries. Germany has been the most successful. There the workingman's insurance has attained the dimension of a gigantic social institution. Dr. Frederick Zahn of Munich, Director of the Bavarian Statistical Office, in a recent address, gave the following interesting figures: Out of 16,000,000 laborers in Germany, 14,000,000 are carrying sick insurance, and 15,700,000 invalid and old age policies.

In the past twenty-five years over one billion six hundred million dollars have been paid in benefits. In addition, prophylactic measures are provided for.

Only those familiar with the necessities for correct data in health work appreciate the immediate and imperative need for statistical information. Records of births and deaths and of supplementary details form a basis for advancement. Without such data, the sanitarian gropes in the dark. Yet no request from the health department is so lightly treated. Reform in this can be wrought slowly. Appropriations to pay registrars and enforcement through the courts are the means for the inauguration of a more perfect system.

One of the hygienic essentials in this country is education in the relative values of food products. The phenomenal growth of the urban population which has reduced the number of producers and the almost universal practice of adulteration make imperative the enforcement of stringent laws and instruction in the nutritive value of classes of foods and the economy of selection.

The campaign for a supply of clean, pure milk in many centers has grown out of the effort to lower the infant mortality rate. It has stimulated inquiry and supervision of other food products which is encouragingly prophetic.

Hygiene in its application to personal and community life is essentially preventive. This idea is not sufficiently understood to be taken at its real value; curative measures the people commend, but possible calamity seems remote, therefore, prevention does not appeal. It is this concept of the collective mind that lies back of the extravagant parsimony universal in health appropriations. It also explains public apathy and indifference.

The most practical means for sanitary progress are two, education of all the people in the primary truths of hygiene, and the application of the science through governmental agencies. These are so closely related that they are practically inseparable, but logically may be differentiated.

Hygiene is an organized science; its principles are rational and demonstrable; its application will bring returns economic, ethical and spiritual. This must be acceptably taught to the people by methods suited to the present state of the public mind. Conviction that will lead to action is the end to be sought. Education will create a public sentiment persistent and insistent for measures promotive of public good. Concomitant with this effort, in fact a part of it, the various units of government should be executives in the establishment of hygienic measures and the abolition of insanitary conditions. When people believe that the eradication of typhoid fever and hookworm disease is more important than high or low tariff; when they become convinced that malaria is a national disgrace and uncleanliness a relic of barbarism, there will be money and judicial decisions for the elimination of these defects.

Fortunately, these are the views of an increasingly large number of people. There is a health awakening. The principles of the science of health are every day becoming concrete in laws, and habits of thought and living. It is the conviction of the progressive minority that a Nation's first duty is to conserve and protect its citizens, to develop a community of efficient men and to minimize natural disadvantages. Further, that collective intelligence must plan for the preservation of the people and the perpetuity of the State, and in so doing must recognize public health as fundamental, both in the simple phases and in its comprehensive aspect. (Applause.)

President WHITE—The next subject to be discussed is by one who employs labor in the State of Indiana, and who is a large employer of labor. His subject is "The Duty of the Employer." I now take pleasure in introducing to you Dr. Edward A. Rumely, of Laporte, Indiana.

Dr. RUMELY—Mr. President, Ladies and Gentlemen: Four generations ago, there were but three millions of Americans scattered along the Atlantic Seaboard. Back of them was a vast virgin continent, the richest the white man had ever found in the long migration upon which our race started ages ago. The American continent was rich in timber, in the soil fertility of its vast valleys and prairies, and rich beyond measure in the superabundant deposits of mineral wealth. The first settlers were few in number; they brought with them but few tools and little wealth that today we would call capital. It was the natural and proper thing for them to set to work to gain, with the least possible labor, the great natural wealth that the virgin continent treasured for them.

They killed the fur-bearing animals, felled the trees to export lumber, dug in the quickest way the mineral wealth of the land and started to grow such crops as would carry to market the greatest value from the fertility of the virgin prairies. Wheat was easily transported, and each bushel contained from twenty to thirty cents of soil value. Hence with wheat our prairies were taken under cultivation, and from the returns of the wheat crops cities and railroads and homes were paid for.

Only today, when the average yield per acre has gone down from forty to thirteen bushels are we beginning to see clearly that by this process we have been drawing heavily upon our soil capital.

While the population was small, labor was difficult to secure. Cities had to be built, roadways opened, railroads constructed, rivers bridged, and a continent brought under subjection. The process of the past four generations was possible only because our fathers economized their own labor and created, as fast as possible, the values they needed to barter off into the markets of the world for capital from the superabundant natural wealth that surrounded them.

Today, we are mining our iron, copper, lead and other metals more rapidly than any other country in the world. The pioneer farmers who worked the soils of the south with tobacco and of the east with wheat, can no longer move off to the west, when, having exhausted the fertility of our lands, they find farming no longer profitable. The hundred thousand vigorous Americans who went last year to Canada with energy, capital and American tools are a concrete evidence that we have reached the end of the course which we have been traveling.

The whole country has been startled by the warning of the far-sighted men, and now the demand for conservation of our natural wealth is becoming more and more insistent. We have been made to realize that every child born brings a mouth that must be fed, a body that must be sheltered and clothed, but no increase in natural wealth. We must still learn that every child does bring two hands which can work, and which, when highly trained and backed by scientific knowledge, can create untold values. Stated otherwise, we must care for our increasing population, not by increased exploitation of our natural stores, but by providing abundant work for skilled labor.

AMERICAN FARM MUST BE FACTORY—NOT A MINE.

Our agriculture has been a process of mining. The farm must now furnish a field for the profitable employment of skilled labor, for the use of capital, and the application of the principles of scientific management, becoming thereby a workshop instead of a mine.

In order to sell the labor power of our people, we must encourage the development of all secondary industries. By “secondary industries” I mean those industries which take raw materials that are largely the product of crude machinery and unskilled labor, and add to them in a large measure labor and capital values.

The agricultural implement manufacturer purchases steel and iron at approximately one cent per pound, and by further refinement creates implements worth eight cents to twenty cents per pound. The automobile maker takes lumber and iron, worth from two cents to four cents per pound, and produces a car worth from thirty cents to one dollar per pound, while the same materials, worked up into cash registers, typewriters, etc., would be worth from $3 to $10 per pound, and in watches from $50 to $5,000 per pound.

CREATE VALUES FROM LABOR.

We began by cutting the maple tree into a cord of wood, worth from three to seven dollars, and each tree furnished material for one day’s work. This same tree—if sawed into lumber—is worth twenty dollars and would furnish employment for one man for three or four days. If quarter-sawed and more carefully treated, it might be worth forty dollars and would furnish employment for more skilled and better paid workers and for a period of from ten to twelve days. And this same lumber, in a furniture factory would produce furniture worth from $100 to $500 and would furnish employment directly and indirectly equal to from six months to one year’s work for one man.

The whole range of values in this series, from the seven dollars’ worth of cord wood or $500 worth of manufactured goods, depends upon the degree of refinement extended to identically the same raw material through the quality and quantity of labor employed upon it, the capital expended and the application of greater scientific knowledge to the processes of production.

The secondary industries that we must now begin to encourage are characterized by a wide variety of work. They have different standards, are not easily susceptible to organization on a large scale, and hence politically have never acted as a concerted and effective force. The National Association of Manufacturers has been held together largely by an exaggerated emphasis upon the struggle against trades unionism. This ideal of strife with labor is no longer sufficient, and many believe that much more can be gained by co-operating with labor to build up the productive power of our people.

SECONDARY INDUSTRIES AND CONSERVATION.

Today, the interests of the secondary manufacturer coincide closely with the demands of the conservation movement, and with the best interest of the Nation. The secondary manufacturer needs a permanent supply of raw materials. It is to his interest to see that coal, lumber, iron, electric power generated from our waterfalls, and every other raw material of manufacture be permanently available at reasonable prices. Where undue monopoly of the power of such raw materials exists, the secondary manufacturer will be acting in accordance with his own enlightened interests if he helps to restrict and regulate by political action. Reckless exploitation, leading to exhaustion of any natural store, threatens the very existence of his business.

In order to produce in large quantities, the secondary manufacturer must sell into broad markets; must use freely and extensively the transportation systems of the country. He realizes that the development of railroading in the United States (which surpasses that of any other country in the world, and has knit together a population of a hundred millions with great buying and consuming power into one homogeneous market) is one of our great national assets. On the basis of this broad market, quantity manufacture can be developed as nowhere else in the world.

President WHITE—Before introducing the next speaker, I will read a letter from Dr. Charles A. Doremus, of New York, whom we expected to be here.

NEW YORK, September 30, 1912.

Mr. J. B. White, President of the Fourth National Conservation Congress:

Dear Sir—Much to my regret I am prevented from attending the sessions of the Congress, though appointed to represent, as a member of its Committee, the American Electrochemical Society.

One of the matters detaining me is work in connection with the American Museum of Safety, which is doing progressive work to conserve human life. There are now twenty-two such museums and their beneficial influence is being felt here and abroad. The large corporations have been enlisted in the work of accident prevention and allied topics and the recent congresses, the Eighth International Congress of Applied Chemistry and the International Congress of Hygiene and Demography, have awakened great public interest in all that pertains to the preservation of health and life.

May the Congress over which you have the distinguished honor to preside still further enlist our people to safeguard not only our material wealth but the people themselves.

I have the honor to be,

Yours very respectfully,

CHARLES A. DOREMUS.

President WHITE—I now have the pleasure of introducing to you Dr. J. N. Hurty, of Indianapolis, President of the American Public Health Association, and Indiana Health Commission, who will speak on "Conservation of the Human Race."

Dr. HURTY—Mr. President, Ladies and Gentlemen:

High authority says we are only fifty per cent. efficient; that we live out less than one-half the natural duration of life, that we consume twice as much food as is needed to maintain efficient life, that we waste as much as we use, and that one-half of all human beings born either die before reaching maturity or fall into the defective, delinquent or dependent classes. In these facts we find reasons why we waste the major portion of all our resources and call it development. In these facts we find reasons for the existence of robber taxation and predatory business. For, a people who waste themselves, will, of course, waste their natural resources. Therefore, the first, the most important, the fundamental conservation, is the conservation of human efficiency. A people who cannot be brought to a realization of the fact that they lead only half lives, and, who realizing, will not end, will show the nations-to-come what fools the present mortals were.

LENGTH OF LIFE.

Length of life is a resultant of strength. "Honor thy father and thy mother that thy days may be long in the land the Lord thy God giveth thee." It is an honor and it is a strength, for a nation to have a low sickness and a low death rate with their consequent lengthened average duration of life. In India, the average length of life is twenty-five years, in the United States, forty, in England, forty, in Germany, forty-three and in Sweden, forty-five. The natural duration is one hundred years. Metchnikoff, after thirty years of study of disease and death says, only a very few die natural deaths, most of mankind commit suicide. That is, most people do not know how, or will not, conserve their vitality, and thus results a greater or less period of disability and inefficiency with premature death. Nature does no fooling, she has her laws and they are enforced up to the handle.

VITAL ASSETS.

Comparison of vital and physical assets as measured by earning power, show that the vital are three to five times the physical. The facts show that there is as great room for improvement in our vital resources as in our lands, water, minerals and forests; and furthermore, this improvement must come first for through human life only is natural conservation possible. The dead past may bury the dead, but living and strong men, not the weakly and sickly, must do the work of Conservation.

ILLNESS.

From our vital statistics, which constitutes the bookkeeping of humanity, we learn that fully 100,000 people in Indiana are sick at all times, 25,000 of whom are consumptives. Not less than half of this is preventable, and three-fourths may be prevented by strong effort. Eighteen experts in various diseases as well as vital statisticians, have contributed data on the ratio of preventability of the ninety different causes of death into which mortality may be classified. From this data according to Fisher, it is found that fifteen years at least could be at once added to the average lifetime by practically applying the science of preventing disease. More than half of this additional life would come from the prevention of tuberculosis, typhoid fever and five other diseases, the prevention of which could be accomplished by purer air, purer water and purer milk. Let the business men, who are in the saddle and who run our affairs, thoroughly consider this. They surely know that disease and premature death are drags to business. Fifteen more years of life to each citizen means an enormous increase in the strength and happiness of the people, with consequent betterment to business.

Minor Ailments must be thoroughly considered in any steps toward the conservation of vitality. They are far more common and farther reaching than is generally realized. They are chiefly functional disorders such as of intestinal canal, heart, nerves, liver, kidneys, etc. These disorders are gateways to the more serious disorders. Those who neglect colds, or what seems to be colds, will prepare the tissues of the respiratory tract for pneumonia and consumption.

Benjamin Franklin, wise and practical, successful as merchant, scientist, and statesman, said—“The having of colds is a great drawback. I notice when I have one my efficiency is greatly decreased. Thought, judgment and understanding are clouded. Furthermore, I notice that colds follow excess in eating and drinking and the much breathing of bad air. They are quite unnecessary.” The losses due to mistakes in business and in the general conduct of life on account of minor ailments cannot be estimated except perhaps as time lost. A study of the matter shows that the time lost cannot be less than four days annually to each supposedly well man. Applying this to the wage earners of Indiana, counting one wage earner to each five people, making 500,000 in all, and we have to pocket an annual loss of 2,000,000 days or 5,470 years. In dollars, counting the average wage at $500 per annum, the loss amounts to $2,735,000 annually. This is certainly a prodigious loss to suffer in Indiana because of minor ailments, all of which can practically be avoided by proper public and private hygiene.

Neurasthenia, so common in the United States, is one of the most serious and insidious introductions to grave disorders, which may be due to depraved nutrition, to needless worry, or failure to have adequate recreation.

Patent Medicines. A source of drug habit, ill health, disease, inefficiency and race poisoning, militating against business is the horrible patent medicines. Medicines at their best, given under skilled medical direction are very dangerous things. (Applause.) The drug addicts, made so by a certain kind of practitioners, by self doctoring, and the taking of patent medicines, are numbered by hundreds of thousands. A large proportion of drunkards are started on their way by taking tonics. It is mostly the alcohol in tonics which produce the seeming improvement and which give temporary relief, but which invariably make the last state worse than the first. Alcohol, and all other drugs, are more dangerous than dynamite, and trade in them should be restricted more severely than trade in dynamite. (Applause.) The earth has been ransacked for drugs to cure. Everywhere we see emblazoned advertisements of medicines which the ad says will cure every disease from corns and ingrowing toenail to syphilis and gonorrhea; and yet, sickness and disease grow apace with our civilization. The world has been fine-combed from the equator to the poles for a something with which to bring health and prolong life; and lo, and behold, like the blue bird, these blessings are in every household patiently waiting to be called. At present, we are in the patent-medicine stage of ignorance, from which we must emerge before real conservation of human life and energy can be realized. (Applause.)

SCHOOL HYGIENE.

In conserving vitality, the child must have physical defects removed as far as possible, then he must be brought up amidst healthful surroundings and itself trained in all that conserves health. This great State of Indiana has already taken steps in this direction. The 67th General Assembly ordained that the schoolhouses hereafter built shall be sanitary in all particulars. This means, that waste of money and waste of child strength and happiness, shall cease in this fair State so far as this one matter goes. The same assembly has given permission to school authorities to institute medical inspection of school children that they may be relieved of morbid physical conditions which cause pain, inefficiency, illness and early death. It was a marked forward step to grant this privilege but it was a mistake of the Legislature in favor of loss of vitality not to make this practical care of children compulsory. Physical strength is the fundamental requirement for the making of children into educated and moral citizens. There is now a world-wide movement led by Switzerland and heathen Japan to save children and make them strong. A Japanese physician traveling in this State said—“We have relatively fewer short graves in our cemeteries.” The intelligence and business sense of a community could be accurately measured by determining its relative number of short graves. Youth is the time to serve the Lord. We must train the body in youth as well as the mind or the opportunity to conserve vitality is largely lost. A far better business scheme than securing more factories would be for the business men to turn their attention to the conservation of human vitality. The returns would be immense, failure to score in such an effort is impossible.

Hygiene has been permitted to extinguish cholera and yellow fever, and by the grace of private benefaction it will soon banish hookworm disease which now incapacitates 2,000,000 people in the South. And may God hasten the business men to permit hygiene to banish those twin leprosies, syphilis and gonorrhea, which are important factors in the causation of insanity, crime, and pauperism, and which so fearfully wreck the lives of so many innocent women and children as well as wreck the lives of the guilty. (Applause.) Syphilis and gonorrhea are responsible for the existence of a large proportion of defectives of various kinds which fill our institutions. Let hygiene drive these plagues away, and, Indiana, instead of building another insane hospital, for another million dollars, which she must shortly do, could donate one of the five now existent to educational use of some kind. (Applause.) I strongly advise Indiana to listen to the health cranks if she wishes to save health, time and money.

SAVING VITALITY.

"*Strength, Endurance and Fatigue*, are the three great elements to be considered in conserving life. The measure of strength is the force a muscle can exert once, the measure of endurance is the number of times it can repeat an exertion. Fatigue is caused by fatigue poisons, which must be removed from the body during rest, principally during sleep.

Anything, therefore, which reduces strength and lessens endurance and prevents removal of fatigue is inimical to vitality conservation."

SCIENCE OF LIVING.

The science of living begins at the mouth. Barring the taking of drugs, as a man eats and digests his foods so he is. Owing to drug taking and errors in human feeding, disease is latent in man at all times. Only a few escape sickness and pain and die natural deaths. This is not as nature would have it. Josh Billings, recovering from heart trouble caused by the excessive use of tobacco said—"Nature made us all right, we make fools of ourselves." Other drugs which are of almost universal use and which affect heart, nerves or efficient elimination are coffee, tea, spices, cocaine, morphine, chloral and alcohol. (Applause.) All of these are drugs, and all are poisons, and all more or less disturb the vital functions, reducing vitality and efficiency.

Any departure from unstimulated nutrition works harm. Stimulated nutrition is unnatural, and perforce, is opposed to strength. Immoderate eating—feasting and gluttony—reduce vitality and induce disease with its consequent inefficiency. A very old adage says—"Most men dig their graves with their teeth." The old time author of this was striving for the conservation of human vitality. Immoderate amounts of nitrogenous foods, exemplified in white of egg and lean meats, cause auto-intoxication. They do this by undergoing putrefaction in the digestive tract, thus making toxins, which in turn being absorbed into the body, cause the following train of ills which results in loss of vitality and efficiency. Some of the auto-intoxication or over-eating ills, are—biliousness, coated tongue, foul breath, clammy hands, clammy feet, dry lusterless hair, putty complexion, dulled hearing, dulled vision, dulled taste, dulled smell, early loss of memory, loss of continuous thought and attention, headaches, vertigo, dyspepsia, loss of strength, rheumatism, insomnia, fugitive pains and aches, hysteria, nervousness, nightmare, irregular heart, shortness of breath, brittle nails, dry harsh skin, cancer and premature old age of the doddering and slobbering kind. (Applause.)

Until we learn and practically apply the science of living we cannot attain over 50 or 60 per cent. efficiency and must continue to live lives of sickness, pain and disease, and die before the natural duration of life has one-half expired; and if this does not hinder and delay the conservation of natural resources nothing will.

"*Over-fatigue*, is a cause of loss of vitality. The present working day from a physiological standpoint is too long. Over-work better expressed by the term over-fatigue, starts a vicious circle leading to the craving of means for deadening fatigue, thus inducing drug habits and drunkenness."

"Experiments in reducing the length of the working day show a great improvement in the physical and mental efficiency of laborers and results in an increased output sufficient to pay the difference. However, the great justification of the shorter day is found in the interests of the race and nation, not the employer. Public safety requires, in order to avoid railway collisions and other accidents, the prevention of long hours; lack of sleep and undue fatigue is quite as great as the waste from serious illness. A typical succession of events is, first, fatigue, then "colds," then tuberculosis, then death. In order to prevent in the beginning this increasing line of destructive agencies, undue fatigue must be prevented."

HEREDITY.

Vitality largely rests upon inherited qualities. A child born of weak parents, those parents having received their weakness by inheritance, will itself be weak in the same way. Idiots breed idiots. Whatever improvement the child may enjoy, must rest upon its inherited foundation. If a child inherits brown eyes they must stay brown, no amount of cultivation may change their color, but inherited weak sight may be improved to a greater or less degree. Two forces, therefore, control vitality, namely, conditions preceding birth and conditions during life. In other words, the foundations of vitality are wholly inherited, and may be cultivated to the degree the inherited foundations will permit.

A perfectly sound physical and mental inheritance is rare and is the greatest of all assets. The highest development of a nation will begin when the human law conforms to God's law of development and parenthood is denied to defectives. Prisons and asylums are now sufficiently numerous, as it is evidence of defectiveness of the masses to conduct our affairs so as to necessitate their increase. Indiana now has five great insane asylums, each representing about one million dollars, and there are enough insane in jails, poorhouses and in homes to fill another one. Our population increased 16 per cent. in the last decade and insanity increased 29 per cent. There is a business problem for you.

To go along in the future as in the past, permitting, even fostering the production of the hereditary insane, of the hereditary pauper and criminal, of the hereditary idiot and feeble-minded, and then building great palaces in parks to care for them, will mean we have not the common horse-sense necessary for the proper conduct of our affairs. (Applause.)

HYGIENE.

We must look to hygiene, the science of health, to conserve human vitality. The term includes every necessary force to prevent disease, to increase strength and endurance, and to prevent the production of the unfit.

The ponderous and oppressively costly courts have been grinding for centuries and crime increases. Punishment and fear of punishment restrain evil doing, but does not eradicate the tendency to evil. This and other defects we must, as far as possible breed out of the race, and science can find a valid answer for every objection which obstructionists can raise to this proposition. Fostering insanity, crime, pauperism and imbecility, is not evidence of understanding and of high ability.

The divisions of hygiene are: Federal, State, Municipal, Institutional, School, Domiciliary and Personal.

Hygiene not only makes for greater physical strength and endurance but it makes for greater moral strength. It is the essence of charity, kindliness, patience and truth.

When, through hygiene, defectives, delinquents and dependents are no longer propagated, when simplicity and frugality of living are achieved, voluntary celibacy and voluntary childlessness will become discreditable, and sickness, disease and premature death will disappear before temperance and sanitized homes.

President White—This admirable paper causes me to say to every one here that they cannot afford to go away and not deposit a dollar with the Secretary for the book of the Proceedings of this Congress. The book of these admirable and practical addresses should be in every home, should be in the library. I hope that every one will leave their address, will register, and receive as soon as they are published a copy of the Proceedings. (Applause.)

It was Louis D. Brandeis who said a year or two ago that the railroads of this country could save a million dollars a day with practical economy and with good system. He got that idea from and quoted Mr. Harrington Emerson of New York City, who will now address this audience upon “The Rescue of the Fit.”

Mr. Emerson—Mr. President, Ladies and Gentlemen: There is a growing clash between employer and employe. The old order is passing away and the new order has not yet come in. The millions lost in strikes are forever wasted. This direct waste due to supposed conflict of interest is one of the great losses. The other is more serious. Not one man in ten is in the place in the world best fitted for him, not one place in ten is filled by the best man in the world for the job. When the job is not bossed by the right man and when the man is on the wrong job there is a waste whose magnitude is incalculable. It is to mitigate, palliate, obviate, these two great sources of waste that on a large scale a new plan is being put into operation. The theory that underlies it is founded on principles, not on empiricism or on tradition or on rule of thumb.

It is theory that has given us the best designs for steam turbines, gas engines, dynamos, aeroplanes—it is theory that gives us this plan of the *Employment Department*.

What is the theory?

All manufacturing costs fall under three divisions: Materials, Labor, Equipment Charges.

Materials means all materials, whether for manufacture or operation.

Labor means all personal service or personal charges, whether direct, indirect, supervising or managing.

Equipment charges are made up of taxes, insurance, depreciation and interest on investment.

Although these three classes of expense are so different there are some general economic laws which apply to all of them and it is quite certain that what we have learned to accept as to materials, may have some lessons applicable to personal service and to equipment charges. When our building materials consisted of prairie sod the problem was simple, we picked out the best sod in sight, plowed it up, hauled it to one side and erected it into walls. When the task is to build an automobile the handling of materials is not so simple.

In automobile plants the engineering department designs what is wanted, then draws up specifications, precise and scientific specifications; steel that will test under tension or torsion so many thousand pounds, steel balls, that are so round, so hard, so even in size, bronze, that is so resistant, copper that is so pure, etc.

The purchasing department then calls for tenders or for bids. Samples or specimens are submitted for test and these go into the testing laboratory where they must come up to specifications. The purchasing agent says: How good a wire can you sell me for $0.10 a pound? What will the price be on wire testing 200,000 pounds?

The materials having been tested and bought are put into the storehouse under a competent storekeeper. It is his business to see that they do not spoil, that they are not wasted or stolen. He issues only on requisition, the requisition specifying the proper quality and quantity. When the materials go into use they are continually inspected during the progress of the work.

There is therefore an inspection department. Engineers have learned that it is not the price of materials that counts but the quality. As quality goes up quantity goes down and price goes up but not as fast as quality. Although steel wire is dear and cast iron is cheap, we build bridges out of steel wire. Although we can buy carbon steel for $0.14 a pound, we pay $0.60 a pound for high speed alloy steel because it works faster and so much more powerfully that it would be cheap at $800 a pound if we could not get it for less.

As to complex modern materials we need therefore an engineering department to design and specify, a testing department to test and analyze, a purchasing department to buy at the best price and on the best terms, an inspection department to watch results from day to day, hour to hour; a storekeeping department to hold and to conserve, to issue carefully and economically.

Modern personal service is more complex than modern materials. How can we afford to omit as to personal service any of the safeguards found necessary as to materials? These necessary safeguards we apply through a very highly organized employment department directed and managed by specialists of the higher class and a corps of assistants.

In the employment department all these methods so necessary as to materials, we apply also to personal service control, whether we are securing a factory superintendent or a shoveler of sand. First of all an organization is outlined. It is evident that to perform certain kinds of tasks there is only one best organization. Battleships are a modern development, they have been slowly evolved. America started it when the Confederate Government sheathed the Merrimac with railroad rails and sank all the wooden ships. As the London Times editorially said, “The Merrimac made all the navies of the world obsolete.” Great Britain, Germany, France, Italy, have helped develop battleships, but the organization controlling every battleship in the world, whether Japanese, Chinese, Russian, Turkish, Chilean or American, is substantially the same. An American officer could be transferred to a foreign navy and find himself at once. Naval organizations the world over are interchangeable. The ordinary manufacturing concern has no standardized organization, it has generally grown like Topsy. Positions are ill defined and generally worse manned. The first duty, therefore, of a modern employment department is to outline the organization, the one best organization for the business in hand.

Its second duty is to specify the essential and required qualities for each position.

There are three different ways of filling positions:

1. To have on one's hands some incubus, a king's son or a king's mistress or some political henchman, and to create a position for the incubus to fill, "duke of this" or "countess of that," or a fat contract on city work. In England this is called "finding a berth for a friend"—*a berth—a place in which to fall asleep*.

2. The second way, and the more usual one, is to see a real vacancy and to shove a friend into it, hoping he will make it a go. The man and the job stand as good a show of fitting each other as a man would of getting the right clothes by drawing a suit in a raffle. It was Roosevelt who saw a vacancy in the Presidency, grabbed Mr. Taft, shoved him into the place, and now declares he does not fit. Personal liking is not the proper basis for a Presidential preference.

3. The third way to fill a definite vacancy is to find the man fitted for the place, and, after test, put him into it, even as we find a suitable wire for a bridge and put it in.

If we have a locomotive of definite design and we need an exhaust nozzle, there is only one design of nozzle that will answer. So if in the organization there is a position to fill, the best man for that position must have certain qualities and not have others, not every man, not the convenient man in ten, probably not one man in ten thousand is the man for the place. The employment department seeks diligently for the right man, the man who combines experience with aptitude. If it had to choose it would prefer the man without experience but with all the aptitudes to the man of experience without aptitudes. The man with aptitudes can learn quickly, reliably and fast; the man without aptitudes can never be anything but a misfit. Therefore the employment department having secured a number of prospects, carefully tests the most promising.

The old-fashioned plan is to ask a few questions, secure a few recommendations, take a look at the man, and if a hunch is felt that he will do, accept him. I know all about this plan, for I have tried it for twenty years, and in some years it has cost me $50,000. The plan does not work. I received the best set of recommendations I ever saw about a sea captain, and when we entrusted him with a $140,000 steamer he deliberately wrecked her in order to make some graft out of the repair bills.

That the man was a scoundrel was written in large type all over his face, but in those days I could not read plain print and I was better fitted, and that was *not at all*, to navigate the steamer than to select a captain.

When I taught in college I got an inkling of the right way. I taught German, and at the beginning of the year my classes were filled up with sixty students, and at the end of the year there were only twenty left. I worked on the theory that there was no profit to any one in making a bluff at studying German. It was either worth while or it was not. If worth while, learn German; if not worth while, don’t waste time on it. So I weeded and weeded my German garden until only those were left who could really learn. They learned to know German as well as they knew English. The weeding process was hard on me and hard on the misfits, hard on the good students. I gave an immense amount of rough effort to no purpose in an absolutely useless attempt to make silk purses out of sows’ ears. Then sows’ ears might have made good mince meat, but the carving and slashing I gave them hurt them to no purpose. My time was taken up on rough work until the misfits and the good students failed to receive the specially skilled attention and help their progress required. After a couple of years of this I tried a new plan. It was evident that any students who did not know English, English grammar, English spelling, English pronunciation, were not fit to study German, so I examined all applicants as to English, but I gave those who failed a week’s test, lest some genius should by chance be overlooked. I never found the genius. Under this plan I started out with a class of twenty-five instead of sixty. I gave my time to those who could profitably make use of it, and not to those who could not, and every one of the twenty-five learned German.

A man or woman can be tested in five minutes for fundamental aptitudes and traits of character as easily and reliably as I tested the prospective German pupils. It would take one too far to go into the whole subject of character analysis. A great composer like Mozart, Beethoven, Wagner, can originate music, but there are thousands who can learn to play it well. So with character analysis. It requires a special and rare gift to uncover the lessons written in the coloring, in the texture, in the shape of head, in the expression of the face, of the body, of the hands, in the clothes, in the personal tricks of habit, to cross-check these tests by others as the answers to test questions, but this knowledge has been so formulated that all can learn. So instead of trying to play bumbly puppy by ourselves, of missing the accumulated researches that have been going on all over the world, of repeating all the mistakes that others have made, we do as the Japanese did when they adopted the British navy, the German army and the American schools as models. We are advised as to our employment department by a specialist of the highest skill in character analysis, in all problems relating to the handling of people. The tests are rapid but they are many and they interlock so far as to be conclusive. A man can lie with his eyes or with his lips or with his body, but no man is skillful enough to lie at the same time with eyes and lips and hands and body.

After men have been tested they are employed, not before, and they are only employed because they have the qualities that fit them for a particular place. They may be at the time only 30 p. c. men, they may be succeeding an 80 p. c. man, but the great fact is that the 30 p. c. man can and will become a 100 p. c. or a 110 p. c. man, while the 80 p. c. man is perhaps in reality an overstrained 70 p. c. man. Starting with the best of human material years are not lost gradually collecting it. Not only are the unfit excluded, but what is very much more important, the *fit are rescued*, they are given opportunity, they jump at once into the places they can fill instead of waiting for years.

What is the unnecessary cost to a business of a 30 p. c. man compared to a 100 p. c. man?

The hourly costs of a man are: His hourly wage, the hourly machine charge, the hourly overhead charge. These three items will easily average $0.70 an hour in a machine shop. If the man works at full efficiency he gives us in a year 2,700 hours of standard work in 2,700 hours of actual time at a total cost of $1,890. At 30 p. c. efficiency, it will take 9,000 actual hours, costing $6,300 to deliver 2,700 hours of standard work. The added expense due to inefficiency is $4,410 for a single worker.

Efficiency does not mean strenuousness. The fluttering rooster is strenuous, but he makes little progress; the eagle flies efficiently, covering miles of country, yet never moving a wing. The Chinese coolie on his river treadmill is so strenuous that he wears himself out in a few years. As a producer of power he costs $1,300 a year for the horse-power hour you can buy from Niagara for twenty dollars. The chauffeur of an American automobile gliding along at forty miles an hour, carrying six passengers, is efficient, not strenuous.

It is evident that under this modern employment plan the rate of wages per hour ceases to be a critical question. The efficient man, like steel wire and high-speed steel, is always worth more than he would think of asking for his services.

The requisition calls for a man with certain qualities—it never calls for a man at $0.20 or at $0.30 or at any other rate per hour. Fixed rates per hour are obsolescent when one man turns out the work for $6,300 and the other man turns it out for $1,890. For $1,890? No, he does not. We do not ask him to; we can not secure and hold any 100 p. c. man for the wage rate in the $1,890. We pay the man more, we gladly pay him more, we pay him as much as we must to secure him, but he is cheap at almost any price.

The man who receives a salary of $60,000 a year is expected to profit his company to the extent of $6,000,000 a year; the man who works for a dollar a day is always a loss, a severe loss, and, therefore, we try to eliminate him by the substitution of a machine. The fight against a machine is to carry on a losing fight against the whole current of the age. To put the worker in a position for which his aptitudes qualify him is to double, treble his value, and everybody is best fitted for something. A group of children were playing automobile; one was the engine, another the chauffeur, others the passengers. A little tot far behind was hurrying along. What are you doing? I am playing automobile. What part are you? I am the smell. In a watch there is not a useless piece. In a perfected organization there is not a useless man; there cannot be an unqualified man without endangering the whole. As I write, 20,000 mill hands are rioting at Lawrence, Massachusetts, because somebody has blundered, because some position had been badly filled.

The wage question is ethical. The workman is worthy of his hire, but also man does not live by bread alone. The man scientifically selected is 100 per cent. efficient because he likes his work, is fitted for it and it likes him; it is no longer a drudgery, it is a pleasure. We are rescuing the fit for the work which by nature's right belongs to them and under this plan there are few unfit.

In years gone by when a beef was slaughtered much of the carcass was wasted—the horns, the bones, the hair, the hoofs, the blood, the offal went to waste. Now nothing is wasted; everything has its use, and the offal we return as fertilizer to the soil is of greater perennial use than the tenderloins and sirloins which find their way to the tables of the rich.

We have in the past treated men as if they were coal, a raw product only fit to burn, or as the German soldiers pathetically called themselves, in 1870, mere cannon fodder. But mere coal contains ammonia, beautiful dyes, strange and powerful medicines, as well as heat units.

Everybody is normally good for something, and if fitted to the right place is worth more than he now is.

At Seattle a boy of 17 was excluded from the high school because he could not learn their lists of English kings or American Presidents, but that boy went out and when I met him he had grabbed the evaporative power of the sun and was propelling a boat with it on the waters of Puget Sound. He was fit, more fit in a mechanical way than any other boy I ever knew.

It is to this Rescue of the Fit that I look forward for the great uplift of American industries, the great increase in happiness and the great elimination of strife.

It is being put to a practical test in a plant employing 200 men and it is working.

This is my message to you.

President White—There is nothing further on the program for this afternoon, and we will therefore adjourn until 8:00 o'clock this evening, when Dr. Wallace and Judge Lindsey will speak in Tomlinson Hall.

SIXTH SESSION.

The Congress assembled at Tomlinson Hall, at 8:00 o'clock p. m., and was called to order by President White.

President White—The delegates, visitors and citizens of this city have a rare treat in store tonight in the program that has been published, and I have a rare honor in introducing the speakers. It will be a red-letter day in my life, and I know it will be in yours.

I now want to make this audience acquainted with "Uncle Henry" Wallace. I would be glad if everyone could know "Uncle Henry" as I have been fortunate enough to know him. He has been an inspiration to every young man and every farmer and all who have known him in the State of Iowa for the past twenty-five or thirty years. He loves to sit down in his office, or study—and I have been there to see how he works—answering letters that the farmers from all over the country write him, and who look to "Wallace's Farmer" as a source of profit and information upon every subject that affects the home. He comes close to the home, close to the family, to the fireside, answering all their questions and telling them just how they should do this or that, and all in that fatherly, kindly, brotherly way, so that he is referred to by everyone who knows him as "Uncle Henry." He is going to talk to us tonight upon "Human Efficiency," and he will speak from a very practical standpoint, for he has had experience all along the line.

I now take pleasure in introducing to you Dr. Henry Wallace, of Des Moines, Iowa, former President of the National Conservation Congress. (Applause.)

Dr. Wallace—Mr. President, and Members of the Congress: It might not be amiss, before entering into a discussion of the subject proper, to recall the different subjects which have from year to year engaged the attention of the Conservation Congress, and to show how the choice of the subject for each different Congress was the natural and logical result of the discussion of the preceding Congress.

The first Congress was called, and the Congress itself was organized, as a forum in which the leading men of the Nation could discuss the problems raised by the Conservation Commission, appointed by Theodore Roosevelt at the suggestion of Gifford Pinchot, then holding the position of Chief Forester in the Department of Agriculture. His position as Forester enabled him to see the terrific waste going on in the management of our forests, and the various means by which the government forests were passing into the hands of individuals, subsequently to be wasted for private gain. He saw clearly that unless our forests were conserved and managed as are the forests of all other civilized nations, soil erosion would render future forest growth impossible, would fill our rivers with silt, dry up the streams in summer and convert them into raging torrents in winter, depriving us of water for irrigation, and diminishing in value the water power, or white coal, on which future generations must largely depend for power and transportation.

A forum was greatly needed in which the questions raised by this fearless idealist—to whom the Conservation of our resources for future generations is both wife and child—could be openly and fearlessly discussed by leaders and in the hearing of the American people. When the First Conservation Congress was called to meet in Seattle in 1909, naturally the main topic for discussion was the Conservation of the forests and of the water powers, which were then fast passing into the hands of great corporations.

By this time the public conscience was aroused. The people of the United States began to see clearly that we dare not go on in the future, looting and wasting our natural resources as we had done in the past. They began to realize that the generations of the unborn had rights in the oil, the coal and other minerals in that portion of the public domain that we had not as yet recklessly thrown away, or allowed to be stolen from us under forms of or in defiance of law. So the Second Conservation Congress was called in St. Paul, in 1910, as a forum in which the leading men of the nation could thresh out the problem as to whether these resources to which the American people at present held title should be administered by a Congress chosen by the people and speaking for the people, or whether they should be administered through an act of Congress by the several States in which the Government property happened to be located.

The historian of the future alone will be able to measure the beneficial results of the fierce conflict between those who would despoil these resources for private gain and those who would conserve them for future generations. We can, however, see some of the results in the change in the policy of our national administration, in the vigilant watch now maintained by the present Secretary of the Interior; by the success which crowned the efforts of Mr. Pinchot and others who kept constant watch over bills intended, by means of concealed jokers, to break down the fixed policy of the Government; and by the veto of the President of vicious bills which, notwithstanding the utmost vigilance, were enacted by the last Congress. This watch and guard over the heritage of the unborn could not have been maintained successfully, had it not been for the white light thrown upon the problem by the Second National Conservation Congress.

I was, unexpectedly to myself, chosen President of the Conservation Congress at the close of the St. Paul meeting; and with the consent and advice of my executive committee, in making out the program for the 1911 meeting in Kansas City, fixed the attention of the American people on the necessity for the Conservation of the fertility of the soil, and the development of a better social and family life among the tillers of the soil.

The time had come for the American people to understand that the rapid and regular advance in the cost of living was due mainly to the terrific waste of the fertility of the soil, that had been going on for more than a hundred years. It was time for the farmer to learn that he was not in a position to throw stones at the lumberman who had wasted our forests, or at the mine owner who is wasting one-third of the coal in the process of mining; that he, while a sharer in the cheapness of the products of forest and mine, had himself been mining the fertility of the soil, stored for his benefit through countless ages, and selling it at the bare cost of mining; and in doing so had built up cities the world over, which must cry for bread when the fertility of his soil became exhausted.

It is too early yet to measure the full results of this Kansas City Congress. This should be noted, however, that, whether the result of the discussions of this Congress or not, the people of the United States have shown an interest in agriculture and the maintenance of soil fertility which they had never shown before. Bankers, railroad officials, capitalists are beginning to see that unless the farmer receives encouragement and efficient aid, this nation will soon cease to be a factor in supplying other nations with food, and will gradually become a consuming instead of a producing country, so far as the products of the soil are concerned. We are beginning to see that unless a more satisfactory social life is established in the open country, the increasing disparity between rural and urban population must continue and the cost of living must go on increasing, and with it increasing discontent and social disturbance.

My successor and his executive committee, with their wide experience in practical affairs, saw clearly that if we are to restore fertility to our wasted soils, if we are to do anything worth while for the Conservation of our resources of any kind or character, there must be an increase in the efficiency of the individual. They therefore wisely chose the subject of “Vital Resources” as the main center around which discussion must revolve at the present Conservation Congress.

The subject of Vital Resources opens up a very wide field for investigation and discussion. Various subjects in the group have been discussed and others will be, by specialists who have given their particular subject years of conscientious and close study. So when only last week I was urged to make this address instead of discussing a minor phase of the subject, there was nothing left for me but a general discussion of the subject of Human Efficiency.

Man, after all, is the biggest thing on this planet. The farm people are always bigger than the farm. No matter how rich by nature the farm may be, it will lose fertility if the farmer is not big enough. The first-class farmer will take an inferior piece of land and in time bring it up to his own measure. If the farmer does not fit the farm, it will in time come up to or decline to his measure. The average production of the soil is the expression of nature’s opinion of the fitness of the man who tills it for a term of years. The most severe condemnation of the American farmer is the fact that, with some of the richest soils in the world, he has so wasted its fertility that he is crying out for commercial fertilizers; while the “heathen Chinee” has farmed for at least forty centuries, and has maintained his soil fertility without the use of commercial fertilizers.

If any great business has attracted attention by its success, one always asks: Who's the man or men behind it? The greatness of this nation is measured not by its soil, its mines, its forests, its water powers, but by the efficiency of its people. This is true of all nations. The cynical Bismark, who always cast covetous eyes on Holland, is said once to have remarked that the way to redeem Ireland was to transport the Dutch to the Emerald Isle and transport the Irish to Holland; that the Dutch would make Ireland an earthly Paradise, while the Irish would not keep up the dikes except with the help of the Germans, who would in that case soon have a seaport.

The only way by which you can restore the wasted fertility of the soil and the waste of our forests and develop properly our mineral resources; the only way in which we can as a nation take the place to which we are entitled—that of leader in the world's trade and commerce—is by increasing to the utmost limit human efficiency, physical, mental and moral. These three are ineradicably linked together, because they are integral parts of every human being. We can not develop fine human beings physically without the development of the intellect and the soul; nor can we develop either the intellectual or the moral to the limit without taking care of the body.

If we are to have the maximum of efficiency in the man, the child must be well born, must be free from incurable diseases, mental, moral or physical. To every generation of human beings is given by an allwise Ruler the power to foreordain the character and quality of the generation to come. The coming generation is as helpless in our hands as clay in the hands of the potter. By marriage parents decree the personality of their children. By "personality" I mean the inherent tendencies—physical, mental and moral—which, when developed wisely or unwisely, make or mar the character. In that little pink lump of humanity—the pride of the father and the joy of the mother—are bound up in various combinations the incidents, passions and capacities of the parents. It is this which gives its awful sacredness and tremendous possibilities to marriage.

The State by the extent to which it discourages and represses vice and crime, by the extent to which it prevents and controls disease, by the extent to which it encourages the marriage of the fit and prevents the marriage of the unfit, foreordains the character of the next generation. I know that I am approaching ground but little trodden, in which many fear to tread, and to tread on which is by many deemed sacrilege. But if we are to be a virile nation, strong in body, in intellect and in morals, the truth must be told fearlessly; and there is no more fitting place to tell it than where the people of this Congress are making a study of our vital resources.

To put the matter with brutal frankness: The State must soon determine whether the hardened criminal shall be allowed to take an active part in foreordaining the character of future generations; whether the manifest degenerate, whether that degeneracy be the result of being badly born or of vice or crime, shall be allowed to breed degenerates; whether those afflicted with incurable and transmissible disease shall be allowed to transmit them to a helpless posterity.

In order that the State may act wisely, it is time for a most thorough and searching investigation of existing conditions, material and moral, which lead to crime; the extent to which criminal tendencies are transmissible—criminal tendencies, mark you, for crime itself is not transmissible; what proportion of our crimes are due to intemperance, and to what extent the unbalanced state of mind which makes self-control impossible, and leads to crime, is due to inheritance. It will no longer do to say as some do: that intemperance, by killing off the unbalanced and weakling, rids society of an encumbrance; nor that nameless diseases weed out of the race those unable to maintain self-control. While all this is in a certain sense true, it furnishes no argument for abating zeal in repressing these crimes against humanity. That terrible saying of Anne of Austria: “God does not pay at the end of every week, but at last He pays,” finds striking illustration in the fate that sooner or later befalls the intemperate and the impure.

On one subject there is no need of any investigation. We must either adopt such measures as will insure as far as possible that the coming generation shall be well born, or we will compel our posterity to pay the price, as we are paying it now. We stand before the world today convicted of having more murders, more suicides and far more lynchings in proportion to our population, than any other civilized nation on the face of the globe; and also with having, speaking generally, by far the most corrupt city governments. Is it not time for us to investigate and see why we thus stand condemned in the eyes of the nations, and to what extent we are breeding crime, the crime that is our disgrace? For be assured that we must in all cases pay the price, not in cold cash alone, but in blasted lives and ruined homes and a lower degree of human efficiency. If we are to be a great nation, worthy of our blood inheritance and worthy of our material resources, our children must be well-born.

If we are to secure that measure of human efficiency that will enable us to make full use of our inheritance, whether of blood or material resources, we should see to it that the coming generation is not merely well born, but well fed. The farmer is wise in that he takes special care of the young things that come on his farm. He builds a lamb creep, that the young lamb may get feed denied its dam. He sorts his pigs into convenient sizes, and shuts them out of the feeding places until the feed is properly placed, and then lets them all in at once, so that they may all have equal opportunity. He does not allow the weanling colt to take its chance with the selfish and unprincipled horses in the stalk field. He protects his colts and gives them food "convenient" for them. If an unruly beast in his stock yard tyrannizes the young and robs them of their food, he does the sensible thing. He dehorns the unruly. He will tolerate no oppression about his farm. In this he is wise; wise, because he knows that if he fails to do this, the red flag of the sheriff will sooner or later stand above his door, and the farm will be sold to some man who will handle it more wisely.

The State has a similar responsibility for taking care of the young. Whatever may be their endowment by nature, that is, by birth, they need the nurture which is necessary to bring out and develop fully the gifts of nature. The State should smite anything that stands in the way of the proper nurture or feeding of the young. If the State is to prosper, it must protect the weak against the encroachments of the strong; and of all classes, the children of the State need its protection most.

If organized capital provides so little pay for labor, that the laboring man can not properly feed his children, then the State should dehorn that organized oppressor, as the farmer dehorns the unruly bull or boss cow. No profits to the individual or the organization, even though they be members of the State, can compensate for the robbery of the children of the State. If the State on investigation finds that the money that should purchase food for the young goes into the till of the publican, then the State should smite the publican in its wrath—not the individual publican, who perhaps may feel that he is earning his bread in the only way for which he is fitted, but the system which makes it necessary for the prosperity of the producer of intoxicating liquors, to corrupt so many hundred of our youth for every thousand dollars of invested capital. (Applause.)

If we have a system existing, whether in the State or the Nation, which can thrive only on the debauchery of the young and on the robbery of the child, by taking that which should go for food to support it, then it is time that the State and the Nation should control it to a point where it can neither seduce the young or rob the child; and that point is suppression. We must do that or do worse, namely, pay the price. We are in fact paying that now. The individual who will not keep account of his expenses is in danger of bankruptcy, no matter how great his resources; and the State which refuses to count the cost of any institution or system which tends to debauch morals, and corrupt the young, is on the way to destruction. For there is no avoiding the payment of the cost, whether we keep account or not; and that cost is not merely the dollars and cents, but starved children, blasted lives, broken hearts, ruined homes, increase of poverty and an increase of criminality, which is beyond the possibility of mathematics to compute. The State can afford to tolerate nothing whatever that stands in the way of proper nurture of the young; nor can it safely endure anything which tends to dwarf them physically, mentally or morally. (Applause.)

The State, however, will always succeed best by removing the causes that lead to improper nurture, or to the formation of vicious or criminal habits. The State can not endure poverty, grinding poverty, among any class of its people; nor can it endure having its children poorly housed. The slum is the enemy of the State and of every citizen of the State. The vice and crime of the slum reach out to the west end or the east end or the avenue, or wherever the wealthy and prosperous congregate, thus saying to all men: We are brothers. The poverty-stricken may well say: If you will not give us our rights, if you grind our faces, we will not merely levy toll on your pocketbooks, but we will infect you with our vices.

If we are to have the highest efficiency in the next generation, the State (and by the State I mean the government, whether State or National) must see to it that infancy is protected from the abominations of soothing syrup, and “sleep-easy,” that usurp the place of the catnip tea and other herbs which soothed infantile pains in the days of our grandmothers. It is useless to expect efficiency, if we pour into the innocent lips of unsuspecting childhood the habit-forming drugs which benumb the brain, stifle sensibility, and lay the foundation for incurable vices when the babe has grown to manhood. Let us get back to the ideals of the ancient psalmist, who, contemplating the future of the chosen people, uttered the prayer that "our sons shall be as plants grown up in their youth, and our daughters as cornerstones fashioned after the similitude of a palace." That is, a plant carefully cultivated, spreading freely, its roots drawing sustenance from the soil beneath, its leaves drawing sustenance from the air and sunshine, bracing itself against the storm; the daughters the cornerstones of the home, with all the adornment that we bestow on a palace fit for the abode of royalty. Let us go back to this ancient ideal, if we are to be a happy people, whose God is Jehovah.

If we are to maintain human efficiency, the State must lay a heavy hand on the venders of impure food. After what Dr. Wiley has told you, there is no need for me to enlarge on this cause of inefficiency. Suffice it to say, there was a time in the memory of some of the older men, when there was no pure food question. Our oatmeal, our cornmeal, our flour came from our own farms. There was no shorts in our buckwheat, no white earth in our flour. If our meats were tainted, it was due to our own negligence. In these latter days we have become by force of circumstances more completely "members one of another," drawing our food from all parts of the habitable earth; and hence the State must protect us from imposition. If we are to reap the benefits which come from the modern system of division of labor, we must not quibble about the expense involved in enforcing honesty in those who feed us.

If we are to have efficiency in the generation now entering upon the stage, or in the one to follow, we will need to make radical changes in our system of education. No matter what the natural endowment, it will be comparatively inefficient unless properly developed. Education does not consist of putting in but of drawing out. Culture is simply the proper development of the gifts of nature. All children are born with the capacity for doing, and doing well, some small part of the work that needs to be done in this great world of ours. This capacity is usually indicated by a strong preference for that kind of work. The capacity for doing is largely a matter of inheritance, and education is simply the development of this capacity. No education which fails to develop what is in the child is worth having; but no matter what may be the natural endowment, the capacity to govern in State or Nation, or to build a road, or to plow a straight furrow, or polish a pin, every child must have put into his possession the tools by which he can secure that education which will fit him for his life work. He must know how to read, that he may be in touch with his fellow-man. He must know how to write, that he may communicate his thoughts to other men. He must know how to reason, that he may put this and that together and draw conclusions. These lie at the foundation of all education.

Some education is acquired in mastering the "Three R's," namely, the power to observe—to see things—to tell what is seen and to draw conclusions; but the "Three R's" are, however, merely the tools by which we ourselves afterwards acquire an education. In spite of all the money we spend on rural education (in my State from 42 to 50 per cent. of all rural taxes), our children neither read well nor write well nor reason well. How can they when our rural schools average twelve pupils, most of them less than ten, and often five, three, or only two or one pupil, and are taught mainly by persons themselves but poorly educated, and who are teaching simply to acquire the experience necessary to secure a position in a city school. Neither the reading nor the writing nor the arithmetic of these schools has any connection with the farm nor any relation to farm life, nor is the teacher as a rule in sympathy with that life. Yet this is all the education that 90 per cent. of the farmborn will ever receive.

Little education this for the mighty task of feeding the world at prices that those not on farms can afford to pay. If the farm boy was so thoroughly drilled in reading that he could read to himself with understanding and read to others with expression, if he could express his thoughts so clearly and fully that the dullest could understand, if he could see things as they are, and tell accurately what he sees, he would in time without further teaching become a leader of men.

The farmborn, however, usually fares better than the townboy in the race of life. In growing up in the open country he learns what books can not teach—the know-how, so far as farm operations are concerned—and needs but to learn the reason why. The townborn, as a rule, has no opportunity to acquire the know-how by following the occupation of their parents; and hence much of his school life is spent in acquiring information which, apart from its educational value, is of no sort of use to him in after life. What the farmborn need, if they are to be efficient in life, is an opportunity to learn in a secondary school in the open country the reason why. What the townborn need is secondary education which will fit them for the work they are to do. If our farmborn are to be efficient, they must have centralized schools taught by teachers who have selected teaching as their life work and are paid accordingly, and thus be able to acquire in the open country a secondary education that will enable them to see clearly the reason why they should plow, or sow or feed. If our townborn are to be efficient, they must have in addition to a thorough mastery of the "Three R's," which is the birthright of every child, such training as will fit them for their life work.

The misery of our system, whether in town or country, is that it assumes that the chief end of man is to figure in some one of the so-called "learned" professions. So the high school is keyed up to the standard of admission to the college and university. The grade school exists to qualify pupils for admission to the high school. Hence the surplus of doctors without patients, at a time when humanity is learning how to avoid needing a doctor; of lawyers when men are fast learning to keep out of law.

In short, the end and aim of all education in the future must needs be efficiency in the line of the chosen vocation. The great lack of our present system is the failure to give the child a complete and thorough mastery of the tools by which any education worth while must be acquired: the ability to read with understanding, to express itself, whether in speech or writing, so that all may understand; the ability to see what is to be seen and tell it in plain English, and to put this and that together and draw a just conclusion.

I need not say that no training for efficiency is complete that does not involve the ethical as well as the intellectual and material. This is a Christian nation, and the ethics of Christianity should be taught in every school as well as in every home. We may not, and should not teach the dogmas or doctrines of any sect or denomination. We must forever keep separate the Church and the State; but underlying all these creeds and denominations there is an ethical standard which all but the criminal or would-be criminal accept; and this should be taught, because it embraces our highest ideals of manhood and womanhood and citizenship.

The crimes of which we are rightly ashamed are due largely to the fact that the jealousy of the churches toward each other has heretofore prevented the teaching of ethics to the children in our schools. Without the practice of ethics, without the striving to realize moral ideals, there can be no moral efficiency, and without moral efficiency intellectual efficiency may become productive of evil instead of good. An educated brain without an educated conscience is a source of danger to the public welfare. It is high time for the churches and all good people to get together and agree on ethical standards to be taught in every school, that will put moral as well as intellectual training in the coming generation.

I have touched merely the high places of the subject of human efficiency. I have endeavored to say that if the generation which is to follow us and carry on our work is to be efficient, the children must be well born and well fed, protected from the vampires that endeavor to suck their lifeblood, and must have an opportunity to develop their natural capacity by an education and training—physical, mental and moral—that will enable them to do the world’s work with profit to themselves and their fellow-man. (Applause.)

President White—In Denver, Colorado, some twelve years ago, there was found a friend for children; there was found a judge who believed that in the child brought before him for some breach of law, he could see something divine, that he could see the soul, the germ of the future man, the germ of a future life—something to redeem. He believed he could see why Christ said, “Suffer the little children to come unto Me, and forbid them not, for of such is the Kingdom of Heaven.” He had faith in the child, and when he found it necessary, in the way of discipline, to send them to the reform school, he placed faith in them; he told them to go out to the reform school, and he would be up to see how they were getting along after awhile. He developed character from the start, and it soon became noised around. In every paper, in every magazine, all over the world the name of Judge Ben B. Lindsey, of Denver, Colorado, was known. (Applause.)

I now have the pleasure to introduce to you Judge Lindsey, who will speak on the subject “Is the Child Worth Conserving?” (Prolonged applause.)

Judge Lindsey—Mr. Chairman, Ladies and Gentlemen: I am sure I appreciate this very cordial and kind reception, because it shows your appreciation, not of the speaker, but of a cause—in which I have only had a small part—that has come close to the heart of the people of this Nation in the last decade or more, and well it is that the great National Conservation Congress should not overlook the welfare of the child.

I was delighted to note in the splendid address which we have just listened to by Dr. Wallace, that this important matter of the child, the youth, furnished its principal theme, because without childhood there is no manhood; without the conservation of childhood there is nothing else to conserve.

But my friends, I am in rather a difficult position tonight. I am reminded, indeed, of an experience I had with a boy about twelve years ago in my court. It was before the days of the Detention Home School. One day I was down in the jail, and they told me they had a boy there that they thought belonged to me—that they could do nothing with him. So I went with one of the men down the long corridor to a cage, that would have been a disgrace to the king of the jungle, and there in a heap on the floor was a boy. I recognized him as a boy who had been missing from the truancy department for about a month. It seems that he had read in a paper that some man in New York had lost his boy—about the age of this boy—so he framed up a story and went to the police and told them he was the missing boy. There was a reward offered for the lost boy, and there was an argument between the lieutenant and one of the sergeants as to who saw him first—and in trying to straighten this out they discovered the fraud. I said to him, “Harry, what did you mean by this?” He replied, “Judge, dey spoiled the dandiest bum I ever thought of.”

I asked him where he had been for the last month, and he said he had been living in a piano box. I went with him, and he showed me the piano box, and as we came up a dog, a faithful, friendly dog, jumped out and recognized the boy. While we stood there I heard some boys just around the corner of the alley, talking about someone. Some of them insisted that “he only gets full once a month,” while some one said “he gets full once a week.” For a moment, I forgot the law of the “gang” in regard to “snitchin’,” and I said, “Harry, who are those boys talking about?” There had been a boy before me for intoxication, and I wanted to find out if this was the same one. So I said, “You slip around there and find out who it is they are talking about, and you’ll save a lot of trouble for me—and for somebody else.”

The boy hesitated a little, torn between loyalty to the judge and loyalty to the “gang,” but finally he went. In a little while he came back, and looking into my face without the least change of expression, said: “Judge, dem fellas is talkin’ about the moon; some says it gets full once a week, and some only once a month.” Then, after a minute, he continued, “Judge, you don’t always find out everything you want to.”

Some time ago, I was informed by my friend, Mr. Shipp, that I was to read a report at this Congress, but I found when I arrived that they had me down for an entirely different subject. However, both my report and the subject assigned me concern that very important subject, “the child.” Many things have transpired during the last year to give us hope and courage in the work for the child, and after all there is one thing that marks this century of ours, makes it distinct from all other centuries, and that is the fact that it is the century of the child. Indeed, it seems we are to realize the promise of holy writ, that a little child shall lead us. We are beginning to see, through the misfortunes of the child, through their tears and sufferings, many of the causes that are not only responsible for the troubles of children but for the troubles of men. For, after all, there is no child problem that isn’t a parent problem—a problem of the home. And when we get back to the problem of the home we are, of course, face to face with all the great social, economic, industrial and political problems. Even the political parties are at least beginning to understand that if they would meet with the approval of the people they must concern themselves more about the problems of humanity; that they must present real remedies which promise an immediate check for the terrible waste of life, of energy and of power, which is going on in this nation. That while it is important to conserve our material resources—the power of our waterfalls and the verdure of our forests—we must also conserve our human resources. After all, these great assets of the nation are very closely linked together. The strength and future well-being of the race itself must depend in a large measure upon the Conservation with which this organization has so well concerned itself during the past decade. It is only a natural and to be expected step in the evolution of its work that it should equally concern itself more directly with the human beings who will not be able to profit from its work unless their welfare is also conserved in other directions.

It has long been the opinion of specialists and social workers in this country that the national government itself was not doing all that it should do for the welfare of the children of the nation. Largely because of the failure of any government agency directly responsible for such work, the various methods of dealing with the many-sided problems of childhood were more or less in a state of chaos. The matter was first brought to the attention of Congress through a bill formulated and agreed to by the various child saving agencies of the State of Colorado, and introduced in Congress by Hon. John F. Shafroth, the present Governor of Colorado, in 1902, providing for a government bureau that should directly concern itself with the welfare of children. This effort was followed several years later by other child saving agencies in the introduction of what is now popularly known as "A bill for the establishment of a children's bureau." This bill was free from some of the objections of the earlier bill which included governmental protection for dumb animals, as well as a special bureau for the welfare of children. Great impetus was given to this final effort by former President Roosevelt, who called the White House conference on dependent children that met in Washington City on January 25 and 26, 1909, when a resolution was adopted recommending the enactment of the then pending measure.

The National Consumers' League in its Tenth Annual Report presented, perhaps, the ablest summary of all those presented concerning the necessity for such a bureau. This summary pointed out many items of information that ought to be valuable concerning the children of the Nation—information that, as amazing as it may seem, was practically impossible to be had in this Nation of ours concerning its children:

> 1. How many blind children are there in the United States? Where are they? What provision for their education is made? How many of them are receiving training for self-support? What are the causes of their blindness? What steps are taken to prevent blindness?
>
> 2. How many mentally subnormal children are there in the United States, including idiots, imbeciles, and children sufficiently self-directing to profit by special classes in school? Where are these children? What provision is made for their education? What does it cost? How many of them are receiving training for self-support?
>
> 3. How many fatherless children are there in the United States? Of these, how many fathers are dead? How many are illegitimate? How many are deserters? In cases in which the father is dead, what killed him? It should be known how much orphanage is due to tuberculosis, how much to industrial accidents, etc. Such knowledge is needful for the removal of preventable causes of orphanage.

4. We know something about juvenile illiteracy once in 10 years. The subject should be followed up every year. It is not a matter of immigrant children, but of a permanent, sodden failure of the Republic to educate a half million children of native English-speaking citizens. Current details are now unattainable.

5. Experience in Chicago under the only effective law on this subject in this country indicates that grave crimes against children are far more common than is generally known. There is no official source of wider information upon which other States may base improved legislation or administration.

6. How many children are employed in manufacture? In commerce? In the telegraph and messenger service? How many children are working underground in mines? How many at the mine's mouth? Where are these children? What are the mine labor laws applicable to children? We need a complete annual directory of State officials whose duty it is to enforce child-labor laws. This for the purpose of stimulating to imitation those States which have no such officials, as well as for arousing public interest in the work of the existing officials.

7. We need current information as to juvenile courts, and they need to be standardized. For instance, no juvenile court keeps a record of the various occupations pursued by the child before its appearance in court beyond, in some cases, the actual occupation at the time of the offense committed. Certain occupations are known to be demoralizing to children, but the statistics which would prove this are not now kept. It is reasonable to hope that persistent, recurrent inquiries from the Federal children's bureau may induce local authorities to keep their records in such form as to make them valuable both to the children concerned and to children in parts of the country which have no similar institutions.

8. There is no accepted standard of truancy work. In some places truant officers report daily, in others weekly, in some monthly, in others never. Some truant officers do no work whatever in return for their salaries. There should be some standard of efficiency for work of this sort, but first we need to know the facts.

9. Finally, and by far the most important, we do not know how many children are born each year, or how many die, or why they die. We need statistics of nativity and mortality.

The American Federation of Labor, the labor unions, and, of course, practically all of the social workers of the Nation have united through every means in their power to create the sentiment that has finally resulted in a Federal children's bureau.

This, then, is the most significant and, at the same time, the most hopeful single item of accomplishment for the conservation of the childhood welfare of the Nation for the past year. Next in importance to the establishment of the bureau itself is the appointment by the President in the person of Miss Julia T. Lathrop as its chief. Her long and devoted service with Miss Jane Addams at Hull House in Chicago, her well-known interest and experience in the sociological work that has occupied so many years of her useful life have especially equipped her for this work. While, up to the time of the establishment of the children's bureau, we were rather lagging behind the European nations, the various national and international conferences held throughout Europe during the past year have been greatly stimulated by the example our Government has set in establishing this special work for the Conservation of the Nation's best asset. It is hard, therefore, to estimate the far-reaching influence of this wise and generous step on the part of the National Government. It was my privilege, with others, to attend sessions of the congressional committees and speak in behalf of the National Children's Bureau, and my enthusiasm is just as great as it ever was for that important step, but I am not one of those who have believed that when we establish the bureau we have done all that we can do as a Nation to conserve the welfare of the children. No doubt the bureau will accomplish much through such an educational campaign as it may be able to conduct, and the gathering together of very important information upon subjects that at present are left largely to conjecture, and concerning which we shall still be left very much in the dark. But I wish to predict that its chief service in the end will be, as I hope it will be, to point out some of the needed changes in social, economic and industrial fabric that must be made if we are going to truly conserve the interests of the child. A program of social justice, definitely proposed and persistently carried out will in the end do much more for the welfare of the children of the Nation than all the bureaus that we can establish.

The agitation carried on principally by social workers, juvenile courts and probation officers, for the past ten years in behalf of what is popularly known as "Mothers' Pensions," has begun to bear fruit. As far back as 1899, a few of the States recognized the principle that it should share with certain homes the responsibility for the education of the child by not only providing free schools but also by providing aid for certain needy parents of children in order that the children could have the educational advantages afforded by the State. The demand for an extension of this recognition of the principle has met with response during the past two years in the State of Missouri and the State of Illinois. While the Missouri law was the first definite mothers' pension law, so-called, to become effective, it differs from the Illinois law in that it is limited to certain large cities. A somewhat similar law in Illinois has now been in force for a little more than a year. It is much broader in scope. Generally speaking, these laws vest power in the juvenile courts, after proper hearing, to direct the authorities dispensing public revenues, to pay to the parents—generally the mother—of dependent children a sum sufficient for the mother to care for the children in her own home, where the conditions are such, of course, as to justify keeping the child in its own home. It is assumed that the judge will act with wisdom and discretion and not abuse the power vested in him for the protection of dependent children. As a rule, this confidence in the court is not misplaced. But I am strongly opposed to legislation of this kind that is not carefully hedged about with such safeguards as to avoid possible abuses under it. For it is the abuses of such laws that furnish ammunition to its foes. This is not an objection to the principle of the law or a criticism of those who are entitled to so much credit for its passage. Most any kind of a law to start with establishing the principle should be more than welcome. The safeguards needed must largely develop in the course of practice and experience under it, when they may be added by suitable amendments from time to time—not an uncommon history of most all legislation of this character.

In many States, as in Colorado, where we have on several occasions attempted to secure legislation of this kind, we have met with failure for several reasons. The need has not seemed to legislators to be as acute as in States with more congested populations, in large cities like New York and Chicago. And in many States the laws already on the statute books have been fairly sufficient for their needs. Not only in Colorado, but many other States, to my personal knowledge, in exceptional or proper cases, mothers have been pensioned by the county commissioners, or assisted under school laws to such an extent that the lack of a more definite law upon the subject has not been seriously felt. But any State, in which there is a city of over 100,000 population or a considerable number of small cities of over 25,000 population, if it would truly conserve the welfare of its children, should not hesitate to adopt definite and effective legislation of this character.

But signs of opposition to such legislation are by no means lacking. It has been denounced in some quarters as paternalistic—socialistic, and entirely beyond the province or within the power of the State. But the time has long since passed in this country when there should be any serious question of not only the power but the duty of the State when it comes to the protection of its children. I say “its children” because the State is the supreme parent—the over-parent. From one viewpoint the State is superior to the natural parent. It says to the parent, “If you neglect your child you forfeit your right to its custody.” This is a just power to be wisely exercised. It is primarily for the welfare of the child. Because of the natural ties of love and affection that are supposed to exist between parent and child it is assumed by the State that the best place for the child is in its own home with its own father and mother. This is a wise balance for this rather exceptional power of the State. The State wisely recognizes that the home is the foundation of society, and since it is in the interest of the State to keep the child in the home, as one of the very best methods of preserving the home, the first duty of the agents of the State should be to bring about that result in every case possible. In fighting for the child the State is only fighting for its own preservation.

Another prime duty of the State is to compel the father to support the child and also to support the wife, not so much because it is the wife but because it is a woman who is the mother of a child, or may be the mother of a child. One great weakness in the nonsupport laws of the various States and, at the same time, a danger in the mothers’ pension acts of the various States is the lack of a practical system of operation and enforcement that will not permit the father to shirk—that will hold him to a strict accountability to his duty to the State, namely, to support the child. The child is the State and the State is the child. The man or woman, therefore, who does most for the child does most for the State. As a part of every nonsupport law and mothers’ compensation act should be provisions for workhouses where fathers who wilfully and without excuse refuse to perform their duty to the child should be committed.

Failing in the last Legislature in Colorado to get any legislation for the relief of needy mothers, our people have appealed to the people under their rights to initiative laws, for what we term a Mothers’ Compensation Law, rather than a Mothers’ Pension Act. We think that the difference is more than a mere haggling over terms.

The State maintains a standing army for its protection. Soldiers fighting its battles, or standing ready to fight its battles while performing that function, are paid—compensated. They receive money, food and clothing from the State. When the fight is over, when they have retired from service, in their old age they are pensioned.

In a different capacity, but none the less important and effective, do mothers of children serve the State. They do not face death on the field of battle, but they go down to the gates of death and bring back their children. The perils and hardships that soldiers endure in times of war are more than equalled by the struggles of hundreds of thousands of mothers fighting the enemies of the State that killing competition and the injustices of present economic conditions have raised up in its path. In fighting these enemies to save their children to the State these women are more serviceable soldiers of the State even than those sons they reared, who may have died on the field of battle.

The term “pension,” therefore, is a misnomer. It is confusing. It interferes with a real understanding of what this fight is all about.

It might not be a bad idea to consider pensioning mothers, as we pension soldiers after the battle is fought, after they have gone through the valley of the shadow, after they have slaved, and toiled and suffered and reared their children to manhood and womanhood, to guarantee them a peaceful, happy old age by providing a “pension.” But while they are engaged in the service of the State, in saving the State by saving the child, I insist, where it is necessary to enable them to do their part in the battle, they should be paid—they should be compensated.

I insist further that the compensation should no more be in the interest of the mother than it should be in the interest of the soldier, except as a means of preserving the home and the State, except as in the interest of public morals and for the prevention of poverty and crime—all of which is necessary to save the State.

Maternity is more than a prompting of nature. It is a patriotic duty to the State. As in the case of the patriot who enlists for the war, of course it should be voluntary and in accord with social and religious custom. But a wilful evasion of so plain a duty should be visited with the same contempt that meets the deserter from the ranks. As the profession of the soldier is no more the business of the individual without the part and duty of the State, neither is the perpetuation of the race wholly the business of the individual. And of course it is the duty of the State to see that those individuals responsible for the race should perform their duty. There must be laws recognizing the man as the breadwinner and the mother as the home maker.

The man must be held strictly accountable to the State for the support of the woman he has chosen to be the mother of his children. And this must be primarily not so much in the interest of the woman as in the interest of the children she bears or is expected to bear. If the man fails in his duty he should be compelled to perform that duty where it is possible to compel him to do it. If that is not possible, then the State itself must assume the burden. If the man has wilfully shirked it must provide workhouses in which he can be made to perform the duty he has voluntarily undertaken.

But, at whatever expense or hazard, the State must see that the child is protected. This is impossible unless the mother is protected.

The State has no right to scold women for race suicide when the State itself is responsible for race suicide. The father would have just as much right to scold his child for stealing when the neglect of the father was responsible for the thefts of the child. The State has just as important a part in this problem as the individual. The individual must do his duty, but the State must see that the conditions are such that it is possible for the individual to perform his part. If the struggle for bread makes maternity a tragedy instead of a blessing, it is the duty of the State to reverse the conditions and make maternity a blessing instead of a tragedy. (Applause.)

In conclusion, I want to utter a warning. In standing for the policy of the State to guarantee compensation to the mothers of children, the State becomes responsible in a measure for every child coming into the world. The next logical step will be for the State to demand a right to say who shall and who shall not be the fathers and mothers of its children. It follows that the Mothers' Compensation Act is only a part of a new code now in process of development in which the State shall become more and more responsible, not only for the children who are born into the world, but for the kind of children that are born into the world and the parentage of those children. It is all a part of a wise system of laws the purpose of which shall be as far as proper and possible to exclude the unfit from the rights of parenthood.

The revival of that interesting cult of eugenics now attracting so much attention, the demand for the teaching of sex hygiene and the agitation of kindred subjects now going on throughout the whole civilized world, is simply a response to the growing need and the growing demand that the State should become the over-parent of the race.

It is impossible in the time allowed for this discussion and the subjects that such a report should occupy to do more than discuss one or two of the recent activities in behalf of the Conservation of the child life of the Nation. Much excellent work has been done by other organizations, some of which, because of the limitations mentioned, it may be impossible to refer to. But I must especially commend the work of the Men and Religion Forward Movement and the excellent report of its Boys' Work Commission. After all, the work for the boy is necessarily in a large measure also a work for the girl. This report ably discusses the religious needs of the child; the message of Christianity to childhood; the essential principles of organized work with children in the church, the Sunday-school and local organizations, and the relation of these organizations to the home and the child and to social and sex hygiene.

Of similar importance is the laborious, able and excellent report on the safeguarding of adolescent youth, prepared for the International Sunday-school Association under the direction of Mr. Wilbur R. Crafts and his committee of assistants.

Dr. Wallace spoke of the need of moral education, and I heartily agree with him; but what are you going to do in the case of a bright boy who knows more about politics than he does about Sunday-school? I have a boy like this in my mind. He knew the ward boss, knew all about him—his authority over the dives and all that. But I thought he needed moral training, so he was induced to go to Sunday-school. I saw him afterwards and asked him what he learned in Sunday-school. "Aw," he said, "it's a place where all the little kids go and gives up a penny, but they don't git nothin' back." "But you learned something, didn't you?" I asked. "Yes," he said, "they learned me about the angels; they learned me they had wings like chickens, but they didn't learn me whether they laid eggs or not."

I agree with Dr. Wallace that we need to change our methods. Of course Dr. Wallace has had a different experience from mine. He has children. I have children—a thousand of them—but they belong to other folk.

There is no more important subject than the safeguarding of childhood and youth against the moral perils of the modern community. Under this head the important matter of regulating dance halls, skating rinks, moving picture shows and various places of amusement is becoming more and more one of the serious problems of community life.

The able reports of one or two vice commissions of the large cities, notably Chicago and Minneapolis, have added much to the literature and information valuable to those who are interested in conserving and protecting the moral welfare of the nation's youth.

Let me say here that I was in a city where they had such a vice commission, and one of the officials told me the number of women who had been forced into prostitution, or had been forced half-way there. I asked him the ages of these women, and he said practically they were all between eighteen and thirty-five, and on looking up the statistics we found that this number of women thus forced into this unholy life was 10 per cent. of all the women between eighteen and thirty-five in that city. It is a frightful thing, my friends, but if these things exist, if they are facts, we are false to our children and false to our country if we try to blind our eyes to these facts. It is our duty, and as Dr. Wallace has said, there is no place in this country where these things ought to be more freely discussed than in a Congress like this.

The child welfare exhibit, beginning with that of Chicago and duplicated in a measure in other large cities, is one of the most notable contributions in recent times in the great work of conserving the welfare of childhood.

The wider use of the schools in its more careful regard for the physical welfare of children must also be added to the hopeful signs of the times. The terrific waste in money, energy and effort that is going on in the cities because of the many boards controlling such activities as schools, playgrounds, social centers, public libraries, art galleries and public baths promises to be largely avoided by a consolidation of these activities under the control of one board with the schools as a great social center, to which is added its neighborhood dance hall, public baths, public library, public assembly hall, public playground and social center under one single authority, such as a board of education and recreation that promises to avoid the bickerings, difficulties, expense and waste that is the outcome from duplicated boards. Activities that are largely educational and concern the city's youth, now largely under a half-dozen boards or authorities, should be brought together under one general authority. An amendment to the Constitution of the State of Colorado proposing such a consolidation and the use of the schoolhouses as polling places and for the discussion of governmental, social and political questions during campaigns, is to be voted on at the November election.

And now, my friends, in conclusion I want to say that one of the prime duties of the Nation—its duty to the child—is to extend to the women the same rights as the men, that they may go to the polls and vote on these measures. (Applause.) This is not politics, Mr. President, it is a plain, economic proposition, because I believe the women of this country are awake to the needs of the children, especially those in the centers of population, and when they are given this right they will pass laws that are necessary to bring about right and justice for the women and children of this Nation. (Applause.) I would not have my position today but for the fact that women vote in Colorado. (Laughter.) Either the bosses, the machine or the gang would have got me long ago. Why? Because I went beyond the court into the swamp lands, not beyond the city, but within the city, and showed up the ghosts of poverty and crime and the relations between a certain type of lawless big boss and vice. And when the mother could see that the protection of her boy and her girl from vice depended upon clean politics and righteous laws, then, my friends, the change began to come, and it is coming in our State as in every other State in this Nation—then began a reign of truth and right and progress. (Applause.) And when the women of our city understood what machine rule meant, they rose in their might, with the ballot in their hands, and put an end to machine rule in that city.

I remember a little boy that belonged to one of our debating clubs on the west side, who was very much disturbed over the making of some new laws. He came to see me, and when I asked him what he wanted, he pulled a piece of paper out of his pocket, on which was written: “Resolved, That all kids over ten years of age shall have a right to vote for the juvenile judge.” “And then,” said little Benny, “when we gets that law through the bosses will never get you, and we kids will get justice.”

But it was not necessary.

I remember we had a little fellow who was quite a fluent speaker, and finally one of the bosses began to get alarmed at the effect this boy’s talk was having. The boss said to him, “You have a lot of mouth, but you have no vote.” Quick as a flash came the answer, “I haven’t got a vote, but I would have you understand that in the State of Colorado my mother has a vote, and my sister has a vote, and she married a fellow and he has a vote, and they will see that he votes right.”

And the boy’s prediction was more than verified, for when the votes were counted the majority was on the right side, and the people who were working to relieve poverty and the suffering of children had won by ten thousand majority.

So I feel we must have the women with us in this struggle for the rights of childhood in this Nation, and with that right guaranteed they will bring about sooner than any other agency some changes for good in this Nation. If we are to save the child we want to save the State, for the child is the State and the State is the child. Take care of the child and the State will take care of itself. (Great applause.)

Miss Adeline Denny (in the audience) moved that a rising vote of thanks be tendered Judge Lindsey, who is in favor of women as well as children. The motion was carried and the Chautauqua salute given.

President White—The Congress now stands adjourned until 9:30 o’clock tomorrow morning.

SEVENTH SESSION.

The Congress convened in the Murat Theater, on the morning of October 3, 1912, and was called to order by President White, at 9:45 o'clock a. m.

President WHITE—We are a little late in gathering this morning. The meetings last night were rather strenuous. There were meetings in two different places, and the one I attended had seventeen or eighteen hundred in the audience, so we know we have a large attendance. The idea of having sectional meetings is a good one, because it enables discussions at greater length upon special subjects that concern different people interested. Day before yesterday, we had three meetings going on at the same time. Then we have an illustration of what is needed in the way of civic reform and good government over at the State House, and none of us should miss this. It is going on all the time. It appeals to the eye, and we can see at a glance so much that is needed in this battle for reform.

I have some announcements to make before I call upon the first speaker. I have a telegram from Mrs. G. H. Robertson of Jackson, Tenn., and one from Anna Caroline Benning of Columbus, Ga. These telegrams contain greetings, and also suggestions as to the next meeting place of the Congress. We are glad to have suggestions as to the next meeting place, but under the Constitution the Executive Committee takes up this matter for consideration, and they have three or four months to do it in.

JACKSON, TENN., October 2, 1912.

President National Conservation Congress, Claypool Hotel, Indianapolis.

Mothers and teachers of Tennessee interested in conservation of childhood beg the National Conservation Congress to hold its next meeting in Knoxville. This will mean much to Tennessee. We hope you will see that Knoxville is, all things considered, the place of all others for you.

From President Congress of Mothers,

MRS. G. H. ROBERTSON.

COLUMBUS, GA., October 2, 1912.

Mr. J. B. White, President Fourth National Conservation Congress, Indianapolis, Indiana.

Please greet the officers and members of the Fourth National Conservation Congress for me and tell them that illness prevents my attendance, and say for me the disappointment is great, for my heart is in the work.

ANNA CAROLINE BENNING.

President WHITE—We also have a report from Col. M. H. Crump of Bowling Green, Ky., which will now be read:

REPORT OF THE COMMITTEE ON ESTABLISHMENT OF A NATIONAL PARK TO INCLUDE MAMMOTH CAVE.

Immediately on notification of appointment by President J. B. White, the committee (Dr. Henry S. Drinker, Hon. William P. Borland, Mr. Gifford Pinchot and Col. M. H. Crump, with Dr. W. J. McGee as Chairman) organized by correspondence and proceeded to work through both individual and collective action. Largely at the instance of Col. Crump, Hon. Robert Y. Thomas, Jr., a Representative from Kentucky, introduced a bill (H. R. 1666) providing for the establishment of a National Park to include Mammoth Cave, and this was duly referred to the House Committee on Military Affairs. Before this body your committee appeared at formal hearings on February 1, February 5 and May 3. Early in February a similar bill was introduced in the Senate by Hon. William O. Bradley and referred to the Senate Committee on Public Lands; before this body also your committee (through Col. Crump and the Chairman) appeared at a formal hearing on February 6. Both before and after these hearings members of the committee personally presented the matter before members of both branches of the Federal Congress; Dr. Drinker by correspondence, since he was out of the country until too late to attend the early hearings.

Your committee have to report, with regret, that while the requisite early steps looking toward the desired legislation were taken, the bills have not yet been reported from the Congressional Committees and probably will not be during the present session. Accordingly, we recommend that this be considered a report of progress; that the National Conservation Congress be requested through its Resolutions Committee to once more urge on the Federal Congress the eminent desirability of creating a National Park to include the Mammoth Cave, and that an appropriate committee be created through the National Conservation Congress of 1912 to continue action in the premises.

Respectfully submitted,

W. J. MCGEE, Chairman.

MALCOLM H. CRUMP.

President WHITE—This report will be turned over to the new President.

It is now my pleasure to introduce to this audience a gentleman from the Pacific Coast who has long been an active worker in the cause of Conservation, especially in the conservation of forests. He is well known to all on the Pacific Coast and to every man in the Central and Eastern States. He is President of the National Lumber Manufacturers' Association, and he will treat the subject of Conservation from "The Lumberman's Viewpoint." Major E. G. Griggs, of Tacoma, Washington. (Applause.)

Major GRIGGS—Gentlemen, Members of the Convention: I want to voice the sentiment of the lumbermen of the country particularly in approving the action taken by this Congress in allowing us to have our own conferences in reference to the interests in which we are vitally concerned, together with the general meeting. I think that has been one of the best features of this Congress.

The objects of the National Conservation Congress are so clearly exploited in the Second Article of our Constitution that I believe a repetition of them is clearly in order that we may keep them uppermost in our minds:

"(1) To provide a forum for discussion of the resources of the United States as the foundation for the prosperity of the people, (2) to furnish definite information concerning the resources and their utilization, and (3) to afford an agency through which the people of the country may frame policies and principles affecting the wise and practical development, conservation and utilization of the resources to be put into effect by their representatives in State and Federal government."

I have attended all of these Congresses and have been wonderfully impressed with the zeal and interest manifested in these proceedings. The vital questions considered are touching the popular chord and its effect is vibrating the length and breadth of the land.

Some are drawn here by one interest and some by another, but all recognize the wisdom and need of arousing our people to a consideration of the resources of our country and their proper utilization. In the cauldron of our national development, mix a little philanthropy, patriotism and politics and you can stir up the most phlegmatic of our citizens.

To my mind, the great results we wish to secure in this Conservation effort can only be realized by directing the attention of the millions who do not attend these annual meetings to the importance in our State and national life of the subject-matter we have under consideration.

The vast majority of the American people wish to see general prosperity and proper utilization of the resources of the country, regardless of the political ambitions of any individual or party. Conservation will only be realized when it takes such a strong hold of the people that any man or set of men will sink to political oblivion if they do not promote its cause.

Three years ago we were somewhat startled by the announcement, I think from the originators of this movement, that the electric companies had combined to control the water powers of the country. Today I come from a State where a stupendous amalgamation of capital has recently combined the hydro-electric plants of the Puget Sound basin. Not that this is detrimental to our development, but that the acquiring of these perpetual rights and control of natural resources should be well considered by the people and subject forever to their supervision.

The cupidity of capital will only be curbed by the assurance to the long-time investor that the Government is behind the investment and the people will not forever back the investment unless they are in on the deal.

Our country is comparatively new and we need to encourage capital and labor in every way to develop the latent resources, but we want to make better trades than we have made in the past if we wish to hold the respect of either.

The old saying that "Uncle Sam is rich enough to give us all a farm" sounded siren-like to all, and was necessary to encourage the settler, but there is a limit to even Uncle Sam's patrimony—and irrigation costs money. No doubt the State of Washington, which I represent, has profited as much as any other by the liberal policy of the Government, but there forest reserves have been declared and in lieu of worthless timber tracts scrip has been issued to bona fide settlers or original grantees, and for this same scrip some of the choice timber lands of the country have been exchanged. I conclude that "David Harum" is discounted in a trade.

The lumber manufacturers of the country have in recent years been in the limelight of trust investigators, and to what purpose? If to foster the political ambitions of some demagogue, I am sure it will fail. There may be organizations back of labor and capital which come under the ban of the law, but when such general conclusions are drawn, as in the Missouri Ouster Case, that the National Lumber Manufacturers' Association is an unlawful combination in restraint of trade, as President of that association I "deny the allegation and defy the allegator."

There is too much of loose talk in censuring the efforts of associations generally. The very theories you as Conservationists are advancing are uppermost in the minds of association workers. And the greatest development in forest Conservation and fire protection has its origin and support from these associations. We have something to conserve and are not mere theorists. With rising values of timber and utilization of lower grades of lumber, the product of the entire tree will be saved.

This is where the shoe pinches. It is going to cost more money to conserve. The low grades of lumber, slabs, and waste from a mill must bring enough money when sold to pay for the labor expended in saving them. Then, and then only, will they be saved.

Trees can only be reproduced on soil suitable for that purpose and for no other. The timber crop is the process of years of growth, and annual taxes, perpetual fire risk and the desire to use the land for more frequent crops are the deterrent features of reforestation. We only need to look abroad, where common lumber brings the price of mahogany in this country, to realize that an article to be saved and reproduced must have commercial value.

Your great centers of population in the East and Middle West are today beginning to realize that happiness, health and long life of the people will be your greatest commercial asset. The country is becoming aroused to the needs of forest, lake, stream and fresh air to build up American citizenship. We in the West, like the pioneers who have worked their way across the American continent, do not appreciate our own resources until we realize the vast sums being appropriated in your dense centers of population to reinstate in a measure the surroundings in which we revel.

Population, transportation and ability to pay are all determining factors in our national development. It takes something more than philanthropy to meet a payroll or pay the grocer, and too little heed is given the trials and privations of our pioneer life in some of our theories.

We lumbermen of the West Coast, where transportation charge alone equals more than the original cost of our lumber to you, are sometimes rebuffed in our efforts to conserve, where of necessity the waste is large.

We are not slow in the West and South in developing the use of wooden block paving, in establishing creosote plants to prolong the life of our product, but in our recent attempts to get the consumer to use odd and short lengths to prevent a waste in our mills of 2 per cent. of our planing mill product, we are balked in our efforts and forced to the burners with a lot of trimmings.

I have just read the following from an address delivered by Joseph B. Knapp, assistant district forester in the United States Service, which bears out my contention:

"Coast lumbermen a few years ago unitedly endeavored to introduce the use of flooring, ceiling, finish and other planing mill products in multiples of one foot from three feet upward. At this time the United States Forest Service made an investigation of the waste due to manufacturing planing mill products in multiples of two feet. We found this waste to be over two per cent. of the material run through planing mills in Oregon and Washington, or the equivalent of the yearly growth of wood on approximately 30,000 acres of good timberland. The consuming trade refused to accept odd lengths and after a conscientious attempt on the part of lumber manufacturers, it was found necessary to discontinue the manufacture of odd lengths over ten feet. It is therefore seen that the useless waste in the manufacture of lumber can not always be attributed to the lack of a desire on the part of the lumber manufacturer to introduce economical practices. It remains for the ultimate consumer of our timber products to determine in what form these products shall be supplied to him, and therefore conservative lumbering and close manufacture are dependent as much upon the layman as upon the manufacturer."

Our British Columbia neighbors are keenly alive to their timber interests and their forest service is alive to the situation. Mr. Benedict, assistant forester of British Columbia, in a recent address stated that in British Columbia, on a very conservative estimate, after eliminating waste land, rocky mountain slopes and peaks, they had 65,000,000 acres capable of producing merchantable timber and valueless for any other purpose.

"The productiveness of this land in timber will vary from 1,000 hard feet per acre per year in particularly favorable localities on the coast to 25 or 50 board feet per acre per year on the mountains of the interior," he says, "but I am confident that the average yield will amount to 100 board feet at least. This gives an annual production of 6,500,000,000 feet.

“Allowing for a temporary overproduction of lumber brought on by the desire of the holders of timber limits to realize on their investment as quickly as possible, it will be seen that the stand of mature timber will last from fifty to seventy-five years. At the end of seventy-five years, when this mature timber is cut, the present stand of second growth timber will have matured so that the annual production can be maintained perpetually at 6,500,000,000 feet. All this provided the present stand of mature timber is preserved from destruction by fire and likewise that the second growth is able to escape fire and grow to maturity.

“The stake then for which the forest protection force is working is an annual crop of 6,500,000,000 feet of timber, worth to the Government, say, $6,000,000, and to the community $100,000,000. To win the stake fire must be kept out of the area of 100,000,000 acres, or a block of forest 400 miles square. The problem, both on account of the immense area, the variety of causes of fire, the absence of means of transportation and communication, and the present sparseness of population, is a most difficult one to solve. The safe harvesting of the annual yield will require, besides the expenditure of large sums of money, the good will of every citizen in the Province. However, everything favors the satisfactory working out of the problem.”

I quote the above to prove that we are not alone in our efforts to conserve and provide for the future of our country.

Our associated efforts are being extended continually along the lines of economy in manufacture, in the matter of standard grades and sizes, inspection and insurance. Where is the commodity that can be intelligently transported and marketed without a thorough knowledge of both production and consumption? I now claim, and always have claimed, that associated efforts to disseminate this information and collectively endorse projects financially and otherwise to promote the study of forestry and lumbering are the highest types of Conservation of the Nation’s resources.

In the great State of Washington, which is now furnishing more lumber than any other State in the Union, and where the lumber production is the chief industry of the State, we are vitally concerned in our legislative work, and concerning our Workman’s Compensation Act I wish to bring to this particular Congress a special message. I believe this act emphasizes the benefits of co-operative effort in conserving human life and in protecting the breadwinner, upon whom depends the life and happiness of so large a population.

With an industry affecting throughout the United States over 45,000 sawmills and 800,000 employes, regardless of families dependent on them, you will agree with me that we are all vitally interested in workmen's compensation.

In a recent Bulletin of the National Lumber Manufacturers' Association, Mr. Bronson wrote as follows:

"Thirteen States have adopted workmen's compensation acts, and all have become effective since September 1, 1910. All but one of these laws are optional, the exception being the Washington law, which is compulsory, and which, according to the brief experience had, seems to be the most satisfactory both to employers and employes, saving the employer all expense for industrial insurance, and saving both employer and employe all court costs and giving to the employe the full compensation provided by the law without any deduction for lawyers or fees.

"The thirteen States which have adopted compensation acts are California, Illinois, Kansas, Massachusetts, Michigan, Nevada, New Hampshire, New Jersey, New York, Ohio, Rhode Island, Washington and Wisconsin. In all of these States the common-law defenses are in whole or part wiped out where the employer does not come under the compensation act."

The law in the State of Washington has now been in operation one year. To those of us who have operated under it, it has already proved what its advocates claimed the most advanced piece of legislation enacted—and we have woman's suffrage, too.

It is surprising what an effect it has had in clearing the court docket of damage suits and in paying to the injured at a time when the money is needed all the award without the intervention of a third party. With a carefully selected commission of three, responsible to the Governor, and non-partisan in character, we have launched a statute that, regardless of any improvements which may be determined or defects disclosed, has improved the industrial atmosphere both for employe and employer.

The State appropriates a fund of $150,000 to administer the act for two years. The risks are classified and rates assessed, and quarterly payments called for as required.

The statistical records being made and knowledge gained by the Commission in administering this act will be invaluable and it has already brought to the attention of loggers, lumbermen and all manufacturers the loss of life and limb incidental to the business, the benefits of factory inspection as a further prevention and the fixing of responsibility as to accidents.

I secured from the Commission the attached reports, which I shall not read in detail, but which show detailed comparisons and some interesting figures. (See reports, A, B and C, appended to this paper.)

Our commissioners have been called on many times to address conventions and congresses regarding the law, and I can best state their views by quoting from an address of Commissioner Pratt at our recent Logging Congress, and from a statement issued by Commissioner Wallace. Commissioner Pratt says in part:

"The Workman's Compensation Act has been in operation now for nine months, and those of you who are actively in business in the State of Washington are more or less familiar with its results.

"In the first place, it is compulsory alike on the employer and employe. The employer has to pay into the accident fund a sum of money based upon a percentage of his payrolls, and the employe must accept the awards of the Commission allowed for work accidents in lieu of his right to sue at common law, subject, of course, to a right of appeal on the amount awarded.

"All the extra hazardous industries of the State are divided into forty-seven classes, each class with a fund of its own, and the accidents arising in that class shall be a drain only upon that fund. As the payments for work accidents deplete the fund in each class it provides for monthly assessments to be made to recoup each class. The payments out of these funds are only for work accidents, all the cost of the administration of the law being paid out of the general taxes of the State. For the first twenty-two months of its operation an appropriation of $150,000 was made.

"It is unlawful for the employer to deduct any portion of the premium paid into the accident fund out of the wages of the employe. It provides for penalizing any establishment which from poor or careless management is unduly hazardous by raising its rate. If an employer, besides employing men in extra hazardous employment, employs men in non-hazardous employment, the premium shall be paid only the payrolls of the extra hazardous work, but the employer and the non-hazardous employe may elect to come under the act and both shall receive the benefits of the act.

"As each class must pay for only such accidents arising in that class, and as assessments are made only as the funds of that class are depleted, there are but two things that govern the cost of this insurance: the amount of the awards and the number and seriousness of the accidents.

"As I have said, the classes that the loggers and lumbermen are most interested in are seven, ten and twenty-nine.

"The rate for class seven is 5 per cent.; for classes ten, 2½ per cent., and twenty-nine is 2½ per cent. All operations in which those present are interested in these last two classes take the same rate. Class ten is by far the largest class we have, and as it covers several distinct operations it has been divided into four different subdivisions or groups.

"10.1 covers logging and logging operations of all kinds.

"10.2 covers sawmills and lumber yards, etc.

"10.3 covers shingle mills and operations connected with a shingle plant.

"10.4 covers mast and spar manufacture, stump pulling, land clearing, etc.

"We have had the following table compiled of comparative risks of wood-working industries:

Class Number.	Av. Number of Men Employed.	Number of Time Awards.	Number of Workdays Lost.	Workday Lost per 1,000 Men.	Number of Dismem-berment Awards.	Dismem-berment Awards per 1,000 Men.	Number of Deaths.	Deaths per 1,000 Men.
7	4,120	172	5,862	1,402	17	4.1	7	1.69
10.1	12,801	440	14,926	1,166	35	2.7	22	1.72
10.2	17,770	763	14,941	841	51	2.9	7	0.39
10.3	5,565	221	5,766	1,036	56	10.1	0	0.00
10.4	381	50	1,144	3,003	5	13.1	1	2.60
29	3,787	156	3,368	888	24	6.3	0	0.00
	44,424	1,802	46,007	1,035	188	4.2	37	0.83

"This table is not as nice a one as I should have liked to show this Congress of Loggers. It shows where the great harm is being done in class ten and it shows which is the greater risk and what part of the class should be charged a higher rate than the other part. Furthermore, not only are we keeping a strict account with each class, and division of a class, but we are keeping a strict account with each individual operator and in the end will publish an account of just how many accidents each firm or corporation has had, just how much has been paid out for them in awards or pensions, for injuries to their workingmen.

“Now, what are we going to do to prevent this loss of life and limb? In the first place, there has been a Labor Commission since 1905 and the mills and factories have been subject to inspection and have been forced to put on safeguards. The loggers have steadfastly refused to allow any inspection laws covering logging to be put on the statute book of the State. Logging is a hazardous life at the very best and calls for strong, dare-devil men, and men who are willing to take chances. Danger is always present and men become so used to it that they get careless. This, however, is no excuse for needless loss of life and limb.

“Once more I want to urge upon the lumbermen of all classes the necessity of more rigid inspection; to have some one about the plant whose sole duty is to see to it that every machine is safeguarded the best that possibly can be, and that safeguards are kept in place. It will be money in your pockets if you want to put it on such a mercenary level as that.

“Also I want to urge that a movement be put on foot that our colleges and universities establish chairs of logging and safeguarding engineering, so that our young men, just fresh from school, shall have a better knowledge to start with on these subjects than did their fathers. Many and many a man-killing machine is used just because some one has not invented a better one.

“The report of the expense of the Commission shows that the total amount expended to July 1 out of an appropriation of $150,000 was $87,062.14, and that the proportion of expenses to the amount of business done is 11 per cent., a showing so much below what it costs casualty companies merely to solicit their business as to be notable.

“The president of one of the casualty companies, while I was in New York, showed me their experience, which showed that the cost to them for the last year was 51 per cent. of the premium. As you see, our cost is about 11 per cent. Of course, we do not have to solicit, nor do we have so large a force in the field for adjustment.

“The Commission, is keeping well inside of its appropriation, as the allowable average expenses for twenty-two months would be $6,818.18, while the actual average has been $6,620.16.

"Other details of the financial report are as follows:

Total receipts, accident fund				$699,508 72
Total expense				86,062 14
Total fund				$785,570 86
Cash in fund,	36	per cent.	$281,993 32	
Reserve fund,	20.5	per cent.	161,154 49	
Claims paid,	32.5	per cent.	256,360 91	
Expense,	11	per cent.	86,062 14	
	100	per cent.		$785,570 86

"We are executing the law, backed by the State of Washington, and there is less quibble in settlements made by our Commission than there would be by an adjustment made by a casualty company. Neither do we have to pay any attorney's fees, as the Attorney-General's office has to attend to all this part of the work for us.

"Since the first of October, there has not been a case filed in any court in the State for damages done to any workman who came under this act.

"This has been a great relief to our courts, and in time will be felt in reduced taxes. The cheapening of our court costs and the removal of all personal liability suits should work a reduction of costs to the general taxpayer.

"One of the features of the old common-law system was the ambulance-chasing lawyer, whom we all know. This gentleman is practically out of business as a result of the Workman's Compensation Act, but is undertaking to find some activity in the industry of appeals. Out of over 6,000 claims passed upon, only twelve appeals have been filed, one of them from the Imperial Powder Company, to interpret the law, one to determine the scope of the interstate commerce law, one filed by an insane claimant, and several that are in the process of adjustment and dissolution.

"One appealed case has been tried in court and the court sustained the Commissioner's finding as far as temporary total disability was concerned, but found the claimant entitled to compensation for permanent partial disability, remanding the case to the Commission for additional compensation, which was promptly awarded. Had the Commissioner been in possession of the facts, the award for permanent partial disability would have been made without appeal."

Commissioner Wallace makes the following pertinent statement:

“The Washington State insurance system has succeeded beyond the best hopes of its friends and sponsors. In this act, one of the youngest States is giving the older commonwealths another example of a wise and progressive law. The State’s control over public utility corporations, giving the suffrage to women, eight-hour laws for underground miners and women wage-earners, full crew law for railways, and other laws enacted during the past four years in the interest of labor deserve full praise and should not be forgotten in the triumph of our compensation act.

“The compensation act has thus ushered in an era of publicity regarding the appalling maiming, dismembering and killing of workmen in the mines, mills and workshops of our State. The great question just now becomes not what we can give to pay for pain and suffering and even death, but how can we best safeguard those who toil. This will be real progress; compensation must ever be mere apology.

“Concerned as we have been as to how the little home flock could be kept together when the breadwinner was stricken down in his endeavor to make an honest living, and thinking in terms of dollars and cents how much it will take to keep the wolf from the door during these times of industrial disaster, we may have overlooked the fact (or was it because we were not familiar with it?) that, according to the best authorities who have made accident prevention a scientific study for a number of years, 75 to 90 per cent. of the accidents that occur are preventable.

“Our law has been widely commended and is in reality the best compensation law in the United States today. It has been rarely condemned, save by those who profited by the old legal system. It has shown the great waste of human energy, manhood and womanhood—wastes which reflect discredit upon this young and virile commonwealth—and as these things begin to be understood by the people they will insistently ask, what can we do, not only to preserve the mineral, the timber, and the water-power resources of our State, but what can we do to conserve our greatest asset—human life?”

I am confident this Congress will endorse the sentiments expressed and I only wish to add the employer and employe, State official and private citizen, voice the same sentiments and desire to give them widest publicity.

REPORT A.

INDUSTRIAL INSURANCE COMMISSION OF WASHINGTON.

Statement of Condition of Accident Fund, September 1, 1912.

Class.	Total Amount Claims.		Estimated Reserve on Approved Claims.	Balance in Fund.	Deaths Requiring.		Occupation.
	Paid In.	Paid.			Pension.	No Pension.	
1	$19,350 71	$6,400 60		$12,950 11		6	Sewers.
2	17,839 53	3,860 85	$3,481 60	10,497 08	1		Bridge and tower.
3	6,343 73	1,653 90		4,689 83			Pile driving.
4	1,996 85	463 80		1,533 05			House wrecking.
5	70,194 23	18,207 10	20,008 25	31,978 88	6	2	General construction.
6	52,990 61	9,178 86	2,063 95	41,747 80	3	5	Power line installation.
7	84,249 20	28,886 99	17,863 22	37,498 99	8	4	Railroads.
8	30,745 73	7,982 15	1,180 07	21,583 51	1	2	Street grading.
9	6,340 35	2,173 30		4,167 05			Ship building.
10	266,461 72	167,741 35	95,777 56	2,942 81	33	29	Lumbering, milling, etc.
12	6,432 63	1,642 25		4,790 38			Dredging.
13	16,371 87	2,775 43	7,574 72	6,021 72	4		Electric systems.
14	26,817 34	8,120 36	1,266 00	19,430 98	1	1	Street railway.
15	4,275 45	1,284 21	754 52	2,236 72	1		Telephone and telegraph.
16	82,110 27	33,794 11	28,020 37	20,295 79	11	6	Coal mining.
17	14,800 60	4,786 55	2,352 55	7,661 50	1	3	Quarries.
18	6,368 70	4,808 75		1,559 95			Smelters.
19	7,098 59	717 36	2,903 83	3,477 40	1		Gas works.
20	1,202 20	405 00		797 20			Steam boats.
21	8,319 89	4,576 43		3,743 46			Grain elevators.
22	7,656 94	2,370 60		5,286 34			Laundries.
23	4,152 43	543 40	2,805 92	803 11	2		Water works.
24	8,084 75	5,826 45		2,258 30		1	Paper mills.
25	1,489 06	402 65		1,086 41			Garbage works.
29	27,134 69	16,760 72		10,373 97			Wood working.
30	789 83			789 83			Asphalt manufacturing.
31	7,051 68	1,580 73	842 08	4,628 87	1	1	Cement manufacturing.
33	11,289 16	1,536 30		9,752 86			Fish canneries.
34	28,349 76	15,404 90	3,156 32	9,788 54	1		Steel manufacturing, foundries.
35	6,216 34	1,395 05		4,821 29		1	Brick manufacturing.
37	9,857 48	1,828 73	3,295 17	4,733 58	1	1	Breweries.
38	3,812 52	1,114 95		2,697 57			Textile manufacturing.
39	2,627 55	415 49		2,212 06			Food stuffs.
40	2,149 77	203 55		1,946 22			Creameries.
41	6,516 49	1,297 80		5,218 69			Printing.

42	9,885 96	5,485 86	3,888 34	511 76	1		Longshoring.
43	4,584 50	2,494 25		2,090 25			Packing houses.
44	1,412 96	680 25		732 71			Ice manufacturing.
45	445 14			445 14			Theatre stage employes.
46	463 27	1,908 95		1,445 68	7	1	Powder works.
47	632.44	39 75		592 69			Creosoting works.
48	1,000 16	83 95		916 21			Non-hazardous elective.
	$875,913 08	$368,833 68	$197,234 47	$309,844 93	84	63	

F. W. HINSDALE, Chief Auditor.

REPORT B.

INDUSTRIAL INSURANCE COMMISSION OF WASHINGTON.

Statement of Expense Account for the Month of August, 1912.

Mileage—	
Commissioners	$50 00
Auditors	98 23
Railroad Fare—	
Commissioners	20 35
Auditors	195 88
Hotel—	
Commissioners	97 10
Auditors	447 90
Incidental Expenses—	
Auditors	9 15
Salaries—	
Commissioners	900 00
Auditors	2,260 29
Physicians	406 50
Office	2,067 06
Miscellaneous—	
Stationery	256 81
Postage	322 61
Telephone	66 30
Telegraph	8 24
Office supplies	208 97
General expense	60 20
Rent	110 00
Total	$7,585 59

F. W. HINSDALE, Chief Auditor.

REPORT C.

Olympia, Washington, August 31, 1912.

Industrial Insurance Commission, Olympia, Washington.

Gentlemen—Herewith statement of claims handled by this Department during the month of August, 1912. Also, the number handled during the period from October 1, 1911 to August 31, 1912.

Claims Received.

	Month of August.	Total to Date.	
Accidents reported	1,374	10,586	
Files incomplete		1,471	
			9,115
Files complete	1,455		
Monthly payments continued	262	1,972	
Claims reopened	8	129	
			2,101
	1,725		
			11,216

Claims Disposed of.

	Month of August.	Total to Date.	
Finals	1,097	5,255	
Monthly	262	1,972	
Fatal	36	214	
Total permanent disability		2	
Rejections	78	324	
Suspensions	46	281	
No. claims	198	1,420	
Total disposed of			9,468
	1,717		
Claims in the work			1,748
Total			11,216

Respectfully submitted,

J. F. GILLIES (Signed),

Claim Agent.

President WHITE—This is certainly a very important paper. I want to say here that tomorrow, in Kansas City, Mo., a committee from the organized labor interests, and a committee from the manufacturers will meet to discuss a proposition to prepare a bill for presentation to the next Missouri Legislature that shall be fair alike to employer and employe, in regard to compensation for injuries. It has worked well in Washington, it is humane, and it does shut off the dishonest, shyster lawyer who means to get three-fourths or more of the award for the injury, and gives it all to the person who is injured, without any attorney's fees. (Applause.)

I will take just a moment at this time to appoint the Nominating Committee:

George E. Condra, Chairman; E. T. Allen, H. A. Barker, Mrs. Marion A. Crocker, E. G. Griggs, Mrs. Elmer E. Kendall, Henry Wallace, and N. P. Wheeler.

This committee has the duty of considering and nominating the officers for the next Congress. They will have a couple of days for the work.

At the first day's session, there was a report on the program from the National Congress of Mothers, which was to have been presented by Mrs. Orville T. Bright, of Chicago. Through an unfortunate misunderstanding, which was not the fault of Mrs. Bright, she was not here on the first day. We are glad to have the report at this time. I now take great pleasure in introducing Mrs. Bright. (Applause.)

Mrs. Bright—The one object for the conservation of all the material resources of a Nation is for the use, comfort and benefit of the homes of the people.

It would be of little importance what became of forests, lands, waters, minerals or food were there no men, women and children to use and enjoy them.

Therefore, at the very heart of this Conservation work should be the two departments covering homes and child life.

It has been a source of encouragement to see that men who are leaders in many great developments of our land, have given definite place to the study of the conservation of the home.

There is need for it if America is to be the greatest of all the nations, for with its wonderful natural resources it can only be as great as the quality and character of its people.

Great minds are needed to think and plan with wisdom and unselfishness for the America that is to be, for the protection of homes that are to shelter and nurture the men and women who a few years hence will take our places.

The United States has its Departments of State and War and Navy. It has not yet seen that the greatest questions it has to meet are the protection and care of the American people and American homes.

The U. S. Department of Agriculture is educating the farmer to make the most out of his land. It gives him information concerning the soils, the rotation of crops, the protection against the many enemies of plant life, the care and feeding of stock and poultry. It protects the forests and the fisheries. All these things for the service of man have received the guardianship of the Government.

Homes are just as important as farms, and there is just as great need of proper consideration for their elevation and protection as there is that of farms and stock and forests.

The protection of infant life is of more value, even in a pecuniary way, than the protection of the cotton crop, yet three hundred thousand babies die annually whose lives might be saved if the United States gave the same careful, intelligent information to the mothers as it does to the farmers.

The annual sacrifice of three hundred thousand American citizens from preventable causes is a waste far too great not to receive governmental consideration. Time need not be wasted on compiling statistics. There is need for prompt and decisive action to prevent this needless sacrifice; it means that each year the possibility for at least one hundred thousand homes of American citizens is cut off. That means a serious loss to this Nation and one for which immigration can not compensate.

The wonderful advance in agriculture can be paralleled in human culture if the same methods are used. The trains that go through the country for agricultural demonstrations should carry instruction to both men and women on home-making and child nurture. The list of valuable educational pamphlets published and sent free of postage should include instruction in child hygiene and sanitation.

There is today a need for a Home Department in the National and State Governments that is equipped to study the home problems of America and meet them as only can be done by thorough study and knowledge of conditions, their causes and remedies. The sacrifice of infant life is a small part of the waste that undermines the homes.

Juvenile crime, its causes and treatment are of more vital moment than the boll weevil or the chestnut blight, for the possible good citizen transformed into one who is a menace and expense to society is a great waste.

There are countless organizations which give material and charitable relief. There are few which give the help that will enable the average home properly to guide and train the boys and girls who are wayward, or will help parents to learn efficient methods of child nurture. The home has the greatest power over human life and human character. Too long has it been left to chance and ignorant experiment to make it efficient in its work, stable and permanent.

The home is founded by the marriage of a man and a woman. It is a matter of grave concern to the Nation when divorce breaks up one in every twelve homes, and leaves the children bereft, not only of a normal home but deprived of a true conception of what marriage and parenthood should be. The conservation of the home requires that serious study and work be done to change this condition in America. It can not be done by legislation alone, though one of the greatest needs today for the protection of the home is Federal law governing marriage, divorce and polygamy.

It is a serious menace to the home when forty-four States may make as many different laws as they choose on a subject which is the foundation of the Nation’s future.

That a man may be legally married in one State, and that such marriage is illegal in a State adjoining, that divorce is easy in some States and difficult in others, that polygamy is permitted to continue in some States, and that freedom to spread the cult is allowed, have all been undermining influences in the God-given standards of marriage, home and parenthood.

The Government has found it necessary to assume jurisdiction over interstate commerce, railroads and express companies. It is of even more vital importance that it should have jurisdiction over marriage, divorce and their violations. In addition to this, there is need for definite plain teaching of youth in regard to the true high ideal of marriage, of parenthood, and the making of a home. This would prevent a large proportion of the divorces. A standard should be set in regard to the home, and boys and girls should get that as part of their education.

Ignorance of hygiene is responsible for the drawbacks and failures of many homes. It is inexcusable that any boy or girl should be permitted to reach manhood and womanhood without a clear knowledge of personal hygiene, sanitation, and food values. This knowledge is essential to good home-making and good parenthood, and is equally necessary for men and women.

Congestion in cities should not be permitted. In the seaport cities many immigrants from other lands have not the means to go farther, and if they had the means, do not know enough about the country to place themselves where their qualifications would fit best. The cry against immigration is one with which I can not sympathize. The Americanizing of the immigrant should be placed in other hands than the politician's, who uses him en masse for a manipulated vote.

The special education of immigrant men and women would be an important service to good home-making and the ability to train the children to be useful citizens.

The proper distribution of immigrants by careful information as to opportunities for work and the earning of a home is greatly needed. The proper assimilation of our immigrant population is still in its infancy, but is of vital moment, for they also are the future citizens of America.

The city home of the American citizen should not be left to the will of builders whose only thought is to build houses for sale. Many apartments are built today without the amount of light, sun or ventilation necessary to health. Some cities and towns are realizing the need for regulating this.

The Conservation of the home demands that every State should have requirements as to building homes. The problem of a comfortable home for the family with a moderate income is a serious one today. Few cities or towns are giving the thought necessary to make a city of good homes for the average family at prices possible for them to pay.

The country home is equally in need of study and help. The opportunities for social life and educational advantages equal to those given to the city home should be supplied. That means larger appropriations for schools, the employment of the best teachers, the consolidated school, the use of the schools and churches as centers of educational and social life, the making of good roads between home and school and church and market place.

The Government Department of the Home should take all these things into consideration. It should bring to the overworked farmer's wife better household facilities and more help. The greatest drawback to country life today is the overworked wife, who can not get needed help and who goes beyond her strength in cooking and doing housework for farm help as well as her own family.

No one who knows of the terrible results of hook worm in the South resulting from the unsanitary, poverty-stricken hovels, where physical weakness had for years sapped the vitality and energy of men, women and children, can gainsay the fact that Government study of the causes and the remedy has done a service of inestimable value to thousands of homes. Seven years' life among those people proved that many of them were in quality equal to the best American stock, but that disease had brought upon them the unjust stigma of laziness with resulting poverty.

The Government could study and publish the results of its investigation, but Dr. Stiles had to get contributions from individuals to do the educational and medical work necessary to uproot this disease. That is not as it should be. The power to help should go with the power to investigate, for the condition was of much wider interest than to the individuals directly affected.

The National Congress of Mothers and Parent-Teacher Associations have made the subject of the home, of parental education and of child welfare its special study and work for seventeen years. It has worked steadily to build up a united system of parent-teacher associations in connection with every school, to bring about the co-operation of home and school in child nurture.

It has required that these associations should be for child study so that parents might have guidance and help in their problems. It has instituted study courses and provided educational material for the parents. It has headquarters in Washington and has valuable co-operation from Government departments. It should be the Homes Department of the National Conservation Congress because its work is well established, covering every State and reaching to other Nations. It is the only national organization whose membership is composed of parents and teachers and whose educational leaders include the greatest specialists in child nurture and child welfare in home, school, church and state.

I would suggest to the National Conservation Congress that it make the National Congress of Mothers and Parent-Teacher Associations the Homes Department, because in that way it will have consecutive work of high standard, and will bring a strength which could be secured in no other way. Co-operation without duplication brings results.

The National Congress of Mothers offers its co-operation in every phase of conservation for which the Conservation Congress was organized. It also asks co-operation of the Conservation Congress in its international work for home, parenthood and child nurture.

It invites this Congress to be always represented at its annual conferences and at the Third International Congress on Child Welfare in Washington, D. C., in May, 1914.

Life, health, character, all depend on the home and its efficiency. To equip every home for efficiency in its special work is the greatest need in Conservation.

President WHITE—That is surely a fine paper, in a holy cause.

The topic of the next section of the program is "Conservation of Human Life." The subject, "Saving Miners' Lives," will be discussed by Dr. Joseph A. Holmes, of Washington, D. C., Director of the National Bureau of Mines. (Applause.)

Dr. HOLMES—Mr. President, Ladies and Gentlemen: Those of you who have endeavored, even in part, during the past month to attend the Congresses in session in the United States, have found the time all too short to make that possible for you to do. These Congresses have covered all subjects. There is a feeling of unrest, a feeling that we have not done in the past the things which we ought to have done, and that it is high time we were trying to find out what are the best things to do. For some two hundred years in the development of this country we have allowed the individual very largely to take care of himself. We started out with the government theory that each man is entitled to "life, liberty and the pursuit of happiness," and the individual was left to himself to accomplish the purpose he had in view. The country developed rapidly through that system, and we built up a great Nation, but we have in the meantime neglected the public welfare. We are in a state of unrest today in regard to the future, feeling that it is time we were doing what we are trying to do—look after the question of the public welfare.

We have had, furthermore, a period of most rapid progress. When you, Mr. President, and others of us here, go back today to the schools where we went years ago, we hardly know the place. Buildings have changed, new ones have been built, and the teachers of today are different from those we were accustomed to. We see the great system of transportation built by the railroad men of today; we see the means of communication, some fifteen million miles of telegraph and telephone lines, enabling the people to talk with one another.

Yet even that is not speedy enough, so we are using the wireless. And so along all lines of industry, we have developed at a tremendous rate. But it has been a one-sided development, and now we have come to look particularly at the other side—the public welfare, and we are trying to find out what is best, from the experience of all countries, so that the American people may do the best that can be done for the welfare of this country.

It has therefore come about, in connection with this one-sided development, that we have lost sight of that great subject which we have for consideration today—the conservation of human life. We have been too busy to think about it. We have jumped on and off street cars and railway trains; we have slipped on our waxed floors; we have met with all sorts of serious accidents in our fast automobiles and flying machines; yet we ask, "Why are these miners so careless as to kill themselves, and these railroad employes?" And we are just as bad as they are. So let us not talk about careless railway employes, or careless miners, but stop to think what we can do to help the entire situation. Let us ask ourselves the question, "Am I my brother's keeper?" and there is but one answer, and that is in the affirmative.

I want to call attention to the fact that we have in this country two great foundations of industry. We have always considered agriculture as the great foundation industry of the country, but we have another—mining. The savage did not need any mines. He only wants a limited amount of material for clothing, and a large amount of material in the way of something to eat. He does not need the great modern appliances which we have today. When we drew up the Constitution of this country we did not think the mining industry of much importance, and it was not possible to anticipate the great complex social fabric which we have in the United States today. A man said to me the other day that he thought if Thomas Jefferson could see the things that we are asking of this great Federal Government, he would not know what to do. My own judgment is that he would advocate the doing by this Federal Government of all the things that this great American people demand that it do today. (Applause.) The trouble is that while we were making this tremendous progress, all of the people were not keeping pace, and perhaps it is well that this is true, because if there was nobody to hold back we would not only progress too rapidly, but progress in too many one-sided ways.

We recognize, furthermore, that while agriculture has made tremendous strides, and in large measure because of the investigations conducted under the Federal Government, other branches of industry have made rapid strides, but they have been forced to one-sided development in order to keep pace. It needs, then, the great co-operating influence of some great force like the Federal Government to help keep the industries from becoming one-sided.

The mining industry touches us on every hand, and today in a great hall like this, where you can find materials from every part of the world, you will find they came from the mine, or were manufactured through the agency of the products of the mine. We can not do anything on a large scale today without the aid of this great mining industry. During last March, the English people awakened to a realization of that fact. They did not consider mining as one of the great fundamental industries, but the stopping of the coal mines for four weeks stopped all the industries of the British, and they came to the conclusion that the very life of the nation was in danger by the cessation of coal mining.

Mining and Conservation should be linked very closely together. Men realize the fact that with agriculture, the resources increase year by year. We increase the fertility of the soil by taking the nitrogen from the air, and from that we get the crops, so that the wealth of the country based on agriculture is easily predicated. The mining industry is just the reverse. We started in this country with greater mineral resources than we will ever have again. Furthermore, in agriculture we have the healthiest vocation known, while mining is the most dangerous industry in the world.

Now, this mining industry has increased so rapidly that we have not been able to take care of many of the difficulties that have arisen, nor do we have a realization of how rapid that increase has been. We have increased in forty years from less than a ton to every man, woman and child, to, in the last year, six tons. Forty years ago a pound of iron, as compared with thirteen today for every man, woman and child. And so it has been with the great industries—they are increasing so much more rapidly than the population that it is hard to tell what has become of this increase, and one of the questions is, can this increase continue? Some of our great statesmen in Washington who have been fighting this Conservation movement, say it can not continue. The fact that the mining industry has nearly doubled every ten years, they say can not continue. But no man today would say that this country will not continue to grow, and as it grows this great mining industry will increase also. We are just entering upon our development. We are just beginning to export the products of our mines, so when we ask the question whether this great Nation will continue to grow, and this industry will increase, there is but one answer, and that is in the affirmative. We ask another question—are these resources inexhaustible? And there is but one answer, and that is in the negative, because we are now beginning to see the end of some of these resources. Shall we curtail the development of an industry like this and not supply the needs of the people? Our politicians ask this and expect us to answer in the affirmative, but no conservationist answers it that way. We say, no, the needs increase and we must meet the needs.

What can we do to perpetuate the welfare of the country? There are but two things we can do, and they are fairly easy to do. Use more and more efficiently all the resources, and prevent unnecessary waste. Now, in connection with this wasteful use of our resources, you say, after all, is there any great waste? What can we do to stop it? Only a few years ago the State of Indiana thought its natural gas was gone, so it passed laws forbidding the waste of natural gas; the Supreme Court of the United States confirmed such an act in regard to coal—after the coal was gone. One of the Supreme Judges said that a man who owns a coal mine had a legal right to destroy it if he wished to. But in the State of New York one of the associate justices overruled the Supreme Justice, and in every case the Supreme Court of the United States, as well as the Supreme Court of the several States, have shown a desire to keep pace with the progress of this country in interpreting the Constitution of the United States for the permanent future welfare of the people of this country. (Applause.) There are a good many signs of improvement, not only in what the Federal and State Governments are doing, but in what private individuals are doing.

Only yesterday, I went through the great plant at Gary, and I found the United States Steel Corporation was using two million horse-power developed from gases from its own operations, which only a few years ago was allowed to go to waste, and that power is not only operating all the machinery of that company, but is supplying the power for other industries in the immediate vicinity. I found that the slag coming from the furnaces, which in many great manufacturing sections of the country we see piled up in great, unsightly masses, is all being converted into cement, and that cement is being used by the people of this country. And so we find an interesting situation—that the steel being manufactured by that plant is likely soon to be a by-product, and not the main product for which the plant is operated.

And so it is when we watch the great industries of this country. Under this great spirit of Conservation individuals are meeting the Federal Government and State more than half way, and they are finding what is the greatest basis of permanent success—that it pays to conserve our resources. And when that great company does any mining for ore in the lake country, instead of burying the materials which they cannot use today, they are laying that material to one side, so that just as soon as it becomes useful it will be immediately available for preparation for that purpose.

Out of five hundred million tons of coal mined last year, we wasted, by leaving it underground, no less than two hundred million tons. Meanwhile, if we could have exported that coal to Central or South America and brought back from these countries raw materials which we could use in manufacture, it would be something worth doing; but to waste it entirely is nothing more than a discredit to this nation. But what are we going to do about it? The coal operator cannot change the situation, because he is doing the best he can at the price he gets; the miner cannot change the situation, because he is doing the best he can at the price he is paid. It is not simply a question for chemists and engineers—it is a problem for statesmen, and the statesman is the man who must remedy the economic conditions.

To come to the main subject of the Conservation of life, the greatest loss of life we have in mines is in the coal mining industry. I want to say in connection with this, that a careful study of the situation for the past several years has led me to believe that the coal operator in the United States is just as humane and just as anxious to conserve the life of his men as the coal operator in any other known country. (Applause.) Furthermore, that while it is true that of the miners, less than half read the English language and 75 per cent. are non-English speaking and know little or nothing about the laws regulating the principles and purposes of a great country like this, yet they are no more careless in mining because of that fact than are the miners from England and Wales who come here after long experience in mining and knowing perfectly our language and customs. These men are up against a condition that they cannot remedy, and while I do not say that they are doing the best they can under the circumstances, I think they are more and more coming to do the best they can, and I believe we will have more and more effort on the part of both miners and operators to do what is right. We have developed so rapidly in the past hundred years that we have not stopped to think of human life, and we cannot expect these reforms to take place without any effort on our part. There is recognition on the part of both miners and operators, that I am my brother’s keeper, and it is a most encouraging sign.

There are these two great reforms in connection with the mines of this country—safeguarding the lives of miners and improvement of conditions under which they labor, and the stopping of waste of our essential resources. The Federal Government is trying to get at the actual information, they are trying to conduct investigations in an impartial manner, and they want to bring about a condition acceptable to both miner and operator. We have suspicion on the part of the operator of the miner; and suspicion on the part of the miner of the operator; and suspicion on the part of other parties in reference to both. What we want to do is to have a condition in this country so that the miner and operator, co-operating with each other, can work together and bring about these great reforms that are needed.

This general welfare clause of the Constitution, which was regarded as an agreement with the devil, is today our great saving clause for getting things done by the Federal Government. The Federal Government, Mr. President, has waked up long ago to what it ought to do for agriculture, and in the next few years it will conduct investigations far more extensive than today—it will submit remedies brought together from the experience of all mining countries of the world, and it will lead in this great movement for a general improvement of conditions. But after all, what may be done by the Federal Government will depend upon what is done by the Federal Congress. There is where we must do our work, to make them appreciate the difficulties of a great industry like this, and the correctness of this clause.

I want to say a word in behalf of these miners. As I said before, more than half of them cannot read the English language. Under the rules and regulations we have permitted these men to come into the United States, and when they come it is interesting to see how they appreciate becoming an American citizen. I talked to a Lithuanian who had only been in this country a few months, and I said, “Are you not very lonely?” and he said, “Yes, but I am an American.” (Applause.)

These men are here, and we have done mighty little for them. We cannot wonder that they segregate in their rooms at night, after working in the mines all day, and read Socialistic literature which comes from their country. We do mighty little to encourage them to learn the English language; we do mighty little to encourage them to enter into the spirit of true America; we have neglected them all too long—and then we complain that they are not American citizens. I appeal to you as citizens of the United States and of the State of Indiana, to see that everything that is possible is done to make good citizens of these men. Get legislation under which they can work, and the safety problem will take care of itself. (Continued applause.)

President White—The next subject for consideration is “The Prevention of Railroad Accidents,” by Mr. Thomas H. Johnson, consulting engineer of the Pennsylvania Lines, West. I take pleasure in introducing Mr. Johnson. (Applause.)

Mr. Johnson—In approaching this subject it will be well to get our viewpoint adjusted to a true perspective and just proportion. Accidents on railways which result in death or injury to persons, are all reported to State and National officials, and when the statistics for the year are compiled and published the total figures are startling, and suggest that the transportation business of the country is conducted at a fearful sacrifice of life and limb. It should be remembered, however, that in no other line of the Nation’s activities are similar complete statistics available.

The only data at hand to show the relation between the numbers killed and injured on railways, and those occurring in other lines of action, are found in a pamphlet issued by the city of Chicago, entitled “Report of the General Superintendent of Police,” from which the following table is taken:

CHICAGO, ILLINOIS.

Accidents Reported by the Police Department, Year 1911.

	Fatal.	Non-fatal.	Total.
Steam Railway Accidents	187	554	741
Street and Elevated Railway Accidents	106	3,646	3,752
Accidents Caused by Teams and Vehicles	135	2,812	2,947
Accidents Caused by Falling from Windows, Scaffolds, Porches, etc.	149	2,680	2,829
Bitten by Dogs	4	1,281	1,285
Injuries by Personal Violence	177	2,729	2,906
Overcome by Gas, Smoke or Heat	189	653	842
Scalded or Burned	81	216	297
Various Other Causes	193	1,945	2,138
Total	1,221	16,516	17,737

From this it will be noted that of 1,221 fatal accidents in 1911, only 187 occurred on steam railways, and of 16,516 nonfatal, only 554 are charged against them. Comparing the total accidents, it will be seen that five times as many persons were killed and injured on street railways as on steam railways, four times as many by teams and vehicles, four times as many by falls from windows, scaffolds, etc., and again four times as many by personal violence. Even the dogs come in for having done 73 per cent. more damage than the steam railways. Taking the last two items together, it appears that Chicago’s vicious dogs and more vicious men are nearly six times as destructive of life and limb as are the railways.

While the foregoing figures are for the city of Chicago only, they are indicative of the fact that throughout the country the number of accidents on railways is a mere fraction of those occurring elsewhere, and this fact has been recognized by the accident insurance companies when they issue policies calling for double compensation if the accident occurs while traveling in steam or trolley cars.

If the grand total of accidents on railways appears so startling when presented in concrete figures, what would it be if equally complete figures could be had for the other types of accidents classified in the Chicago report?

And now having cleared the atmosphere in that respect, we will proceed to consider the railway accidents on their own merits.

The Interstate Commerce Commission issues a series of quarterly bulletins of railway accidents. They also issue an annual report of general railway statistics, in which a summary of statistics of railway accidents was included prior to 1910, but which has since been admitted as an unnecessary duplication. The statistics of the annual report have been compiled on a somewhat different basis from those of the bulletins, and the two sets of figures cannot always be reconciled. In compiling the following tables the annual reports prior to 1910 have been followed as being the final word of the Commission.

It should be noted that the statistics of railway accidents are divided into two general classes:

First. Accidents due to the movement of trains, engines or cars, which may properly be called “transportation accidents.”

Second. Accidents not connected with train or car movements, such as happen to shopmen, warehousemen, trackmen handling material, etc., such as are equally occurring in other industries, and which are more properly classed as “industrial accidents.”

This discussion will be chiefly devoted to the first class, as being distinctively “railway accidents.”

PROGRESS IN THE PAST.

The loss of life from railway accidents began with the day of the opening of the first railway in England, in September, 1830, on which occasion a prominent citizen, a member of Parliament, was knocked down and fatally injured, sending a thrill of horror not only through the great throng of spectators, but also throughout the civilized world. That unfortunate accident was not due to any defect in track or equipment, nor to any fault in the operation of the train. It was due to the victim's failure to appreciate the danger attending the then new and novel mode of transportation, and inadvertently putting himself in a position of danger. It was the forerunner and prototype of many thousands of others which have since occurred through carelessness and sheer recklessness of the victims, and which the railway companies are powerless to prevent.

But as railways multiplied other accidents occurred, which were due to defects of one kind or another in track and equipment, or to inadequate rules governing train movements, and the duties of the several employes. Each accident has been carefully studied as to its cause, and, so far as possible, remedies have been applied. Thus the immense system of transportation as it exists today has been a gradual development from crude beginnings. The light iron rails inadequately secured at the joints have been replaced with heavy steel rails with effective joint fastenings. Train movements have been safeguarded by a well-digested system of rules, uniform on all railroads; by standard forms of train orders with all ambiguities of language eliminated, and by block signals, interlocking and automatic couplers, air brakes and other safety devices. Stoves and oil lamps, with their menace of fire, have given way to steam heating and electric lighting. The inflammable wooden cars are being replaced with steel equipment. In fact, there has been a steady progress from the beginning in the effort to reduce the danger to life and limb.

But accidents continue to happen, partly because the rapid growth of traffic and the demand for greater speed are creating new conditions, partly because materials have hidden defects and the human machine is not infallible, and partly because discipline has been largely subverted through the attitude of the brotherhoods of employes.

In order to show in a general way what has been accomplished, the average figures for the five-year period from 1889 to 1893, inclusive, have been compared with the corresponding figures for the years 1907 to 1911, inclusive, with the following results:

Ratio of passengers carried to one killed has increased 35.5 per cent.

Ratio of employes to one killed has increased 54.7 per cent.

This shows a very decided gain in the twenty-two years covered by the record.

The number injured cannot be compared in the same way, for the reason that in the later years the reports include large numbers of minor injuries of a more or less trivial nature, which were not included in the earlier reports, but which the Interstate Commerce Commission now requires to be reported, thus swelling the number injured out of all proportion to the earlier reports. Under the present rules, if a passenger lets a window sash bruise his finger, and it is brought to the attention of any of the train crew, it must be reported, and enters into the final statistics with as much weight as the loss of an arm or a leg.

CAUSE AND PREVENTION.

In the Accident Bulletin for June, 1910, pages 10 and 11, there are given detailed statistics of twenty-six "prominent train accidents" with the causes of each. They embrace thirteen collisions and thirteen derailments, resulting in sixty-two killed, 306 injured, and a property loss of $261,584. The causes assigned may be grouped under fifteen heads, as follows:

Excessive speed, 5; ran by meeting point, 2; failed to flag, 5; disobeying orders, 1; misunderstanding orders, 1; failure to receive orders, 1; conflicting orders, 1; signal light out and engineman failed to stop, 1; broken rail, 2; explosion of boiler, 1; spreading of rails, 1; washout, 1; trestle failed, 1; insufficient ballast, 1; defective temporary junction of new and old rails, 1. Total, 26.

These fifteen assigned causes may be summarized thus:

Failure of persons, 18; failure of boiler, 1; failure of track and structures, 7. Total, 26.

Of the seven failures of track and structures, the two cases of "broken rails" and one "washout" may be considered unavoidable. The remaining four cases in that group, viz., "spreading of rails," "trestle failed," "insufficient ballast" and "defective temporary junction of old and new rails" were preventable, and could have occurred only from neglect of those charged with their care and maintenance.

The one case of "explosion of boiler" may have been due to defective material, or to negligence of the engineman.

We find, therefore, that in this group of accidents, twenty-two were preventable, three unavoidable and one doubtful.

Of the unavoidable, the “washout” may be dismissed as being beyond the control of human agencies, but the “broken rail” calls for further consideration.

Rail failures are generally due to chemical or physical defects, not entirely under control of the manufacturer, and not discoverable by inspection of the finished rails. Under the present practice the manufacture of rails is watched at the mill by the railway company’s inspectors. Specimens from each heat or melt are tested under a weight of 2,000 pounds falling fifteen feet to twenty feet. If the test piece breaks the steel is regarded as too brittle, and the rails from that heat are rejected. If it does not break, but the deflection exceeds the prescribed limit, the steel is too soft, and those rails are accepted as seconds, to be used only in yards and side tracks. All test pieces which do not break under the foregoing drop test are then broken and examined for internal defects. If defects are found, further tests are made, and the heat rejected in whole or in part, on the extent of unsoundness disclosed.

But herein lies a difficulty. Internal defects can only be found by breaking the rail. A rail broken is past usefulness. Hence that form of inspection cannot be applied to every rail; and as we can only test a limited portion of each heat, some defective rails must inevitably be passed and get into track. Complete statistics of all rail failures on a large proportion of the railways of the United States have been collected by the American Railway Engineering Association for several years past. These reports have been collected and classified as to the several causes, the results being printed in the publications of the Association. They show that the rails which fail annually are less than one eighth of one per cent. of the rails laid. This indicates fairly successful inspection, and would be quite satisfactory were it not that a single failure may result in such horrible consequences.

Five years ago (1907) as the result of several conferences between a committee of the American Railway Association and the rail manufacturers, a systematic study of the subject was undertaken, with a view to ascertaining the cause, and if possible, the prevention of rail failures. This research work was placed in charge of the Rail Committee of the American Railway Engineering Association, who engaged the services of a competent expert, who devotes his whole time to the work, furnishing freely of their materials and facilities at the mills. The line of investigation includes studies of the effects of variations in composition; in time in the bath; in time in the ladle; in manner and rate of pouring; in size of ingot; in rate of reduction at each pass; in temperature of the metal when rolled; in the effect of different alloys, etc. The field of investigation is broad and complicated. Much progress has been made, but much remains to be done. It is hoped, however, that success will ultimately be reached, and the rail failures in service be reduced to the lowest possible minimum. Certainly the railway engineers and the manufacturers are making every effort to accomplish that result.

Of late the adoption of some form of automatic stop has been suggested, and more or less urgently advocated. But let us consider: Referring again to the list of causes of the twenty-six accidents, such a device would have been called into play only in one case, that of running by a signal when the light was out. It could have had no influence on any one of the other twenty-five cases. Furthermore, it has been the experience the world over that emergency devices, resting in "innocuous desuetude" for long intervals of time, usually fail to work when the emergency arises. It may be said that it should be some one's duty to see that the apparatus is kept in working order. Very true. But therein is a reversion to ultimate dependence on the human factor with its attendant weakness and frailties.

The foregoing list of accidents embrace only a few of the more prominent "collisions" and "derailments." But there are other forms of accident, as shown in the following statistical tables copied from the Interstate Commerce Commission Annual Report for 1909:

ACCIDENTS RESULTING FROM THE MOVEMENT OF TRAINS, LOCOMOTIVES, OR CARS.

Interstate Commerce Commission Annual Report, 1909.

KIND OF ACCIDENT.	Employes.							
	Trainmen.		Switch Tenders, Crossing Tenders and Watchmen.		Station Men.		Shopmen.	
	Killed.	Injured.	Killed.	Injured.	Killed.	Injured.	Killed.	Injured.
Coupling or uncoupling	137	2,271	4	35	2	2	1	17
Collision	205	1,973	1	10		1	1	23
Derailments	184	1,186		10		2	1	6
Parting of trains	7	233		2		1		

Locomotives or cars breaking down	9	159		2				6
Falling from trains, locomotives, or cars	295	4,433	1	56		30	2	65
Jumping on or off trains, locomotives or cars	84	4,135	6	64		24	4	59
Struck by trains, locomotives or cars	243	577	72	79	21	25	41	89
Overhead obstructions	47	775		6				4
Other causes	133	13,376	9	243	2	121	14	465
Total	1,344	29,118	93	507	25	206	64	734

KIND OF ACCIDENT.	Employes (Continued).							
	Trackmen.		Telegraph Employes.		Other		Total.	
	Killed.	Injured.	Killed.	Injured.	Killed.	Injured.	Killed.	Injured.
Coupling or uncoupling		7			11	50	155	2,382
Collisions	18	132		7	27	163	252	2,309
Derailments	13	64			10	117	208	1,385
Parting of trains	1	2			1	12	9	250
Locomotives or cars breaking down	2	10			1	1	12	178
Falling from trains, locomotives, or cars	13	159		7	36	234	347	4,983
Jumping on or off trains, locomotives or cars	16	130		13	22	261	132	4,686
Struck by trains, locomotives or cars	353	412	8	12	187	345	925	1,539
Overhead obstructions		4			5	20	52	809
Other causes	25	882		34	83	1,340	266	16,461
Total	441	1,802	8	73	383	2,542	2,358	34,982

KIND OF ACCIDENT.	Passengers.		Other Persons.					
			Trespassing.		Not Trespassing.		Total.	
	Killed.	Injured.	Killed.	Injured.	Killed.	Injured.	Killed.	Injured.
Collisions	69	2,379	13	49	25	447	38	496
Derailments	17	2,426	32	69	6	287	38	356
Parting of trains		47	3	3		13	3	16

Locomotives or cars breaking down		2		1	1	4	1	5
Falling from trains, locomotives or cars	37	425	413	732	13	72	426	804
Jumping on or off trains locomotives or cars	81	1,503	445	1,688	11	120	456	1,808
Struck by trains, locomotives or cars;								
At highway crossings	2	3	112	211	621	1,619	733	1,830
At stations	30	67	365	334	66	183	431	517
At other points along track	1	12	3,371	2,037	79	143	3,450	2,180
Other causes	12	2,715	190	635	47	1,030	237	1,665
Total	249	9,579	4,944	5,759	869	3,918	5,813	9,677

Referring to the column of totals under the head of “Employes” you will note the large number of killed and injured in coupling or uncoupling cars; this in spite of the fact that all the equipment is fitted with automatic couplers, intended to prevent just those accidents.

The next two items, “Collisions” and “Derailments,” are also large, both as to employes, passengers and others, and we have already seen that in the former list eighteen out of twenty-six were due to “failure of persons.” Referring again to that list it will be further seen that sixteen of the eighteen were due to failure of the persons in charge of the trains, which justifies us in assuming that a similarly large proportion of these totals are due to like causes.

Please note also the large numbers, running through all these classes of persons, opposite the items “Falling from trains, locomotives or cars” and “Jumping on or off trains, locomotives or cars.” These may all be charged to the carelessness of the victims.

So, also, those “Struck by trains, locomotives or cars” nearly all of these are chargeable to the fault of the parties themselves.

“Other causes” are also prolific in casualties, but the data at hand does not disclose the extent to which they are chargeable to carelessness of victims or others, to preventable or to unavoidable causes.

Your attention is also directed to the very large numbers of killed and injured while “trespassing” on the railway property. Some of these belong to the great army of tramps infesting the country, but the largest part are people of the communities along the lines, who persist in using the tracks as a public thoroughfare. In most of the States there are laws on the statute books which are adequate to prevent this if duly enforced, but it seems impossible to get such enforcement. On the lines with which the writer is connected, efforts have been made in the past to break up this practice, but without success. Parties arrested by the railway company’s police and taken before the local magistrate have been released without punishment or only assessed a nominal sum to secure to the magistrate his fees. A rigid enforcement of these laws, and similar action as to jumping on or off locomotives and cars in motion (as is done in Europe) would eliminate approximately one-half the total killed and one-fourth the injured.

Here is a field in which the railways alone are helpless, but where much can be accomplished by legal enforcement, supported by strong popular approval. Without the latter, little aid can be expected from the average country justice or city magistrate.

INDUSTRIAL ACCIDENTS.

The first man to publicly call attention to the need of organized effort in this direction was Mr. R. C. Richards of the Chicago and Northwestern Railway, in his booklet “Railway Accidents, Their Cause and Prevention,” published in 1906. The seed thus sown has taken deep root, for at the present time nearly every leading railway in the country has an organized safety committee, whose duty it is to make regular periodic inspections to see that work places and tools are in safe condition; that yards, tracks, stations, buildings and grounds are clean and properly lighted; that shop machinery is protected by safeguards over gearing and other exposed moving parts, and that men are taking proper precautions for the protection of themselves and others. They report upon conditions which they feel can be improved, investigate accidents with a view to preventing repetition, and recommend improved methods of work to reduce risk of accident.

The Northwestern, after the first sixteen months, showed a decrease of 23.7 per cent. in deaths, and 29.8 per cent. in injuries, compared with the previous period of the same length. On the Pennsylvania Railroad the result of the first eleven months was a decrease of 63 per cent. in the combined number of deaths and serious injuries. These results are most gratifying, and demonstrate the usefulness of such close inspection and watchfulness.

IN CONCLUSION.

Accidents due to washouts, and to hidden defects in material are in the main unavoidable, though the former may sometimes be avoided by increased care and watchfulness during and after storms, and it is hoped that the latter may be materially reduced through the investigations now in progress in steel making.

Accidents due to imperfectly maintained track can be avoided by better maintenance, or by reducing speed to correspond to the conditions of the track. Speed and track conditions are inter-dependent factors.

Accidents due to jumping on or off trains in motion, and to trespassing, can be and should be eliminated by a rigid enforcement of existing laws, or the passage of new ones, if those on the statute books are found to be inadequate. As already stated, this would save one-half the annual deaths and a large proportion of the injured.

Substantially all of the casualties in coupling and uncoupling cars are due to carelessness of the men themselves, and the same may be said of most of those due to falling from or being struck by trains, locomotives or cars. It is difficult to suggest a remedy for this, or to formulate a course of procedure to reform the men in this respect. Recent and prospective legislation affecting the employers' liability will not be conducive to increased carefulness, but will rather tend to foster carelessness.

Train accidents due to error, negligence or incompetence should be corrected by proper discipline. But the administration of discipline is restrained and obstructed by the brotherhoods, whose officers claim the right to be present at all investigations, and the discipline ordered must meet their approval. They contest suspension and dismissals by appeals to higher officers who have no personal knowledge of the men, and use every means at their command, even to threatening a strike, to prevent the order from being carried out, often with success, all of which is subversive of discipline.

It is not a comforting thought that, when you, here assembled, disperse to your homes, some of you may place your lives in the hands of a man who is retained in the service through intimidation, rather than fitness and merit.

There can be no remedy for this while unprincipled demagogues and politicians, catering for votes, continue to appeal to class prejudice, and while the sympathies of the people, public officials and arbitrators seem to be arrayed against the railways.

President WHITE—We have now something else very interesting, and the next speaker will only keep you fifteen minutes. I now take pleasure in introducing Mr. A. B. Farquhar, of York, Pa., who will speak on “Vital Statistics and the Conservation of Human Life a National Concern.” He knows his subject; he knows it by experience; he has been through it; and he has met the classes, met the conditions he speaks of. He has a message to give you that is well worth hearing.

Mr. FARQUHAR—Mr. Chairman, Ladies and Gentlemen: Yes, I have been interested in this subject for perhaps seventy years of the seventy-five years of my life. I am very much interested in the work done in Pennsylvania. I shall refer to that, as is proper in a man sent as a delegate to this Congress by the Governor, and I believe it is a good example to other communities.

Vital statistics are usually assumed to cover only the number of births and deaths occurring in a given territory within a given time, a subject not attractive to the general reader, but this address will be devoted more particularly to the objects for which and the agencies by which such statistics are assembled, which is far more important and interesting, especially as it includes the social questions they resolve for us. “It is sometimes said figures rule the world; but this at least is true, that they show how it is ruled.” To this saying of a wise man may be added that they also show how it might be or should be ruled; they best illustrate “philosophy teaching by example,” because most precise and definite in form of presentation. They are of most use when applied to the most important interests of mankind, and have no higher function than in bearing their part in safeguarding the nation’s health.

For its vital statistics the Federal Census Bureau has always had to depend on data collected by local agencies, and of the imperfection of those agencies, and especially the large territory for which there were none—no attempt to keep the official record of births and deaths, it has loudly complained. Notwithstanding the commendable efforts that have been made throughout the country to supply deficiencies by State legislation, so much remains yet to be done that a census report, as late as 1907, showed less than half the population of the country, and only a third of the States in number, within the registration area. But the movement has been forward, and it is gratifying to note that the most significant step in advance was made by Pennsylvania, in a law creating a state department of health and fixing its duties in 1905. Until then that commonwealth was said to have “the poorest registration of any of the Eastern States,” though its first law for the purpose had been passed fifty-four years earlier; but “it had in 1906, the first year of the operation of the new law, an effective registration of births and deaths practically as complete as that of any registration State in the country, and far superior to the majority.” The best point about the law of 1905, and its most significant difference from that of 1851, is that it is executed. Obedience is no longer optional, but compulsory. Authority under it is centralized in the hands of the Governor, Attorney-General and Commissioner of Health, and practically for most purposes in those of the Commissioner.

The total appropriations for work under this department since 1905 have been $9,286,080. The number employed by its various divisions is 3,625. Of this number there are 1,170, nearly one-third of the total force, who are local registrars in the vital statistics service.

With what is so large a force to occupy itself? The 1,170 registrars receive all birth and death certificates and issue all burial permits (to which registration is a prerequisite), and the bureau has also charge of marriage certificates, filed with it by the clerks of county courts. The medical inspection division establish quarantine under direction of the county inspectors, see to placarding houses and disinfecting them after cases of communicable disease, guard against the sale of milk from premises where any such diseases are found, and represent the department in co-operation with local health boards. Supervision of the medical inspection of schools forms also an important part of the duty of these officers, some 300,000 children having been examined during the past school year. At the free tuberculosis dispensaries, with which the department has provided the large centers of population, the indigent receive free medical advice and necessary supplies. The commissioner has supervision, by the act creating the health department, of all systems of public water supply and of public or private sewage disposal. Detailed plans must be filed with the department, and no new construction can be done until the Governor, the Attorney-General and the Commissioner have approved the plans. The biological products division distributes, through 656 stations in all parts of the commonwealth, free antitoxin to the poor. The stations are located as impartially as practicable.

What has been accomplished by all this equipment, discharging all these functions, cannot be completely told; a few figures may be given, with a result here and there, and the rest left to estimate of probability. For example, the statement that 6,724 patients were admitted to the Mont Alto Sanatorium in the four years 1907–1911 certainly indicates the magnitude of the problem, and the importance of giving it the best attention we can. It is perhaps a little more significant that 58,004 patients have been treated in the department's tuberculosis dispensaries since they were organized. The activity of the sanitary engineering division is clearly shown in its recorded count that up to June, 1912, 40,447 private sources of stream pollution had been abated on notice from the department. One hundred and eleven modern sewage-disposal plants have been built or are in process of building, 306 municipal and private sewer systems are under construction in accordance with plans approved by the health department. Ninety-seven modern water filtration plants have been or soon will be constructed under State approval. It is worth while to connect with this fact another even more gratifying: the death rate from typhoid fever in Pennsylvania, which was, in 1906, 565 per million inhabitants, had fallen to 206 per million in 1911. As a final instance, the death rate from diphtheria, a little over 42 per cent. in untreated cases, has been reduced in the average of the 35,111 cases treated with antitoxin between 1905 and December, 1911, to 8.07 per cent., or less than one-fifth. Further, a certain district having been set apart for the trial of 5,000 units, instead of the usual 3,000, as an initial dose of diphtheria antitoxin, the death rate in that district has now shown a reduction to 4.22 per cent.

This story is not told for the mere satisfaction of praising our Keystone State or its faithful and capable public officers, though for that, too, it affords opportunity. Its function is to point a moral, to indicate a course of treatment of the subjects of vital statistics and public health, which, as Pennsylvania's experience leads me to believe, may well be applied to a wider field than Pennsylvania. It is not by accident that the association of statistics of births and deaths and marriages, with a State office for the promotion of public health, has come into favor at the same time in so many parts of the country. The force of example is something, to be sure, as is also the circumstance that a physician is usually at hand, when a birth or death occurs, that he is apt to know what there is to tell about the occurrence, that he is apt to know how to report, and that the State health office is one to which a physician might naturally address himself. But more important than these considerations is the value of birth and death records in the conservation of the people's health. From the greater or lesser number they show the favorable or adverse effects of accompanying conditions can be judged, and a conclusion reached as to how such conditions should be regulated. Nor could any condition be more important to regulate than those affecting health. The people’s health is its most precious asset. Dr. Wiley says he “would rather be a strong, vigorous man without a dollar than a sickly millionaire,” and thus indicates the pecuniary value of health to an individual. Multiplying that value by the number of the population, the amount becomes fairly appalling.

We have a department of agriculture expending vast sums—nearly fifty millions in the last decade—in improving the soil, improving the growth of vegetation, improving the health of animals, and no department to do anything to improve human health. We spend $700,000,000 a year for past and imagined future wars, and pay no attention to the 700,000 calculated above—a larger number dying every year, unnecessarily from disease, than bullets have slain since the continent was discovered. As we are reminded by Dr. Dixon, our Pennsylvania health commissioner, we are spending millions a year for the protection of our forests and water supply and other natural resources, but it is no credit to our intelligence that while guarding these material interests we allow man himself, without whom all else is worthless, to remain unguarded.

Yet it is a mistake to say that we do and have done nothing; what has been done is greatly to the credit of mankind, only it has not been enough. Jenner’s discovery and his application of it has left no excuse for smallpox anywhere. The president of the board of health in Mexico assured me that compulsory vaccination had freed his city of smallpox; and the Japanese health authorities, since their enforcement of compulsory vaccination, have ceased altogether to look upon the presence of smallpox as a source of danger. It is no longer a scourge in the Philippines and Cuba. Similar to the work of Lister in antiseptic surgery is that of Pasteur and Koch in various germ-diseases, of King and Carroll and Lazier in mosquito-transmission of infection. With the elimination of the Stegomyia mosquito, yellow fever is no longer dreaded; Havana and the gulf ports are as safe as anywhere; and the construction of the Panama Canal has become possible—as, but for the discoveries by Carroll and Lazier (or their rediscovery of Dr. King’s discovery) it never could have been.

From the brilliant successes attained in the directions just indicated, we seem to see that the most important thing for us is to know; we are to find our safety in knowledge. When we know that malaria is inoculated by the bite of the mosquito Anopheles, and yellow fever by the mosquito Stegomyia, that typhoid fever is fed to us, in a large proportion of cases, from the feet of the house-fly, that the fearful bubonic plague is inoculated by the bite of a flea infesting the rat, we have already traveled more than half way to deliverance. We can drive off the mosquito, or, by oiling the puddles, prevent her from hatching; we can "swat the fly," or abate the manure-heaps and other filth from which it draws its unblest being; and, if we can not catch the flea, we can make war upon its host, the rat. If, as is computed, within the last 2,000 years 2,000,000,000 people have fallen victims to the bubonic plague, it is enough to justify wholesale enlistments in a grand rat-hunt.

Half a century ago people were afraid of night air, and closed their windows at night. It is hard to guess how many lives might have been saved by opening those windows. We are told that the average duration of human life has doubled in the last 200 years. Whatever gain there has been is due, more than anything else, to more knowledge.

The case of pure air as against contaminated air is but one way of putting the general case of cleanness against foulness. Bad air has the same vices that attach to dirt in other forms; one of the uses of more knowledge is to be able to detect dirt in all forms, however concealed or disguised; and another is to discover the best means of sweeping it away. Our ancestors used to drink water from pools and wells that were sinks of organic filth, to worship in churches built over an array of corpses in all stages of putrefaction, to wear the same suit of leather clothes, day and night, till they fell apart or the wearer outgrew them—all because they knew no better. They had no conception of the disgust with which such habits were to be regarded by a more educated posterity. Now the golden rule of health is "Wash you—make you clean!" It is not enough to make, or even to keep, the children's faces clean; we must look no less to the cleanness of the lung passages, of the alimentary canal—yes, of mind and heart also.

Morally and esthetically, there is nothing in relation to which the duty to be clean is more stringent than the reproductive function. The source of the greatest work in all God's creation, the human race, ought more than all else to be pure; and the necessary condition of our endowing the earth in coming ages with a better human race than it now has, or has ever had, is that we provide that coming race with the best kind of parentage. The quality of the next generation is determined by the quality of this generation; it will be in most respects as we make it, clean if brought forth in purity, foul if engendered in foulness. And the truth so strikingly evident in the moral and esthetical view is even more clear in the view we are here taking, that of the race's health. To sexual impurity, by the testimony of the best physicians—the illustrious Dr.

Osler for instance—more physical degeneration is due than to any other one cause. Dr. Prince A. Morrow, president of the Society of Sanitary and Moral Prophylaxis, estimates the number constantly ill from syphilis in this country—although that number has of late been considerably reduced—as still no less than 2,000,000. The syphilitic poison is communicated by inoculation—a contagion that has no danger for us so long as held at a respectful distance; and the essential point in guarding against it is to preserve that distance. Like the venom of the rattlesnake it is best known in a knowledge of its lurking places. It was first recognized in Europe, some time in the fifteenth century; and it came from the Orient, not of its own initiative, but because Europeans went after it and fetched it. Similarly now, a man does not have it unless he goes after it. There is nothing in the whole range of human disorders that shows more emphatically than this, the feebleness and inadequacy of the best possible cure as compared with prevention. Knowledge seems all that is needed for complete prevention; any young man, having more than the resolution and self-control of an infant or an idiot, ought to require nothing more than an elementary acquaintance with a few facts that should be at the command of every instructor of youth, to insure his leaving the syphilis and gonorrhea factory permanently alone. If their baleful function were made clearly known to those who most need to know it, the entrance door to every such temple of moral and physical ruin would carry to the eyes the sign that greeted those of Dante: "All hope abandon, ye who enter here"—a prospect whose unrelieved blackness looks even darker when contrasted with the brilliant glory of the hope relinquished. It is a law of our human constitution that the richest, deepest, keenest joys that life has for us are those that come from the contrast of two sexes. Even when that contrast is hostile, there seems to be some pleasure in it; but immeasurably more when it is an incident of ardent attraction. Byron in one of his earlier poems thus puts it:

"Devotion wafts the mind above,
But Heaven itself descends in love;
A feeling from the Godhead caught,
To wean from self each sordid thought;
A ray of Him who formed the whole;
A glory circling 'round the soul!"

It is too well known that the poet's own loves, in after years, were not always of this ideal quality; but no one ever better set forth the exalted possibilities of the sex sentiment, to which the continuance of life on earth is due. But the worst, we are often reminded, is the corruption of the best, and it is another possibility of the same sentiment that it may urge a man to blast his whole future by incurring an incurable disease, and sadder yet—too often to involve others, tender and innocent lives, in his own condemnation. If more knowledge can ward off such a grisly fate, it is surely inhuman cruelty not to supply that knowledge, however disagreeable the duty may appear. When clearly seen as a duty it will be no longer disagreeable.

While making this call for more knowledge of vital truths primarily on account of the young men, since it is in the vast majority of cases the man who tempts, the man to whom the outcast woman owes her fall, it would be the wildest folly to stop with one-half of the rising generation. The future of the race is too dependent on its mothers to excuse or permit the neglect of any preparation of them for motherhood, which health in its fullest sense may demand.

Most of the great questions of health in its widest sense, of health as a public concern, resolve themselves into resisting the entrance of this or that species of bacterial germs into the body. The essential distinction between Mother Earth, that bringeth forth flowers and fruits, and grass for our herds, and dirt or filth, the especial opprobrium of the hygienist, is that the latter carries germs of bacteria. Cleanness, in the hygienic sense, is freedom from pathogenic germs; and when the doctors tell us that the marked improvement in health conditions recently observable in Germany and Switzerland, and pre-eminently in Sweden, is due to their exceptional attention to cleanliness, they use the term with particular reference to the provoking causes of preventable sickness. Not only is the death rate from the acute diseases in those lands rapidly falling off, but diseases of the chronic class are beginning to yield to the inculcation of better habits among the people.

We are by no means without instances in this country, of death rates reduced by preventive methods, as shown for young children in our largest cities after the introduction of pasteurized milk. Deaths have been thus spared for that peculiarly helpless class of sufferers, to the extent of fully 50 per cent. in some districts—in large measure through the well-directed activity of one public-spirited New York merchant. But we have much to do in other lines, and we have only begun to free ourselves of the typhoid fever incubus. As late as fourteen years ago there were 11,000 cases of that infection in the camp at Chattanooga, with 800 deaths. In the entire Spanish war the deaths of our soldiers from diseases, it was calculated, were thirteen times as many as from wounds in battle—the diseases mostly, like the Chattanooga typhoid, of the preventable kinds.

Loss of life by preventable accidents, on railways, in factories and mines, is too closely associated with that by diseases to be here omitted, though entitled to much fuller treatment than we can here afford. The deathroll from this cause is still disgracefully large in this country, far surpassing any country of Europe; but there are signs already of diminution. For instance, one steel factory, reporting 43 accidental deaths among 6,000 employes in 1906, showed only 12 fatalities in a payroll of 7,000 in 1909, safeguards having been introduced in the meantime. This instance is very good, so far as it goes, but we need to make much more progress in the same direction.

What we want is systematic effort, by some powerful consolidated agency, to promote the conservation of human life. We have no need to find fault with any of the organizations now engaged in furthering that end, several of which are doing good work. We may gratefully acknowledge the aid of the various medical societies, “regular” and “irregular”—though we take the liberty of wishing that they might fight the common enemy a little more and each other a little less. We may also welcome the assistance of the life insurance companies, notably the Equitable and the Metropolitan, whose managers clearly realize how their interests are involved. Whatever lengthens the average term of human life is a factor operating to increase their dividends and to reduce the cost of insurance to their policyholders. It is worth while to note, at this point, that the majority of life insurance officers are strong advocates of the formation of a national bureau or department of health.

Still more do we owe to the activities of State and municipal boards of health, which do more good because they have more power. Where properly supported they have done a great work, at obstructing the spread of epidemics by quarantines and other methods of isolation, at curing pollution of water supply, at instituting improved sewer systems, at bettering the general food supply by inspection of markets. You have just heard a condensed account of the activities of one of our best State health departments, that of Pennsylvania. You will infer from what that department has done in seven years what might be done by a national bureau or department, with powers and field of operation extending over the entire country.

The movement for a bureau or department of health, national in its scope, has been most actively advanced in Washington by Hon. Robert L. Owen, Senator from Oklahoma. His bills call for a department, and he gives strong reasons for the view that such an organization would, while that of a bureau would not, suffice for the national governmental activities in behalf of the public health. President Taft strongly urges a “Bureau of Public Health,” and plainly intimates a preference for the bureau plan. The “Committee of One Hundred on National Health,” formed by the Association for Advancement of Science, in 1906, with Prof. Irving Fisher as its president, originally contemplated a department whose head should be a member of the President’s cabinet, but it has in its recent publications adopted the alternative phrase “bureau or department,” which course is here followed, because there is manifestly nothing to gain by keeping up a contest on the point. The memorial prepared by the committee of one hundred proposes for a national department of health certain functions, as follows:

1. Administration—Including the national quarantine work, and whatever regulation of interstate commerce might affect human health, such as meat inspection and enforcing the food and drug act.

2. Co-operation—The work of assisting State, county and city health agencies, after some such fashion as the National Department of Agriculture co-operates with State agricultural colleges and institutions.

3. Research and Investigation—The work of obtaining needed scientific information concerning the cause and prevention of diseases that now shorten or impair human life; this would include a study of accidents, of poisonous manufacturing trades, of hygienic conditions in schools, etc., just as yellow fever was studied in Cuba, as the hookworm is now to be studied under private endowment, as the work of the Pasteur Institute was conducted under French government support.

4. Education—The work of supplying to the country scientifically established data on matters pertaining to health, such work as is done by the “publication division” in most of our governmental departments; thus rendering available for practical use the work of research and investigation. The countries in which is found the most rapid reduction of the death rate are just those (Sweden for example) in which the spread of a knowledge of hygiene is widest.

Of these functions the mere statement is a most powerful argument for the bureau or department suggested. It only remains to remove a few misunderstandings. One objection, for example, is powerful in many minds—that such a centralized office must necessarily be the organ of a particular medical school, and must so give that school—the one denominated “regular,” for example—an unfair advantage, unsuited to a government of liberty and equality. To this it may be frankly replied, that the primary objects of the new office being the four just stated (administration, co-operation, investigation, education), it would aim to collect and diffuse the greatest attainable amount of accurate knowledge on the subject of health; and that if it found a larger quantity of better knowledge in one school than in another, it would be false to its trust if it did not spread that knowledge accordingly. Personally, the writer finds it hard to believe that it could treat a school that taught the unreality of disease, or the surpassing value for all kinds of disorders, of drugs, of a narrow range of characteristics, on an exact equality with schools that deal with facts as they find them; but he heartily agrees that the citizen ought to enjoy the liberty of choosing his own medical advisers, so far as he does not endanger life or health by so choosing.

There are other objections to organized national work for health, many of them from a so-called National League for Medical Freedom, the most active workers in which have been shown to be interested in one or another kind of proprietary medicine, backed by some “mental healers,” and by associations of druggists who object to the “pure food and drugs act” of 1906. Several homœpathical State societies have repudiated that “league for freedom,” and have emphatically attested their approval of the proposed bureau or department of health; this, notwithstanding their well-understood grievances against “regular” practitioners. Some of the best informed among the osteopaths and the Christian Scientists are pronouncing similarly; and so, if the disavowals keep on, the League of Medical Freedom may soon be left with only those who seek freedom to dope their victims with drugs that enslave; stupefy them—infant and adult—with opium and thinly disguised alcohol, and generally to reverse the progress of a century. But, since it is estimated that $75,000,000 a year are expended by our fellow-citizens for patent medicines, it is easy enough to see how they must regard a national department which is to improve the sanitary conditions of the country, show people how to care for health, stop the sale of poisonous nostrums and impure foods, and end the career of opium under the name of “soothing syrup.” Their profits would be gone, and of course they disapprove and protest.

Altogether, the cause of a national bureau or department of health is commended, both by those who favor and those who oppose it. It could not ask better advocates than the distinguished men who heartily favor it, on the congressional or the collegiate stage; nor more suitable adversaries than those constituting the League for Medical Freedom.

President White—This is a most valuable paper, and it will be printed, together with the other papers and addresses of this convention. Every one should avail themselves of the opportunity to subscribe for this book, which costs one dollar.

I will now introduce to you the gentleman who kindly gave his hour to Mr. Farquhar. He is Mr. Reginald Pelham Bolton, of New York City, who will speak to you on "The Prevention of Elevator Accidents."

Mr. Bolton—The preservation of human life and the protection of our fellow-creatures from physical injury, claim prior consideration over conservation of mere materials.

Any form of danger which results in the destruction of life, and exhibits a tendency toward increased developments, invites our systematic investigation. Ameliorative measures, if undertaken in advance of the growth of an evil, are of double value. To one phase of the subject, of the conservation of life, I desire to direct your attention.

The increase of fatalities and injuries resulting from the extensive use of passenger elevators has become sufficiently marked to deserve careful attention by those who are concerned with the benefit of our fellow-citizens. Complete statistics as to the number of accidental occurrences in and about elevators of all classes throughout the country are not available, but an estimate based upon such official returns as relate to labor alone, indicate that the annual total is now probably in excess of seven thousand, of which probably three-fourths are of a preventable character.

From small beginnings, the roll of such accidents reported by the New York Department of Labor, which it is conceded do not cover all such occurrences, rose in 1909 to a total for five years of 1,600 injured persons, of whom 198 were killed and about 298 permanently disabled.

The Wainwright-Phillips Commission of the New York State Legislature reported in 1911 a list of injuries and deaths, in the three years 1907 to 1910, affecting 1,108 persons, of whom 106 were killed and 241 were more or less seriously and permanently crippled. In addition, no less than 200 persons fell down hoistways, of whom 43 were killed outright and 19 permanently injured.

These occurrences took place only on elevators in industrial establishments, and are only those which have been officially reported.

The Industrial Commission of the State of Wisconsin reported for the ten months, September, 1911, to June, 1912, thirty-nine accidents in and upon elevators, and fifteen more due to falls down elevator shafts; all occurring in establishments of various industries. Accidents occurring in transportation were 195, so that the relation of elevator accidents and falls was 28 per cent. of transportation.

That such accidents are duplicated outside the limits of observation of labor departments is indicated by an examination of the reports of the New York county coroners, which show about one hundred deaths annually from elevator accidents in the county of New York only. In the year 1911, in the Borough of Manhattan, there were reported sixty-eight fatalities in connection with elevators, about two hundred permanent injuries, and probably about three hundred more may be estimated as having sustained lesser injuries.

The fact that accidental occurrences in or about elevators are thus found to be deplorably numerous and increasing is not to be taken as a reflection upon the general security of elevator travel. Their number is relatively small in comparison with the vast number of persons utilizing these appliances. One express schedule elevator handles about 700,000 persons per annum. Further, by far the larger number of mishaps are not due to failure or fault of the elevator itself, but occur in and about the entrances of, or in the hoistways of such apparatus, from persons falling through unguarded openings into elevator shafts, and of course a number are due to the recklessness and incompetence of employes and operators.

It remains the fact, however, that a large part of these occurrences are unnecessary, just as was found to be the case with many of the forms of danger to life and injury to limb which attended the operation of freight and passenger trains prior to the adoption of certain of the safety appliances and methods which have been brought into general use on railroads, as a result of the concentration of public attention upon the subject, and legislative action based thereon. Similar attention and action with the compilation of statistics upon the subject will undoubtedly result in diminishing the number of fatal and injurious occurrences connected with elevator operation.

Some loss of human life and injury to the person may to some extent be regarded as an unfortunately inevitable accompaniment of all forms of motive apparatus, and the complex conditions of modern existence have not only increased this liability by demands for more rapid movement of all forms of mechanical transportation, but the vast increase in the usage of appliances has introduced new elements of danger.

In no class of transportation are the effects of haste and crowding more apparent and dangerous than in the modern means of vertical transportation, use of which is now made by all classes of people. Liability towards accidental occurrences in elevators, therefore, affects the whole public, and it is needless to dilate upon the general concern in, and economic loss resulting from deaths or injury of any member of the community. It may be conservatively estimated that the economic value of the mere services of persons killed in and about elevators, based upon life expectancy, and the loss of time of those injured, would annually exceed the cost of equipment of all passenger and freight elevators with modernized safety appliances.

There are some features connected with elevator accidents which call for consideration and rectification. These have grown up around the development of the appliance in a manner somewhat peculiar to it. The elevator is a transportation apparatus which is for the most part privately operated and owned. Unlike the railroad, it is not regarded by the law as the apparatus of a common carrier. Unlike the road carriage or car, it is not operated upon the public highways. Unlike the machinery of a factory, it is not utilized exclusively by employes.

Its development and use have been, perhaps, too restricted to require the attention of such legislation as has been rather freely applied to the other classes of appliances engaged in transporting human beings.

It has therefore come about that the legal status of the elevator is in a very indefinite condition, its public regulation is generally local and therefore at best erratic, and the liability for the security of its occupants is as varied as the legal practice and rulings of different States.

The results are unfortunate to all concerned except perhaps that part of the legal profession which concerns itself with the prosecution of claims for injuries. Only two States, Pennsylvania and Rhode Island, have adopted legislative provisions, of limited character, relating to elevators. The former State provided so long ago as 1895 a requirement for automatic locking devices on all passenger elevators, thus being the pioneer in this direction. The State of Rhode Island by its general law, Chapter 129, requires all elevators “to be equipped with safety appliances to prevent the starting of the elevator car in either direction while any door opening into the elevator is open."

The State of Wisconsin, by its Industrial Commission law, Chapter 485, of 1911, placed in the hands of that body general power to require safeguards "in all places of employment," but it does not appear that the powers of the act extend to every class of building in which elevators may or can be employed. Other efforts have been made to effect legislation in the same direction, but have so far failed of enactment.

A bill was introduced in the House of Representatives December 12, 1910, by Mr. W. Bennet, requiring all elevators in the District of Columbia to be provided with gate and car interlocking devices, which bill did not become law. A bill was introduced in 1911 into the Assembly of the State of New York amending the labor law in the direction recommended by the Wainwright-Phillips Commission, and empowering the Commissioner of Labor to require automatic door-locking and car interlocking on all passenger elevators in factories. Senate Bill 911 and Assembly Bill 329 of 1911 were designed to require in general terms the use of "such safety devices as will prevent accidents to persons getting on or off elevator cars and from falling through open doors into the elevator shafts."

The attention of the American Museum of Safety has been directed for some years towards the accomplishment of some amelioration of existing conditions, and that humane organization made a strong effort to arouse public interest in these measures and to secure their enactment, but without success.

The subject has received some sporadic attention by several public associations, including the National Civic Federation, the American Association for Labor Legislation and the New York Association for Labor Legislation, but without effective results.

With the foregoing exceptions, the obligations of an owner of a building, as regards the security of an elevating appliance, are practically limited to a compliance with the then existing local regulations to the purchase of a device commensurate with the existing state of the art, of a design made by a reputable concern, and to the employment of reasonable care in upkeep and operation.

No legal obligation appears to lie upon an owner to alter or modify the appliance in conformity with greater knowledge of the art, or to add to it greater means of security. Until some unfortunate occurrence has taken place, an owner of property naturally feels unwilling to embark on such expenditures. The present system of liability insurance rather tends to such a situation, as an owner has no inducement in the form of reduced premiums, to expend money upon desirable safeguards. If the liability corporations should concede a substantial reduction of premiums, in connection with appliances dealing with a certain proportion of the risks attending elevator operation, much could be accomplished without the aid of special legislation.

While the law-making powers do not hesitate to direct such measures to be taken with and upon the property of common carriers, they seem to regard the operation of a practically public conveyance within private property as a privileged possession and hesitate to enter the castle of the owner and involve him in enforced expenditures upon a privately operated appliance.

Yet an elevator, whether used for the purpose of the carriage of goods, of tenants, of employes, or of visitors to a building, is a common carrier earning a profit, even if indirectly, for it is as much a source of revenue as is the machinery of a factory around which many enforced safeguards have, by legislation, been thrown.

If, therefore, the owner of a building installs elevators for the convenient carriage of tenants and visitors within his property, he does so because the apparatus enhances the value of that property, and that enhancement is largely due to the public use of the appliance, in which use the unknowing users have some right to legislative protection from results of ignorance or incompetence, of neglect or parsimony.

It has taken a long time for this view of the matter to become even partially recognized, even in the city of New York, in which the use of elevators has multiplied beyond all conception of what seemed probable twenty-five years ago. The number of passenger elevators in the Borough of Manhattan alone, now exceeds nine thousand, and these increase annually by about five hundred new machines. The estimated number of freight elevators, none of which under present circumstances are subject to official inspection, is not less than ten thousand.

The regulations regarding elevators in Manhattan, commencing with feeble beginnings, have advanced under the careful direction of the present Superintendent of Buildings of Manhattan, Rudolph P. Miller, C. E., into the field of interference with private control, and the department is compiling further regulations which will go a long way towards the protection of the public in safeguarding the elevating apparatus they are compelled to use. The Manhattan regulations, while in themselves excellent, are directly applicable to passenger elevators only with such freight elevators as are within the same shaft enclosure as a passenger elevator. They require the operator to be of reliable and industrious habits, not less than eighteen years of age, with at least one month's experience in his duties.

A number of known elements of unsafe character are prohibited and some constructive features of value are insisted upon. No provision is, however, made for automatic interlocking of gates and car movement, nor are projections in the shaft prohibited. Some good, detailed regulations and suggestions have been issued by the Wisconsin Labor Commission, but these and other State and local regulations could be substantially increased in value, by a thorough technical investigation and settlement.

Some improvement of deficiencies in apparatus existing prior to these rules has been effected by requiring safeguards to be applied upon any alteration or large repair work being sanctioned. This course has brought about the addition of speed safety appliances in a number of old installations where this elementary security was absent.

Later regulations will, in similar manner, require carefully conducted tests of all machines whether new, altered or repaired. Many minor matters of security are or will be thus provided for, yet the limited powers of a bureau can but at best halt in dealing with the entire problem. And when the regulations of Manhattan are made, as they should be, the best possible, it is regrettable that in another city or even in another borough of the same city, the same desirable conditions will not apply.

Yet the security of an elevator requires the same measures of attention, in one State as in another, as much in the merest hamlet as in the great metropolis.

The use of elevators is now widespread through all States, and in all classes of buildings, affecting the convenience and security of all classes of persons; and calling for the establishment of well considered and equalized regulation in every part of the country.

It speaks volumes for the sense of responsibility of our leading manufacturers of elevators, that among all the tens of thousands of machines turned out by such concerns as the Otis Elevator Company and their competitors, accidents due to the physical breakage of the machinery of elevators should be in number only what they are, when they include the failures of machines built in days when the industry was small and the art far less understood than it is at present.

When we reflect upon the fact that the passengers carried in elevators in the city of New York far exceed in number those carried on all the surface and subway lines, we may the more appreciate the point to which I desire specially to direct your attention, namely, the desirability in the public interest of State regulation, and as far as possible, uniform regulation, of the security and operation of elevators. The local regulations may be left to care for details of installation but the State authority is necessary to require elevators to be not only modern but progressively modernized appliances; that no antiquated and essentially dangerous apparatus shall be continued in use, and that necessary safeguards and properly qualified operators shall accompany their operation.

The State may further require that in excessively tall buildings, where the elevators constitute the only practical means of egress in emergency, there shall be a proper sufficiency of such appliances capable of removing the occupants within a reasonably safe period of time.

The limitations of the carrying capacity of an elevator are now well understood, and the safety of operatives in high loft buildings and of tenants in loftier “tower” office buildings, demands that the parsimony of owners and the ignorance of architects should not be allowed to restrict the exit of occupants of such buildings. A second elevator, in the Triangle fire disaster, would not only have saved its capacity in human occupants, but would have averted the fatal overcrowding of the single car which rendered it practically of no avail.

Many loft buildings of twelve stories and some even exceeding twenty stories are in existence in which the elevator accommodation is utterly inadequate for the removal of occupants of upper floors in a reasonable time, in case of emergency. The effectiveness of exterior “fire escapes” and of crooked interior stairways, especially for great heights, is now known to be strictly within certain limitations, and elevators have on many occasions demonstrated their value in the saving of life in panic and fire.

Office buildings are constructed thirty and more stories in height, without fire escapes and with winding stairways which are useless in emergency, and with such limited elevator capacity as would not remove the tenants in less than thirty minutes.

A most important and desirable subject for general action is afforded by provisions for safeguarding elevator gates and doorways. In and about these orifices, as previously observed, a large proportion of unnecessary accidents and fatalities occur. The unlatched door, the open gate, the absence of inner gates, the projecting sill, and the slippery tread, are fruitful causes of deplorable injuries and have caused the unnecessary loss of many precious lives. The proportion which this class of occurrence bears to the total is evidently large. An analysis of a list of four thousand accidental occurrences shows the following proportions:

	Per cent.
Getting on or off cars	58
Falling through unguarded openings	20
Fractures and fall of cars, only	17
Mechanics making repairs in shafts, etc	4
Unexplained	1

A number of devices have been developed during recent years, which have overcome objections to their use in the past, whereby the gates of elevators must be securely locked and fastened before the car can be moved. Six of such devices are approved for use in the State of Pennsylvania. It would seem that so simple a feature eliminating the essential danger surrounding the operation of a car moving vertically between floors in a shaft would long ago have been demanded by every form of authority.

With other engineers, I was at one time opposed to the use of such appliances on the ground of their uncertainty. But the growing volume of fatalities directly attributable to the lack of such safeguards, together with radical improvement in their construction, now demand the opposite conclusion.

There has been particular objection in some large cities to the application of devices for locking the gates, on the ground that the speed of operation on rapid schedule service would be retarded and inconvenience and overcrowding would result. In order to satisfy myself upon this point, I made this year a series of comparative trials of elevators equipped with one such appliance, the Clarke automatic safety devices, and found that no such loss of time in service actually resulted. On the contrary, a trial of the elevators in the Atlantic Mutual Insurance Company’s Building, 49 Wall street, New York City, and in the Hotel Imperial, showed that the operators made better time with the device in service, as they were compelled to make more exact landings and thus avoided much of the time frequently wasted in reversals of the car movement.

Under the present circumstances, therefore, it seems that the proper time has arrived for action in this respect, and that the example set by the States of Pennsylvania and Rhode Island may be embodied in careful legislative requirements in other States, which would, at some expense, it is true, to private owners, safeguard the public from those peculiarly present dangers which have taken such unnecessary toll of human life and limb, in the ghastly entanglement between the gate or doorway and the moving car, or the dreadful fall through the opened gate.

It would be very desirable, if, in the investigation of this subject, and the preparation of legislation to deal with it, competent technical and legal ability were employed, as the subject is of a technical character. Some of the legislation already in existence has been worded in so ill-considered a manner, as to give the impression that it was phrased in order to prevent the recovery of damages by injured persons.

The expression of your interest in this matter will tend to strengthen the hands of those who are seeking at present, by the limited means available, to enforce good methods of installation, proper safeguards and proper operation. It will also aid our great manufacturers, who lead the world in the design and construction of these truly American appliances, in securing the proper surroundings and proper care they are constantly urging for the appliances they construct, and will aid humanity by averting some unnecessary wastage of the health and lives of our fellow creatures.

Following Mr. Bolton’s paper he presented the following resolution:

> Whereas, The number of accidents and fatalities attending the operation of elevators is increasing, many of which are of a preventable character;
>
> Resolved, That the National Conservation Congress recommends to the Legislatures of all States an official investigation of this subject, and the enactment of such provisions as have been adopted by the States of Pennsylvania and Rhode Island.

President WHITE—This is important, and if there is no objection it will be handed to the Resolutions Committee.

Mr. FREDERICK KELSEY (Orange, N. J.)—I would like to offer this resolution:

> Whereas, Under the laws of the District of Columbia and some of the States, fictitious and fraudulent overcapitalization of corporations is permitted; and
>
> Whereas, Under the operation of these promoter-made laws enormous and widespread losses to innocent persons all over the country and throughout the civilized world have resulted;
>
> Resolved, That this Congress earnestly favors the amendment of these laws and calls upon the President and the United States Congress to enact such legislation affecting the incorporation and control of corporations as will bring the creation and conduct of these creatures of the State back to the moorings of common honesty.

I would like to say that, like most of the previous speakers, I have given this subject very careful attention. I was chairman of a committee, a civic and economic committee of our State, which committee spent eight months in considering this subject, and I want to say that you cannot appreciate the widespread loss, the injury, the injustice of improper concentration of wealth that has been the direct outgrowth of these laws in our own State and other States of the Union.

President WHITE—The resolution will be referred to the Resolutions Committee.

The Congress now stands adjourned until 2:30 o'clock this afternoon.

EIGHTH SESSION.

The Congress assembled at the Coliseum, at the State Fair Grounds, Indianapolis, on the afternoon of October 3, 1912, and was called to order by President White.

President White—This Conservation Congress was to have been addressed today by the Governors of two of the States. I am very sorry to announce that Governor Hadley, of Missouri, is unable to attend.

This Congress is greatly honored today. The city of Indianapolis is greatly honored today. The State of Indiana is greatly honored, and I personally am greatly honored. I feel honored in having the privilege of presiding over a meeting at which our distinguished guest is to speak.

He who causes two blades of grass to grow where only one grew before is a public benefactor. He who with one talent helps one child, one boy, to rise to manhood and usefulness, is a great and useful citizen. He who is fortunate enough to possess ten talents and who is an inspiration to thousands of the youth of the land, who has planted in their minds and in their hopes the desire to become great and useful in this world, to become great and good, efficient citizens—he is the greatest of all.

He is the Governor of a great State, and has inspired the citizens of mature age to a better government for the people and led them on to a greater field of usefulness. We feel perfectly safe in trusting him. To whatever position duty may call, whatever fortune may trust him with, the people will be safe under his guidance. (Applause.)

I feel unworthy to present to this audience one who has been the leader in so many good works, one who has been a practical conservator of human effort, but I take pleasure in introducing to you as the speaker of this day one who has come here to get closer in touch with the Conservationists of the United States, to gather from this audience an inspiration as to the great force of Conservation which is to lead the world—the Hon. Woodrow Wilson, Governor of New Jersey. (Great applause.)

ADDRESS BY THE HON. WOODROW WILSON, GOVERNOR OF NEW JERSEY.

Mr. Chairman and Fellow-Citizens: It is with genuine pleasure that I find myself in this place, facing a company of men and women who are devoting themselves to so disinterested a cause as that to which this Congress is consecrated.

Your chairman has stated in exactly the terms of my own thought, the errand upon which I have come. It would seem presumption upon my part to instruct this Congress, or to attempt to instruct it in the means of Conservation. I have come here, as he has said, to share in the inspiration of the occasion, to gather into my own thought an impression of the men and women who are working for these great objects in the United States. When I was on my way out here, and was thinking of this occasion, I prepared my talk on the conservation of our natural resources. When I arrived at the station, I was told to change the subject, that was not what the Congress was, this year, devoting its particular attention to, but to the conservation of the vital energy of the people of the United States. I had thought that I would have to apologize to you for wandering off before I had finished my address, into that very topic, because it seems to me that the more broadly we view the field of obligation, the more clearly it will appear to us that our duty is only done in respect to the laying of the foundation, when we have conserved the natural resources of America, for those natural resources are of no consequence unless there is a free and virile people to use them.

We are in the midst of a political campaign, and most of the audiences that I have faced have been political audiences. I want to say very frankly to you, that it is a comfort to me to face another kind, because, in a campaign, we take politics, as it were, to the people, but on this occasion the people of the United States are bringing to us the great forces of their thought.

A congress like this means something more vital, in some aspects, than any of the ordered efforts of political parties; for here are represented the men and women from every quarter of the Union, come together to speak that great volunteer voice of America, which is the atmosphere of politics, which creates the environment of the public man, which is the independent conscience of a great people asserting itself and instructing those who serve it, what their lines of best service are.

All voluntary effort distinguishes a free people from a people that is not free. An effort, an organization, that comes about whether the politician wants it or not, is the kind of effort and organization which shows that the people are ready to govern themselves and to assert their own opinions, whether the men in the public eye now consent to be their servants or not. (Applause.)

I have often made this boast about America, that, truly as we love our own institutions, proud as we are of the political history of America, if you could imagine yourself absolutely forgetting the documents upon which our constitutional history rests, over night, in the morning, we could make a new Constitution; we would not lose our self-possession, we would not lose our long training in self-control; we would not lose our instinct and genius for self-government. Strip us of one government, and we would make a new America in which we would shine as much as we did in the old. (Applause.) If that be not true, then it is not America, for America consists in the independent and originative power of the thought of the people. And so, when men and women from every part of the country gather in a great congress like this, to speak, not of matters of interest so much as of matters of duty, you realize in a gathering like this the vitality of the heart as well as of the mind of America, and men of every sort must give heed to the utterances of gatherings of this kind.

I know that there are some persons who come to these gatherings representing only themselves. I know that a gathering of men interested in a special cause is a great magnet to the crank. I know that all sorts of people, with special notions of their own, come sometimes to exploit them; but, after all, we ought to be very tolerant even of them, because some of the finest notions in the world have lived for a little while very lonely in the brain of a single man, or a single woman, and it is only by the tolerance of preaching that they get their currency, and finally get their imperial triumph by conquering the minds of the world, so that it is these voluntary contributions of thought, these irresistible currents of national life that are the most vital part of every people’s history. That is the reason I say it is a comfort to face an audience that I am not trying to persuade in regard to anything, but with which I am trying to get in sympathy, in order to share the great force which they represent.

It would be almost like assuring you that I was a thoughtful and rational being to say that I am in profound sympathy with the whole work of this great Congress, and that I am in particular sympathy, in keenest sympathy with that part which affects the conservation of the vital energy of the people of the United States. (Great applause.)

We have prided ourselves, ladies and gentlemen, upon our inventive genius; we have prided ourselves upon the ability to devise machines that can almost dispense with the intelligence of man. We have become a great manufacturing people because of this genius, because of our ability to draw together not only the tangible machinery of great enterprises but also the intellectual machinery of great enterprises, and we have been so proud of the mere multiplication of the resources of the Nation, so proud of its wealth, so proud of the ingenious methods by which we have increased its wealth, that we have been sometimes almost in danger of forgetting what the real root of the whole matter is.

I say, without intending to indict anybody, that it has too often happened that men have felt themselves obliged to dismiss superintendents who overtaxed a delicate piece of machinery, who have not gone further and felt obliged to dismiss a superintendent who overtaxed that most delicate of all pieces of machinery, the human body and the human brain. (Applause.)

If you drive your men and women too hard, your machinery will presently have to go on the scrap heap. If you sap the vital energy of your people, then there will be no energy in any part of the life you live, or in any enterprise that you may undertake. The energy of your people is not merely a physical energy. I am glad to say that the great State of New Jersey, which I have the honor to represent, has been very forward among her sister States in attempting to safeguard the lives and the health of those who work in her factories, and in all the undertakings which are in danger of impairing the health. I am glad to say that our Legislature has been to a very considerable extent, though not so far as it ought to be, thoughtful of the health of the children, thoughtful of the strength of women, thoughtful of the men and women together who have to breathe noxious gases, who are exposed to certain kinds of dust bred in certain manufactories, which dust carries congestion and danger to the lungs and to the whole system—we have been thoughtful of these things, but after all, we stand in exactly the same relation to our bodies that the nation stands to her forests and her rivers and her mines.

I have no use for my body unless I have a free and happy soul to be a tenant of it. We have no happy use for this continent unless we have a free and hopeful and energetic people to use it. I know that I have sometimes spoken of how foreigners laugh at Americans because they boast of the size of America, as if they had made it, and we are twitted with a pride in something that we did not create. We did not stretch all this great body of earth and pile it into beautiful mountains and variegate it with forests from ocean to ocean, and they say, "Why should you be so proud of what God created? You were not partners in the creation?"

But it seems to me that it is perfectly open for us to reply, "Any nation is as big as the thing that it accomplishes, and we have reason to be proud of the size of America, because we have occupied and dominated it." (Applause.)

But we have come to a point where occupation and domination will not suffice to win us credit with the nations of the earth or our own respect. It was fine to have the cohesive and orderly power to plant commonwealths from one side of this great continent to another. It was pretty fine, and it strikes the imagination to remember the time when the ring of the ax in the forest and the crack of the rifle meant not merely the falling of a tree or the death of some living thing, but it meant the voice of the vanguard of civilization, making spaces for homes, destroying the wild life that would endanger human life, or destroying the life which it was necessary to destroy in order to sustain human life; and that the mere muscle, the mere quickness of eye, the mere indomitable physical courage of those pioneers that crossed this continent ahead of us, was evidence of the virility of the race, and was evidence also of its capacity to rule, to rule and to make conquest of the things that it needed to use. But now we have come to a point where everything has to be justified by its spiritual consequences, and the difficult part of the task is that which is immediately ahead of us.

Until the census of 1890, every census bureau could prepare maps for us, on which the frontiers of settlement in America were drawn, and until that time there had always been an interspace between the frontier of the movement westward and the little strip of coast upon the Pacific, which had been occupied, as it were, prematurely and out of order.

But, in 1890, it was impossible to draw a frontier in the United States, it was impossible to show any places where the spaces had not, at any rate, been sparsely filled, sparsely occupied by the populations that lived under the flag of the Union. It was about that time, by the way, or eight years later, that we were so eager for a frontier that we established a new frontier in the Philippines, in order, as Mr. Kipling would say, "to satisfy the feet of our young men."

But the United States, ever since 1890, has been through with the business of beginning and now has the enormously more difficult task before it of finishing.

It is very easy, I am told, though I have never tried it, roughly to sketch in a picture, that all the students in art schools can make the rough sketch reasonably well, but they almost all, except those who have passed a certain point, spoil the picture in the finishing. All the difficulties, all the niceties of art, you have in the last touches, not in the first, and all the difficulties and niceties of civilization lie in the last touches, not in the first.

Anybody with courage and fortitude and resourcefulness can set up a frontier, but we have discovered, to our cost, that not many of us can set up a successful city government. (Applause.) Almost all the best governed cities in the world are on the other side of the water; almost all of the worst governed cities in the civilized world are in America. And the thing that is most taxing our political genius is making a decent finish, where we made such a distinguished beginning. We show it. You can feel it under you as you traverse a city; you can feel it in the pavements. They are provisional, most of them, or have not been laid at all and in jolting in the streets that are not the main thoroughfares of an American city, you feel the jolt of unfinished America. We have not had time, or we have let the contract to the wrong man. (Great applause.)

But, whatever be the cause, we have not completed the job in a way that ought to be satisfactory to our pride. You know that we are waiting for the development of an American literature, so I am told. Now, literature can not be done with the flat hand; you can not write an immortal sentence by taking a handful of words out of the dictionary and scattering them over the page. They have to be wrought together with the vital blood of the imagination, in order to speak to any other reader except those of the day itself. And, as in all forms of art, whether literary, or musical, or sculptural, there is this final test: can you finish what you begin? I believe, therefore, that the problem of this Congress is just this problem of putting the last touches on the human enterprise which we undertook in America.

We did not undertake anything new in America in respect of our industry. You will not find anything in the way of industry in America which can not be matched elsewhere in the world. If the happiness of our people and the welfare of our people does not exceed the happiness and welfare of other people, then, as Americans, we have failed; because we promised the world, not a new abundance of wealth, not an unprecedented scale of physical development, but a free and happy people. (Applause.)

That is the final pledge which we shall have to redeem, and if we do not redeem it, then we must admit an invalidity to the title deeds of America.

America was set up and opened her doors, in order that all mankind might come and find what it was to release their energies in a way that would bring them comfort and happiness and peace of mind. And we have to see to it that they get happiness and comfort and peace of mind; and we have to lend the effort, not only of great volunteer associations like this, but the efforts of our State governments and national government, to this highest of all enterprises, to see that the people are taken care of, not taken care of in the sense that those are taken care of who can not take care of themselves, because the best way to teach a boy to swim is to throw him into the water, and too much inflated apparatus around him will only prevent his learning to swim, because the great thing is not to go to the bottom and many of the devices by which we now learn to swim make it unnecessary to swim, because you can stay on top just the same, and I, for my part, do not believe that human vitality is assisted by making it unnecessary for it to assert itself. On the contrary, I believe that it is quickened only when it is put under such stimulation as to feel the whip, whether of interest or of necessity, to quicken it. But the last crux of the whole matter comes here: I am not interested in exerting myself unless the exertion, when it is over, brings me satisfaction.

If I have to work in such conditions that, every night, I fall into my bed absolutely exhausted, and with the lamp of hope almost at its last dying flicker, then I don't care whether I get up in the morning or not; and when I get up in the morning, I do not go blithely to my work. I do not go to my work like a man who relishes the tasks of life. I go there because I must go, or starve, and there is always the goad at my stomach, the goad at my heart, because those dependent on me will suffer if I do not go to my work and the only way I can go to my work with satisfaction is to feel that, wherever I turn, I am dealing with my fellow-men, with fellow-human beings. So that we must take the heartlessness out of industry before we can put the heart into the men who are engaged in the industry. (Applause.)

The employer has got to feel that he is dealing with flesh and blood like his own and with his fellow-man, or else his employes will not be in sympathy with him and will not be in sympathy with the work, and a man who is not in sympathy with his work will not produce the things that are worth using.

All the stories we tell to our children about work are told of such men as Stradivarius, who lingered in the making of a violin as a lover would linger with his lady; who hated to take his fingers from the beloved wood which was yielding its music to his magic touch. In all poetry and song since, Stradivarius has been to us the type of the human genius and heart that is put into the work that is done without attention and zest.

We point to some of the exquisitely completed work of the stone carvers of the Middle Ages, the little hidden pieces tucked away unseen in the great cathedrals, where the work is just as loving in its detail and completeness as it is upon the altar itself, and we say this is the efflorescence of the human spirit expressed in work. The man knew that nobody, except perhaps an occasional adventurer coming to repair that cathedral, would ever see that work, but he wrought it for the sake of his own heart and in the sight of God. And that, we instinctively accept as the type of the spiritual side of work.

Now, imagine, ladies and gentlemen, imagine as merchants and manufacturers and bankers, what would happen to the industrial supremacy of the United States if all her workmen worked in that spirit. Would there be goods anywhere in the world that could for one moment match the goods made in America? Would not the American label be the label of spiritual distinction? And how are you going to bring that about? You are going to bring it about by such work as this Congress is interested in and the work which will ensue, because the things which you are discussing now are merely the passageways to things that are better.

Just so soon as you make it a matter of conscience with your legislatures to see to it that human life is conserved wherever modern processes touch it, just as soon as you make it the duty of society to release the human spirit occasionally on playgrounds, to surround it with beauty, to give it, even in the cities, a touch of nature, and the freedom of the open sky, just as soon as you realize and have all of society realize that play—enjoyment—is part of the building up of the human spirit, and that the load must sometimes be lifted, or else it will be a breaking load, just as soon as you realize that every time you touch the imagination of your people and quicken their thought and encourage their hope and spread abroad among them the sense of human fellowship and of mutual helpfulness, you are elevating all the levels of the national life, and then you will begin to see that your factories are doing better work, because, sooner or later, this atmospheric influence is going to get into every office in the United States, and men are going to see that the best possible instruments that they can have are men whom they regard as partners and fellow-beings. (Applause.)

I look upon a Congress like this as one of the indispensable instruments of the public life. Law, ladies and gentlemen, does not run before the thought of society and draw that thought after it. Law is nothing else but the embodiment of the thought of society, and when I see great bodies of men and women like this, running ahead of the law, and beckoning it on to fair enterprises of every sort, I know that I see the rising tide which is going to bring these things in inevitably. I know that I see law in the making; I know that I see the future forming its lines before my eyes, and that, presently, when we come to an agreement, and wherever we come to substantial agreement, we shall have the things that we desire. So that, for a man in public life, an assemblage like this is the food of his thought, if he lend his thought to what his fellow-countrymen are desiring and planning; and all the zest of politics lies, not in holding things where they are, but in carrying them forward along the lines of promise, to the place where they ought to be. (Applause.)

You are our consciences, you are our mentors, you are our schoolmasters. The men in public life have only twenty-four hours in their day and they generally spend eight of the twenty-four in sleeping—I must admit generally to spending nine—and in what remains they cannot comprehend the interests of a great nation. No man that I ever met, no group of men that I ever met, could sum up in their own thought the interests of a varied nation. Therefore, they are absolutely dependent upon suggestions coming from every fertile quarter, into their consciousness. They are subject, or they ought to be subject, daily, to instruction. A gentleman was quoting to me today a very fine remark of Prince Bismarck's. He was taxed with inconsistency, with holding an opinion today that he had not held yesterday. He said he would be ashamed of himself if he did not hold himself at liberty, whenever he learned a new fact, to readjust his opinions. Why, that is what learning is for. Ought any man to be ashamed of having accepted the Darwinian theory, because he did not hold it before Darwin demonstrated it? Ought any man to be ashamed of having given up the Copernican idea of the universe? Ought any man to be obliged to apologize for having yielded to the facts? If he does not he will sooner or later be very sorry, because the facts are our masters, and if we do not yield to them, we will presently be their slaves. I suppose if I chose to assert the full consistency of my independence I would say that I was at liberty to jump from the top of this building, but just as soon as I reached the ground nature would have said to me, "You fool, didn't you ever hear of the law of gravitation? Didn't you hear of any of the things that would happen to you if you jumped off a building of this height? Suppose you spend a considerable period in a hospital thinking it over," and it would be very impressively borne in upon me what the penalties of ignorance of the law of gravitation are. Now, it is going to be very impressively borne in upon the public men of this country if they ignore them what the laws of human life are. As Dr. Holmes used to say, "The truth is no invalid. You need not be afraid; no matter how roughly you treat her, she will survive, and if you treat her too roughly there will be a certain reaction in your own situation which will be the severest penalty you could carry."

I come, therefore, to Indianapolis today to put my mind at your service, merely to express an attitude, merely to confess a faith, merely to declare the deep interest which must underlie all human effort, for, when the last thing is said about human effort, ladies and gentlemen, it lies in human sympathy. Unless the hearts of men are bound together the policies of men will fail, because the only thing that makes classes in a great nation is that they do not understand that their interests are identical. (Applause.)

The only thing that embarrasses public action is that certain men seek advantages which they can gain only at the expense of the rest of the country, and when they have gained them those very advantages prove the heaviest weight they have to carry, because they are then responsible for all that happens to those upon whom they have imposed and to those from whom they have subtracted what was their right.

So that the deepest task of all politics is to understand one another; the deepest task of all politics is to understand everybody, and I do not see how everybody is going to be understood unless everybody speaks up, and the more independent spokesmen there are the more vocal the Nation is, the more certain we shall be to work out in peace and finally in pride the great tasks which lie ahead of us. (Great and prolonged applause.)

NINTH SESSION.

The Congress reconvened at 8 o'clock p. m., in the Palm Room of the Claypool Hotel, and was called to order by President White.

President White—This is the evening session of the National Conservation Congress. I foresaw what was coming a long time ago when we began to prepare a program. I knew there would be a large number of ladies here, because they were getting very enthusiastic. I knew they would want section meetings for themselves to talk over matters of vital interest and plan how they were going to work for Conservation in all its departments, vital, social and political.

I felt that I was not capable and I did not know of any man who was capable of presiding over a large number of women, who sweetly and persistently know what they want and are bound to get it. (Laughter and applause.)

I was invited by the lady who is going to take charge of this meeting to attend the convention of the General Federation of Women's Clubs at San Francisco, and right there I decided that Mrs. Moore should preside at this Congress at some one of its meetings, and I politely told her so at that time. I did it in justification of her rare ability displayed upon that occasion, and, selfishly, because I knew I was too timid to rule on points of order where there were so many women. (Laughter.)

I take pleasure in introducing Mrs. Philip N. Moore, of St. Louis, former President of the General Federation of Women's Clubs, and a member of the Executive Committee of this Congress. She needs no introduction, as you all have met her many times. I now turn the meeting over to her good graces and good will.

Mrs. Moore—Mr. President, Ladies and Gentlemen: During the year that I have had the pleasure of working with the presiding officer of this Congress, it has been his gracious courtesy during the whole time to the woman who was on the Executive Committee that has induced me to accept the position he has given me tonight.

Many of you will remember that four years ago, when the Governors were called to the White House in Washington, to discuss the natural resources of our country, the only woman's organization that was represented at that time was the General Federation of Women's Clubs, through its President. From that time to this, the Conservation Congress has recognized this organization as being very much interested in the conservation of the natural resources of the country as well as in the conservation of human life through its public health department, through its industrial and social conditions and through its home economics, four of the strongest departments of the General Federation. I am, therefore, very proud tonight to accept the courtesy of the presiding officer of the Congress.

While we are waiting slightly for the first speaker of the evening, I have asked the next one upon the program to take her place. I am sure it will be just as much of a pleasure to you, and I am sure it will be a pleasure to her, to take the earlier place upon the program.

We are all very much interested, as men and women, as fathers and mothers, in the Children's Bureau which has been created this past year, and we were very much interested in the possibility of a woman being made chief of that bureau. There never was a question in our minds but that it should be the very best person that could be found, whether man or woman. But the fact that there was a woman who by education, training and experience was fitted to take this place has been a pleasure to all who are interested in that special development. The fact that she has looked into the life of children from birth through childhood, with work and play and home and school as they have applied to the life of the child, will be of the greatest benefit to us all through these future years.

I am very glad to introduce to this representative audience of the Congress, Miss Julia Clifford Lathrop, who is Chief of the Children's Bureau of the United States Department of Commerce and Labor.

Miss LATHROP—Madam President, Ladies and Gentlemen: I need not explain to a Congress interested in Conservation why the representative of this new Bureau should be here and should wish to speak about the Bureau itself.

When I was first honored with this appointment it was suggested that the Bureau should be staffed with women alone, and I was asked what I thought about it. I said I should be very much embarrassed; that I had never known any children who had not two parents, and that I felt that if there had been intended a division of that sort the Lord would have communicated it long ago. I thought it would be presumptious for me to begin, so the Children's Bureau has on its staff both men and women and, perhaps, I may as well begin by saying something about that staff, and about the organization of this new Bureau. And, first of all, perhaps I may forestall a criticism which is likely to come before very long that we are rather dilatory and are not accomplishing very much, by reminding you that the Bureau did not go into operation last April, when the President approved the bill, but only on the 23d of August, when the appropriation became available, so that really the Bureau is just forty-one days old. It has a staff of fifteen persons, and it has to spend for this first year a sum aggregating about thirty thousand dollars. Its province, as the law says, is to inquire into and report upon all matters pertaining to the welfare of children and child life. You can see for yourselves whether that is a big job and whether the army really seems adequate for immediate performance of the contract.

It is, of course, enormously important that a Bureau of this kind, undertaking a sort of work which, after all, is in some respects new, should be composed of people who have very much at heart the welfare of children; who have, even as much as that, the scientific training and wisdom which is necessary if we are to make an appeal to people's emotions and sentiment. So the staff of the Bureau is composed of people who have been selected from various departments of the Federal Civil Service as having, particularly, acquirements in science, as statisticians, and in other respects particularly fitted, as we believe and as my superiors believe, to do the work of this growing Bureau.

The Bureau has this great general object. Now, it is a question how to take hold of this great task, where to begin, but the law itself does give some hint—it enumerates certain objects which we shall discuss in detail as time goes on.

The first of these is infant mortality and the birth rate, and after that follows various subjects, such as juvenile courts, or the care of children in regard to diseases and accidents which may befall them, the regulation of their labor, legislation affecting children and all matters pertaining to their welfare.

It is all very well for those of us who are doing all sorts of volunteer work, as most of us are, to begin on the problem of helping people at any point where we can take hold. We do not have to know any great fundamental facts; to know that babies need care, and that children ought not to go to work when they are too young, to know that children need to go to school, need to be healthy, need to be happy, and that they need recreation just as much as they do education and that the two are part of the one same great sort of development—all these things we have a right to begin on anywhere we can. But when the Federal Government takes hold, I think it somehow promises a sort of basis to all the rest of us, and it seems as if it were its business to see where there was the most fundamental point to begin its work. When we come to look at the question of dealing with children, we are constantly faced by the fact that we do not know how many children there are; we do not know how many children are born and die in this country. We do know once in ten years how many children exist at a particular moment, and by that decennial information we know that the Bureau has to deal with about thirty-six per cent. of the total population of this country directly, that between thirty-five and forty per cent. of the population of this country is under sixteen years of age, which seems to be the age of the end of childhood, just by common acceptation; at least, at that age in many of our States children are permitted to become independent workmen. So you see we have a very large number of children with whom we have the right to deal directly, and, as I tried to show a moment ago, we think we have a right to deal indirectly with all their parents. We think the whole country is a good deal our province in prospect, but we cannot be satisfied with this decennial knowledge of how many people there are in this country. What we want is a great, democratic continuing public edition of "Who's Who in America." We want to know day by day the advent of every citizen into this country. We want, in fact, in a phrase which is not particularly exciting, "birth registration."

First of all, we want it because we want to know, and we want to know for various reasons, which I think do not occur to most of us every day. In the first place, we want to know because, unfortunately, a great many babies come into this world under circumstances which do not give them the best chance in the world. If the advent of a little child could be at once communicated to doctors and nurses where doctors and nurses are not taken for granted, it would be possible to prevent the risk of that blindness which sometimes overtakes newly-born children; it is possible to establish the health of the mother and child together, so that it may have the best chance in the world; and you all know how throughout this country, even in our remoter counties, there is coming to be a great and splendid health service. I think we cannot be too delighted with what the Red Cross Society, with what many similar societies are doing in the way of rural nursing. I think if Florence Nightingale could look out over America now, she would think we are beginning to realize her noble words about health and nurses.

Now, those are perhaps the most important reasons why we want to know the advent of every child, so that we can help that little child, and help his mother and keep her alive, because it is a very terrible thing that out of all the children who are born into this country a very large number—two hundred thousand, some people say, and three hundred thousand, some people say—die before they are twelve months old, and more, a third of them die before they have been in the world a month. I want us to consider this for a minute, not as an economic problem, but what it means in the old fashioned terms of human suffering, the agony and loss of family happiness and joy, that two hundred thousand little babies should die and leave the hearths to which they come every single year of the world in this country. And when we think that already, doctors tell us, we know enough so that that waste is very largely the fault of our carelessness and selfishness and greed, it makes us blush to think we are not all working hard to save the lives of these children.

Now, there are great efforts already undertaken to save the lives of these little babies, in which many of you are already engaged, and we may well hope that such societies as this Conservation Congress and the Congress for the Prevention of Infant Mortality will before the next decennial census occurs have made a great difference in the number of children who are born, only to die.

But, suppose that a child lives, is there any real sense in his having a birth certificate, or is that just some abstract notion of the statisticians, who get all the certificates and have punching machines and a great many mechanical contrivances for numbering and making computations out of figures? I think you will be surprised to find how much human connection it has.

In the first place, as to this very Bureau for which I am venturing to speak, we are told to find out about the diseases of children and about the birth rate of children. How can we know about the birth rate unless we know how many children are born and die? In Washington, there was set up a wonderful placard on the wall to show the birth rate in that city, and there were columns of red and green and other colors, and you just knew, humanly speaking, that the birth rate was fluctuating that way, and you talked with somebody and you discovered that this birth rate was fluctuating as one set of gentlemen or another was electing the health officer.

So now we want to have some authoritative way of truly finding out how many children come into the world in order to know what the birth rate is, in order to know how to study the diseases of children, and then, when children grow a little older, we want to have a public record of their births so that we may know when they are entitled to go into the schools to begin on that system of care and culture for which the public schools stand, and then beyond that, when they are older and the time comes for their advent into the army of work to which we hope we all belong, then we want to know that those who are less favored are not hurried into that army unduly. How much it would simplify the problem if we had not to trust to all sorts of chance ways of proving a child's birth and if we had a public record of it.

Have you ever thought that we are the only great nation which does not know how many children are born into it, and which does not do its children the dignity of putting their names down in a public record? All Europe has a public registry, and why? Because it has a standing army and wants the names of its boys for conscription. Now, in a country of peace, aren't we to have any victories for peace? Are not we to recognize a child as having any dignity to be a peaceable citizen and not either a target for a gun, or the man behind the gun?

I think you will, perhaps, be interested if I venture to tell a story of a neighborhood in which I have lived long, an illustration of how a birth certificate is a good thing. A little while ago, a family came over to Hull House for some help. They were awfully poor. The oldest girl, who was at that time the breadwinner of that family, was out of work on account of the garment workers' strike. There were eight children. They had come over at the time of the earthquake in Messina. The father had been entombed and his mind had almost succumbed to the fearful experience. He was always thinking the walls were coming on him and he was not in a very good frame of mind to be a successful breadwinner. So they got into difficulties and asked the Charities Board to help them. There was another younger girl and they thought they had better get a work certificate for Giovanna, but the truant officer said she looked too young and couldn't have it, and she was sent back to school. And then they got a little more desperate and they tried again to have poor little Giovanna go to work. The Charities Board, who were helping the case, thought they had a right to dictate a little as to how they should help, and they just wrote to the City Hall in Messina, and the City Hall in Messina sent back a very prompt letter showing how old the children were, and showing that the daughter who had been at work for two years was really about fourteen years old and had been working that time illegally, had been cheated of two years of school, where she might have learned good English and learned American housekeeping and had a better chance to earn more money the next two years. The other little girl was still younger. So the people in Hull House and the people in the Associated Charities and the factory inspector made a veritable cordon around this helpless family and demanded they be sent back to school. The oldest girl went back very unwillingly. She said indignantly, "Me go back to school, me big enough to be married." She was very hurt and humiliated. I am not sure we did right about it. Giovanna was confiscated and sent back to school for two years. This family did not have a fair chance over here just because the factory inspector and school authorities, not having any birth reports over here for children, followed the usual system of guessing and did not think to take advantage of what Messina, notwithstanding the earthquake, had to offer from her very responsible records.

Has it ever occurred to you that to very many of our foreign residents a birth certificate for a child would be an absolute asset?

In Chicago is a very prosperous and highly respected man. He came from Germany when he was a baby of four years, with the family. His father was never naturalized and the man himself never was challenged in his right to vote. He grew up and attended our public schools, and all that. He accumulated a fortune and went back, as many people do, I suppose rather proud, to see the old country and friends who remained there and with whom he had kept in constant communication, a prosperous and splendid example of what America could do for a man. He had been in Berlin for about two hours when the police were on his track merely because he was a German citizen and must serve in the army. He telephoned to a lawyer friend and asked him what could happen. He said, "There is just one thing that can happen, and that is that you get out of town." So, two hours after he arrived at Berlin on this triumphal journey, he left very actively, and he is said never to have heaved a sigh of such joyful relief as when the crossed the boundary into France. That is an example of people really knowing where they were born and being able to prove it.

I suppose the reason we have not been more eager about our birth rate is because we have not thought anything about it. We think a great deal about writing the baby's name in the family Bible and christening it in the church we attend, but somehow we have forgotten this larger, more fundamental thing. The advent, of every citizen of this country ought to be on the books of the commonwealth.

There is a very good story, which belongs to your own Dr. Hurty. I do not know whether you all know it, or whether I dare tell it, but I will presume that this audience is largely made up of visitors, and steal his story. In this State, Dr. Hurty is authority for saying there was a farmer who had a ne'er-do-well son and a granddaughter, and when the farmer came to die he wished to leave the farm to the granddaughter, but he left the use of it to the son until the granddaughter should arrive at the age of twenty-one. When the girl, as she thought, was twenty-one, she claimed her inheritance, but the other side said she was only nineteen. She went to the Bible, where her name was written down, but the leaf was torn out, and the court was very much perplexed. It came to be a serious legal question, and finally a neighbor recollected that the grandfather had had a very remarkable calf born on the same day with this little girl, and he said he knew the farm books kept by the grandfather would record this pedigree. So the farm books were looked up and the birth of the calf was discovered and the birth of the girl was established. (Laughter.) You all remember how George Bernard Shaw warns us against placing confidence in the *deus ex machina*. He says you cannot presume on things being some miraculous way you would like them to be, and so we cannot presume on grandfathers always keeping herds of cattle. (Applause.)

I am perfectly sure, as I have said before, as I had the honor of saying over at San Francisco before the General Federation of Women's Clubs, that if the women of America wanted birth registration they would get it in a twelvemonth. Now, it sounds so very remote from putting down the baby's name in the book.

In the State of Indiana you have a very good law, Dr. Hurty tells me, and all that is necessary is for the women of Indiana to say that they want the names when their children are born recorded in the public records of Indiana. In 1910, when the last census was taken, all that we know about the births in this country was what we learned from eight States, the New England States, Pennsylvania and Michigan. Not your State, or mine, Illinois, was deemed worthy to be considered at all. So far as the general government was concerned, for anything it knew, nobody had been born in either of these States in ten years. In the next census year, I hope very much in a great many States in this country, perhaps in all the States in this country, we shall be able to be recognized by the general government as having been born and as having been born very accurately, so that we will be worthy to be counted, as much so as if we lived in Boston and Massachusetts, which, they are always telling us, are the most accurate State and city.

Of course, the Bureau cares for a great many things besides the registration of births, but I hope I have made it plain that we should ask that we be allowed to get a method of acquiring steady, constant and reliable means of legal proof as to the children who enter this Nation, because it is the dignified basis for a governmental Bureau, which I believe is destined to grow to proportions which none of us can measure, which shall continue long after all of us are gone. No other bureau in the world makes so tremendous an appeal to the emotions and sentiment—a children's bureau, a bureau to concern itself with the life and happiness of the children of a great nation, and the more appealing it is, the more must it be founded upon facts which will bear the very closest scientific scrutiny. What the Bureau will be doing years from now I do not know. I know what it must do now. The law is very distinct about some of the things it must do, and by implication many of the things it cannot do. It is a bureau to gather information and to publish it as the secretary of the department under which it exists may direct. It can publish in any way which the secretary deems best. There are a great many different ways of publishing facts. We are learning to publish facts through the sort of thing you have in the State House here and other exhibits, through the appeal to the eye. In this way thousands of those who cannot study very carefully or cannot read a table to save their lives may understand, and I hope it is with some of the simpler methods of popularizing things that this Bureau may begin to make itself useful.

The Bureau, although it is a different type from all the other work of the government in a certain sense, after all, is not so isolated as we might think. There is a Bureau of Labor, which has studied much the labor of women and children. There is the Census Bureau, of which I have spoken. There is the Bureau of Immigration and of Education, and the Bureau in the Department of Agriculture, which has concerned itself very much in the South with those very interesting and productive efforts for better farming, which have begun their activities by stimulating tomato-canning among the girls. All of these things, part of them purely educational, part of them a matter of direct work, are things which we shall not do over again, from which we hope to learn very much.

There have been a good many anxieties about this Bureau, many people have thought it was a mistake. Some people have said, “Ah, well, you are going to center everything about children away off there in Washington where there will be a government with a lot of very comfortable clerks sitting about in offices and writing down figures about children instead of doing things for children and you will palsy local effort.” If the Bureau does that it is a failure. What the Bureau must do is to stimulate and help local effort. It must gather facts and try to present them so convincingly and simply that they will be useful and stimulate many to activity.

Then there has been a great dread lest the Children’s Bureau might interfere with parental rights, lest the Bureau might seem to override the dignity and privacy of homes. I do not believe the Bureau will ever do that, because I know that the people who care most about the Bureau are people who realize that the welfare of the child is measured by the welfare and the wisdom of its parents, and that the way to help the child is not to take him out of the family, but keep him in it and help the parents to help him. And the Bureau will do its work with a fine respect for parenthood. And perhaps I cannot better close, since this is a woman's meeting, and we may well be generous to the gentlemen scattered here and there, by a story of a man, a father.

Not long ago I went to a meeting in Chicago, at which there were many delegates from the foreign colonies in that city. It was a representative meeting standing for about one hundred thousand residents of that foreign town. It was really a meeting of protest against threatened restrictions which many of us thought very ill-advised and cruel, which were to be applied to immigration. A man rose who belonged to a foreign colony which we are accustomed to regard as especially dull and illiterate, and he told very simply how that colony had come from a people who had been oppressed, the study of its language had been forbidden, reading and writing had been forbidden, and in a way, a certain illiteracy and dulness had been forced upon them; and he told so simply with what ardor they came here where there was freedom, where there were schools. I shall never forget how simply he said, "I am a father, and, like every father, I want my child to go higher than me."

That was the simple but overwhelmingly eloquent expression of a man whose English was very broken, but who, after all, spoke exactly the great impulse which has controlled all of us since the beginning of that wonderful seventeenth century when parents began to come over here. And, as I heard him speak, I thought that whether it was those who came in the cabin of the Mayflower, or those who sank in the steerage of the Titanic, they were all moved by that same mighty impulse, that the next generation should have a better chance than they had.

Now, this Bureau must move forward if it is to be useful in the same spirit in which families move forward, in which the race moves forward, to give the next generation a better chance than this has had. I thank you. (Applause.)

The CHAIRMAN—Those of us who heard Dr. Wiley, the other evening, give his impressions, may be interested in giving to Miss Lathrop another fact which will prove the value of birth registry. Dr. Wiley said that no one across the water could marry unless he could prove that he had been born. It would be impossible for many to marry in this country, if that were the case here.

We have always admired the way the Daughters of the American Revolution have taken the history of our country, have looked up the old stamping grounds and marked them, and have taught the children in schools the traditions of the country, to honor the makers of our country and to make them good American citizens. But we are really more pleased that the Daughters of the American Revolution have recently taken up more modern things, and that they are preserving the resources of the children. The speaker has been very much interested in modern life, in community life for the rural life of our country.

As a loyal Daughter, I have great pleasure in introducing to you Mrs. Matthew T. Scott, of Washington, D. C., President General of the Daughters of the American Revolution.

Mrs. Scott—Madam President, Members of the Fourth Conservation Congress: Among the many opportunities for service, which today are open to women in this country, there are three to which I wish to call attention for a few moments this evening. The first is that of the unrealized possibilities of the home life of the nation. If we only were endowed with a larger share of that priceless attribute—the constructive imagination—we should be able to see the untold resources which still lie latent, waiting only to be discovered, developed and enjoyed in the mysterious precincts of that laboratory of the soul—that forging-room of character, that fountain-head of those subtle forces which add temper and edge and distinction to our ordinary human attributes—that civic and social Holy of Holies—which we call Home.

And let us remember that the sources of our country's permanent prosperity and glory lie not in the form of our government, in the wisdom of its administration, nor even in its written laws and constitution, but deep in the intellectual and moral life of society, and potentially in those nameless influences, radiating from the women who give its halcyon charm to hearthstone and library and to all the intimacies and inspirations of the home. For, after all, it is the home—the sanctuary to which we women must hark back—the home, with its *sanctity*, which is the palladium, the corner-stone, the key to the arch, of all that is most precious in the life and destiny of America.

Again, let us never forget, that to us women—the home-makers of our land—as never before in the world's history, is entrusted the healthy development of the social and moral fabric of society in our country, in the innumerable and intricate complications of this Twentieth Century civilization. A distinguished educator has recently said:

> "At the present time the world is awakening to the teachings of the old prophets. Now, as then, the morals and ethics of a nation are just what the wives and mothers, the home-makers of the land, make them."

Again, the home is also the place where the future citizens of this nation are to be trained. The place that fosters patriotism, obedience and love, reverence for authority, the finest elements of character. Some day the present generation will have to hand this country over to the sons and daughters who are being trained by fathers and mothers of today to administer the affairs of the home, in preparation for the larger field and wider duties of government. It is well for youth to learn that honest toil is never hopeless or degrading. It is well for youth to be at one with Nature and to learn of her; to know and feel the joy there is in bountiful, glorious Nature; to be familiar with her song—the ripple of the river on its stones, the murmur of trees, the rhythm of the sap that rises in them, the thunder in the hills, the stars shining in perennial beauty, the song of the thrush, and the carol of the lark; to watch the sun in its course and learn the dim paths of the forest.

"It is the song of infinite harmonies."

The man, woman or child of vision responds—perhaps, all unconsciously and inarticulately—but responds like a vibrating chord to the note of these melodies, that should be part of the charm of the home-life of the farm.

There can be no disputing the fact that a goodly number of American women are wonderfully successful home-makers. But at the same time, it must be admitted, that a large number of our household mistresses must plead guilty to the charges of extravagance—technical ignorance of household economy—and a considerable degree of all-round inefficiency, both as housekeepers and as home-makers, for the terms are not synonymous.

It is a commonplace among sociologists that in most well-to-do American homes enough is wasted in the kitchen alone to keep a French family in comfort. We are also wasteful of light and heat, and, above all, of our time and energy. Our country is in dire need of a woman who will do for the home what a distinguished inventor and public benefactor of Philadelphia has done for the factory—that is, introduce an “efficiency” system, which will do away with our present waste of both money and time, and increase the quantity and quality of the actual output—not only of creature comforts, but also of artistic attractiveness; and of that indefinable atmosphere of peace and restfulness, which, of all the by-products of home life, is certainly the pearl of greatest price. The time will surely come when both mistress and maid will prepare for their life’s work—as home-makers—with the same care and enthusiasm that men now put into the work of perfecting themselves in their various trades and professions. Home-making, like piano-playing, is an art—to succeed in which requires something more than temperament. Until the technique has been properly mastered, temperament has little opportunity to manifest itself to advantage.

A generation ago home-making and farming were occupations that anyone with a mediocre intelligence and a reasonable degree of industry was considered sufficiently equipped for. But today these two avocations occupy a secure and increasingly important place among the learned professions.

Agriculture, “dignified by the ages, as old well-nigh as the green earth itself,” has become a scientific profession alluring to men and women of brains and culture, who quickly become enthralled by its ever-expanding and fascinating possibilities. In every State in the Union we have magnificent agricultural colleges and schools of domestic science, in which are being prepared for their respective careers thousands of prospective farmers and thousands of prospective housewives. Moreover, several bills have been pending before Congress which provide for the widest possible dissemination of instruction in agriculture and domestic science (including the pure food problem) among the rural population of every county in this Nation.

This is a glorious work. The proposed instruction in agriculture is something which, as a farmer, I am particularly enthusiastic about. Yet I feel that quite as important as this will be the educational facilities in the household arts and in the highest home-making ideals, which are to be placed within reach of every housekeeper and every prospective housekeeper in this land. Just as agriculture is the basis of all our material prosperity and power, so the home is the perennial and sacred source from which emanate those potent, ennobling and refining influences, which slowly and silently have lifted man out of past savagery, and will yet, we trust, lift us out of our present state of semi-civilization—with its class war, political and business corruption and industrial brutality—on to higher and even higher stages of moral, intellectual and social development.

This is my idea of the relatedness of Conservation to the home. Is there any question that this is truly Conservation—its essence—in the minds of any member of this convention?

The second realm of opportunity which I want to point out to you is that which spreads out before us in a bewildering splendor of promise, in connection with the schooling of the young, as related to the home. We are all aware that the large majority of our common and high school teachers are women. In many of our States women vote for members of the School Board, and if a majority of them really wanted this right, there is no doubt that they would secure it everywhere. In this event it would be a comparatively easy matter for them to formulate and carry through policies of their own. Thus from the cradle to the university the education of the children is potentially in the hands of the women of this country.

This is a power which the priests of various religions frequently have endeavored to obtain on the grounds that if they were allowed to control and dominate the child’s mind during its formative period, their influence upon its after-life would be dominant and enduring. I wonder if we realize what almost unlimited power over future generations is thus entrusted to our hands. Are we, as women and mothers, exercising that power with an adequate sense of the responsibility which it places upon our shoulders?

There are now thousands and hundreds of thousands of our sex who are pining for something to do, which they can feel is entirely and splendidly worth their while. How fortunate it is, that here, already at hand, is a task which Nature, and “Man, the tyrant,” are agreed is peculiarly adapted to our particular tastes and talents. But what are we doing about it? Little as a sex, I fear, that is either significant or creditable. When not merely in a few isolated cases, but as a class, the women of America decide what they want in the way of education for their children, if they want it badly enough, there is no earthly power which can stand between them and their splendid ideal goal.

But this means work, persistent, intelligent work. First of all, in the matter of self-education, and, secondly, in that of carrying on an aggressive campaign for the education of our own sex, and, if possible, of the other sex as well.

I am beginning to get deeply concerned, not about the lack of adequate opportunities for service on the part of women, but about our failure, so far, to measure up to the incomparable opportunities which are already ours. If there is any subject in which we, as women, ought to be intensely and intelligently interested, it is in this subject of education—not in the academic sense alone, but in the broader view of character-building—upon a proper understanding and handling of which depends the very future of civilization.

This, I take it, is truly Conservation work and when thoroughly grasped, will as truly mark milestones of progress in our lifetime, as those we may leave behind in material form.

The third of these brilliant avenues of possible social service, which open before us in beautiful vistas of alluring opportunity, is one which is involved in the purchasing power of women. As a general thing, men are the wage-earners and women are the wage-spenders for the home. Nearly all of the household expenditures of the family are made by the wives and mothers of the race. It is a sad commentary upon our business ability, and our rudimentary sense of social solidarity, that so few of us have any realizing sense of the potential power over the business and industrial world, which is inherent in this our position as buyer, or spender, for the family.

I call your attention to the fact that if the women of America would pool their purchasing power, and, resisting all the blandishments of the “bargain counter” and the “sale”—based on sweat-shop labor—would demand pure goods, made and sold under sanitary and salutary conditions—more could be accomplished for the moral and material uplift of the factory-worker and the saleswoman than by the enactment of a volume of restrictive statute, the breaking of which we thoughtlessly connive at, and practically become a party to, in our mad scramble for cheapness at any cost of human degradation and wreckage.

A superb organization, known as the “Consumers’ League,” has come into being, for the express purpose of enabling men, as well as women, to utilize their purchasing power in the great work of raising the standards of the business and industrial worlds, both as to the purity of the product offered to the public and the fairness of the treatment accorded to employes.

Of all the splendid “movements” and “causes” which today invite our co-operation and support, this is one of many, which seem to me to fall naturally within our province, as wives, mothers and home-makers. It is the principle underlying this great crusade of the “Consumers’ League” and like organizations which appeal to us. As a matter of fact, this is a work for the betterment of women in the business and industrial worlds, and as a consequence improvement in the home, which we women cannot avoid doing, without definitely and publicly shirking our heavy economic and moral responsibilities, as family purveyors and budget makers. Or, in other words, as domestic chancellors of the exchequer.

Far be it from me to say that the members of our sex may not some day decide to undertake, in addition to their other duties, the heavy responsibilities of the voter and political worker. Perhaps it may transpire, that upon our planet the true super-man is woman, and that she is entirely capable of doing the man’s work as well as her own. But, in the interim, until this fact has been satisfactorily demonstrated, let us devote ourselves whole-heartedly to what is more particularly woman’s work; to those delicate and difficult tasks for which man’s clumsy fingers and prosaic processes of reasoning are unfitted and wholly inadequate. And, above all, let us be quite sure that we do our especial work—at least as well as he does his—before we insist upon taking a hand in his activities and improving upon his methods of performing his highly useful, if somewhat less exalted, functions.

It may seem in these lines of work—somewhat unique—and hitherto undefined as belonging to the realm of Conservation, that I am departing a long way from the usual addresses on that subject. But I ask your careful consideration of this subconscious knowledge of every woman's breast, that at least every issue and question I have referred to has its foundation, in the broadest and deepest sense, in the life and action which center in the home.

In the ways which I have so hastily outlined and in other, and perhaps better ways, that may not yet have occurred to us, our great work of Conservation is destined to continue its triumphal march upward and on—in the name of the great principles upon which it is founded, and in the name of patriots, living and dead, who have labored and sacrificed to make of this, our fatherland, what, under God, it is, has been, and ever must remain—the greatest nation on earth. Because, beneath the ample folds of its unconquered flag, there live more free, happy and God-fearing people than upon any other part of the habitable globe. (Applause.)

The CHAIRMAN—I had been tempted to introduce the next speaker as a charter member of the organization of the Daughters of the American Revolution, but I was told by her that this would be considered antediluvian, so that I have not any right whatever to use the knowledge I possess. I have also been told by her friends, for I am sorry to say that until this meeting we had not known each other, that she is the personification of patriotism.

It gives me great pleasure to present to this audience the Honorary Vice-President-General of the Daughters of the American Revolution, Mrs. John R. Walker, of Kansas City, Mo.

Mrs. WALKER—Madam Chairman, and Ladies and Gentlemen: The term Conversation has become so all embracing, from the viewpoint of a Daughter of the American Revolution, it is as much a work of patriotism as that of our own great organization—the one dealing with the present and the future, the other with the past, the present and the future. The motto of the Daughters of the American Revolution is "Home and Country," and so lofty is its ideal, so practical its work, it will be felt throughout all time, as will this broad, wise work of Conservation. The spirit of commercialism, of money worship, about in the land, is fast sapping the resources of our great country and begetting a selfishness that makes a willing sacrifice of the rightful heritage of future generations. It would seem in the order of things in this work of Conservation, that the men of our land should give special concern to its material needs, its lands, its waters, its mineral resources, and that the conservation of life should appeal as nothing else to woman, the transmitter of life—Life, a priceless boon. We protest against child labor—implore with all the tenderness, developed through mother love, to spare the child in the greed of money getting. Refuse the work of little hands, and little feet, in factories, mills, and mines, and out of your abundance make it possible for them, during the few short years of childhood, to enjoy the freedom of the bird and the butterfly, give them a memory of Nature's blessed joys—God's pure, sweet air; the wayside flower plucked at will, the willow-shaded stream, and all that the sweet breast of Nature offers so freely, without money and without price—to the child of poverty. The Daughters of the American Revolution are awakened to the realization that we, the home-makers, descendants of the woman of the spinning wheel, hold the destiny of a nation in our hands, that we must not only accept but consecrate ourselves to woman's highest mission, the crowning glory of womanhood—guiding the young feet into right paths.

To give patriots to our country, we must rear patriots, train Americans for America. In our great work of patriotic education our aim is to train the youth of our land in good citizenship; teach them to battle for good laws and social conditions, and to be courageous in the fight, daring to do right in both the political and business world—thus honoring his birthright. The Daughters of the American Revolution have gathered the alien into the fold of the children of the republic, to make of them true Americans, do for them the best we know how; and many a lesson we can learn from them of thrift, industry and patience under discouragement. In my own State opportunity came to such men as Carl Schurz and Joseph Pulitzer, poor emigrants, who became pre-eminent in our country's history. The privileges of the American woman go hand in hand with her responsibilities, in her zeal for home and country; she is pointing the way, realizing that our children have a great work before them, a great problem to solve.

The Jewish dramatist, Zangwill, says: "To think that the same great torch of liberty, which threw its light across all the broad seas and lands into my little garret in Russia, is shining also for all those other weeping millions of Europe, shining wherever men hunger, or are oppressed, shining over the starving villages of Italy and Ireland; over the swarming cities of Poland; over the ruined farms of Roumania; over the shambles of Russia. What is the glory of Rome and Jerusalem, where all races come to worship and look back, compared to the glory of America, where all races and nations came to labor and look forward.” America! great charity of God to the human race.

Conservation of life! As I stand before the shafts erected at Arlington and Richmond and read to the memory of sixteen thousand who fell in battle, to the memory of eight hundred unknown dead, my very soul cries out against war. Eight hundred unknown dead! Can you not see the long procession of anguished, broken-hearted mothers, waiting and watching—watching and waiting, and hoping? Our law makers oppose legislative measures advocating universal peace. How can they with our Civil War yet fresh in memory, the nations of the earth yet shuddering over the horrors of the war between Russia and Japan? The heart sickens at the memory of the undying hatred of the human heart; the blood thirst for blood in its brutal frenzy, sacrificing her young men—the hope of a nation—and all for what? One more island, perhaps, or insignificant kingdom. A war involving principle, as our Revolutionary War, hundreds of years afterward excites the most passionate interest and feeling; but wars for power, and possession, the world cries out against. The time has come to sheathe the sword and spare mankind. The vast expenditure of money for more destructive engines of warfare, for the slaughter of men, would go so far in our work for humanity, the helpless, the unfortunate, the struggling. War affects not only those who bear arms, but those who stay at home; the entire country is affected. War retards progress, paralyzes effort; ambition cannot feed a sorrow, hands are listless and lax when the heart is heavy. Mrs. Browning’s Italian mother wails: “Both boys dead, one of them shot by the sea in the East, and one of them shot in the West by the sea. Dead! both my boys. If your flag takes all heaven, with its white, green and red, for what end is it done, if we have not a son?”

On one occasion, a distinguished Confederate general was a guest at our table; he had fought from the beginning to the close of the Civil War. The little boy of the family gazed upon him with awe and admiration. To know and be close to a great soldier, one who had commanded armies and fought many battles, was indeed glory for a small boy. After gazing upon him long and steadily, he startled the assembled company by saying: “General, how many men have you killed?” We gasped in horror, wondering what the reply would be. Quickly the General responded, “I don’t know that I have killed any.”

We read “The Charge of the Light Brigade”; “Scots Who Ha’e Wi’ Wallace Bled,” and other stirring poems of war, and see only the glory of it. Death by shot and shell and sabre stroke is heroic; but the question of a little child startles us with the question of our individual responsibility; we are brought face to face with the words engraven on the tablets of stone, “Thou shalt not kill.”

Universal peace is no longer a dream. The peace court at The Hague is established, and marks an epoch in international law. Let us not cease in our efforts until the pressure of strong public sentiment becomes so compelling, legislation will be favorable. Our country is the beacon light; she stands for justice, for freedom, for God; she is the messenger of the Prince of Peace, is elected to proclaim with trumpet call, peace to all the nations of the earth and the islands of the sea.

I cannot let this opportunity pass without asking this influential body of men to throw the weight of its great influence in favor of another matter taken up by the Daughters of the American Revolution—the desecration of the flag. I was appointed by our President-General Mrs. McLean, to speak on the subject before a committee of the United States Senate, and, with representatives from other patriotic societies, urged legislation upon it. It is a matter of sentiment, but what is life without sentiment? With you men laboring for your country’s welfare, see to it that our country’s emblem is held sacred, shall not be used as an advertising medium by the soulless money-maker, who cares for naught save personal gain, who does not consider that this banner stands for this great country—“your flag and my flag.”

“And Oh! how much it holds,
        Your land and my land
  Secure within its folds,
        Your heart, and my heart,
  Beat quicker at the sight,
        Sun-kissed and wind tossed the
  Red and Blue and White;
        The one Flag—the Great Flag—the Flag for me and you

Glorified all else besides—the Red and White and Blue."

Wherever we fling it to the breeze, it carries a breath of freedom into every land and unto every people. Should we not hold it a sacred thing? (Continued applause.)

The CHAIRMAN—For the past two years the next speaker has been working in the General Federation as Chairman of the Department of Conservation. We have worked so closely together, I, as her adviser, and she doing the large work of the organization, that it is almost like speaking of one's own family in introducing this speaker. I shall not try to tell of her work. We are the very best of Conservation friends to this day—Mrs. Marion A. Crocker, of Fitchburg, Mass.

Mrs. CROCKER—Madam Chairman, and Mr. President and Members of the Convention: Conservation is a term so apt that it has been borrowed and made to fit almost all lines of public work, but Conservation as applied to that department bearing its name in the General Federation means conservation of natural resources only, and that is a field so vast that we have found it all that can well be handled under one head without a chance of neglecting the very principle for which the Conservation movement was established. And then it is always easier to come back to simpler things. I do not mean exactly "simpler," but to those that touch our lives from day to day, of which we may see the effect almost from hour to hour, and therefore it seems so unnecessary to dwell on these things that are far away. The problem of Conservation of natural resources is so wide and far extended that much of it must be solved on great government plans, and that seems to make it even more remote.

Now, we all concede that there is nothing so important as the conservation of life, of health, education and vital force, so closely connected with the life. We all grant that, and it is only because the conservation of natural resources is so closely related to these other lines that it is of any vital consequence. But, with the other side having been so strongly emphasized, and, to my sorrow, a few times I have noticed it even being decried in this conference, it seems to me it has become my bounden duty to emphasize the other side, because if we do not follow the most scientific approved methods, the most modern discoveries of how to conserve and propagate and renew wherever possible those resources which

Nature in her providence has given to man for his use but not abuse, the time will come when the world will not be able to support life, and then we shall have no need of conservation of health, strength or vital force, because we must have the things to support life or else everything else is useless.

Do not think I am pessimistic. I should not feel this so strongly, but I feel that this Congress was originally established for the conservation of natural resources, because the other side had received so much greater recognition and it is naturally nearer to our hearts. You do not know how much harder it is to appeal to people for these far-away things than to those that are so near and dear to them, and the things they can take hold of in an animate way.

I would like you to review with me just a few of the natural resources and the result of their Conservation, or the result of a lack of Conservation.

We will begin with the forests, because in our natural conservation we consider that the foundation of the fundamental principle of the conservation of natural resources. And what does the forest for us? What is the purpose of the forest? Why must we have them? Well, the forest makes soil in a way; that is, it makes humus matter, which is so large a portion of the soil that it may well be termed the soil. The forest is the only crop that grows that gives to the soil more than it takes from the soil. It also conserves the mineral in the soil that it takes Nature ages to produce by its slow processes of disintegration, and at the same time prevents the filling up of reservoirs, lakes and streams, and to that extent prevents the pollution of the waters. The forest is a great health resort, and why? Because it actually purifies the air. Its action is just the reverse of animals. It gives the air what we need and takes from it that which is detrimental to our health.

We must look a little into plant life and see what nature does that we may fully appreciate that point. I cannot take time tonight because of the late hour to go into the whole life of the tree, but I will say that its principal constituent is carbon, and it takes from the air the carbonic acid gas which is so detrimental to human beings and to all animals. It has a way of converting it into its own life blood in combination with the sap taken up from the roots, by the marvelous process in the leaves, by this little understood substance called chlorophyll, that has the power of converting this poisonous substance for us into the life of the tree, and then taking so much from it and giving it to the soil. That is a most important factor which is so often overlooked.

Then the forest is valuable as a wind shield for crops. And for the wood supply. Wood is demanded in all the industries or the arts, for almost all things we use.

These are the fundamental things the forest does for us. Are we not working for conservation of strength and health and human life when we are working for the forest?

While the General Federation takes up many phases of water Conservation, perhaps I may just say that we have irrigation, drainage, waterways, the deep canals for transportation, we have water power, which is the coming thing. This is something to be conserved, and which conserves our coal, which conserves the purity of our atmosphere by not having all the gases turned into it by the burning of the coal.

All these things it does for us.

And then the very last and most vital is the pure continuous supply of water, which all human beings and which all animals demand. It is, next to the air we breathe, the most important factor in animal existence. Are we not working for health, for strength and for life when we are working for this pure plentiful water supply, and does not that come pretty near working directly for conservation of human life? Have you anything you can bring forward that touches much more nearly the health, life, strength of human beings, the child, than this same conservation of water, which is a natural resource?

The soil is indirectly our staff of life. From it does not come our bread? Must not this seed fall into the ground, spring from the earth and be protected until it reaches maturity, and we have food? Many other instances might I bring forward had I time.

Then the animal kingdom is much more nearly related to human existence than we would think at the outset; but when we come to look more deeply into it we find this close relationship.

I so often come up against the saying, “Oh, I am so much interested in human life. I have no time, no thought, no desire to give to the animal kingdom. It is all right enough for you sentimentalists, but I am not interested." Yes, but even from a selfish point of view, if we do not care at all for any suffering, or anything which may come to the animal kingdom beside ourselves, it is of economic value to us.

I will choose but one example of the animal kingdom, and that is the birds, because it is said that all vegetation from the earth would cease if the birds existed no longer. It is very interesting to know that Longfellow appreciated this economic value of insectivorous birds long before there was any movement on foot for bird protection, and I wish you would all read the poem, the last of the Wayside Inn stories.

This very conservation of bird life is one of the things that is the great new problem of conservation of natural resources, and one in which you women take a hand and have the real control. I know you have heard so much about that I am not going to give you statistics as to what the birds do for agriculture. I am going to ask you a personal favor: that this fall when you choose your fall millinery, will you not think of your Chairman of your Conservation Department of the General Federation, and I beg you choose some other decoration for your hats. This is not sentiment. It is pure economics. You have no idea what you do when you wear these feathers, until you think really deeply into it, and I am not speaking of the egret, of the paradise feather, wholly, but of the less choice feathers. There is only one exception to this rule, and that is the wearing of the ostrich plume. That is a legitimate business and one to be encouraged. There is no reason why we should not use ostrich plumes if so we deem it best, but in regard to everything else in the way of feathers, let us turn over a new leaf for the fall. Will you not spread this gospel, not only to yourselves, but all the other women need to be asked to do the same thing? There are so many other articles, all the jets, the laces and ribbons. Will you not consider those things, even leaving out the sentiment?

I might cite for you many examples where conservation of natural resources works for the betterment of the human race, but I have just brought up a few of the most important.

Now, I want to say just a few words about the way to go to work to do some of these things. I will not go into the larger fields of forestry, or even into shade trees, except to emphasize the fact that while the shade tree is a very important one, and especially in the cities, we must never lose sight of the larger fact that after all it is not forestry, it does not stand for that, and that our arbor day, where we plant the one tree, should extend far beyond that. But I think one of our primary ways of working is to begin with the school, perhaps begin with the normal school. Many of the States have made great progress in that. I really have not the record of Indiana in that regard. I may be carrying coals to Newcastle to bring up this subject in Indiana. My own State, Massachusetts, stands very high in this line. Still I know there are many States that need this message. There is a great work to be done with the children, in making the school garden, and then the home garden; to teach the children to know what the soil is made of and how it should be treated, to make them love the growing flower and to make them respect the property of others. There we are laying the foundation of things for the next generation.

I know perhaps of no better book on the subject than that fine book for children, "The Land We Live In." I sent a copy into each State of the United States last year, with a request to each of my State chairmen that she do all she could to introduce that work into the libraries of her State, and the schools, feeling sure that if every child could read that book or hear it read, he would have a different idea of the natural resources and the need of natural Conservation. Some of the States have hundreds and thousands of copies of this book, and I am sure it is doing a great propaganda work.

I am going to tell you a little story of how I became interested in these things. It was before I was out of school myself, although pretty nearly so. It was when the welfare work began of taking the children out in the country from the slums in the north end. I was personally acquainted with one of the teachers, who was among the first to take the children out in the fresh air to breathe and see the grass and flowers and trees that they had never seen before. One little boy, after he had looked around in amazement —it was in the fall of the year—saw the bright red apples on the trees, and he looked up and said, "Apples on trees, by God!"

It is overwhelming, isn't it? I don't wonder that you gasp at it. But look a little more deeply into it and see the pity of it. That child had been born and bred in the slums of the north end of Boston and actually had never seen apples on trees. He had seen apples in barrels. How did that poor child know that they did not grow in barrels? No, it had never occurred to him. They did not teach, in those days, the principles of horticulture in the schools. Was it not pathetic? Doesn't that teach a lesson? That has come home to me many and many a time. I actually believe that was the foundation of my interest in Conservation. I think I was born with a love of the soil. And the story of the boy added to that, made me feel that I must know something about nature, about the fundamental principles, about the other side of life, the vegetable kingdom that supports the human life. Those two things combined taught me a lesson that I never, never could forget, and I wish you would think them over.

I will say to you this one message, while you are working for this thing of prime importance, the conservation of life, for which this Congress has stood at this fall meeting, do not forget that the conservation of life itself must be built on the solid foundation of conservation of natural resources, or it will be a house built upon the sands that will be washed away. It will not be lasting. I thank you. (Great applause.)

President WHITE—I want to have read into the record of this evening's proceedings, by title only, a paper which was intended to have been read by Mrs. Elmer Black, of the International Peace Congress. She was expecting to be here and was on the program originally, but we learned that she could not get back from Europe in time to be present. She sent on her contribution in the way of a paper. It will be published in the Proceedings. The title is "War is the Policy of Waste—Peace the Policy of Conservation." (For Mrs. Black's paper, see Supplementary Proceedings.)

President WHITE—I wish to say further that your very gallant Sergeant-at-Arms, Col. John I. Martin, wants to address the ladies for just three minutes.

Col. MARTIN—Madam President, Ladies and Gentlemen: For the very cordial manner in which you have carried out the suggestion made by our popular, esteemed and whole-souled President of the National Conservation Congress, the Hon. J. B. White, that I briefly address this association, and for your kind invitation, I return my most profound thanks.

Nowhere in this wide and extended country can there be found a grander association of noble, unselfish women, planning, acting, counseling upon the great subject of conservation of human life than this organization under whose auspices we are all assembled this evening. Nowhere can there be found an institution more efficient for good, more blessed in all its labors of love and humanity, more universal in its application to the advancement of love and sympathy, stimulating education, encouraging enlightenment and scientific and humane development and morality, than an institution of the character of this band of noble women, engaged in such a magnificent undertaking as your association promulgates. Fully appreciating the fact that as the world grows better and people become more educated and more honest in their endeavors to espouse the cause of the weak against the strong, and the right against the wrong, then such organizations as the Ladies' Auxiliary of the National Conservation Congress will be heralded as the very acme of perfection along the lines of the contemplated work in which you are now engaged. (Applause.)

In all ages of the world chivalry has yielded to feminine beauty, patriotism, loyalty and devotion, and I am sure that our popular President, Captain White, his efficient officers and all the members of the National Conservation Congress are at all times ready to listen to advice and counsel from the fair sex, and to surrender with wise discretion to all her laudable undertakings. (Applause.) Wherever cheeks have turned pale with waiting, weeping and watching, there was woman's presence to cheer, to comfort and to save, and in her garden of the sun heaven's brightest rose is yet to bloom, and when it comes it will be the bright-hued mission of a heavenly charity. The poets have sung no truer rhyme than that inscribed by one of your own number:

> "Woman, not she with trait'rous lips her Savior stung,
> Not she denied him with unholy tongue,
> She, when Apostles shrunk, did dangers brave,
> Last at the cross, and earliest at the grave."

God Almighty, in his crowning work of creation, gave woman to man, made weakness her strength, modesty her citadel, truth, gentleness and love her attributes, and the heart of man her throne. (Applause.)

The Chairman—The meeting will stand adjourned.

TENTH SESSION.

The Congress convened in the Murat Theater, on the morning of October 4, 1912. It was called to order by President White.

President White—We will put things through on the ten-minute plan this morning, so as to give every one a chance who has a place on the program. Today we have reports from the committees, and elect our officers. We can then get ready for another Congress, for we are all going into the field, we are going to work for Conservation, and the whole country is going to take it up. We will give them the text, and the press will take it up, the politicians will take it up, and we will each be a committee of one to go forth through the country and make this Conservation idea a potent force that will change and correct legislation for the benefit of all the people. (Applause.)

Mr. A. B. Farquhar, who was to speak this morning, spoke yesterday, and therefore his address, for which a great many expected to be present, will be printed and you will have an opportunity to read it. Every one should subscribe for as many copies of the Proceedings as he can afford, for distribution among friends. It is without doubt going to prove to be the greatest book on conservation of human life that has ever been written. These papers are scholarly, and they are true, and the truth will prevail if we can only get people to read and to think. We want to give you all an opportunity to subscribe for this publication, which will be published as soon as possible, and will only cost one dollar, and those who pay this dollar will be entitled to membership in this organization next year, so that if your Mayor, or your Governor, or your civic body does not reappoint you, you are sure of membership next year, because you have paid in your dollar and subscribed for the book.

Dr. Livingston Farrand, of New York, will now speak to us on "The Problem of Tuberculosis."

Dr. Farrand—Mr. President, Ladies and Gentlemen: The problem of tuberculosis in the United States is simple in its outlines. Stated in their lowest terms, the figures which describe it are sufficiently impressive and appalling. Increasing experience and added knowledge serve only to confirm earlier estimates and to emphasize the seriousness of the situation which confronts us. The vital statistics of our country are notoriously faulty and incomplete, but the lesson they teach must arrest the attention of every thinking citizen.

According to the census of 1910, treating the non-registration area on the same basis as that from which mortality reports were recorded, there were 150,000 deaths from tuberculosis in that year. This is, of course, an under statement by many thousands. Rigidly conservative estimates agree that the mortality from tuberculosis in this country is at least 200,000 each year and very probably considerably more. Let us for the moment, however, deal with the demonstrable facts and not enter the field of estimate.

The real problem is not the number of deaths from tuberculosis, but the number of living cases of the disease. In calculating this different methods have been employed. For many years, the ratio of three living cases to each death was used as an index of the situation in any community. It was quickly realized by those familiar with the situation that this proportion was far too low, but with our almost total lack of registration, figures to demonstrate the discrepancy were not available. With the improvement in recording the facts of disease in certain typical centers of population, it became certain, however, that a ratio of five to one was not only conservative but below the truth.

More recently records of great value have been obtained which confirm the convictions of experts and allow still sharper definition of the problem.

It has remained for the city of Cleveland to work out during the past two years a system of tuberculosis registration and administration which is undoubtedly the most complete in the country for a community of its size and complexity. Without going into details of method, notification, and registration have been brought to such a point in Cleveland that of all the deaths from tuberculosis now occurring approximately ninety per cent. have been previously recorded and under observation by the Department of Health, before death is reported. This is an achievement for a city of its population of extraordinary significance. There are in round numbers something over 700 deaths a year from pulmonary tuberculosis in that city of 600,000 inhabitants. There are in register and under observation at this time approximately 4,600 cases of tuberculosis. Allowing for the ten per cent. in the mortality not reported before the death, it is obvious that the number of living cases is over seven times the number of deaths and with slight allowance for the very large number of active cases in any community which have not yet come to diagnosis, we can demonstrate in that city a ratio of eight living cases for each death.

It is singularly fortunate that this demonstration has taken place in a community of sufficient size to include the problems in some degree of all our larger cities and to be regarded as reasonably typical of the situation throughout the country. It has also been shown that except for certain centers, where the problem of congestion is extraordinarily prominent, the rural situation in the United States does not differ appreciably from that obtaining in all cities and towns so far as the presence of tuberculosis is concerned. I have no hesitation, therefore, in asserting that we must from this time on raise our figures and use a ratio of at least eight to one in calculating the prevalence of tuberculosis on a basis of the recorded deaths from that disease.

Apply these figures to the country. The Bureau of the Census indicates 150,000 deaths a year. On this basis we have 1,200,000 living cases of tuberculosis. Let us not forget, however, that 150,000 recorded deaths is far below the actual number, for it is easy to show in most of our communities that many deaths properly to be assigned to tuberculosis are reported under other terms, and the area of the United States from which no statistics are forthcoming includes precisely those States where the mortality is high and the prevalence of tuberculosis demonstrably widespread. We are still absolutely certain that the mortality from this disease is at least 200,000 each year, and the number of living active cases more than a million and a half.

Such, numerically speaking, is our problem. What are the efforts for its solution?

Since the discovery of the bacillus as the cause of this disease in 1882, an organized campaign has gradually been developed. The inferences from the discovery of the cause were perfectly inevitable and indicated the lines of operation. It became entirely clear that tuberculosis, being due to a specific germ, was infectious, and it was equally clear that the bacillus and its life history being known, the disease was theoretically preventable. Here, too, the outlines of the campaign are simple, even though the details of operation are varied and the end in view baffling to attain.

It was inevitable that the first sporadic efforts based upon slight experience should have been more or less random, and that years of trial and proving should precede the establishment of definite method. Some degree of order is, however, emerging, and we are witnessing an increasing clearness of purpose and definition of attack in the preventive movement against tuberculosis which is now sweeping over the country and the civilized world.

While recognizing the unfortunate complexity of the social conditions whose maladjustment is perhaps the chief underlying factor in the problem, while recognizing fully the obligation to lend all possible aid to the betterment of those conditions, the administration of the campaign against tuberculosis has still conceived its specific task to be a direct attack on the sources of infection; this, because experience has indicated such procedure to be the best and most feasible means of prevention. As the logical conclusions of laboratory discovery and clinical experience began to express themselves in organized movement, it was recognized that the preliminary task in prevention was one of education; an education which should impress upon the public mind not only the fundamental facts that tuberculosis is infectious and preventable and the methods of its infection and prevention, but an education that should bring about an improved knowledge of public and private hygiene, and particularly an education which should create a public sentiment which could appreciate conditions and would support and even demand those measures which expert advice and experience might indicate as necessary. This educational propaganda, now so familiar, has been in the United States the particular province of private organization. The union of professional and lay effort in this latter day crusade has been one of the most inspiring of social phenomena and has already resulted in accomplishments of imposing dimensions.

With our political organization such as it is, this enlightened public sentiment is an absolute essential if the responsibility for the situation is to be an official one, and not left for the suggestive and stimulating but less final and efficient efforts of private philanthropy.

The insistence upon official responsibility has been made an essential point in our American campaign and toward its intelligent acceptance by public authorities all efforts are directed. As may well be appreciated, the attainment of this desired end is slow, even though ultimately inevitable.

In planning the campaign, an ideal program was not difficult to lay down. It included as fundamental:

1. The education of which I have spoken, not only as it applies to tuberculosis but as contributing to the solution of that problem of misery which is, after all, the chief problem of the day and which reduces in the last instance largely to terms of good or ill health.

2. Enactment or enforcement of protective laws of which the basis was that notification and registration agreed upon as preliminary to official knowledge and control of the situation.

3. Adequate institutional provision for all classes of cases; the sanatorium for the curable; the hospital for the advanced and hopeless, and dispensaries for early diagnosis and as centers for that all-important field of action, education and treatment in the homes of the poor.

The developments of the years have not served appreciably to modify the main features of this program. Emphasis has shifted from time to time and will continue so to shift, but the fundamentals remain more firmly established than ever.

In developing the movement in this country, the most effective means of stimulating action in our various communities has been the voluntary association for the prevention of tuberculosis. In organizing these societies the local community has been recognized as the essential centers of action. The effort has been made, therefore, to obtain in every community of considerable size an organization embracing elements both medical and lay which shall charge itself with the task of securing adequate official treatment of the tuberculosis problem as it there presents itself.

In many of our commonwealths such organizations can best be brought about through the action of a State society, whose special function becomes one of organization and of securing desired legislation. In other cases the initiative is local in origin. Where State societies exist, these act as co-ordinating agents for the affiliated local societies, and the National

Association for the Study and Prevention of Tuberculosis acts as a clearing house for them all.

It will be seen at once that such organization is but preliminary, and would be entirely futile, did it not result in preventive measures of a definite sort. There is, however, no other index equally valuable of the vigor and growth of this movement in the United States. Speaking from the national point of view, the organized campaign in this country has been in existence exactly seven years. In 1905 there were in the entire country but twenty-one of these societies, while at the present time there are no less than 660, working in co-operation and presenting a united front to the enemy. There is no considerable area that does not contain some such center of intelligent action.

The carrying out of the program outlined a moment ago is the special function of the organized movement. In the development of this program it is historically interesting that it was institutional provision for tuberculosis that first obtained support. It was the sanatorium for the cure of curable cases with its peculiar appeal which first engaged attention. From our present point of view, it was perhaps not the logical beginning, but it was certainly the obvious and perhaps the most fortunate point of attack. The sanatorium with its promise of restoring to a wage-earning capacity those unfortunates who formerly had been regarded as doomed to a speedy and inevitable death, was peculiarly fitted to arrest public attention and to engage public support.

As the movement for sanatorium establishment developed momentum, attention turned to the need of special dispensaries as logical centers of preventive work. Time will not permit even an outline of this phase of the problem. Suffice it to say that with the first general survey of the movement in the United States, six years ago, there were in the country but eighteen dispensaries exclusively devoted to tuberculosis. There are today more than 400 such foundations and their number is increasing at a rapid rate. All those who deal hand to hand with the problem become impressed at once with the fact that tuberculosis is pre-eminently a disease of social life, of living and working conditions. In the absence of adequate institutional facilities it is unavoidable that the problem should be attacked in the homes and workshops of the people, and with such weapons as may be at hand or which can be devised. With early diagnosis and careful instructive nursing supervision, much can be done even in the distressing conditions which characterize the crowded and poorer quarters of our great cities. The center of activity in this field is everywhere the dispensary, and the elaboration of its function to include supervision in the homes of indigent patients has been one of the most interesting and important of recent developments.

The third and possibly the most important aspect of institutional provision was the last to be taken up with energy. Every survey of our equipment during recent years has served to emphasize the shocking lack in our facilities for the care of advanced cases of tuberculosis. It has become increasingly evident that as centers of infection the consumptive in the advanced stages presented the most serious problem. Equipped as we were, with a healthily growing movement along educational, sanatorium and dispensary lines, the time seemed ripe for a vigorous attack on this point of weakness. The result has been that during the last four years there has been a concentration of energy in this direction and a notable advance has been made. Without pausing to specify various kinds and degrees of hospitals and sanatoria for the treatment of tuberculosis in the United States, it is encouraging to note that we now have over 500 in the country, as compared with 111 seven years ago. The number of beds contained in these institutions is approximately 30,000, a number small when compared with the need, but encouraging when compared with the situation but a few years since.

The third feature of the program already mentioned, that of legislation, is less susceptible of numerical expression, but it is in many ways the most fundamental and most significant of advancing intelligence. The principle of compulsory notification and registration has been insisted upon from the outset, and it has now come to be fairly generally accepted in all parts of the country. With few exceptions the more important States provide for registration by enactment either of the Legislature or of the State health authorities. In most of our larger cities local regulations are also on the statute books. Unfortunately the enforcement of these regulations is far behind their expression, but the situation is rapidly improving, and the example of such cities as New York in initiating the principle, and of Cleveland in demonstrating its possibilities, is of inestimable value.

In dealing with the question of public hospital establishment, the best adapted political unit has caused much embarrassment where a given community is not large enough to support an independent institution. Federal provision is agreed upon as being out of the question. The State as such is in most instances regarded as having the same limitations to a lesser degree as the national government. It is fairly generally accepted that where the municipality is of sufficient size it should accept responsibility for its problem. In those sections where communities of lesser population are the rule, the county is now in the focus of attention.

Little difficulty has been encountered in procuring the necessary legislation for local and county institutional provision. We have now reached the point where the possibility of mandatory State legislation is being considered with care and some favor. In this connection one should note the recent passage by the Legislature of the State of New Jersey of a law which undoubtedly represents the most advanced legislation in the United States and probably in the world. Without going into details, the law in question provides for the establishment of special tuberculosis hospitals in all the counties of the State, for the payment by the State of a certain sum ($3.00 per week per patient) toward the maintenance of such hospitals, for the compulsory segregation in such hospitals of dangerous and incorrigible cases of the disease, and for the general supervision of these provisions by the State Board of Health, though the primary responsibility is placed upon the local health officer. This legislation is of the highest interest, not only in its promise of results, but as an enactment into law of principles formulated as necessary by expert experience even though in advance of public appreciation.

Reaching into every field of social activity as this campaign must do, it is inevitable that new phases of importance should successively make their appearance and demand attention. I should say that perhaps the most striking is the essential importance of the child in the tuberculosis problem. With improved methods of diagnosis and wider facilities for examination, there has been shown a prevalence of tuberculosis in children of school age that is most alarming. It is a conservative statement that there are today in the public schools of the United States 100,000 children who will die of tuberculosis before they reach the age of eighteen if the present rate of mortality be continued. A very recent estimate presented by the United States Bureau of Education states that at least 15,000,000 children now in attendance in the schools of the United States are in need of a physician's attention, and that of this number 1,000,000 have or have had tuberculosis. It has become clear that if our educational campaign in the interest of preventive medicine and public health is to achieve success, the attention must be concentrated upon the coming generation rather than upon those who have already passed their years of plasticity.

We see, then, on every hand the tendency to attack the problem in the schools, and this not only by the establishment of provision for open air teaching and the improvement of the undernourished and the predisposed, but upon insistence of regular and intelligent instruction as to the prevention of disease.

Such in its general outlines is the plan by which we are working. With such a situation and with such a campaign what then is the outlook? I have little sympathy with the enthusiasm which deals in specific predictions or which assigns a date for the practical achievement of theoretical possibilities. It is perhaps inevitable that an impatient public should demand results before definite results can be forthcoming. There is, on the other hand, a corresponding obligation for conservatism in expression when indicating probable or even possible results. A drop in the mortality curve of a slowly developing and slowly progressive disease such as tuberculosis, is not a matter of months but a matter of years. It is unjustifiable to expect results from the specific campaign against tuberculosis in an observable diminution of mortality for some years to come. I believe, however, that we have reached a point where our equipment is such as adequately to test our basis of operations and to warrant an optimism as to the future if our reasoning and method be correct.

Believing as we do, that the soundness of the procedure is certain, it would seem reasonable to expect a response in the mortality tables within five years, and that ten years should afford indisputable proof.

There is, of course, no doubt that tuberculosis is diminishing and has been diminishing for a generation. This decrease is not to be assigned to the specific warfare against the disease, but is doubtless correlated with other factors. It is uncertain whether we are to assign as its cause the general improvement in public hygiene or whether there may be perhaps an acquisition of immunity gradually extending through the civilized world. In my own judgment this decrease in the prevalence of tuberculosis is associated with the improvement in hygienic conditions which has been so marked during the last fifty years. I believe we are justified in expecting an acceleration in this diminution as a result of the specific measures now being adopted not only here but in Europe. While we cannot interpret them with confidence, there are already appearing certain figures of possible significance. It should not be forgotten that the first result of all concentrated activity and interest is a greater accuracy in mortality and morbidity statistics, and that an actual decrease in tuberculosis might appear in official reports as an apparent increase in the disease.

Taking all these factors into account and viewing the situation candidly and with all the precautions possible, I do not hesitate to assert that optimism as to the future is justified, and that the end of the present decade will witness the beginning of another drop in the mortality curve comparable to that which was seen in the closing years of the last century. (Applause.)

President White—Dr. W. C. Mendenhall, of the United States Geological Survey, at Washington, was expected to be here this morning to speak upon the subject of "Water as a Natural Resource." He is unable to be present, and Mr. Jacob P. Dunn, Secretary of the Indiana Historical Society, will now have ten minutes to discuss "The Conservation of Navigable Streams."

(For Mr. Dunn's paper see Supplementary Proceedings.)

President White—The subject of the next address is "Social, Industrial and Civic Progress." It is to be a review of fifty years of what has been done in labor economics, by one who has given a great deal of study to the subject, Mr. Ralph M. Easley, of New York City, Chairman of the Executive Council of the National Civic Federation.

Mr. Easley—Mr. President, Ladies and Gentlemen: In view of the fact that the work of the Committee on Civics was not outlined at the time it was organized, and as it was the desire of the national officers of the Conservation Congress that its work should not duplicate nor overlap the work of other organizations, the mapping out of a practical program for the committee was deferred until this meeting of the Congress.

Recognizing this situation, the officers of the Congress suggested that, as Chairman of the Committee on Civics, I should briefly review the progress that has been made by others in this country along industrial, social and civic lines. This seemed to me wise because at a gathering of this kind, which has discussed conditions that call for improvements, it might be helpful to note what progress our country has already made along these lines. To look back adown the slopes we have so painfully and undauntedly climbed in advancing to our present plane of material and moral welfare, far from inspiring us with a smug complacency, should heighten our resolves and give renewed energy and freshness of spirit.

Another reason for accepting the suggestion is that I had just read in an English newspaper a sweeping and vitrolic criticism of our social and civic conditions. Our unkind critic spoke of us as a people so utterly bound up in the worship of the "almighty dollar" that we had lost whatever social vision might have illumined the minds of our fathers. To all sense of social righteousness we were as a people pitiably indifferent. In mill, factory and mine our working people slaved; in tenement and farmhouse our poor lived, little if any better than the poorest of Europe's poor; our sick and otherwise helpless were scarcely given a thought. Politically we were rotten to the core, statesmanship and graft going hand in hand.

That, in short, ours was a dog-eat-dog civilization, and that the only direction in which light might be seen breaking was in the "fact that making headway among the wisest and most far-seeing Americans was the conviction that American institutions were a failure!"

The editorial concluded with the statement that if any one considered that view a biased one, all such skeptical readers need do was to acquaint themselves with the writings and speeches of American sociologists and magazine writers or to converse with any of that "dwindling proportion" of our well-informed citizens to whom human values are not a mere academic phrase or an abstraction.

It is unnecessary to point out that our English critic might have used his columns to better advantage if he had differentiated between the sociologists and magazine writers who seek our country's good and those who seek only its destruction—a very important differentiation to make at this time.

In fact, our critic may be a Socialist, who is only passing along to England the general cry of the pessimists of this country, that “whatever is, is wrong”; and that there is a great unrest in the industrial world which will, sooner or later, burst out in volcanic force and engulf us in a terrible cataclysm—all of which is unspeakable rot.

I think I am in a position through the organization with which I am connected (composed as it is of the representatives of the great labor, agricultural, manufacturing, banking, commercial, educational and professional organizations) to know something about this “great unrest” upon which the Socialists and other radical writers and speakers declaim so much, and I can assure you that the only unrest in the industrial and social fields that I can discern is that wholesome, normal unrest which comes through the education of the people, and therefore a better understanding of their rights as workers and the translation of that knowledge through the labor unions and other social and economic organizations into concrete demands for better living conditions.

But let us take a birdseye view of the situation and see whether we are advancing or going backward. I think you will agree with me that the following bare outline of a few of the important achievements and the work now being done by organizations and movements of public-spirited citizens is inspiring and encouraging.

Let us start with the industrial gains.

The American Federation of Labor and the railway brotherhoods have in the past twenty-five years secured better wages and working conditions for millions of wage-earners and the eight-hour day for hundreds of thousands, and they have developed a system of collective bargaining and methods of conciliation and arbitration that are reducing the number of industrial disturbances. To get a clear idea of what this means in terms of progress, let us consider that while in the past six months 500,000 coal miners and their employers have made contracts covering wages, hours and conditions of employment for a term of four years; all the railroads east of Chicago are arbitrating their differences with their thousands and thousands of engineers, trainmen, conductors and so on; the hundreds of thousands of carpenters, bricklayers, painters, plasterers and others of the thirty-five crafts involved in the building industry have made contracts with associations of builders all over the land from Maine to California; while the publishers of the great daily newspapers throughout the United States have made a five-year contract with their printers, pressmen, stereotypers, etc.; and the street railway employes in many great cities and many others of the 135 crafts belonging to the American Federation of Labor have made satisfactory contracts with their employers—I say, let us consider that while this is what is going on today in this country, we shall not have to go very far back into history to find the time when it was a penal offense for a man to join a labor organization, or for workers to ask collectively for an increase in wages, and to find that, while we are now legislating in the interest of the employe for a minimum wage, at that time the effort of legislation was for a maximum wage in the interest of the employer.

In the meantime, the State factory legislation has revolutionized the methods of sanitation in the workshops of the country and is safeguarding better and better the lives and limbs of the workers.

Employers are making increased provision for the welfare of their employes through sanitary and safe work places, opportunities for recreation and education, model homes rented or sold, and relief funds for sickness, accident and death benefits, as well as old age pensions, all affecting millions of railroad, factory, mine and department store workers.

The National Child Labor Committee has led a campaign that in ten years has secured wholesome legislation in practically every State in the Union, reducing hours of labor, prohibiting children under fourteen years of age from working in factories, mines and mills, and preventing night work for women and children in many places.

The tenement house reform movement in New York alone, where the problems are greatest, has made seventy-five per cent. improvement in fifteen years; and as an example of the growing recognition of big business of its social responsibility, it may be pointed out that when the Supreme Court upset the Tenement House Law, and by a decision wiped out all that had been accomplished in twelve years through the tenement house agitation, the allied real estate interests in New York joined with the tenement house reformers in securing the passage of a State law and a city ordinance correcting the defects.

Amazing in magnitude and usefulness are the health organizations, public and private, devoted to securing more efficient methods of sanitation and the prevention of disease, recent statistics in New York City showing as a result of such work that the mortality rate has decreased fifty per cent. in fifty years.

There are various national and local organizations devoted to the protection and education of the millions of immigrants from all parts of the world who have landed on these shores in the past ten years, and whose assimilation and adaptation to American standards and conditions have constituted one of the problems of the age.

There are thousands of non-sectarian hospitals and institutions for the scientific care of dependents, defectives and delinquents.

Splendid work is being done by the great charity organization movement which is teaching independence and thrift through its penny provident societies, and which has organized some of the most important preventive and remedial agencies.

The National Federation of Remedial Loan Societies covers twenty-eight cities, where societies lend money to the poor at reasonable rates to protect them from the loan sharks, the New York organization alone having a fund of millions for this purpose. A rapidly increasing number of large employers have changed their attitude towards their employes, in that they now aid instead of discharging those who incur debt—the latter policy having played directly into the hands of the loan sharks.

The National Association for the Promotion of Industrial Education has brought the manufacturers' associations and the labor organizations into harmonious support of the measure providing a federal appropriation of $5,000,000 for industrial education of the young workers in towns and cities, whether in factories, stores or offices, and including domestic science for the girls. The measure also provides an equal amount for the sons and daughters of the farmers.

The tremendous program of constructive work undertaken by the United States Bureau of Labor and the Bureau of Mines in the interest of the workingmen and by the Department of Agriculture for the farmers should alone silence our English scoffer. The recent establishment of the Children's Bureau is an achievement of which humanitarians may well be proud.

The public school system and other free educational institutions enable the children in this country today to receive twelve times as much schooling as their grandparents—a tremendous factor in our advancement of itself and one that readily accounts for much of the unrest without which no progress could come.

The universities, especially the State institutions, have in the past ten years enlarged the scope of their work to such an extent that many of them can be classed as leaders in what are termed the "uplift movements" of the day. A complete catalogue of the public work done by the University of Wisconsin alone would be a revelation.

The Playground and Recreation Society of America and other recreation movements are assisting in the development of children's playgrounds in parks and schools and are bringing health and good cheer to congested centers.

The Association for Labor Legislation is working jointly with the American Medical Association to safeguard wage-earners against occupational diseases.

The American Bankers' Association is organizing a movement to help the farmers of the country develop idle land in the effort to decrease the cost of living.

One of the most encouraging signs to those who are alarmed over the high cost of living, and that is about all of us, is the recognition by the farmers, State agricultural colleges and railroads, of the necessity of introducing up-to-date methods for raising and marketing grain, live stock, fruit, dairy produce, etc. Only last week I read the announcement of a convention called in Kansas, where three thousand delegates will meet to consider this very question of improving the methods of farming. These delegates will represent not only farmers but also the bankers, merchants, wage-earners and all divisions of society.

It would take a volume to describe even in outline the great social and economic reforms being promoted by Mr. Andrew Carnegie, Mrs. Edward H. Harriman, Mrs. Russell Sage and Mr. John D. Rockefeller, whose

$60,000,000 gift covers the promotion and development of the high school system in the Southern States and the promotion of higher education throughout the United States, while his Sanitary Commission has discovered and is eradicating the hookworm disease in the South. The Carnegie Institute of Washington, with an endowment of $22,000,000, was founded to encourage in the broadest and most liberal manner investigation, research and discovery, and the application of knowledge to the improvement of mankind, while the Carnegie Foundation for the Advancement of Teaching, with its $15,000,000 endowment, provides retiring pensions for the teachers of universities, colleges and technical schools. The Russell Sage Foundation, endowed by Mrs. Sage with $10,000,000, has, for its purpose the improvement of social and living conditions in the United States of America.

There are the tremendous achievements through the institutional work of the churches of all denominations. Three-fourths of the efforts in the live churches of today are devoted to material welfare, as is evidenced by the especial care of the orphan, the sick and the poor on the part of the Catholic Church; the great Hebrew philanthropic and educational agencies; and such single illustrations as the social work outlined in the handbooks just issued by Trinity and St. George’s parishes in New York—the former being a revelation to those who believed that the millions of Trinity Church were being used only for commercial profits.

The Young Men’s Christian Association, with its tremendous energy and enthusiasm, while organized primarily to promote the spiritual growth of young men, has lately, under its “physical and social well-being” clause, gone into the field of industrial betterment with conspicuous success.

The Men and Religion Forward Movement and the Federation of Churches, representing many million members of Protestant denominations, have recently adopted broad programs of industrial and social reform.

There are the movement to suppress the social evil, known as the Federation of Sex Hygiene; the Anti-Tuberculosis Society, with its wonderfully comprehensive and successful efforts in fighting the great white plague; the Red Cross Society which, in addition to relieving distress in great disasters, has fostered with marked success annual competitive drills of “first-aid” crews from the mines; the Boy Scouts of America, inculcating patriotism and good citizenship; the National Consumers' League; the New York Museum of Safety and Sanitation; the Prison Labor Reform Association, and hundreds of other organizations and movements devoted to human betterment too numerous even to mention by title.

And last, but not least, there is the educational work being done by the National Civic Federation through its Departments on Conciliation, Compensation for Injured Workmen, Industrial Welfare, Pure Food and Drugs, Reform in Legal Procedure, Regulation of Interstate and Municipal Utilities, Regulation of Industrial Corporations and Uniform State Legislation.

As much of the work of the various departments of the National Civic Federation called for uniform State legislation, a special department was organized to co-operate with the Commissioners on Uniform State Laws.

The importance of uniformity to all business and commercial institutions is clearly recognized, when we consider that our larger corporations—such as the railroads, telegraph, insurance, banking and trust companies and, in fact, so far as taxation is concerned, all manufacturing concerns whose plants are in different States—are subject to forty-eight masters, each with a mind quite different from that of the others. The "interminable" law's delay, the clashing of the States upon the question of regulation of corporations and combinations, the diversity of State laws on ordinary commercial matters, such as warehouse receipts, bills of lading and negotiable notes, the urgent need for a uniform bill on compensation for industrial accidents, all give emphasis to the need for uniformity. But even this chaotic legislative situation shows encouraging signs of clearing up.

So much for progress along industrial and social lines; but we have made and are making just as great progress in this country along other lines that affect the general welfare of the people. And also our ethical standards and our aspirations are conspicuously higher. For instance:

Within the past twenty years there has been a most remarkable gain in the popular concept of the relation of industrial, railway and municipal utility corporations to the public. The large corporations called trusts have been taught even in the past five years that they must recognize certain "rules of the game" that give their competitors a chance, and what is wholesome about this from the ethical standpoint is that they now admit the justice of these changed conditions.

The abolition of rebates and free passes and the placing of railroad, telegraph, telephone and express companies absolutely under the regulatory power of the Interstate Commerce Commission are so far-reaching that the benefits to the people are impossible to measure. From federal regulation of railroads, it was only a step to State regulation of street railway, gas and electric light companies.

The idea that railways or big corporations are masters of the people has been dissipated.

Today, through insistent demand of the people for publicity, it can be said that the big business of the country is being done behind glass doors. The improved methods of doing business adopted by banks, trust companies and insurance companies during the past five years would alone justify this statement.

In practically five years, thanks to the great educational work of the National Conservation Congress, there has been a complete transformation of the public mind in the matter of proper control of our natural resources, such as our public lands, timber and water power. It was not many years ago, when I was living in the West, that it was considered a smart thing to “grab off” all public land that one could get hold of. This was generally accomplished by taking land in the name of your mother and father and all your children, past, present and future, and it was not bad form even to use your neighbor’s name in taking up claims. I found my own name had been used in three or four different counties by some of my ambitious neighbors.

Politically speaking, we have progressed from the state where our elections were great public scandals and where primary elections were “free-for-alls,” with no legal status whatever, to a day when, thanks to the Australian ballot law, ballot-box stuffing is practically unknown and primaries are generally so conducted that the voters control.

Campaign contributions that were largely responsible for corruption in politics and legislation are now by law made public to the world.

The initiative, referendum and direct primary have been adopted in some form in two-thirds of the States and in over two hundred cities the commission form of government, often with a recall attachment, has been adopted. These measures, whether they prove to be practical reforms or not —and there are many who doubt that—undeniably testify to the paramount power of those agitating for a so-called “progressive program,” they all being opposed by what are termed the “reactionaries.”

The civil service, from being a thing detested by nearly everybody twenty years ago, is so popular today that political parties are vying with each other to see which can include the largest number of civil employes. The President has just ordered the 35,000 fourth-class post-masters be taken from under the political brokerage offices of the Congressmen and placed under the civil service law.

The government of cities, which has been the burning shame of this country, as it was in the early days of every other country, is slowly but surely becoming more decent and effective. The work of the National Municipal League, the hundreds of local municipal reform associations, and the National Bureau of Municipal Research with its local bureaus, furnish abundant evidence of the truth of this statement. The Bureau of Municipal Research is not only making an exhaustive and painstaking analysis of administrative methods in many large cities, and installing more up-to-date and efficient systems, but it also has prevailed upon the Federal Government to have a similar investigation made in its various departments. It has, in addition, organized a training school to meet the demand for municipal experts.

The administration of justice and the influence of wealth upon the decisions of the courts have been revolutionized in the past ten years. It used to be charged that the criminal courts convicted only the poor and released the rich, whereas today the penitentiary that has not a half dozen or more bankers or rich malefactors within its walls is the exception. There is no man or corporation so powerful today as to be immune from attack by the government when violating the law.

The American Bar Association and the National Civic Federation are jointly working to bring about a reform in legal procedure which will wipe out unnecessary delays and cost in litigation, thereby opening the courts more freely to the wage-earner.

Five years ago there was no such thing as a Pure Food and Drug Law. Today there is a federal act which has been made the basis of legislation in thirty-five States, and in another five years it is likely to be practically impossible for misbranders or adulterators of food and drugs to live outside of our penal institutions.

The rural free delivery, the postal savings bank and the parcel post are all great advances from which the farmers largely benefit.

The building and loan associations and savings banks, unknown in early days, are great aids to wage-earners.

In other words, reform is writ large over all sections of the country and all classes of society. There are:

Over two thousand boards of trade and chambers of commerce, at least half of whose efforts are directed towards municipal and industrial reforms, and the other half to commercial reforms;

Thousands of church societies and committees aiding in the improvement of industrial, social and political conditions in their respective localities;

Thousands of women's clubs, representing over two million of the brightest and most energetic women of our nation, devoted to securing civic improvement, factory legislation and reforms in public schools, to spreading information upon social hygiene and domestic science and working for the protection of women and the redemption of unfortunate ones;

Thirty thousand labor organizations, whose purpose is not only to secure better working conditions, better wages and a shorter workday for wage-earners, but also to lift them to a higher plane of citizenship, and

Millions of farmers who, through granges, alliances and institutes, are working not only to improve the home life on the farm, but to educate their children in the use of better and more scientific methods of production.

Pretty fair, is it not, for a people whom our English critics and our American Socialists say are bereft, or almost so, of a social sense?

And it must also be kept in mind that this resumé does not refer to progress in science, invention and the arts, nor is attention called to the fact that never before in the history of this country were the basic conditions better than they are now, despite the fact that a national political campaign is supposed to be on.

But while the progress made has been so tremendous that we do not realize it, on none of these lines is it contended that anything near the ideal has been reached. There are yet very many black places and perplexing problems demanding attention on the part of those who love their fellow-men. But the same courage, intelligence and humanitarianism that have accomplished so much will not now falter, but will press forward.

Many in this audience may conclude that I am unduly optimistic and that I am able only to see the good, but I can assure you that I know something as well about the ills of society; for instance, I could cite from the records of the Welfare Department of the National Civic Federation alone a catalogue of industrial horrors showing where greedy and thoughtless, if not unfeeling and criminal, employers are grossly and outrageously mistreating the wage-earners in their employ, paying them atrociously low wages, working them excessively long hours and giving no consideration to the comforts or decencies that a humane employer would furnish. But also from that same record I could show that all such evils are being met by other employers, justifying the belief that, through education and proper agitation, the remaining sore spots can be removed. Last year one great corporation alone spent five millions of dollars in betterment work, including a gradual shortening of the working time in its plans for improving conditions, and several large corporations, operating night and day, have gone from two twelve-hour shifts to three eight-hour shifts without decrease of pay.

As a concrete and striking example of the power of agitation and education, there can be no better illustration than the present widespread sentiment in favor of legal enactments requiring compensation to injured wage-earners in lieu of the old employers' liability system. Through the work of the National Civic Federation and co-operating bodies, this complete reversal of policy has been brought about in four years, fourteen States having already passed workmen's compensation laws. The legislation, both Federal and State, which is now being secured, makes the industry bear the burden, while before the wage-earner took all the chances, did all the suffering and, if, after long-drawn-out litigation, he finally got anything in the way of damages, he had to give up fifty per cent. of it to the “ambulance chaser.”

I am happy to state that a movement is now on foot to make a painstaking inquiry into the progress made during the past fifty years in the directions indicated, with a view not only to discovering the good, but also to ascertaining what social and economic ills remain to be eradicated, and to propose, as far as possible, practical remedies therefor.

It is believed that a movement which will recognize the good and sincerely seek to remedy the wrong would be more effective in accomplishing reform than one designed only to tear down and destroy.

It were well, and with this suggestion I conclude, if at all future gatherings of this great organization some such counting of the milestones passed were to be made a feature. There is good reason for this. There are among our ninety millions of people many who, strange as it may seem, interpret such occasions as this as diagnostic of a body-social sick nigh unto death as the result of neglect. They do not know—and the fault is not wholly theirs—that the patient, far from being in extremis, is in better condition than ever before, that what to them is a death chamber consultation is merely an evidence of periodical stock taking in terms of social health and welfare. (Applause.)

President White—This is certainly a truthful resumé. It is well for us all sometimes to stop and “count our blessings.” (Applause.)

We will now listen to Dr. Burton J. Ashley, of Morgan Park, Illinois. His subject is “Disposition of Sewage,” a very interesting aspect of Conservation.

Dr. Ashley—The universal aim of every one is to succeed. Success in anything depends, it is aptly said, on one’s ability, reliability, endurance, and action—four personal requisites, the absence of any one of which means failure. Ability and reliability are personal qualities, while endurance and action, two of these four requisites, are physical endowments dependable on one’s health. Accepting these statements as correct, then half of our successes is dependable on personal health and one-half on personal quality.

If man’s successes are equally as dependable on health as on his mental or acquired qualities or abilities, then we must draw but one conclusion, viz., that as much attention should be given to the maintenance of a healthy body as to the use and maintenance of our mental and moral capabilities.

Healthfulness depends in part on cleanliness, the state or quality of being clean. Health is natural. Disease is unnatural and is the result of some known or unknown transgression of natural laws. Dirt and disease have always been good friends. Disease is always most flourishing when it has dirt and filth for company; and to be dirty or filthy is to transgress nature’s efforts at keeping the body well.

Water and food are essential to life. Consume them and the liquid waste produce is sewage filth. To man the foulest and most repulsive dirt or filth is that of his own daily making, and well that it is, for it contains the most poisonous substances that exist and civilized humanity everywhere is increasingly directing its efforts to accomplish its destruction in the most sanitary and economical way possible. Modern methods employed by cities or lesser municipalities to disposal of their liquid filth is that of establishing systems of underground drains called sewers, into which such liquid filth is discharged.

The first well designed sewerage system to be adopted in the United States was built in Chicago about the year 1855.

The modern water-closet was not evolved until early in the last century, and in consequence of which evolution water carriage as a means of conveying sewage away logically followed its introduction. Former designs of sewerage provided for drains that would accommodate both the storm waters as well as the sewage. This method is commonly known as the “Combined System,” but when the employment of this character of sewage disposal created nuisances, the demand arose for the abatement of said nuisances, and it was then that civilization faced sewage purification in some form as a remedy. Storm waters are only dirty waters and not, strictly speaking, polluted waters, for merely dirty water will not create an offensive nuisance and requires no purification, while polluted water does. So the “separate system” of sewers was then evolved, namely, where one system conveys the storm waters and a separate system the sanitary sewage, for, inasmuch as only sanitary sewage needs purifying, therefore works of smaller capacity are needed than would be required were the large and unsteady volume of storm waters to be also subjected to the purifying process.

Many experiments have been made and varying forms of sewage purifying plants have been built during the last half century, and out of the many failures there have been evolved a few processes of purification which have proven fairly successful, but from an economic standpoint as well as from a physical one much yet has to be gained.

The broad irrigation of land with raw or crude sewage has been tried out and its limitations discovered. Although physically successful when properly administered, this form of disposal has been found to be expensive. Existing costly land values are usually exceedingly against the adoption of this form of disposal. The Broad Irrigation Plant of Berlin, Germany, with her 43,000 acres of land, is a notable example of the continuance of this form of disposal, but while this scheme has through its years of usefulness sometimes shown profits and sometimes deficits, the profits have never been large enough to pay the interest and sinking fund charges on the capital expended on the purchase of farms.

The method of purification now in general use is what is broadly called the biological method, wherein nature’s own mysterious forces, viz., putrefaction and nitrification, are encouraged usually by first impounding the sewage and then nitrifying the impounded liquid in a filter bed, so called.

This form of sewage purification is found to require a minimum amount of manipulation or labor as compared with some of the other forms.

The plea for sewage disposal is that it enhances life by preventing disease. The United States Conservation Commission reported that eighty-five per cent. of typhoid and malaria are preventable.

The sewage disposal problem is by no means an easy one, for every case being a law unto itself is sure to present a greater or less number of physical conditions that may not be found in any other case. Sewages differ in their composition as people differ. Some sewages are easily controlled and gotten rid of, while other sewages are stubborn to almost refusing to be subdued. The sanitary engineer in arriving at determination is obliged to previously dig deep and acquaint himself with existing conditions before he can safely conclude upon designs or measures or means that will bring successful results. The sanitary engineer’s practice is therefore much like that of a physician who considers symptoms before offering a diagnosis or prescribing a remedy.

Contrivances that have worked successfully in England have often proven to be failures in the United States. The character and composition of sewage abroad differs widely from the composition of our greatly diluted sewage here. Latitude, quantity of contained manufacturing wastes, character of water supply and numerous other components all combine to make the art of sewage disposal a problem. For instance, when the water supply is what is commonly termed “hard” undue collection of scum or mat is almost sure to form in biological tanks, and this is only one of the innumerable vexatious enigmas that confront the engineer or biologist. Pioneer practitioners have frequently undertaken the solution of sewage disposal questions when not qualified for such duty, largely in answer to the urgent request of an impecunious public with the usual disappointing results. But the value and possibilities of health Conservation have now been brought to that degree of successful accomplishment where the demand for specialists in the advancement of this modern art has become enhanced, and the advisability of employing specialists when the nature of the work is of such vital importance as is sewage disposal needs no argument.

Mr. Winslow has told us that a badly constructed or badly operated system is worse than none at all, for it creates a sense of false security and it also breeds a sense of distrust.

The first city to go about the establishment of sewage disposal in a thoroughly exhaustive way was Columbus, Ohio. Its example was shortly followed by the cities of Baltimore and Philadelphia. Very elaborate and exhaustive experiments are now being made by the city of Chicago at an experimental plant costing $60,000, which experiments have already covered a period of over two years, so that when a report shall be forthcoming the character of disposal best suited to that city will be a known factor and such steps as will be taken will be along lines of certainty.

The whole civilized world is or should be deeply indebted to the far-reaching experiments that have been conducted since the year 1887 at Lawrence, Mass., by the Massachusetts State Board of Health. The annual reports of the findings have become classic both at home and abroad. Nor would we forget to mention particularly the fifth report of the English Royal Commission on Sewage Disposal, which, after making exhaustive observation covering a period of four years of the operation of twenty-seven sewage disposal plants in actual daily operation, gave information, the value of which to the sanitary expert and indirectly to the civilized world at large, may not be determined.

At this stage of sanitary advancement our common people should not be further excused for the density of their ignorance regarding the value of preventive medicine as exemplified in clean premises and person and the adopting of respectable sanitary conveniences.

Want of knowledge along lines of modern sanitary advancement is to a very large extent due to the inertness of legislatures in enacting laws to meet the modern sanitary needs. The passing and enforcing of such laws would surely force our ignorance on this subject out of us and place us on a higher hygienic plane, such as has been established by the excellent enactments in a few of our States.

Standing out pre-eminently in this respect are the laws relating to the public health in Massachusetts, with New York following as a close second. One of the most important laws which is the foundation of others in Massachusetts is a provision for the acquirement of land by cities and towns for the purification of sewage. All through the Massachusetts code are to be found an abundance of preventive measures, as well as curative—abatement of nuisances, of offensive trades; establishing water supply and sewage disposal. Then follows a long list of subjects spell as lying-in hospitals, dangerous diseases, spitting, drinking cups, protection of infants, vaccination, quarantine, public school inspection, diseases of domestic animals, hydrophobia, cemeteries, cremating of dead bodies, burials, bakeries, supervision of plumbing, pollution of streams, food and drugs, milk, registry of births, marriages and deaths—not one of which but has its peculiar relation to the producing of sewage, and indirectly with sewage disposal.

As a contrast with the Massachusetts code let me refer to the sanitary laws, or want of them, in the State of Illinois. According to a copy of the public health laws issued for the information of local health authorities and others of this State, there occurs, for instance, but two sections covering the establishing of sewers. Rules and regulations are in evidence for isolating, quarantining, disinfecting and coping with various infectious diseases after they come into existence, but not a statutory provision is to be found establishing sewage disposal, nor for preventing the pollution of streams and lakes. The State Board of Health in this State is well-nigh powerless in taking initiative steps, particularly with regard to sewage disposal and stream pollution. It is high time State legislatures betook themselves to looking more into the all important art of sanitation and its far-reaching results and at once enact laws that will meet the advanced requirements of our daily living, and give such attention to the conservation of health and to the physical welfare of our homes as it in some cases has given to the welfare of the barn, the pigsty and their occupants. Had I the time I could refer to some very astonishing facts that might cause the blush of negligence to come to the faces of our Hoosier legislators.

Ohio has recently enacted a code of plumbing and drainage laws, containing provisions supposed to cover scientific sewage disposal. This code provides for and encourages contrivances that have been most soundly condemned by leading sanitarians both in this country and abroad for a century past.

It was Eugene Field who said:

> “It seems to me I’d like to go
> Where bells don’t ring or whistles blow
> Nor clocks don’t strike nor gongs don’t sound,
> And I’d have stillness all around,
> Not real stillness, but just the trees
> Low whispering of the hum of bees.”

What this tender poet wrote several years ago is increasingly being enacted today by the exodus of the prosperous captains of industry, of commerce and of the professions from their narrow city confines in unneighborly city neighborhoods to well appointed habitations in the outlying suburbs, or in his comfortable summer home up in the mountains or alongside the beautiful waters of some inland lake. These prosperous friends, though removing to the country, are unwilling to yield up any of the comforts and conveniences afforded by municipal service. Sewers usually unavailable in these more or less remote locations causes sewage disposal to become at once one of their most vexatious problems, so here comes a new demand for special skill in aiding our country gentlemen in establishing a satisfactory sanitary service that will tend to his comfort and respectability and prevent a menace to life and health. So all along the line the requirements for the sanitary uplift of home surroundings is widening, and the requirements in the daily living is enhancing, for modern sanitary methods of which sewage disposal is the most important are found to be most effective and therefore more necessary in the conservation of man's most valuable asset—health. (Applause.)

President White—While waiting for committee reports, we will hear from a gentleman from San Francisco, who asks a little time. I will introduce to you Mr. J. P. Baumgartner.

Mr. Baumgartner—I just want to say to you that San Francisco will be in the field at the proper time with an invitation to this Congress to meet in that city in 1915—the year of the Panama-Pacific International Exposition. The State of California has raised twenty million dollars for this Exposition. There will be a million-dollar convention auditorium on the Exposition grounds, and we feel there are many reasons why it would be particularly fitting for this Congress to meet in that city that year. I do not want to press this matter unduly at this time, but I felt I had a duty to perform to tell you that we want you to come to San Francisco in 1915, and that we will extend to you a royal welcome. I thank you. (Applause.)

President White—There is a committee to report at this time. The Chairman of the Executive Committee, Mr. E. L. Worsham, will report on some amendments to the Constitution.

Chairman Worsham—Mr. President and Members of the Congress: The Executive Committee makes the following recommendations for changes in the Constitution of the National Conservation Congress:

That the following be added as Section 3, Article III:

“After a call of the Executive Committee by the Chairman, and after all members of the committee have been notified of the meeting in sufficient time to be present, three members shall constitute a quorum for the transaction of business.”

That Article IV, Section 1, be amended as follows:

“Section 1. The officers of the Congress shall consist of a President, to be elected by the Congress; a Vice-President, to be elected by the Congress; a Vice-President from each State, to be chosen by the respective State delegations; one from the National Conservation Association and one from the National Association of Conservation Commissioners; an Executive Secretary, a Recording Secretary, and a Treasurer, to be elected by the Congress.”

That in Article V, Section 1, the words “during each regular annual session” be stricken out.

That Article V of the Constitution be amended to read as follows:

“Section 4. The President shall appoint a Finance Committee of five, three from the members of the Executive Committee and two from the Advisory Board, whose duty it shall be to plan ways and means of increasing the revenue of the Congress, and to prepare a budget of expenditures. The Chairman shall be a member of the Executive Committee.

“Section 5. The Executive Committee shall appoint in consultation with the Vice-President from the State, a State Secretary whose duty shall be to work with the State organizations for the especial interests of the Congress. Such Secretary shall report progress to the Executive Committee.”

That the remaining sections of Article V be renumbered accordingly.

That Section 2 be added to Article VII, to read as follows:

“The membership in the National Conservation Congress shall be as follows:

“Individual membership, one dollar a year, entitling the member to a copy of the Proceedings and an invitation to the next year’s Congress, without further appointment from any organization.

"Individual permanent, or life membership, twenty-five dollars, entitling the member to a certificate of membership and a copy of the Proceedings and invitations to all succeeding annual Congresses.

"Individual supporting membership, one hundred dollars, or more, entitling the member to a certificate of membership, a copy of the Proceedings, and an invitation to all succeeding Congresses.

"Organization membership, twenty-five dollars, entitling its delegates to the Proceedings and an invitation to the organization to appoint delegates to the next Congress.

"Organization supporting membership, one hundred dollars or more, entitling the organization to appoint one delegate from each State, each of whom shall receive a copy of the Proceedings."

Mr. Worsham—We are proposing some radical changes regarding the membership of the Congress. Heretofore, the personnel of the Congress has varied from year to year, and we have had no way of keeping in touch with delegates who attend. We think it is necessary to place the Congress on a good financial basis, and also to keep in touch with the people who attend from year to year, and we have, therefore, recommended these changes. I move the adoption of this report.

The motion was seconded, put, and declared carried.

President White—I will now call for the report of the Nominating Committee, which will be presented by the Chairman, Prof. George E. Condra.

Professor Condra—Your committee has been working very diligently, canvassing the situation. We have looked over the field, reviewed the work of various persons connected with Conservation, noted their efficiency. We have looked into the future, we have thought of the fitness of certain individuals for the work, and therefore report as follows:

For President, a man who can take up the work where Captain White leaves off—Mr. Charles Lathrop Pack, of Cleveland, Ohio. (Great applause.)

For Executive Secretary, one who has been with the work since its beginning, and has accomplished so much—Mr. Thomas R. Shipp, of Indianapolis. (Applause.)

For Recording Secretary, one who has also been valuable in the work, and has been associated with Mr. Pack and with Captain White—Mr. James C. Gipe, of Indianapolis. (Applause.)

For Treasurer, the man whom the Executive Committee at an earlier Congress gave an earnest invitation to take up this work, that it might be taken care of in a manner befitting this Congress—Mr. D. Austin Latchaw, of Kansas City. (Applause.)

The one who has been nominated for second place, Vice-President, we named because of fitness to serve all phases of the work of Conservation, but especially the conservation of life and the home. Not chosen because she is such a womanly woman; not especially because she has done splendid work for us here, but chosen because she is a great leader and we want her for the work. A person known to most of you—Mrs. Philip N. Moore, of St. Louis. (Applause.)

I do not name the Vice-Presidents of the States, for reasons given in the report of the Executive Committee. I take great pleasure in moving the adoption of this report.

The motion was seconded by Mr. A. B. Farquhar, put, and declared carried.

President White—I now wish to present to you your next President, Mr. Charles Lathrop Pack. (Applause.)

It is with great pleasure that I present to you the President of the next Congress. He is one who is thoroughly in love with Conservation. He is one of those who first studied Conservation. He spent years in its study, and he is, I know, the first American who ever received a fee for scientific forestry advice. He was paid one thousand dollars by the President of the Missouri Pacific Road for his expert opinion. When Mr. Pack returned from Germany, where he had been studying forestry for some time, he was sent for by Jay Gould, who asked him for his expert opinion on some forestry matters. Next morning Mr. Pack found in his box at the hotel a check for $1,000. This was the earliest record of such a fee being paid in the United

States. So, if he was appreciated to this extent by a great railroad president then, we surely can trust him now. We are proud to have him as our President, and we feel he will be a great help to Conservation in the ensuing year. Mr. Charles Lathrop Pack, your new President, will now take the chair. (Applause.)

President PACK—Ladies and Gentlemen: You have a great work before you, not only for the ensuing year, but for all years. The Conservation movement is not one for today, but for all time, and it matters very little the name or the names of the workers in the cause. It matters that you, and every one of you, should have your hearts right and do the right work. Conservation makes for the best use of all resources, and is dead against their abuse. It is your duty and my duty not only to come to these Congresses and confer and talk, but when you go home to be a true advocate of the cause and to be against everything that is opposed to it. (Applause.) Conservation is for men and women, and for one I thank God we have the women with us. (Applause.)

I do not intend to make a speech; I am not a speech-maker. You have plenty of orators. But with your help during the next year, I will try to do my part, and I ask every one of you to go to your homes and come back to the next Conservation Congress with three delegates in place of one. I thank you. (Applause.)

Before we go any farther, I ask you to rise and join me in giving three cheers for that great Conservationist, Captain White.

Three rousing cheers were given, led by Mr. Pack.

Mr. WHITE—Ladies and Gentlemen, Delegates to the Congress, Mr. President: This is glory enough for me. I feel paid for the work I have done in the past year in having the appreciation of such a good class of people. (Applause.)

President PACK—The next speaker on the program is Mr. George M. Lehman, representing the Mayor of Pittsburgh, who will speak to us on "The Investigations of the Flood Commission of Pittsburgh."

Mr. LEHMAN—Mr. Chairman and Delegates of the Fourth National Conservation Congress: It has been the custom in this country to build dams and locks on lower reaches of rivers, for navigation; to build regulating works for forming and maintaining channel depth, etc., and to dredge deposits caused by erosion.

Our country has received large benefit from this process, particularly in certain sections. It would have thrived, however, to a far greater extent and much suffering, involving general living and business conditions, would have been avoided and a better foundation provided for future generations, if, in addition to the above-named developments, attention had been promptly and thoroughly given to the control and conservation of flood water. We have been woefully thoughtless and backward in bringing about a comprehensive treatment of this matter which is of such great national importance.

HISTORICAL AND GENERAL OUTLINE OF WORK.

Pittsburgh having been seriously troubled by destructive floods for over a century, attention was finally directed toward means of alleviation and in 1908 the Chamber of Commerce organized a commission consisting of business men, engineers and other professional men, to ascertain the character and extent of flood damage and make investigations of methods for relief. Later, an enlargement of the commission was made by the addition of city and county officials and representatives of manufacturing and various business concerns affected by floods. The expense of carrying on the work has been borne by public-spirited citizens, including the interests affected by the floods, and by county and city contributions. To this date about $137,000 has been expended.

The work has involved detailed surveys and soundings, within the city limits, of the Ohio, Allegheny and Monongahela Rivers, and a survey of the areas of overflow. The topography was fully developed, and streets, lines of transportation, buildings, etc., located. Extensive topographic surveys were made along the principal tributaries of the Allegheny and Monongahela Rivers for the purpose of determining the possibility of constructing storage reservoirs.

In connection with complete contour maps, diagrams, profiles, etc., made from the above work, studies have been made of the cost and effectiveness of a flood wall, in connection with dredging, in deepening, widening and straightening of the river channel at the city, and the cost and effectiveness of regulating the stream flow by storage reservoirs, located throughout the drainage basins. In addition to the collection of a vast amount of general data, including precipitation, taken from the records of the United States Weather Bureau, the work involved many special studies, among which were forest conditions, geology and stream-flow. For the stream-flow studies, gauging stations were established by the Flood Commission and also a number in co-operation with the Water Supply Commission of Pennsylvania. In the forest studies, the co-operation of the United States Forest Service and of the Forestry Department of Pennsylvania were secured. Valuable stream-flow data have been provided by the United States Geological Survey.

At the beginning of the investigations the matter was treated as of local concern only, but as the work progressed the broad aspect of the problem and its national scope were realized, as it became evident that Pittsburgh’s floods had a direct bearing upon the flood troubles of other communities. Further study disclosed the fact that inseparable from the flood problem was the question of navigation, sanitation, water supply and water power, and that the valleys of the Allegheny, Monongahela and Ohio Rivers could be benefited wherever conditions are favorable for the construction of storage reservoirs. On many of the principal tributaries of the Ohio below Pittsburgh, the topography is favorable for storage reservoirs upon a large scale, and floods could be prevented throughout the Ohio valley by extending the plans of the Flood Commission.

An exhaustive report, consisting mostly of original data, has been published by the Commission, as the result of nearly four years of painstaking work. It is said that this report forms the most comprehensive treatment of a subject of this kind that has ever been carried out. The report contains over 900 pages, including numerous maps and diagrams, and a large number of illustrations, showing flood damage, reservoir sites, forest conditions, etc.

FLOOD DAMAGE.

Pittsburgh, which has a population of 533,905, and about twice as much with the contemplated greater city, is located at the head of the Ohio River and at the confluence of the Allegheny and Monongahela Rivers. The combined drainage area, above the city, amounts to 18,920 square miles. Of the two rivers, 150 miles, directly connecting with the city, have been slackwatered. About 14,000 miles of navigable waterway lies below the city. The National Government, in a few years' time, will have the entire 967 miles of the Ohio River improved.

The tonnage of Pittsburgh, incoming and outgoing, amounted in 1910 to 167,000,000 tons, of which 11,000,000 tons consisted of river traffic. The above total tonnage, which has doubled in the last six years, is twice as great as the combined tonnage of New York, London, Hamburg and Marseilles.

As is frequently the case in communities situated upon the inland rivers of this country, the most important commercial and industrial parts of Pittsburgh are located upon the low lying areas bordering the water. The need of free access to water and of rail and water transportation naturally brings about such development. In fact, on account of the topography, rail communication can in many cases be satisfactorily established only along the stream. Such a condition, however, frequently causes great suffering and interruption to business, involving not only the districts in direct touch with the river, but the whole community.

During the progress of the investigations, it became evident that unless some adequate method of flood relief could be devised and carried out, the larger portion of the flood affected areas could never be properly developed, and the capital invested therein would continue to suffer. The general needs of building operations and of city improvements will of necessity keep pace with the advance of population; and the flood damages, which in their effect involve the home conditions and business life of the entire city and surrounding communities, will become correspondingly greater.

In ascertaining the extent of flood damage to the city, a careful investigation was made of three floods which occurred within a period of about twelve months, from March 15, 1907, to March 20, 1908. In the conduct of this work it was noted that while those coming in direct contact with the floods are alert to the seriousness of the situation during the flood, the matter is, however, after a time almost forgotten; the disposition in most cases apparently being to take the troubles as they come rather than to do anything in the way of even attempting to devise means of relief.

The classification under which this work was done, and the monetary amount of direct losses within the city by the three floods may be given as follows:

Damage to buildings, equipment and machinery	$782,400
Damage to materials	1,698,900
Loss to employer by suspension of business	1,974,200
Loss to employee due to shut-down	1,308,300
Expense of cleaning up	547,400
Charities dispensed and funds for prevention of disease	27,800
Fires uncontrolled through inaccessibility or lack of water pressure	175,000
Total	$6,514,000

It was found that the loss ranges as follows: For the flood of 27.3 feet, $414,700; 30.7 feet, $839,800; 35.5 feet, $5,259,500.

The area comprising the larger part of the mercantile, industrial and railroad interests amounts to about 3,000 acres, 1,540 of which was covered by water during the great flood of 1907, which had a height of 35.5 feet, or 13.5 feet above the danger line. This flood remained sixty-five hours above the danger line of 22 feet. About fifteen miles of river front land are occupied with industrial works of various kinds. The assessed value of real estate as affected by the 1907 flood amounts to about $160,000,000, and a careful estimate shows that this property is nearly $50,000,000 lower in value than it would be if protected from floods. Using the results obtained for the above floods and the flood records for the past twenty years it is estimated that the direct loss to the city has amounted in that period to about $17,000,000, over $12,000,000 of which occurred in the ten years preceding January, 1911.

Based on the assumption that in the next two ten-year periods there will be no increase in number or height of floods over those occurring in the ten years just preceding January, 1911, it is estimated, if protective measures are not provided, that the flood losses at Pittsburgh in the next twenty years will amount to about $25,000,000. As records show, however, that floods are increasing in frequency and height, it is estimated that the losses in the next twenty years will amount to about $40,000,000, or nearly twice as much as it will cost to carry out the flood prevention measures recommended by the Commission.

The Commission did not have resources for securing the amount of damage at the many important points along the rivers, above and below Pittsburgh, but at Wheeling, W. Va., it was ascertained, for instance, that about $1,000,000 was lost during the flood of 1907. Authorities consider that the total loss along the Ohio Valley for the two floods of 1907 amounted to more than $100,000,000. This is indicative of the vast losses occurring annually all over the country.

In addition to many miles of street car tracks, streets and alleys, about 435 acres of railroad and industrial yards were covered, in addition to 17 miles of main railroad, by the big flood of 1907.

At high stages many manufacturing plants must close down. The following is quoted from a report of the American Iron and Steel Association: “Damage to the iron and steel industry unprecedented. At beginning of March, 1907, flood there were forty-four blast furnaces in Allegheny County in blast, and of these thirty-eight had to be banked for an average of two days. Work at most of the sixty-five or seventy rolling mills and steel works was suspended.” Many of the open-hearth furnaces were badly damaged and some of them practically ruined.

FLOOD PROTECTION.

Regarding methods of local treatment, studies and estimates of cost were made of the following: A wall of about twenty-five miles in length to be built in the city along the river fronts; also for deepening, widening and straightening of the river channel by dredging.

The wall, high in places above the river streets, would prevent overflow by confining the floods to the channel. Dredging and removal of obstacles in the channel, bank encroachments, etc., as can now be accomplished, would have comparatively slight effect in reducing flood heights and these means were, therefore, not broadly recommended. Furthermore, these forms of treatment would be of local flood benefit only and communities above and below Pittsburgh would continue to suffer in various ways.

A wall of limited height, however, is really desirable, at least along certain parts of the river. While reservoir control would result in reclaiming considerable areas of land, a wall would provide means for adding to the amount and greatly improve the appearance and usefulness of the banks.

The handling of cargoes, to and from river boats would be greatly facilitated by means of modern devices. Sheds could be constructed along the wall and close to the boats which would lie alongside. Such arrangement would make feasible the bringing directly of river and rail transportation with the great advantage of through rates and routes, a condition which is now lacking at practically all points on American rivers.

FLOOD PREVENTION.

In the treatment of the flood problem, prevention, by the use of storage reservoirs, for the purpose of holding back the damaging part of the flood water, is the rational and comprehensive method, as it goes to the source of the trouble, and extends its benefits throughout the entire river valleys, not only in the form of flood relief, but by improvement of the low-water flow, due to the release of the impounded flood waters during the dry season.

Forest cover is beneficial to some extent in retarding the run-off and in improvement of low-water flow, and the attitude of the Flood Commission is to support such National and State legislation as will tend to preserve and increase the present forest cover. The Commission, however, recommends the use of the storage reservoir system, supplemented by other means where necessary, for the reason that such a system could be speedily brought about. The use of storage reservoirs for flood control is not a new idea in this country and this method is now successfully employed in European countries.

The exhaustive surveys and studies for flood prevention disclosed the fact that forty-three reservoir sites are available in the Allegheny and Monongahela drainage basins above Pittsburgh, and that while not needed for present purposes additional sites are feasible. The forty-three projects would have a total capacity of 80,500,000,000 cubic feet, would cost $34,000,000, and would control about sixty-two per cent. of the total drainage area above the city. After a careful analysis it was found that a less number of reservoirs was practically as effective, under proper manipulation, and a selection was made of the most favorable ones, seventeen in number. These would have a total capacity of 59,500,000,000 cubic feet, would cost $21,700,000, or about $364 per million cubic feet of storage capacity, and would control fifty-four per cent. of the total drainage area.

As a basis, eleven of the principal floods, occurring within recent years, were exhaustively studied and it was found that the seventeen selected reservoir projects would reduce all of them, with one exception, to below danger line. Investigation showed that a low wall built at comparatively small cost along a few parts of the low-lying river fronts could be used in combination with the seventeen reservoirs to prevent overflow by the highest known floods. This combination was therefore recommended, the total cost being estimated at $22,350,000.

Some of the benefits to be derived by preventative methods and stream regulation and development, may be summarized as follows:

> 1. Reducing or doing away with floods and flood damages and their constant menace, thereby encouraging and making possible for present and future generations full development of affected areas.
>
> 2. (a) Improving of navigation, by permanently increased stream-flow in slackwatered rivers, where dry weather flow is frequently inadequate to furnish desired draft, thus providing uninterrupted transportation not only for present business but for future demands. (If the reservoirs were brought up to maximum capacity, that is, above flood control requirements, the low-water flow of the Ohio, at Wheeling, ninety miles below Pittsburgh, would be nearly six times the present minimum, giving an increase in stage of 3.7 feet. One of the largest floods would have been reduced over thirteen feet.)
>
> (b) Making possible slack water on certain rivers, worthy of attention, but now unimproved largely on account of absolute lack of sufficient water.
>
> (c) Reducing velocity of current, due to lowering of high stages, thereby making safer the maneuvering of river craft; reducing wide fluctuations in water levels, particularly at river ports, facilitating thereby the handling of cargoes and increasing clearance under bridges. (Under a certain bridge at Pittsburgh, investigations show that during the past fifty-three years there has been an average of fifty-seven days when the ordinary steamboat could not pass. Had the proposed system of reservoirs

been in operation the water would have been lowered so that there would have been an average of only three days.)

(d) By having the great fluctuations reduced, the erosion of the banks along the bottom lands and at other places would naturally be considerably lessened.

3. Improving sanitary conditions and increasing the quality and quantity of the supply for municipal and industrial purposes. High stages leave deposits on banks, becoming a nuisance to health; and low stages are frequently unable to properly carry away polluted water stagnating in slack water or natural pools.

4. Developing water power, which is feasible under favorable conditions in connection with reservoir systems for flood prevention.

I would call attention to the fact that this brief review upon stream regulation goes far enough to show that so far as damage from floods is concerned alone, the matter is not only of local but of great National concern, affecting as it does railroads and manufacturing interests which supply the Nation. What is true of Pittsburgh is also true of many other river localities, and it is therefore urged that the question be looked at in a progressive manner and that suitable State and National legislation be enacted at the earliest possible moment to provide not only for full navigation requirements, but in addition for flood damage and the combination of needs as outlined in the report of the Flood Commission. It is hoped that this Congress will lend its powerful co-operation in bringing about the accomplishment of this great movement which is so necessary to the public welfare. (Applause.)

President Pack—I am sure we are all indebted for this paper, and to Mr. Lehman for coming from Pittsburgh to present this valuable subject.

If there is nothing more before the Congress at this time we will adjourn until 2:00 o’clock.

ELEVENTH SESSION.

The Congress was called to order by Mr. J. B. White, in the Murat Theater, Indianapolis, at 2:30 o'clock p. m.

Chairman White—It is long past the time for our meeting, but we have not had the last word from Governor Hadley. He wired me night before last of an accident, and that his physician said it would not do for him to come yesterday. Last night we had another telegram, saying he was afraid he could not come, and that we had better not depend on him. I also received a letter. Then I wired him again, but have no reply, so it is barely possible that he will be here in time to speak to us this afternoon. The committee has gone down to meet the 2:50 train. In his letter, he says:

"I want to thank you again for your kindness in giving me such a prominent place upon your program, and were it not for the fact that I know your meeting will be a complete success with Governor Wilson alone, it would be an added regret—my inability to be present."

I know many of you came expecting to hear Governor Hadley, and he certainly will give us a splendid address if he comes. He appointed a commission in the State of Missouri, of which I have the honor of being a member, and we have had meetings at the Governor's mansion, and we are trying to induce the Legislature of Missouri to pass a good law in favor of Conservation of all natural resources. I cannot report as to our progress as I would like, so I will not say anything about what we have done. We know what we are trying to do.

The newly elected President is not here, and he insists that I take his place until he comes. We will now listen to "The Story of the Soil," from one who has given it great thought and attention. He has brought about good results that will be of benefit to the farmer and to every one who lives in the country, and therefore of benefit to all the citizens of our common country. I have pleasure in presenting Mr. H. H. Gross, President of the

National Soil Fertility League, who will speak on “The Story of the Soil.” (Applause.)

Mr. Gross—Mr. Chairman, Ladies and Gentlemen: I am here representing what we think is one of the most distinguished organizations of this country—one devoted to a specific and definite purpose, and that is to secure the application upon our farms of the best methods of farm practices.

In our self-sufficiency we are sometimes disposed to pooh-pooh science. I have heard farmers say, “What do I care about science? I know how to farm. I am a practical farmer.” When I hear a man talk about being a practical farmer, or a practical shoemaker or anything else, I begin to question his knowledge of the art. Reduced to its last analysis, science is simply applied common sense. In other words, to find out the best way of doing anything and then doing it that way.

Scientific farming will increase the output per man, per plow, per mule, per acre, and at the same time it will build up the fertility of the soil. Unscientific methods will wear it out. Millions upon millions of acres of land have been wasted by practical farmers in unscientific farming, by abuse and misuse until the land fails to yield enough to pay the labor of cultivating them. There are millions of acres east of Albany that are not worth today one-fourth as much as they were one hundred years ago.

The soil is our greatest natural asset. It is God’s best gift to man outside of Him who came to save us. It is our duty to conserve this gift as a priceless heritage. In a higher sense the man in whose name the title stands is not the real owner of the land; it is his to use during his lifetime and to pass it on to his successor. It is his paramount duty to turn it over to those that follow him as useful as when he received it. The land is not his except to use, it is not his to abuse. The fertile fields were placed here by God Almighty for the use of humanity for all time and no one has the right to rob the soil of its power to produce and thereby imperil or destroy the birthright of succeeding generations.

Let us look at Europe. They produce two or three times as much as we do upon the same area, notwithstanding their lands have been a thousand years longer under the plow than our own. There must be a reason, and it is that Europe, because of its large population, has been compelled to adopt intensive farming or go hungry. With us it has been different up to the present time. A few years ago, some of we older men can remember the time, when the United States invited everybody to come in and possess the land. An old song says, “Uncle Sam is rich enough to give us all a farm.” Since then our population has increased faster than the farming industry. We are now consuming ninety per cent. of our wheat and ninety-eight per cent. of our corn. The population is rapidly overtaking production. In fifty years our population will be doubled. What shall we do about it? I say to you this, we must do better farming or the people will go hungry. A thousand years or so ago Japan and India were at the parting of the ways—about where we stand today. Japan chose the better part and conserved the fertility of her soil and by intensive scientific culture she has fed her people and has demonstrated that a very small patch of ground indeed is sufficient to support an individual. This has been shown in this country—that one or two acres, properly handled, will take care of a small family. Japan acted wisely and is rich and prosperous today. India neglected her duties and her opportunity and today there are millions starving there on account of the lack of foresight of those people who lived thousands of years ago. Shall we follow Japan or India? There can be but one answer. The intelligence of the American people, the spirit of the age demands that we go forward to attain the highest and best and it is our duty to help to this end.

Denmark, a generation or two ago, was in poverty and distress, its people were crowding into the cities. The government saw something must be done to improve conditions. It wisely decided that agriculture must be encouraged, so it commenced to teach agriculture in the schools. It had its agricultural colleges strengthened, it sent men out among the people as traveling schoolmasters, visiting one community after another. Agriculture was taught in the schools. This helped some, but did not solve the problem. Finally they adopted the plan which we propose to follow, of sending a trained farm demonstrator into every community and stay there, study local conditions, meet the farmers right on the soil, and help them to understand and apply the best methods and get the best results for the time and effort expended. In two generations it brought Denmark from poverty to thrift, and today it is the finest agricultural country in the world. This comes about from carrying the knowledge to the farm home in the personality of the farm demonstrator who helps the farmer apply the best methods in practice.

Wherever the plan has been tried it has succeeded. It is the one plan that has made good, and in my judgment it is the only one that ever will. Now, then, what are we going to do about it? The most important question that has been discussed on this platform during this Congress is the one under discussion now. It is vital, it touches every human interest. The question is, shall we build up our soil and insure the food supply for coming generations, or shall we not? It is a tremendously important question and one pressing for answer.

I am glad to say this to you, that the National Soil Fertility League determined upon a plan, and so far we have had greater success in carrying it forward than we had any reason to expect. Its plan has the approval of nearly every agricultural authority in the land. It awakened a tremendous amount of interest. It shows many people were thinking in a general way that something ought to be done and were ready to rally to the support of any definite proposition that commended itself to their judgment. The National Soil Fertility League, together with the agricultural college men, drafted what is known as the Lever bill, the object of which is to provide for the co-operation of the Federal Government and the several States in carrying forward this farm demonstration plan. Under this bill the Federal Government makes an annual appropriation to every State of $10,000 a year, irrespective of condition; then it makes further appropriations conditioned upon the States furnishing an equal sum beginning with $300,000 and increasing to $3,000,000 in ten years. Except for the $10,000 all the appropriations are prorated among the States on the basis of rural population. Indiana under this plan would get $10,000 right off the reel from the fixed appropriation; it would get $9,400 from the conditional appropriation provided Indiana should furnish an equal sum. So Indiana would get from the Federal Government the first year a total of $19,400. This would go to Purdue University. Next year it would be increased to $28,800 and would go on up to $104,000 from the general government to the State College of Agriculture. In order to get this money Indiana would have to raise $94,000, so that the State would have when the maximum was reached approximately $200,000 to expend for carrying to the farmers of Indiana the existing methods of agriculture and carrying to the farmer’s wife the best they can give her. What a wonderful help this would be.

There are three great needs in the open country. One is better schools. The country schools of today are not worthy of their name. They fail to meet the requirements of the day and generation. The next important need is good roads, and the third is scientific agriculture. Bringing these improvements about will revolutionize conditions. It will raise agriculture to the first place and the highest place in the estimation of the people. It will be the strongest possible magnet to hold the girl and the boy to the farm home. It will make agriculture more pleasant, more profitable and in every way a more desirable vocation.

When I was a boy and went away to school, I entered a class of boys and we were lined up before the principal and each was asked his name and his father's business; one would answer his father was a banker, another a merchant, another a doctor, a manufacturer, and so on. When it came to me, I said a farmer. The boys all laughed and I was obliged to take it. I licked two or three of them afterward to get my standing on the campus.

We used to think that anybody could run a farm. A story is told of a man who had three sons. One was very smart, one was exceedingly good and one was simple-minded. The father said: "Tom is smart as chain lightning; I am going to make a lawyer of Tom. William is about the best boy I ever knew; you can't get him to go wrong; I am going to make a preacher of him. But Jack don't seem to know much of anything, and I will make a farmer of Jack." (Laughter.)

Let me say to you with all possible emphasis that it takes as much ability to run a farm well as it does to run a bank or a factory, and much more than it does to run for office. (Laughter.)

When the Lever bill was introduced in Congress, it passed the committee and was placed on the calendar and was buried there. The question was to get that bill on the floor for a vote. Upon inquiry I found there was only one way to do it in order to get quick action, and that was to get a petition signed by a majority of the members, asking that the bill be taken from its position on the calendar and placed at the head of the list as unfinished business. Mr. Lever secured the required signatures and the bill was thus advanced to the position of unfinished business. The leaders of both parties rallied to its support and the bill finally passed the House by unanimous vote. It is now before the Senate and we want your help to get it enacted into law before the holiday season arrives.

The mind can hardly grasp the benefits that will flow from this legislation. Let me tell you a little of what scientific farming means. Dr. Hopkins, of the University of Illinois, and one of the world's authorities, just told me that they raised on an average ninety bushels of corn to the acre, covering a period of six years, and twenty-three bushels of wheat, average for six years. The Ohio experiment station on wheat for twenty years showed an average of about thirty-five bushels, while the average for the whole country was less than fifteen bushels. Denmark raises forty bushels average, many fields returning sixty and seventy-five bushels to the acre. We must do better farming.

During the ten years from 1900 to 1910 our population increased twenty-one per cent., our meat supplying animals decreased more than twenty-five per cent. We have an unparalleled high cost of living, due to the fact that population is pressing hard upon production. In short, we have too few producers and too many consumers. Increased production is not the only thing necessary. It is quite as important that the farm production shall reach the ultimate consumer from the farm at less than the present cost. Our marketing system is cumbersome, unwieldy, wasteful and burdensome. (Applause.) The woman who orders her supplies over the telephone pays more money and gets less than the one who goes to market. I had the honor of speaking before the National Federation of Women's Clubs at San Francisco on the first day of July. It was the greatest and most intelligent audience I ever faced. They were very enthusiastic and were quick to grasp the points as they were made. This great organization affiliated itself with the National Soil Fertility League, and when they did so we felt it brought to us the greatest assistance that could possibly come. I know of no organization of wider influence than the Women's Clubs of America. I have heard it said, if you want to get anything done to get a woman after it. (Applause.)

We must re-direct our agriculture; we must raise our meat upon the farms. The ranges are gone. The silo, alfalfa and scientific methods make it possible for the farmer to carry at least twice as much stock upon his farm as he thinks he can carry. In the silo the feed is kept practically green and juicy. You get forty per cent. more out of your corn by putting it through the silo than by handling it in the old way. There is no reason why the cost of producing meat may not be reduced practically one-half. The farmer has given and is giving too much thought to how much he can get for what he raises. It is equally important that he raise more. If he wants 2,400 bushels of corn, it is better to raise it on forty acres with a yield of sixty bushels than to raise it on sixty acres with a yield of forty bushels.

Our plan is to bring home to the farmer the best method that has been determined by the agricultural college and experiment station. We want to get the best results from year to year and at the same time build up the soil. This can be done and this is scientific farming. This is what the whole world needs. The colleges of agriculture and experiment stations have gathered a vast fund of knowledge, and if this were put into practical operation it would double the yield of our farms within a few years and give us a large surplus for export and bring money into the country. We would get richer and richer as the years go by. We would largely supply the world with food. Our position in the councils of nations would be paramount. When it comes to the question of peace or war, the country that has the money and the bread basket is ten times more potent than the nation that only has back of it battleships and armies. (Applause.) So I wish to emphasize that the success of this country rests primarily upon the scientific farming of our fields. Let us remember that no country ever became great and remained so that could not furnish its people with an ample food supply at a moderate cost. To that end we are securing legislation that will put the plan in operation. The Lever measure is a simple one, it creates no new administrative machinery; it simply carries to the farmer and puts to work the information and knowledge that the States and Federal Government have been gathering for fifty years. This whole matter may be likened to a great irrigating system. The United States Department of Agriculture is a dam, it has been gathering and has stored up the knowledge—the water. The colleges of agriculture are the main channels for reaching the various parts of the country; but so long as the water is back of the dam it is doing no good; so long as it remains in the main channels it is accomplishing nothing. What is needed is to get the water to the grass roots, or, in other words, our purpose is to get the information to the actual farmer—the man behind the plow.

Fifty years ago Horace Greeley said, “Go West, young man, and grow up with the country.” If he were here today he would say go South and East, for that is the land of opportunity. In my judgment this Congress ought to meet next year somewhere in the South. That part of the land is entitled to recognition, and you will get a welcome such as you never had before.

In conclusion, I wish to urge that you give us every possible support. We need it. It will help you and it will help us. Let us all work together for reviving agriculture. (Applause.)

(A woman in the audience): “Is it true that Congress is investigating this silo business and under the pure food law is it to be condemned? Also, what must we do in Indiana to cultivate alfalfa?”

Mr. Gross—I have not heard anything about the Federal Government condemning silo, and I do not expect to. Inoculate your soil for alfalfa. You had better take this matter up with your people at Purdue. Ask them what to do. They will send you all the information necessary. They will examine into conditions and tell you just what to do. The most valuable crop today, outside of wheat and corn, is alfalfa. (Applause.)

Chairman White—I have been handed a communication, and I wish to say for the benefit of the gentleman who sent it to the chair that it will be referred to the Executive Committee, which takes up matters of this kind. This is the communication:

“You are requested to make a motion that this organization take steps toward publishing a monthly, or quarterly, magazine, to be known as the National Conservation Magazine. If the society is unable to finance it, there is little doubt that the Carnegie Institute or the Sage Foundation would back it.”

Chairman White—I will now introduce a gentleman who will tell you “The Story of the Air,” Prof. Willis L. Moore, of Washington, Chief of the United States Weather Bureau. (Applause.)

Mr. Moore—Mr. President, Ladies and Gentlemen: I have been trying to reason out why the management put me at the end of the program, and I have concluded that they had an idea that along about this time in the proceedings they would need to have some one take the platform that had supervision over atmospheric air of abnormal temperature.

Now, why should one wish to conserve the atmosphere? I shall try to show you that that is one of the assets of this continent, and, I am afraid, almost the only one, that cannot be monopolized. (Applause.) And you will be surprised, and probably doubt my statement when I say that, with all due respect to the matter of conservation of our wonderful mineral deposits, the controlling of the flow of our streams, the preservation of our great forests—all of them important—we have in the atmosphere one of our greatest assets, if not the greatest asset of our continent. Humboldt has said that “Man is a product of soil and climate. He is brother to the trees, the rock and the animals.” All true, but still I would slightly modify that and say that man is largely a product of climate. For it is the action of rainfall, flood and temperature changes that makes soil.

I shall try to show you that it is climatic conditions that produce this wonderful, this powerful, this resourceful composite man called the American.

I am to speak on “The Story of the Air,” but before I elucidate any further, let me give you a little picture of this wonderful ocean, on the bottom of which you live.

In the turbulent stratum in which we live we have vortices in the atmosphere which cause weather. Weather is the result of the motions of air; it is the result of the dynamic heating and cooling of ascending and descending currents of air. If it were not for these vortices cooling the air and heating it, you would have precisely the same temperature on any day of one year as you would have on the same day in another year. You would not have one first day of June warmer than the first day of July, or the first day of December colder than the first day of January.

To demonstrate my first proposition that we have a great asset in the climate of the United States, I call your attention to some of the conditions in Europe. Their great mountain ranges trend east and west; ours trend north and south. Cyclonic storms originate largely from conflict of equatorial and polar currents coming together. The currents of air come together in the lower stratum. In Europe the great mountain ranges prevent that conflict; but not so here, with our mountain ranges running north and south. Here is the great meteorological theater of the world, the region of conflict. What is the result? A people powerful physically and resourceful mentally. An actual air is pure and invigorating.

Now, I just have a thought that may not be germane, but it is upon my mind. I remember some years ago I wrote a report that dealt with the relations of forests and floods. Although from the inception of this movement I have been heart and soul with the people back of it, still because I do not agree with some of my friends on forestry, on the effect of forests on the flow of streams, I was classified as an enemy of the cause. I wish to say that it is a mistake to bring a fallacious reasoning to any good doctrine. I believe it is a positive injury to attempt to sustain truth by falsehood. I do not mean that anybody is wilfully untruthful—no, simply mistaken. There are so many reasons why we should conserve and protect, why we should use wisely our great forest areas, that there is no need to bring to the support of that great project anything like a reason that can be successfully attacked and refuted. I am satisfied, and as time goes on and other investigators come along and go over my data, I am thoroughly well convinced that the forests do not exert a great controlling influence over floods. I am satisfied that the percentage of floods has not increased for the past forty years. When we remove one vegetable covering like the forests, if we go on and plant wheat, or corn, or grass, we simply exchange one form of vegetable covering for another. If we cut the forests away and leave them, they will at once begin the process of reforestation, and within a few weeks the ground is shaded. If you grub out the roots and stumps and plow, you change one form of vegetable covering for another, and the history of the United States, as well as of the world, does not bear out the statement that the floods have increased with the disappearing of the forests; nor has it been shown that any part of the world has been materially changed in its climatic conditions as a result of civilization or the coming of man. But that is no reason why the forests should not be protected and a wise use made of them.

Let us get down to facts. Just so long as the Gulf of Mexico lies down there on the south, and the great Atlantic remains on the east, just so long rainfall in the United States will be as voluminous on the great cereal plains as it was when the first white man set foot on the continent, and in its movement back to the sea the permeable, cultivated soil of the unforested acres will doubtless as well conserve and restrict its flow as the forests. We have over-estimated the effect of the little scratchings upon the earth's surface by the activities of man. The coming civilization of the great West is immaterial in causing an increase in rainfall. When you stop to consider the enormous volume of the atmosphere above the surface, whose vaporous contents must be materially changed and the thermal conditions altered before you can detract from the rainfall, you will realize how absurd are some popular theories. I do not agree. I radically differ from some of my contemporaries in the Department of Agriculture—but people may differ and still be friends. They may differ in regard to the details of a great movement and still not be inimical to its best interests. The man who differs and brings forth the truth is the best friend of the movement, because nothing can stand long that is not predicated on truth.

I am glad to see that in this movement your managers have brought together so many independent lines of human activity. This great movement is only at its inception. I predict that this Conservation Congress will be one of the most potent factors in the Nation for the developing and awakening of the people. You are willing now to have a free forum, to have free discussion by those of differing opinions. And at this time, Mr. President, when there is such great conflict among the forces that make for civilization, we must not only protect ourselves morally and mentally, we must with equal earnestness attempt to conserve and protect the human individual. He is the greatest asset we can have, after all. (Applause.)

A fair wage scale and reasonable hours of labor have done as much to elevate the American citizen and furnish the ties that bind him to home and State as have all the libraries and universities in the land, and I say this without any disparagement of these magnificent institutions for public good. But if you stop to think for a moment, the library can only be used by those who have a reasonable leisure to enjoy it; colleges have closed doors for those who do not receive something more than a living wage. The welfare of this Nation depends not on the accumulation of great wealth in the future; not upon the palaces on Fifth avenue or the villas at Newport. It depends upon the cultivation—upon the high average intelligence and prosperity of those who actually do the Nation's work, whether they labor with brains or with brawn. (Applause.) And right here let me say to you people who are considering these great problems, that we want brawn developed by working hours that shall not warp and distort the image of God; and we want technical and scientific teaching that shall be as free to the sons and daughters of those who work as to these who have their way paid to college. (Applause.) We must lift from the bottom in any great movement; no movement gets very far that is worked from the top down.

So I am glad to see this movement bringing into its counsels those who are affiliated with the great labor movements of organized labor. My sympathies go out to the man who works with his hands, as well as to the man who works with his brain. I thank you. (Applause.)

Chairman WHITE—Professor Moore stated he did not know why he was put down at the last end of the program. Perhaps it is not necessary to remind him that there is an old saying that the best of the wine is reserved for the last of the feast. (Applause.) But where all is good, and where all is best, as has been the case with the program of this Fourth National Conservation Congress, there can be no choice. And again I am going to remind this audience that this Congress is going to prepare a book containing every bit of the proceedings of this meeting, and it will be one of the best publications of proceedings that has been presented by any congress in the land, and I want to impress upon you, delegates and visitors alike, to leave a dollar for a copy of these proceedings.

While we are waiting for a final word from Governor Hadley, I will call upon Mr. Walter H. Page, Chairman of the Committee on Resolutions, who will present the report of that committee, which I hope will be enthusiastically adopted.

Mr. Page read the resolutions (which will be found in full at the beginning of this volume), and moved their adoption.

The motion was seconded, put and carried.

Mr. JOHN B. HAMMOND (Des Moines, Iowa)—I have a resolution to present. It was referred to the Resolutions Committee, but somehow it was lost in the shuffle.

Mr. PAGE—It was referred to one of the sub-committees, and, presumably, was not accepted by the sub-committee. It was not reported to the full committee.

Chairman White—If there is no objection, it may be presented to this body.

Mr. Hammond—

> Whereas, The protection of womanhood and childhood is the heart and center of the Conservation of "Vital Resources;" and, whereas, forty states of the Union have prohibited the maintenance of houses of prostitution, the market places of the white slave traffic and the centers for the dissemination of the most dangerous and revolting diseases; and, whereas, the city administrations of many of the larger cities, in defiance of state law, have set apart districts where the crime of prostitution is tolerated and protected;
>
> Therefore, be it resolved that we condemn such policy of segregation by city officials as contrary to sound public policy and indefensible in morals, and recommend the absolute suppression of the social evil in all its phases.

I move the adoption of the resolution, Mr. Chairman.

The motion was seconded, put, and carried.

Chairman White—We will pass to the next order—the presentation of invitations from the cities desiring the next Congress. This is the usual way. These invitations are not acted upon, because the Executive Committee will take three or four months to consider everything and compare the different cities, looking to the welfare of the next Congress. Mr. Don Carlos Ellis, of Knoxville, Tenn., I believe has something to say.

Mr. Ellis—Mr. Chairman, Ladies and Gentlemen: There is to be held in the city of Knoxville, in September and October of next year, the National Conservation Exposition. Its purpose and nature are precisely parallel with those of this Congress—for the promotion of the development, wise use and conservation of all of the natural resources of this Nation. The Exposition is of national scope, but is to have special reference to the Southern States. There are to be buildings set aside for each one of the five divisions of our natural resources—forests, minerals, soils, waters and vital resources. In these buildings are to be shown, by example, as this Congress has shown by precept, the various results accomplished by Conservation by the Federal Government, the State governments and by private individuals, and the possibilities of Conservation in the future.

The Exposition originated in Washington last February, when a number of the leading spirits of Conservation met in that city and there was formed an Advisory Board composed of the gentlemen whose names I desire to read to you:

Gifford Pinchot, President National Conservation Association, Chairman; Don Carlos Ellis, in charge Educational Co-operation, United States Forest Service, Secretary; Philander P. Claxton, United States Commissioner of Education; Miss Julia C. Lathrop, Chief of the Children's Bureau, United States Department of Commerce and Labor; Dr. Harvey W. Wiley, Director of the Bureau of Foods, Sanitation, and Health, of Good Housekeeping Magazine; W. J. McGee, Soil Water Expert, United States Department of Agriculture; Senator Duncan U. Fletcher of Florida, President Southern Commercial Congress; Logan W. Page, Director United States Office of Public Roads; Bradford Knapp, in charge Farmers Co-operative Demonstration Work, United States Department of Agriculture; Jos. A. Holmes, Director United Bureau of Mines; Representative Joseph E. Ransdell of Louisiana, President National Rivers and Harbors Congress; Senator Luke Lea of Tennessee; Charles S. Barrett, President Farmers' Educational and Co-operative Union.

These various members of the Advisory Board are to represent, in the formation of plans for the Exposition, the various departments of Conservation in which they are acknowledged leaders. They have instructed me, as Secretary of this Advisory Board, to read to the delegates the following letters:

September 23, 1912.

To the Delegates of the Fourth National Conservation Congress, Indianapolis, Indiana:

We, the undersigned members of the Advisory Board of the National Conservation Exposition, take this means of laying before you an outline of the plans and purposes of the Exposition and of respectfully recommending the adoption of the resolutions which will be introduced at this Congress endorsing the National Conservation Exposition.

This Exposition is to be held at Knoxville, Tennessee, in September and October of 1913. It is an outgrowth of the Appalachian Exposition, which has been held at Knoxville for the past two years. Knoxville was chosen as the location of the National Conservation Exposition because the Southern States are in great need of education concerning the proper handling of their great natural wealth; because Knoxville, while in the South, is readily accessible to the entire East; because the State in which it lies is in the transition zone between North and South and has more States bordering upon it than any other State in the Nation, and all the bordering States are southern; because the city is in the center of the region where the National Government is establishing new National Forests and carrying on other lines of work in Conservation to a greater extent than in any other region; and because of the city's preparedness in being willing to turn over to the National Conservation Exposition Company the excellent buildings and grounds which had been acquired for the Appalachian Exposition Company and to raise sufficient additional capital besides. A bill has been introduced in Congress providing for a government building and exhibit at the Exposition, and the Committee to which it was referred has given assurances of a favorable report for a quarter of a million dollars.

The purpose of the Appalachian Exposition was to aid in the development of the Southern Appalachian Region. The new Exposition is a national, not a local project. Its work is to promote the preservation and development of the different forms of natural wealth of the entire country. Its special field, however, is to be the Southern States. The Exposition comes at a time when these States are in the midst of a great awakening. It is to be devoted in an especial manner to assist in this awakening and in directing the course of this awakening toward genuine, permanent progress and highest efficiency. The purposes are parallel with the magnificent undertakings of the National Conservation Congress. The means only are different. To every part of the Nation the Congress is sending its message. The Exposition invites the people of the Nation to view the tangible results and possibilities of Conservation on display. All fields of the Conservation work will be represented, forests, waters, lands, minerals, fish and game, and human efficiency including health, child welfare, education, home economics, good roads, and country life improvement. The Exposition is to be held at a time when special efforts are to be made by such agencies as the southern railroads and the Southern Commercial Congress to direct the tide of passenger traffic through the South. During the same period the city of Mobile, Alabama, is to entertain the Fifth Annual Convention of the Southern Commercial Congress and to hold its celebration of the opening of the Panama Canal, and plans are being made to direct southern travelers of those two months through both Mobile and Knoxville.

Expositions of the past have been commemorative and historical. They have celebrated and glorified past achievements. The field of the new Exposition is the future. It is to tell the progress which we are to make in the coming years, which we are to enjoy ourselves and to hand down to our children. It will be prophetic of the development which is to come and of the permanent enrichment of the country and its people. In the words of the late and beloved Dr. W. J. McGee, “The change thus wrought in the exposition idea is fundamental; the old exposition looked backward, the new looks forward; the old exposition was solely material, the new is essentially moral; the old was a proud boast of achievement, the new a signpost to progress and an assurance of perpetuity. The expositions of the past were as songs of achievement at the end of a good day’s work, the new may well be as living and tangible promises of a still more glorious tomorrow foreordained by the wise action of today.”

GIFFORD PINCHOT, Chairman.

JOSEPH A. HOLMES.

PHILANDER P. CLAXTON.

JULIA C. LATHROP.

CHARLES S. BARRETT.

DUNCAN U. FLETCHER.

HARVEY W. WILEY.

BRADFORD KNAPP.

LUKE LEA.

JOSEPH E. RANSDELL.

MRS. ABEL.

DON CARLOS ELLIS, Secretary.

Mr. Chairman, Knoxville has empowered me to invite to that city, to the Exposition, the fifth meeting of the National Conservation Congress. The National Conservation Congress belongs to the whole Nation, and the Nation is proud of it. For the past four years, since its birth, it has held its meetings in the North and Northwest. The South needs the Congress, particularly at this time, when it is in a phase of its great industrial awakening, and it earnestly urges that the Congress come within its bounds next year. If it should come South next year, there is certainly no more fitting place for its sessions than in that city which has done so much by its own energies and industries for Conservation as has Knoxville. It will be centering in Knoxville in that year and at that time all the forces working for Conservation throughout the United States. Knoxville is a smaller city than others in the South where the Congress might be held, but it is a city of between seventy and eighty thousand people. It has five excellent hotels. Two main railroads run through, and it has shown its ability to handle large crowds of people by the way it has taken care of the Appalachian Exposition for two years, with an average of twelve thousand visitors a day.

The Exposition is moving along parallel lines with the Congress, and it is in a way an offspring of the efforts of this Congress. It has taken up the ideas that have been promulgated by this Congress, and is going to apply them by showing at Knoxville the tangible, visible results of Conservation. The people in that section of country are in great need of instruction along these lines.

The plant already established for this other exposition is valued at between one-half and one million dollars. Already several buildings have been erected, and all this has been turned over to the new Exposition as a foundation.

I have letters with me from the various commercial bodies of the city, and this has been also heartily endorsed by the Governor of the State.

In conclusion, I wish to offer a resolution made by the Advisory Board:

> Whereas, It is the sense of the Fourth National Conservation Congress, assembled at Indianapolis, Indiana, October 1 to 4, 1912, that the National Conservation Exposition, to be held at Knoxville, Tennessee, in September and October, 1913, will be a strong factor in the advancement of the Conservation and wise use of the national resources of this Nation, and particularly of the Southern States; and

Whereas, It is, further, the sense of this Congress that education in the care of natural resources is particularly needed in the Southern States, where the resources are of great value and their development in a period of a great awakening, but their Conservation at a low ebb; therefore, be it

Resolved, That the National Conservation Congress hereby signifies its gratification that the National Conservation Exposition is to take place, and its earnest hope that all persons and institutions interested in the Conversation of any of our natural resources will give to the Exposition their cordial support and co-operation.

I move the adoption of this resolution.

Hon. R. M. Austin, Congressman from Tennessee—I wish to second this as a citizen of that progressive city, and I wish to join in the invitation extended by Mr. Ellis, not only to the delegates to this National Conservation Congress, but also to the citizens of this great capital city of Indiana. I hope this invitation will be accepted and this resolution just read will be passed. We will be happy to see you all when you come to sunny Tennessee, away up in the mountains, and this little city of ours of about eighty thousand people, which nestles at the foot of the Great Smoky Mountain. We will show you the richest mineral and timber section in all the Union. There are ten counties in this Congressional District. Five have coal, six iron, six marble, five zinc, two copper, and the largest amount of hardwood timber now existing on the American continent. It is an ideal location, not only for a Conservation Exposition, but an ideal place for a meeting of this great and useful organization, the National Conservation Congress of America, and we hope you will all come.

Mr. Chairman, we do not intend to open the doors of the Exposition until we know that Captain White, of Kansas City, answers “Present.” (Applause.)

I wish, while I am on my feet, to commend the very excellent report from the Committee on Resolutions submitted by the able editor of “The World’s Work,” Mr. Page, and to say that so long as I am a representative in Congress I shall, by my influence, do all that I can to carry out the principles set forth in these resolutions. (Applause.)

The motion on Mr. Ellis’ resolution was put and carried.

Mr. A. M. Loomis (New York)—I wish, very briefly, to read the action of the New York State delegation, adopted possibly before this matter of the Knoxville Exposition had become known.

> The New York delegation at this, the Fourth National Conservation Congress, wishes to go on record in favor of asking the delegates to this great body to hold the next annual meeting in the East,—to be more explicit, in New York State. There is an urgent reason why the work of the Congress at a point nearer the great centers of the business and wealth of the country, and in the section of the more crowded population would have wider effectiveness, and greater force along lines of practical understanding of its work, and needed legislation in favor of the great reforms for which it stands.
>
> One point in New York State stands out in particular as the ideal place for this Congress to gather, namely Chautauqua, the home of the great Chautauqua Institution, on the shores of beautiful Chautauqua Lake. At this point, in a little city in the woods, are ample accommodations both for meeting places, exhibits, and housing for a gathering of five thousand people. The Assembly houses more than double that number for ten weeks each summer and has an auditorium hardly excelled in America, seating more than eight thousand people, as well as many other halls and buildings for meeting places and exhibits.
>
> This institution stands for all that the highest aims of this Congress point to, in education, morality, and direction of human effort. Its reputation is world wide, and its home offers an ideal meeting place for the Conservationists, ideal in that for which the two institutions stand, and ideal in location, accommodations, railroad facilities and the economy with which a great meeting of this kind could be conducted there.
>
> The New York delegation unites in inviting the Congress to choose Chautauqua, New York, as the place of its next meeting.

Chairman White—The chair, in behalf of the delegates, wishes to thank the representatives from New York who have invited us to Chautauqua, as well as the representatives from Tennessee for inviting us to Knoxville. This subject will be referred to the Executive Committee, who will, in their wisdom, consider it all as it may relate to the best success of our cause.

Is there anything more to be presented at this convention? If not, the chair will state that the Fourth National Conservation Congress is now about to pass into history. Tomorrow will be the beginning of a new Congress—the Fifth Annual Congress, with the new President and new officers in some respects—but with a great many of the old ones, too—and we hope that all who are here will be present at the next Congress, the Fifth National Conservation Congress, wherever it may be held. And in the meantime the work will go on. It will begin tomorrow and continue throughout the year. Everywhere any delegate has influence, the cause will be heard and will be advanced.

The Chair wishes to thank this Congress and its delegates for the kind consideration given him while he has been presiding, and for the support he has received from every one. We now stand adjourned, subject to the call of the Executive Committee. (Applause.)

SUPPLEMENTARY PROCEEDINGS.

FORESTRY SECTION.

Delegates specially interested in Forestry held section meetings in the Turkish Room of the Claypool Hotel throughout the sessions of the Fourth National Conservation Congress. The Standing Committee on Forestry consisted of Prof. Henry S. Graves, Chairman; J. B. White, Major E. G. Griggs, George K. Smith, William Irvine and E. T. Allen. Chairman Graves, being unavoidably absent, delegated Mr. Allen to arrange meeting facilities and represent him in an effort to further the progress of forestry at the Congress.

The first session of the Forestry Section was held on the evening of October 1, with about twenty-five foresters and lumbermen present. (At later sessions the attendance increased to forty.)

Mr. Allen, acting as Chairman, announced that Professor Graves had suggested that such preliminary meeting be called to determine, first, if a section meeting on Forestry should be conducted, and if so, the lines it should follow. Mr. Allen suggested the probable advantage of formulating plans for more systematic forestry work at future Congresses, and of utilizing the opportunity thus afforded to exchange experiences and ideas on legislation, forest protection and educational work. The meeting concurred in this suggestion and determined to hold a series of meetings on Forestry at this Congress.

Second Session—10 a. m., October 2.

Mr. E. T. Allen called the meeting to order, and Mr. D. Page Simons, of California, was chosen secretary. The chair then presented a tentative program for ensuing sessions covering publicity work, co-operation in forest protection, needed forest legislation, and organization for future Congresses. He described the educational work conducted by the Western Forestry and Conservation Association, and read a communication from Professor Graves, United States Forester, emphasizing the need of a propaganda for more adequate and uniform State forest legislation.

Mr. T. B. Wyman, of Michigan, representing the Northern Forest Protective Association, then described the co-operative effort by Michigan lumbermen covering a territory of seven and one-half million acres. He told how they had been enabled to maintain a patrol service and that their association had made a careful study of fire causes. In the campaign of public education, he said, they had utilized modern advertising methods.

Major E. G. Griggs, of Washington, President of the National Lumber Manufacturers' Association, pointed out the necessity of united effort in a campaign of education which would bring about a better understanding, on the part of the public, of all phases of forest industry. He emphasized the need of continuous effort throughout the year, and said that he believed there should be some national frame-work or organization which would unite the foresters and lumbermen for such continuous and concerted action. Major Griggs also praised the work of the United States Forest Service.

Mr. Charles Lathrop Pack, of New Jersey, concurred in Major Griggs' suggestion and said that he believed the Conservation Congress, meeting annually, illustrated the need of a Committee on Forestry, which would be active throughout the year. He said that he believed that other features of the Congress had been much better advertised and organized and that he hoped that before another year the work of the Forestry Committee, particularly, would be on a systematic basis with the necessary funds to carry forward its work.

Chairman Allen pointed out the need of local publicity as was illustrated by the difficulties experienced in obtaining adequate State legislation.

Mr. I. C. Williams, of Pennsylvania, Deputy State Forest Commissioner, said that taxation and not fire protection was the big forestry problem in Pennsylvania. He said that a campaign of publicity for a yield tax measure had been unsuccessful owing to a lack of organization among the friends of the measure to back up the publicity.

Dr. Henry S. Drinker, of Pennsylvania, President of Lehigh University, reported the distribution of a million circulars on forest protection, modeled on those issued by the Western Forestry and Conservation Association. He also endorsed the yield tax principle.

Mr. E. A. Sterling, of Pennsylvania, emphasized the importance of conducting a systematic campaign of publicity which would bring out definite facts. Competent committees, he said, should be in charge of such work so that the publicity would be in effective form and carry weight.

Hon. John M. Woods, Mayor of Somerville, Mass., suggested the danger of relying too much on education and not enough on practical politics. In his judgment, forest legislation could best be furthered by interesting the Governor and the Legislature.

Mr. Henry E. Hardtner, of Louisiana, told of the forest laws of that State and of his effort to secure reforestation.

Prof. F. W. Rane, State Forester of Massachusetts, said that results are a question of enterprising organization and that more system and effective committee work will bring better results.

Col. W. R. Brown, representing the New Hampshire Forestry Commission, said that he believed the American Forestry Association offered facilities for the work under discussion and that means for utilizing them could be devised.

Mr. F. A. Elliott, State Forester of Oregon, then outlined western problems which he showed were peculiarly difficult because of a lack of forest appreciation in a new country. He testified to the efficiency of advertising propaganda to reduce fire carelessness.

Mr. Hugh P. Baker, of New York, said that the Empire State went on the principle that people had to be shown and that, therefore, they were making a feature of demonstration forests and of assisting individual owners.

Mr. P. S. Ridsdale, of Washington, D. C., Secretary of the American Forestry Association, then told of the educational policy of that organization, and said that its magazine was devoting special attention to all practical matters of interest to lumbermen.

After some further discussion along the line of desirable committee action the Chair was instructed, by motion, duly seconded and carried, to appoint two committees, each of which he should be ex officio chairman, as follows: A committee of five on permanent organization, and one of three to represent the Forestry Section in a conference with the American Forestry Association and the officers of the Fifth National Conservation Congress. It was also agreed to appoint a Committee on Resolutions. These committees were appointed, as follows:

Co-operation with Other Agencies—E. T. Allen, chairman; H. S. Graves, and J. B. White.

Permanent Organization—E. T. Allen, Chairman; F. A. Elliott, Don Carlos Ellis, T. B. Wyman, and F. W. Rane.

Resolutions—Dr. Henry S. Drinker, chairman; F. W. Besley, D. P. Simons, P. S. Ridsdale, and H. E. Hardtner.

Third Session—2:40 p. m., October 2.

Co-operative Forest Protection was announced for the topic for discussion.

Mr. Hardtner told of the success of the Louisiana lumber associations in securing legislation.

Mr. Wyman told of the co-operative patrol of the Northern Forest Protective Association, in Michigan, and described briefly their methods and the fire fighting equipment.

Mr. Brown explained the methods of the New Hampshire Timberland Owners' Association. There are four district chiefs, each in charge of a patrol system. They utilize all modern devices, such as telephones, lookouts, tool depots, etc. They have reduced the fire damage one-half at a cost of seven-tenths of one per cent. of the values protected. Mr. Brown urged that the adjoining States should co-operate along boundaries.

Mr. Elliott told of the progress being made in Oregon under their new law providing for syndicate co-operative patrol maintained jointly by the Federal and State governments, the counties and lumbermen.

Mr. N. P. Wheeler told of the fight against forest fires by Pennsylvania lumbermen.

Mr. D. P. Simons described the organization of the Washington Forest Fire Association, which maintains over a hundred patrolmen and protects nearly five million acres. This association also has been very successful in publicity and legislative work.

The report of the Committee on Resolutions was then presented, discussed by sections and adopted. (See resolutions of Fourth National Conservation Congress—Forests.)

Fourth Session—8:25 p. m., October 3.

Chairman Allen reported that the Committee on Resolutions of the Conservation Congress, of which he was Secretary, had endorsed the resolutions presented by the Forestry Section.

Chairman Allen then read the following report from the Section Committee on Permanent Organization:

> Your committee believes that the consensus of opinion of the lumbermen and foresters assembled at the invitation of the forestry committee of the Fourth National Conservation Congress is about as follows:
>
> 1. That the Congress has not so far included satisfactory facilities for securing for forest matters the attention they deserve at such a meeting.
>
> 2. That the facilities to be desired should provide for two main activities:
>
> > (a) The general discussion of forest Conservation needed to bring its importance properly before the public.
> >
> > (b) The meeting for mutual help, in practical constructive detailed work of the men actually engaged in organized forest work.
>
> 3. That unless there is early assurance of such facilities hereafter, the Congress' support from forest interests is in danger.
>
> 4. That private, state and federal forest interests are anxious to support the Congress and in turn to receive all benefit to be derived from it.
>
> 5. That what is clearly needed is a greater recognition of forestry upon its general program and arrangement for sectional forest work outside the general meeting, both to be carefully planned in advance so as to be practical, effective and without lost time.
>
> 6. That probably similar steps should be taken to provide for other branches of Conservation work, so that all may unite in perpetuating the usefulness of the Congress.
>
> 7. That the duty of your committee is to bring about the things outlined above, or at least to suggest some means of doing so.
>
> After careful consideration of what these seven points involve, your committee feels that the very fact that inadequacy in the past has prevented as wide an attendance as desirable, prevents us from conferring at this time as fully with all agencies involved as would be sure to get the best result, and that in particular we are at a great disadvantage in being unable to confer with the executive officers of the 1913 Congress not yet chosen.

For these reasons we recommend as our very best judgment that this meeting correct us as far as may be necessary in stating its beliefs and desires and then leave working out the detail until we can offer the executive officials of the next Congress the courtesy of consulting with them, with the understanding, however, that there shall be no negligence or unnecessary delay and that long before the next Congress all these matters shall be arranged in detail and given the necessary publicity.

Your committee consequently recommends further either that it be given instructions to act as suggested, or that it be discharged and the duties outlined be added to those of the committee of three already appointed to discuss similar questions. We believe that a faithful attempt to work the matter out in this way will be more satisfactory than trying to settle matters at this session. There is ample time if we do not waste it, and less danger of error.

The report was adopted, following the suggestion that the Committee on Permanent Organization be discharged and its duties imposed upon the permanent co-operative committee, including E. T. Allen, Prof. H. S. Graves and J. B. White.

Mr. Allen, being called out to assist in revising the resolutions of the general Congress, asked Mr. Sterling to take the chair, and suggested the reading of a paper sent by Chief Forester Graves, outlining the policy of the Forest Service.

Mr. Graves' paper (appearing elsewhere in the proceedings of the Congress) was animatedly discussed, the meeting without dissenting voice approving the Forest Service policy and deploring any attempt to restrict its operation. Short talks urging its support by all forest interests, State and private, including the Conservation Congress, were made by Z. D. Scott, Minnesota; F. A. Elliott and H. D. Langille, Oregon, and W. H. Shippen, Georgia. A resolution was passed emphasizing the meeting's endorsement of the resolutions commending the Forest Service then before the general Congress (and adopted the following day).

Mr. Langille spoke particularly against the turning over of the National forests to State control and Mr. Shippen of the necessity of Federal control of interstate watersheds.

A discussion of State legislation followed. Mr. Scott described the effort of Minnesota under its new law. Leonard Bronson, Washington, outlined the trend of attempted tax reform, dwelling particularly upon the yield tax system proposed by Professor Fairchild of Yale University, and urged concerted, harmonious effort by all forest States. Dr. Drinker and Mr.

Wheeler reviewed the proposed Pennsylvania law for a nominal land tax and a yield tax from which counties are to be reimbursed for taxes lost during growing period.

Upon motion of Mr. I. C. Williams, Pennsylvania, the meeting went on record as considering tax reform to promote reforestation and better forest management, the most important problem and the one most in need of study and legislation of any before the forest interests of the United States today.

The Forestry Section of the Fourth National Conservation Congress then adjourned, leaving plans for more effective work in 1913 in the hands of the committee of three previously mentioned.

REGISTER FORESTRY SECTION MEETING.

E. T. Allen, Western Forestry and Conservation Association, Portland, Oregon.

Wm. G. Atwood, Chief Engineer L. E. & W. R. R. Representing American Railway Engineers’ Association, Indianapolis, Ind.

Hugh P. Baker, New York State College of Forestry, Syracuse, N. Y.

W. E. Barns, Missouri Forest Service, St. Louis, Mo.

F. W. Besley, State Forester, Baltimore, Md.

F. H. Billard, Forester, New Hampshire Timberland Owners Association, Berlin, N. H.

Leonard Bronson, Manager National Lumber Manufacturers’ Association, Chicago, Ill.

W. R. Brown, President New Hampshire Forestry Commission, New Hampshire Timberland Owners’ Association, Berlin, N. H.

L. S. Case, Weyerhaeuser & Company, St. Paul, Minn.

W. C. Darms, Wisconsin Forest Commission, Wisconsin.

Chas. C. Deam, Secretary Indiana Board of Forestry, Indianapolis, Ind.

Henry S. Drinker, Lehigh University, South Bethlehem, Pa.

F. A. Elliott, State Forester, Salem, Oregon.

E. G. Griggs, West Coast Lumber Manufacturers' Association, National Lumber Manufacturers' Association, Tacoma, Wash.

N. H. Guthrie, Indiana State Forestry Association, Franklin, Ind.

Henry E. Hardtner, Louisiana Forestry Association, Urania, La.

John W. Kellough, Ohio State Forestry Association, Mt. Sterling, Ohio.

H. D. Langille, Oregon Conservation Association, Portland, Oregon.

William R. Lazenby, Ohio State Forestry Association, Columbus, Ohio.

Henry Nelson Loud, Au Sable, Mich.

Frank E. Mace, Forest Commissioner, Augusta, Me.

Mrs. Joan E. Moore, Indiana State Forestry Association, Kokomo, Ind.

John Oxenford, Indianapolis, Ind.

Charles Lathrop Pack, President Fifth National Conservation Congress, 305 Euclid Ave., Cleveland, Ohio.

F. W. Rane, State Forester, Boston, Mass.

P. S. Ridsdale, Secretary American Forestry Association, Washington, D. C.

Z. D. Scott, State Forestry Board, Duluth, Minn.

W. H. Shippen, Hardwood Manufacturers Association, Ellijay, Georgia.

D. P. Simons, Western Forestry and Conservation Association, Los Gatos, Cal.

Geo. K. Smith, Secretary Yellow Pine Manufacturers' Association, National Lumber Manufacturers' Association, St. Louis, Mo.

E. A. Sterling, Forest and Timber Engineer, Philadelphia, Pa.

R. D. Swales, Union Lumber Company, Fort Bragg, Cal.

F. L. Throm, Forester, Wheeler & Desenburg, Endeavor, Pa.

William P. Wharton, Groton, Mass.

N. P. Wheeler, Pennsylvania Conservation Association, Endeavor, Pa.

I. C. Williams, Pennsylvania Department of Forestry, Harrisburg, Pa.

E. B. Williamson, State Foresters Office, Bluffton, Ind.

John M. Woods, Somerville, Mass.

R. C. Young, American Railway Engineers' Association, Chief Engineer Munsing R. R., Marquette, Mich.

THE PRESENT SITUATION OF FORESTRY.

Prof. Henry S. Graves, United States Forester.

A review of the work of forestry in this country during the past year shows that in many directions there has been substantial progress and positive achievement. On the other hand, the continued organized attacks on the National Forest system, and the efforts to break it down or cripple it, present a situation of real danger which the country should realize and vigorously meet. We have before us a task of constructive activity in practical work, extending and building on foundations already laid; we have also the task of preventing a destructive attack upon National forestry.

During the past few years public interest in forestry has been rapidly changing from a mere inquiry in regard to its purpose to a vigorous demand for practical results. This more intelligent public sentiment is now finding its expression in a growing appreciation of the need of better forest laws, greater State appropriations for fire control, and increasing interest in forest protection by private timberland owners. It often happens that public attention is caught only by the most striking new departments and developments, such as a change in public policy or important legislation, while but little is known of the steady advance in applied forestry. The past year has been signalized not so much by new undertakings as by marked accomplishment in the effective carrying out of work previously inaugurated.

PROGRESS IN NATIONAL FORESTRY.

Every year shows increased efficiency in the administration of the national forests. The most conspicuous advance has been in organized fire protection. The disastrous year of 1910 taught many lessons. While that disaster could not have been avoided in the absence of better transportation and communication facilities and without a larger patrol force than the Forest Service could put into the field, it nevertheless showed how, even under the present conditions, the work of protection could be made more effective. Full use was made of the experience gained in that year, and during the past two seasons the loss by fire has been kept down to a comparatively small amount through the efficient system now in force. The problem, however, of fire protection on the national forests is far from being solved. There still remain to be built some 80,000 miles of trails, 45,000 miles of telephone lines, many miles of roads, many lookout stations, and other improvements, before even the primary system of control will have been established. The funds at the disposal of the Forest Service are still inadequate to employ the patrolmen needed to meet more than ordinary emergency. There is even yet danger, therefore, that in the case of a great drought like that of 1910 some fires might gain the mastery and a similar disaster follow.

An account of the progress of the work of the Forest Service in the administration of the national forests would be an enumeration of the different activities in which the work is going on with constantly growing effectiveness. Many of the local difficulties of administration are rapidly disappearing. This is due to the steadily closer co-ordination of the interests of the Government with those of the people living in and using the forests. More and more these people are coming to appreciate that their interests and those of the national forests are one. With a better understanding of the aims and methods of the Forest Service, local difficulties are disappearing and local support of the service is largely replacing opposition. Those who are aiming to destroy the national forest system are not the settlers and others who use the forests, but rather men who seek for their own advantage special privileges to which they are not entitled, and who wish to acquire for little or nothing valuable resources for speculation and personal gain.

During the past year the Weeks law, authorizing the purchase of lands on navigable streams, has been put into effect, and the Government has already entered into contracts for the purchase of 230,000 acres in the Southern Appalachian Mountains, and about 72,000 acres in the White Mountains. These lands are being secured on the most desirable areas, and it has been possible to obtain them for reasonable prices. A special feature of the Weeks law is the co-operation between the Government and the States in fire protection on watersheds of navigable streams. The law provides $200,000, until expended, for such co-operation; but this money can be used only in States which have already inaugurated a system of fire protection under public direction. During the year 1911 there were eleven States which qualified under this law, receiving in the aggregate about $40,000. During the current year sums varying from $1,500 to $10,000 have been allotted to the States of Maine, New Hampshire, Vermont, Connecticut, New York, New Jersey, Maryland, Wisconsin, Minnesota, Oregon, and Washington. There is still sufficient money left from the original appropriation for substantial co-operation during another year. It has been the aim of the Forest Service to spread the money over three years in order that there may be a full demonstration of what can be accomplished, and at what cost. It will then be possible to present to Congress a satisfactory basis upon which to consider whether Federal aid to the States should be continued.

The most urgent need of the national forest work is more ample provision of the funds necessary for adequate protection of the forests against fire. It is especially urgent that the work of constructing roads, trails, telephone lines, and other improvements needed for fire protection be extended much more rapidly than at present.

PROGRESS IN STATE FORESTRY.

A very great obligation rests upon the State governments in working out the problem of forestry. Organized fire protection under State direction, the establishment of a reasonable system of taxation of growing timber, honest and conservative management of State forest laws, education of woodland owners to better methods of forestry, and such practical regulation of handling private forests as may be required for the protection of the public, are problems which require the immediate action of all States.

While no State is as yet accomplishing all that it should, a number of them are making very rapid progress, and are giving as liberal money support as perhaps could be expected under the present conditions. The feature of State forestry which stands out most strongly is that a number of States have gone beyond merely passing forest laws, and have begun to provide the funds necessary to achieve practical results. At last it is beginning to be recognized that the prevention of fire is the fundamental necessity, and that this can be accomplished only through an organized public service. In order to make laws effective there must be adequate machinery to carry them out. The fundamental principle of fire protection is preparation. A forest region must be watched for fires, both to prevent their being started and to reach quickly and put out such as from one cause or another may get under way. The new State legislation recognizes this need, and already there has been inaugurated a measure of watchfulness in the season of greatest danger, through patrol or lookouts under State direction. During 1911, which was a banner year in the enactment of State legislation, laws related chiefly to fire protection were passed by Connecticut, Massachusetts, Minnesota, New Hampshire, New Jersey, Oregon, Washington, and Wisconsin; while Colorado created the office of State Forester. Since the beginning of 1912 Maryland and New York have amended their forest laws, and Kentucky has passed its first complete law.

It is exceedingly gratifying that substantial progress is now being made in the South. Unfortunately, however, none of the Southern States except Maryland has hitherto been able to qualify to receive Federal aid and fire protection under the Weeks law. It is hoped that during the coming year progress will be made in those Southern States in which practically nothing has yet been done.

One of the matters to which the Conservation Congress and all other educational agencies should devote their efforts is to bring about the protection of private lands from fire and the extension to them of forestry methods. While some may say that this is a matter for which the owner is personally responsible, the fact remains that private owners will ordinarily not work out the forestry problem on their lands without the participation of the public in the form of public regulations, co-operation and assistance. This is recognized in some States, but others are doing nothing whatever in this field, and a good many which have made a small beginning are abundantly able to do vastly more than at present. It has usually happened that the securing of good forest laws and the establishment of a State Forest Service has been brought about by the efforts of a small group of interested men, and frequently through the efforts of a single individual who has been able to arouse the interest of the people in his State. Enough States in different parts of the country have initiated State forestry to make it comparatively easy for a State contemplating new legislation to benefit by what has been done elsewhere. All that is really required in the extension of State forestry is to find the man or men in each State who will take the leadership and follow up the matter until the Legislature acts. It would seem that in the heavily timbered States the lumbermen are the men who should be most vitally interested in the conservation of our forests. In some States timberland owners have participated very actively in bringing about State forestry, as for example, in Maine, New Hampshire, Minnesota, and some of the far Northwest States. In other instances the timberland owners have been indifferent and in some instances proper State forestry has failed on account of the attitude of the very men who should be foremost in promoting proper legislation. We need in each State not so much advice from the outside as a few patriotic citizens in whom the public has confidence and who will devote time and real effort to this public task. If the men can be found to do this preliminary work they will have no difficulty in securing competent assistance from other States and from the National Government.

THE ATTACK UPON FORESTRY.

At the same time that forestry has been making steady progress in constructive work and in public esteem, hostility to the national forest policy on the part of those who would substitute private for public control of these resources has become more determined under a new form. The early attacks upon this policy openly sought its overthrow. They came to nothing because the country was emphatically for the forests. At the present time those who attack the national forest policy commonly profess allegiance to the Conservation principle even while attempting to break it down. There is great danger that the public may not understand what is involved in measures whose purpose and inevitable effects do not appear on their face. Two such measures are the proposal to require the elimination from the forests of all lands capable of cultivation, on the plea that this will increase settlement, and the proposal to turn all the forests over to the States in which they lie, on the plea that this will increase their benefits to the people of these States. In both cases quite the contrary is true.

An amendment which was attached to the Agricultural Appropriation Bill last June, and which passed the Senate but was rejected by the House, would have required, had it become law, the opening to private acquisition under the homestead laws of all lands "fit and suitable for agriculture" within national forests, irrespective of their value for other purposes or of their importance for public use. The result would have been not to facilitate but to block agricultural development. It would also have been to transfer to powerful private interests timberlands, water power sites, and other areas, possession of which would tend to private monopoly of resources now under public control.

This measure is not called for in order that agricultural development of lands in national forests may take place. The Forest Service has consistently favored and sought to bring about agricultural settlement of all national forest lands which can be put to their highest usefulness by farming. It urged and obtained, seven years ago, the law which now permits the opening of such land. Under that law about one and a half million acres have been listed for entry by over twelve thousand settlers; and more will be listed as it becomes possible to list the land without defeating the very purpose of the law.

To open land certain because of its superior value for timber, water-power development, or other purposes to be absorbed by speculators or powerful interests would not only defeat the purpose of the existing law but also constitute a breach of public trust and a betrayal of the fundamental principles of Conservation. That principle has often been misrepresented as a policy of present non-use for the sake of future generations. Its true purpose is two-fold: to prevent monopoly of public resources, and to secure their greatest use, both present and future, by scientific development. The national forests are administered with a view to securing, first, use of present resources; second, permanency of such resources; and third, greater and more valuable resources for the future.

Experience has amply proved that the elimination, under pressure, of national forest lands locally considered or alleged to be of agricultural value but in point of fact more valuable for other purposes has led to their early acquisition by timberland speculators, great lumber interests, water-power companies, livestock companies, and others who desire the lands for other ends than agriculture. In 1901 705,000 acres of heavily timbered land were thus eliminated from the Olympic National Forest. Ten years later only a little over one per cent of this land was under cultivation, while three-fourths of it was held for its timber, mainly in large holdings. Other examples might be multiplied. With a mandatory law the pressure for opening land sought under cover of the claim of agricultural value would be well-nigh irresistible in many cases. Local agitation and political influence would in the end break down all effort to maintain public control. Such piecemeal attack on the forests would be made without any opportunity for the public to know what was going on. In the end the dismemberment of the national forests would be effected.

The only safety for the maintenance of the policy which now receives and has long received the overwhelming support of public sentiment lies in a correct knowledge by the public of the actual situation with regard to agricultural lands in national forests. It must be made plain that all but an entirely insignificant part of the national forests is not susceptible of profitable cultivation. The forests occupy the most rugged and mountainous parts of the West. Topography, soil, and climate combine to make them natural forest lands, not potential farm lands. The areas which form an exception to this condition are not over four per cent. of the total; and such areas are now being sought out by the Forest Service and will, under the existing law, be made available for homestead entry as fast as they can be opened without defeating the purpose of the law itself. It is necessary that the country should understand the manner in which bona fide settlement is being brought about in the national forests, and also the motive of those who are trying to break down the system of forest Conservation under the guise of promoting settlement.

There has been during the past two or three years a steadily growing movement to turn over the national forests to the individual States. During the past session of Congress a rider to the Agricultural Appropriation Bill was offered in the Senate, providing for the grant of the national forests to the several States, together with all other public lands, including "all coal, mineral, timber, grazing, agricultural and other lands, and all water and power rights and claims, and all rights upon lands of any character whatsoever." While the amendment was ruled out on a point of order, it received a surprisingly large amount of support.

The proposition so far as the national forests are concerned is to turn over to the individual States property owned by the Nation covering a net area of over one hundred and sixty million acres. This property has an actual measurable value of at least two billion dollars, while from the standpoint of its indirect value to the public no estimate on a money basis could possibly be made. These are public resources which should be handled in the interests of the public. Moreover, the problems involved are such that they should definitely remain in the hands of the National rather than be turned over to the State governments. The property belongs to the Nation as a whole, and every citizen has an interest in it. The Government has already made enormous grants to the individual States, but always to further specific objects of national importance. There should not be a moment's consideration of the proposal to turn the forests over to the States unless it can be clearly shown that the interests both of the States and of the Nation are consistent with such action. In the case of the national forests, public interests both of the Nation and of the States require their continued retention and management by the National Government.

The scope of this paper does not permit a full discussion of this problem. It must suffice to mention a few cogent reasons for government ownership:

1. The property is now owned by the Nation, and should be administered from the standpoint of national as well as of local needs.

2. The problem of protection from fire and of timber production on the national forests is one of national scope and can be properly handled only by the Government; its solution is a national duty.

3. The problem of water control is no less a national duty. Nearly all of the national forests lie on headwaters of navigable rivers or interstate

streams. The Government is now purchasing lands in the East on headwaters of navigable rivers because of the disastrous results to the public which are following abuse under private ownership. It certainly should not part with title to the same class of lands which it now owns in the West. Every interstate stream presents problems which can be properly handled only through the Federal Government. The Government cannot permit the citizens of one State to be damaged by the action or failure to act of citizens of another State. It is of vital importance for this reason alone that property at the headwaters of interstate streams be retained under Government administration.

4. Not only are the interests of the individual States and communities now fully protected, but in many ways far more is being done for local communities than would be possible under State ownership. In the long run, as the timber and other resources are brought into use with improving markets, the States will receive from the twenty-five per cent. of the gross receipts now allowed them and the additional ten per cent. appropriated for road improvements a larger amount than would come in from local taxes under private ownership.

5. The States are not as well prepared, financially or otherwise, to handle the national forests as is the Federal Government. If the forests were owned by the States and handled in the real interests of the public, there would be substantially the same system of administration as today, at a greater aggregate cost for supervision by a considerable number of independent State staffs of technical men. The financial burden would be far too great for the individual States to assume. The result would be either poor administration and lack of protection, or a sacrifice of the public interests in order to secure revenue to meet the financial needs.

6. The successful application of forestry demands a stable administrative policy for long periods. This can be secured far better under National than under State control.

7. A much higher standard of constructive and technical efficiency is possible under National than under State administration. The value of the forests to the public depends directly on the skill with which scientific knowledge is applied to the task of developing their highest productiveness. Both in ability to carry on the research work required for practical ends and

in ability to command professional services of the first order the Government possesses a striking advantage.

8. As largely undeveloped property the forests need heavy investments of capital for their improvement. Their full productiveness can be secured in no other way. The Government is now investing yearly in the forests a considerable part of the appropriation made for them. Even if the States did not seek to make them sources of immediate revenue at whatever sacrifice of their future possibilities, they would be reluctant to expend much for their development.

9. The States both lack the civil service system and standards of the National Government and are exposed to greater danger of being swayed by private interests. In the hands of spoilsmen demoralization would quickly succeed the present high standards of the Forest Service, while the intimate relation of the forests to the welfare of great numbers of individuals would tend to make their administrative control a highly coveted political prize. At the same time the value of their resources would certainly arouse a cupidity which would be exceedingly difficult to control. Scandalous maladministration might easily follow. The Federal Government is better watched farther removed from local influence, more stable, and better equipped with a non-political system and machinery.

The underlying purpose of the proposed transfer of the national forests to the States is really not to substitute State for Federal control, but rather to substitute individual for public control. Its most earnest advocates are the very interests which wish to secure such control. The object of the whole states rights movement as it affects the national forests is to transfer to private owners for speculative or monopolistic purposes public resources of enormous value. Retention of these resources under public ownership is needed to protect the people from abuses which are every day being demonstrated on lands over which the public has already lost control. The proposition is one which the people as a whole would repudiate in an instant if they understood what is proposed. The only danger lies in the fact that some legislation adverse to the national forest system may be passed when the public as a whole is ignorant that it is planned or does not understand the meaning. Vigilance in the defense of its interests and intelligence in the perception of the true character of masked attacks upon those interests are of fundamental necessity if the public is to protect itself.

FOOD SECTION.

The Food Section of the National Conservation Congress met in the Palm Room of the Claypool Hotel on the afternoon of October 1st. Dr. H. W. Wiley, late Chief of the Bureau of Chemistry, as Chairman, discussed the cold storage industry and pointed out that cold storage is a great blessing to the country, in that goods are placed in cold storage that they may be more evenly distributed throughout the year. He showed that there is still room for the investigation of the principles of storage and improvement of the industry. The condition of food entering cold storage is most important.

Frank A. Horne, Chairman of the Commission of Legislation of the American Association of Refrigeration, said there has been a remarkable reversal of public opinion in the last three or four years regarding the place cold storage and refrigerating has occupied with regard to the high cost of living.

He declared the cold storage business has been unjustly assailed, and that a series of investigations and hearings had demonstrated beyond doubt that the popular notion and sensational newspaper attacks were entirely unfounded and erroneous. He said these investigations showed that the cold storage warehousemen performed a useful public function in conserving perishable foods, preventing deterioration and waste by means of a scientific method by which the great surplus production could be wholesomely preserved for later consumption.

Before cold storage came into use a period of flush production meant a glut in the market, and large quantities of spoiled and utterly wasted foods. With cold storage at hand the contrary conditions prevail.

At the general discussion on the subject afterward, Dr. Wiley said the attacks made on the cold storage business five years ago were justified by conditions. He said as a result of an investigation of the business, the cold storage men themselves have joined with the Government to improve conditions.

Charles H. Utley, President of the Quincy Market, Cold Storage and Warehouse Company, Boston, said there would be no occasion for cold storage or the use of any other means for preserving food if human food was not to a greater or less extent perishable. If it were not perishable it would be the practice of every individual to conserve a sufficient amount of food as might be required. No better means of preventing waste of food is known at the present time than by the use of cold storage, and the use accomplishes most desirable results, advantageous to both consumer and producer, by the conservation of food, which is just as desirable as the conservation of our natural resources.

Dr. William A. Evans of Chicago, discussed the capacity of milk for doing harm even when it looks, tastes and smells right. He said milk is a great carrier of disease germs, and that for this reason it should be produced close to the point of distribution. The nearer the baby gets to the cow the more natural it is. Certified milk is all right when it is really certified by a noninterested person, but properly pasteurized milk is probably the safest for babies.

Dr. H. E. Barnard, Food and Drug Commissioner of Indiana, referred to the fact that Indiana was the first State to pass a cold storage law. He introduced resolutions pertaining to the conservation of food, which were unanimously adopted. The resolution follows:

> Whereas, The Conservation of the food products of our country is of the greatest importance to our people, in order that they may have available the maximum supplies of wholesome food; and
>
> Whereas, The subject is deserving of serious consideration, so that production may be encouraged and waste decreased; and
>
> Whereas, The important function of the process of refrigeration is enlarging and diversifying the supply of perishable foodstuffs, as applied in the preparation, transportation and distribution of these goods, thereby giving consumers a larger and more wholesome supply, preventing deterioration and waste, is recognized as being desirable and necessary; therefore, be it
>
> Resolved, That any legislation or administration restrictions or regulations that may be required to properly control the business and protect the public health should be uniform in the several states of the Nation, and be it further
>
> Resolved, That the Congress recommends that the succeeding food committee of the National Conservation Congress be specially charged with the duty of studying the questions involved in the production, collection, sanitary preparation, transportation, preservation and marketing of perishable foods, and to report its findings to the succeeding Congress to the end that such report may be made the basis

of measures to better conserve the perishable foods of the people, to improve the quality of such foods, increase their production, and to promote such relations between the producer, handler and consumer as will bring about a more nearly uniform price through each year.

FOOD CONSERVATION BY COLD STORAGE.

By F. G. URNER, Editor, New York Produce Review.

The conservation of food may be considered from two points of view—first, the safeguarding and preservation of the food currently produced; second, the maintenance of those elements of fertility upon which continuous production depends, and the improvement of methods of production to the end that maximum yield may be realized from the labor and material expended. Both considerations are of the utmost importance in the present conditions of changing relation between the domestic supply of food and the needs of nonproducers. In both progress toward higher ideals is dependent upon an increase of knowledge, and worthy of such educational forces as can be brought to bear by a wise government. In both directions the United States Government, through the Department of Agriculture and otherwise, is endeavoring, by investigation, study and the dissemination of ascertained fact, to foster progress for the common good.

In the United States the development of food production to keep pace with the needs of a population increasing at a rate beyond all precedent, has been crude and wasteful. Beginning with virgin soils the stores of primitive fertility have been drawn upon with little regard for their steady depletion. Methods of careful and conservative agriculture that have been forced upon older communities have been largely ignored until comparatively recent years, when an appreciation of the near approach of the inevitable results of waste has turned forceful educational efforts toward a reformation—efforts which, however, have been handicapped by the necessity of overcoming the prejudice of ignorance and long established habit of carelessness.

Considered broadly, the question of conservation of food is far-reaching and extends to innumerable details. It is the purpose in this paper to discuss simply some of the general principles underlying the subject from the first mentioned viewpoint—the safeguarding and preservation of the food produced—particularly in respect to preservation by cold storage.

It is hardly necessary to enlarge upon the general requirement of food preservation. In northern latitudes, where months of production are, in respect to a large part of the food supply, followed by months of nonproduction, this necessity is evident not only to maintain a satisfactory variety of food but to secure a sufficient quantity. In the United States differences of climatic conditions, although giving an almost continuous production of certain vegetable foods, do not serve to furnish an uninterrupted supply of fresh products of many staple kinds, nor are they sufficient to remove the necessity for utilizing the productive power of the colder regions far beyond the consumptive needs during the comparatively brief seasons of harvest. The practice of food storage from the season of natural production through the season of nonproduction is, of course, to some extent, as old as life itself; but the methods of preservation have shared in the improvements that have characterized a modern civilization. And the development of these advanced methods has brought into the question of food preservation new problems, some of which it is the purpose in this paper to discuss.

Methods of food preservation may be broadly divided into two classes—first, those which accomplish their purpose by changing the physical condition of the food, as by drying, or cooking and hermetical sealing; and second, those which preserve the articles in such manner that, when used, they shall be practically in their original condition. The latter methods depend for their effectiveness upon the provision of such environment as will check or retard the forces of deterioration or decay, and it is in the ability to provide such conditions by an artificial control of temperature and humidity that the preservation of food in apparently unchanged physical condition has been greatly extended.

So long as food products were chiefly preserved from the seasons of production, or maximum production, to the season of nonproduction by the use of somewhat primitive means, and largely by producers themselves, or by methods familiar to the household, the food so held was accepted by the people as a matter of course and recognized necessity. Canned and dried foods were, and are, used with general satisfaction as such; and such staple fruits and vegetables as could be carried in their original condition through the winter months were consumed with a general knowledge of their age, but with a full appreciation of the necessity for such holding and of the comparative excellence of the held goods. Butter and eggs also, when held by producers themselves, even by primitive and inefficient means, were accepted by consumers in seasons of natural scarcity with resignation as to their comparatively poor quality under a general knowledge that nothing better could be expected at prices within common reach.

These conditions remain unchanged today in respect to those forms of preserved food whose character is evident either because of their change of form or because of a popular knowledge that the articles, though indistinguishable from fresh products, must have been held from a crop harvested long ago. But the development of preservation by effective artificial control of temperature has brought some new elements into the situation.

Cold storage has enlarged the number of food products preservable in their original condition and created a new industry; it has largely removed the function of this class of food preservation from the scattered individual producers to large central establishments and thrown the business of accumulating and conserving surplus more fully into the hands of tradesmen. It has permitted the preservation of flesh foods in a raw state which were never before so preservable; and it has so improved the quality of stored products whose current production never ceases entirely that in many cases the held goods cannot be distinguished from the fresh production.

These facts have led to a popular apprehension that cold storage, being utilized largely by nonproducers and necessarily upon a speculative basis, is made a tool for extortion or unjust profits; also that deception is practiced, in respect to foods whose production never ceases entirely, by the substitution of stored food for fresh; and exaggerated statements as to the length of time foods are held in storage have brought in question their wholesomeness and created a popular prejudice.

It is important to know the facts in these particulars so that the true function of cold storage in the preservation of food may be understood, especially because legislative restriction of the industry has been effected in some States and is under consideration in others, as well as in the Federal Congress, in the enactment of which mistaken views have resulted and may further result in public injury.

COLD STORAGE ECONOMICS.

It is a self-evident proposition that, in respect to foods the production of which is seasonal, the ability to preserve a part of the yield to the period of nonproduction lessens waste and permits a material increase of production, thus increasing the available food supply. It is also evident that, supposing all the food produced to be marketed and consumed, an increase in the supply of food tends to a lowering of its average price. Apart from inevitable variations due to climatic conditions the production of particular foods increases or diminishes according to the relative profit realized from that production; and it is evident that a profit sufficient to induce production can be realized upon a much greater output if the period during which consumption is possible can be extended. A maximum production of any food can be realized only when the period of its availability for consumption is constant; and it follows that the maximum production of foods whose yield or greatest yield is seasonal, can be realized only by preservation of a part of the production for use during the season of natural dearth or deficiency which ends only with the beginning of the following period of maximum production.

Upon these simple truths rests the economic utility of cold storage preservation. Practically its benefits in the conservation of food, and in the encouragement of maximum production, are to be gained only through the opportunity for profit, and while the business of carrying foods from seasons of abundance to natural scarcity is open to all it is naturally conducted chiefly by the tradesmen who are permanently engaged in food distribution, and who are most familiar with trade conditions and the varying relations of supply and demand.

An important fact bearing upon the practical use of cold storage preservation as a feature of the distributing business is that no profit can be expected by holding products beyond the succeeding period of maximum production, when prices naturally fall to the lowest point. The variations in selling prices at that period are never sufficient to cover the cost of carriage of goods from a previous season and the lessened value of long stored products in comparison with fresh. There are occasional market conditions which have induced the holding of perishable foods in cold storage beyond twelve months in the effort to lessen a loss, but they are rare and exceptional, so much so that a legal restriction of the period of permissible holding to twelve months would have very little effect upon the inducement to utilize cold storage from a commercial standpoint. But so far as the purely economic interests of consumers are concerned it would appear that no restriction of the period of permissible holding of food in cold storage is either necessary or desirable. The inducement to hold is profit, and profit can be realized only by selling into final consumption. And when goods can be carried to a later date and sold at a higher price it is evidence that the relative scarcity which results in that higher price would have been more stringent had the goods not been so carried. In respect to the time of selling stored foods, therefore, the interests of consumers (as a whole) and of owners of the food, would seem to be identical; for it is the increased public need which results in the higher price, and profit, considering storage operations as a whole, depends upon a correct judgment as to that need.

There seems, therefore, no means by which tradesmen dealing in raw foods can utilize cold storage preservation for their own benefit at the cost of a public injury, but that, on the contrary, the profitableness of holding surplus depends upon the performance of a public service.

The ideal function of cold storage preservation is to carry just such amount of surplus from the time of greatest yield as can be consumed during the later period of relative scarcity at just sufficient advance in value as will cover the cost of carriage and afford a maximum satisfactory profit for the conduct of the business and the necessary investment of capital. But it is impossible that this ideal can be uniformly realized. Even if the operations of storage accumulation and withdrawal for market were uniformly governed in the light of the fullest possible knowledge and with the best of judgment, it would be impossible always to determine the quantity to be stored and the normal price thereof so that later deficiency at corresponding prices would be exactly offset. For the extent of later shortage can never be certainly known and the extent of demand at any particular prices is variable and uncertain. As a matter of fact, these operations of storage accumulation and later output, being carried on by thousands of individual and independent dealers, in the dim light of imperfect knowledge, even as to important statistical facts that might be known, can never result in ideal effects. Sometimes the quantity of certain foods stored at the prices paid proves to be excessive and a part of the surplus, toward the approach of the next flush season, has to be thrown upon the markets at heavy losses; sometimes the quantity put away is insufficient to offset the later scarcity and a part of the surplus, carried late, realizes for larger profits than normal. But these conditions are, to a large extent, inevitable, and while they show that the ideal function of cold storage preservation can not always be realized, they do not materially lessen its value. When a series of years is considered it will be found that the average profits are comparatively small in relation to the risks and the investment involved. And even when, during the flush season of accumulation, prices are sustained above the normal level by an amount of accumulation that later proves excessive, consumers get the surplus later at correspondingly lower prices. The reverse is also true, that when the quantity held is deficient, leading to relatively high prices in a part of the season of natural scarcity, a greater previous accumulation, sufficient to prevent so much advance, would have resulted in higher prices during the previous flush season.

The view that the economic effect of cold storage is to increase production and to lower the yearly average price of food whose production is variable is evidenced by such statistics as are available. In the manufacture of butter, for instance, the months of greatest production are from May to August, inclusive, and the months of usual deficiency are from November to March. In the New York market the average price of creamery butter from May to August during the period from 1880 to 1892, before cold storage preservation was generally used, was 21.9 cents. During the same months in the period from 1902 to 1911, when cold storage facilities were largely available, the average price was 23.4 cents. But while this comparison shows an average advance of one 1½ cents during the four months of normal accumulation of surplus the effect upon prices during the normal season of shortage was very apparent; for in the months November to March in the period 1880 to 1892 the average wholesale price was 34.3 cents, while during the same months in the period from 1902 to 1911 the average for fine fresh creamery was only 28.9 cents, and the average for fine storage creamery 26.7 cents.

THE QUALITY OF COLD STORED FOODS.

The quality of all perishable food products varies according to the methods of their production and the care taken of them during transit from producer to consumer. The more perishable foods, being produced in a very wide territory by a vast number of producers, and usually transported over long distances, are found in distributing markets to be of extremely irregular quality and condition. Usually qualities are best in the seasons of maximum production, and while goods put into cold storage are also of irregular quality most of those intended for long holding are selected, handled and packed with especial care. The effect upon perishable foods of holding in cold storage is various. It is less in respect to those carried hard frozen, as meat, fish, poultry and butter, and upon durable vegetables and fruit, as potatoes and apples, than upon animal products that cannot be frozen, as eggs in the shell. Yet in all perishable foods commonly carried in cold storage, quality, as judged by popular standards, is preservable up to the limit of usual commercial necessity, in a highly satisfactory degree. The more durable fruits and vegetables, carried in properly corrected and controlled atmospheric conditions, after months of holding, are often indistinguishable in point of quality, from those marketed soon after their harvest. Butter carried frozen for months loses very little of its original flavor and character. Poultry, also, if of fine quality and condition when frozen, may be so held for a long period without noticeable deterioration. Eggs in cold storage gradually lose the peculiar freshness of a new laid quality, but under proper conditions they remain sound, sweet and acceptable when carried at about 30 degrees temperature for at least nine or ten months. Scientific investigation conducted by the research laboratories of the United States Department of Agriculture has given no evidence of any effect of an unwholesome character upon the quality of perishable foods held in cold storage up to the limit of usual commercial practice when the products were sound and wholesome when stored.

The cold storage of surplus and the sale thereof in the markets adds not at all to the irregularity in quality of our food supply. On the contrary, the average quality of the supply is improved, for, without the facility of refrigeration, freshly marketed products would inevitably be poorer; they are now often poorer than similar goods of much greater age properly carried in cold storage. Furthermore, the length of time perishable foods are carried in cold storage is, within reasonable limits, no criterion of their quality. Perfect products, properly refrigerated for months, may be, and often are, superior in all the elements of quality to imperfect goods, freshly marketed or held only a short time. Again, because of the very widely spread sources of our food supply, the necessity for collection at innumerable points and transportation over long distances it is hard to say what goods are “fresh.” Even when collected at interior points, transported to distant markets and put into consumption with usual promptness perishable products are often two to four weeks in the transit from producer to consumer, and often under more or less unfavorable environments.

Under these circumstances it is seriously to be doubted that there is any real ethical foundation for the recent demand that, in the sale of perishable foods, there must be a stated distinction between so-called “fresh” and stored products, or for the feeling that consumers asking for broiling chickens in the winter, for instance, are deceived if furnished with acceptable goods frozen six months before. And this doubt is intensified, no matter how scrupulous we may be in standing for truth and fair dealing, when it is considered how difficult will be the enforcement of laws compelling such distinction in commodities of irregular quality and condition whose age and previous environment cannot be known by examination, and in respect to which a comparison of quality is often in favor of the older goods.

The writer’s conclusion from the foregoing considerations, based upon a long and disinterested observation of the practical use of cold storage preservation, is that artificial refrigeration furnishes the most important of all modern factors in the conservation of perishable foods, leading to an increase in their production, and to a consequent lowering of average prices. Also that governmental attention to the industry would be more usefully directed toward providing for continuous and frequent statistical information of the rate of food accumulation and output, to the end that operators may be guided by the largest possible knowledge, rather than toward any undue restriction of the industry or the imposition of costly and difficult requirements which, though seemingly designed to prevent deception, are, upon analysis, found to be unnecessary and impractical.

NATIONAL ASSOCIATION OF CONSERVATION COMMISSIONERS.

This Association, consisting of Conservation commissioners and other persons connected with various departments of State development, held two sessions during the Conservation Congress. Several important subjects were considered, but most of the time was given to a discussion of the work done by certain departments connected with public service.

The first session considered State Surveys, their work and co-operation. As a result of this meeting an agreement was reached as to the order or sequence of the surveys, the object being to secure for the States the largest returns from each survey. The sequence of the surveys and the leading points to be emphasized, as decided by the commissioners, are as follows: (1) topography, (2) structure, (3) drainage, (4) ground water, (5) local climate, (6) soil, (7) plant and animal life, (8) social and industrial conditions. This order is thought to be most helpful so far as the surveys are concerned. It is also the natural order. It was plainly shown that several States have wasted time and money in taking up the various surveys in a way that does not develop these relationships. For instance, some States have started industrial and agricultural surveys before they have mapped the geology, topography and water resources. Such an order does not bring the best results. Furthermore, it is wasteful.

Several prominent directors of State surveys took part in the discussions of this session, among them being Dr. George W. Field, Dr. A. H. Purdue, Dr. F. W. DeWolf, Professor Kay, Dr. C. E. Bessey, Dr. C. H. Gordon, Dr. Frank W. Rane, Prof. George A. Loveland and Hon. J. E. Beal. Among the other speakers were Hon. George Coupland, Mr. Ellis, W. E. Barns, Henry A. Barker, H. E. Hardtner and Dr. H. H. Waite. Dr. David White, of the United States Geological Survey, gave valuable suggestions.

At the second session of the commissioners, the forest laws of Louisiana were discussed by Hon. H. E. Hardtner, ex-Chairman of the Louisiana Conservation Commission. Following this was a general discussion of forest laws and forest management. Hon. W. E. Barns, of the Missouri Conservation Commission, gave a talk on the improvements of lumbering in the South. Prof. Earl O. Fippin of the Agricultural College of Cornell University, Ithaca, N. Y., read a paper; subject, “The Soil Survey as a Means of Agricultural Improvement.” This paper was followed by discussion, in which the value of the soil surveys as it relates to State development, was brought out with considerable detail.

The officers elected for the ensuing year were: President, Dr. G. E. Condra, Lincoln, Neb.; Vice-President, Dr. George W. Field, Sharon, Mass.; Secretary, Henry A. Barker, Providence, R. I.

ACCIDENT PREVENTION SECTION.

On October 2, 3:30 o'clock, the section on "Accident Prevention" held a large and enthusiastic meeting in the Auditorium of the German House. The presiding officer was Mr. M. W. Mix, of Mishawaka, Ind., President of the Manufacturers' Bureau of Indiana. At this meeting the following resolution was adopted:

> Resolved, That we appreciate the efforts now being made by manufacturers of machinery for use in industrial plants to so far safeguard their machines as to minimize the danger of personal injury to workers; and that as manufacturers and individuals interested in accident prevention we recognize the difficulty, and in some cases even impossibility, on the part of purchasers of individual machines to properly attach safeguards; and realize that the original manufacturers of the machines by reason of their wide experience and efficient engineers, are better able to develop and provide proper safeguards for all machinery; and that it is the sense of this meeting that any efforts along this line are highly commendable and will be appreciated by all interested in the Conservation of human life.
>
> Resolved, further; That a copy of this resolution be sent by the President to every machine manufacturer with a request for his co-operation.

CONSERVATION OF WATERS.

Report of the Standing Committee on Waters. W. C. MENDENHALL, Washington, D. C., Acting Chairman.

To the public the Conservation movement seemed to rise suddenly in the last few months of 1908 and the early part of the year 1909, but what the people of the United States was really witnessing then was not so much the origin of a movement as its organization. Through a generation before that time Government bureaus, individuals, and associations here and there had been methodically assembling facts, and those who were familiar with these facts had been reaching conclusions that were oftentimes disturbing in their tenor. These individuals and groups were brought together, their conclusions were given publicity of a most effective type, and what had been scattered and disorganized recognition of a vital problem was given solidarity and nation-wide recognition by the acts of President Roosevelt and Gifford Pinchot in organizing the National Conservation Commission and calling the Conference of Governors. The Forest Service, the Reclamation Service, and the Geological Survey had locked up in their archives the results of decades of research by their representatives, and these results supplied the facts which were the stimulus and the basis for the Conservation movement. Since that first great meeting, the Conservation

Congress, giving official expression to the movement and formulating its doctrines and its platform, has served as a medium for the exchange of ideas among those who are engaged in one or the other of its manifold activities, for the subject-matter of Conservation is as comprehensive as the materials with which humanity deals. Furthermore, the term itself has been impressed upon the public mind. It has passed out of the category of a cult for the few, and has been taken up by statesmen and politicians, scientists and divines, commercial organizations, manufacturing associations, and has even invaded the realm of diplomacy. This proves merely that the seeds were well sown by those who were sponsors for the movement. Their work, the task of focusing public attention upon a theretofore neglected but vital series of problems, was superlatively well done.

Years have passed since that time. It is appropriate that we review the results of those years as it was appropriate in 1908 to make our first inventory of the primary subject-matter of Conservation, namely, our natural resources. The period of initiation, difficult but well performed, is past. There remains a task that will never be finished—the equally difficult and the infinitely slower process of applying the principles of Conservation to our every-day activities. Such an application must be practical, reasonable, and gradual so that modes of life and industrial habits in which change is to be affected can be given time and opportunity to adjust to that change. How well have we, the workers in the ranks for these principles, performed our task? In what fashion has the movement been carried on? What real and creative steps have been taken in the public interest to reserve for future generations without unnecessary suppression of opportunity for the individual in the present or denial of his needs, that share in our natural wealth which should be so reserved?

I shall confine myself to a brief and casual review of that phase of the Conservation movement which deals with the one resource—water. Even in dealing with this one item in the subject-matter of Conservation, I shall have to leave aside for treatment by others, and indeed by other organizations than this, that phase of the problem of the waters which deals primarily with transportation and its allied problems of river improvement and waterway construction. There still remains a broad field, for water is the universal resource. Doctor McGee has estimated that the ultimate control of population in the United States will be exerted by the limitations in its water supply. We cannot say that this limit in population, even though it be placed at from five hundred to one thousand million people, is one that does not concern this generation, for we feel very keenly now in our arid and semi-arid sections the handicap which lack of water places upon our growth. Irrigation and dry-farming methods are attempts to overcome this handicap and forces us to realize that the ultimate growth predicted by Dr. McGee can be reached only through the most careful husbanding of the most universal and important gift of nature—water.

Because the human body, like all other organic structures, is largely water and because all of its nutritive and renewing processes are exercised by the function of water as the solvent of other foods, it has a primary value to man superior to that of any other substance. Its secondary value, scarcely less important than the primary and closely related to it in character, is as an aid in the production of nearly all things which man uses. In the humid regions, the supply is sufficient naturally so that the necessity of water is ordinarily given no more thought than the necessity for air, although without either we should instantly perish. Man’s use of water in crop production, hence, is automatic and unconscious in the eastern United States, but in western part, and especially in the arid districts, he at once becomes conscious of its importance because plans and crops fail without it. He establishes engineering works and conducts it to the land in order that food may be grown upon the land. Here, in the pioneer stages of settlement, comes the first great waste. Water was and too frequently still is carelessly used in irrigation. An equivalent of twenty or twenty-five feet in depth has been applied annually to the land where four or five feet is ample. The excess is sheer waste and in its application the land is ruined. Canals are often carelessly constructed and half of their carrying capacity leaks out before the tract to be irrigated is reached.

As settlement increases and demand becomes more intense, these conditions are improved. Their improvement in our own arid West and Southwest began under the pressure of necessity before the Conservation movement was given a name, but that improvement nonetheless represented the application of Conservation principles and the movement centered attention upon this and similar wastes, made men more generally conscious of them, and stimulated preventive measures. This stimulus, acting upon the public mind, aided many of the Government bureaus that for years had been combating such waste. The Department of Agriculture has a Bureau of Irrigation Investigations, which has systematically studied irrigation methods in the West and Southwest and has published many valuable reports calling attention to the losses of water in irrigation and suggesting methods for its prevention. The Geological Survey in its series of water-supply papers has repeatedly warned communities of the injuries and economic waste resulting from bad management of water supplies. The Reclamation Service, represented in its foundation a branch of Conservation, established and made a practical working idea. Since its foundation it has systematically continued the great work begun by the passage of its organic act in 1902, and is reclaiming, by careful and economic methods, millions of otherwise waste acres in the public land States. It has reached the point where the building of impounding reservoirs and of the canals by which the impounded water is conducted to the lands has been brought to practical completion on many of the projects so that its task is transformed into one of inducing settlement, of inculcating principles of economic irrigation practice in the minds of the farmers; of increasing the duty of water and therefore its usefulness, to the maximum; and of reclaiming through the establishment of drainage systems, lands which have been ruined by over-irrigation under the old systems absorbed by the reclamation projects. This movement is a part of, has aided, and has in turn been aided by the propaganda. It is practical Conservation of a high type.

I should like to diverge here for a moment to a collateral phase of Conservation activity which indirectly bears upon reclamation by irrigation. Our coal land laws provide for the sale of those parts of the public domain underlain by coal deposits at prices of not less than $10 or $20 per acre. Prior to 1906, this law was interpreted as evaluating coal lands on the basis of the thickness, quality and depths of individual beds, and basing sale prices upon these values. Through the fruition of this policy, coal lands are no longer sold at the minimum legal price unless they have minimum values. If coals are of sufficiently good quality and exist in sufficient thickness, they may now be sold at $40, $50, $100, $200, or even $500 per acre. A recent sale in the Rock Springs district, Wyoming, of one section of land at prices ranging from $370 to $410 per acre, netted the Government one quarter of a million dollars more than would have been received under the old policy of sales at minimum prices. This increment of a quarter million goes, like all other receipts from sales of public lands, into the reclamation fund and is there used in the application of water to the arid lands in the West. The Conservation phase of the present coal land policies is thus closely related to the question of waters and their use. The valuation of this natural resource and the sale at valuation prices was one of the collateral movements which stimulated and led to public recognition of the need of Conservation. It is a thoroughly practical application of Conservation principles and is an excellent example of governmental activity in this direction.

In one of the arid valleys of southern California in which irrigated lands bring prices of from $500 to $3,000 per acre and in which the limit to the number of acres to which such values are affixed depends wholly upon the quantity of water available, there has of course been earnest study of every possible means by which this quantity could be increased or made to serve a larger acreage. Here, in 1909, an interesting, practical step in Conservation was taken. Prior to that period water users in this valley who derive an important part of their supply from underground resources which, because of excessive drafts, were becoming depleted, had adopted the unique device of spreading flood waters which would otherwise escape to the sea and be lost, over the rough alluvial lands at the base of the mountain slopes in order that they might there sink and replenish the underground resources. The lands best adapted to this purpose had remained public lands because of their rough and uncultivable character, although adjacent to them were privately owned lands worth many hundreds of dollars per acre. In 1909 a law was passed by which these public lands were set aside for use in the distribution of these flood waters. They are now, and will remain, a permanent public reserve devoted to the conservation of water supplies and the increase of the quantity available for irrigation in a region in which water for this purpose has perhaps a higher value than in any other part of the United States. Here again is an example of practical Conservation work accomplished through the co-operation of private and governmental agencies.

The passage of the so-called Weeks bill in 1911 likewise marks a great advance in the direction of Conservation legislation. This is the bill which provides for the creation of an Appalachian forest reserve by the purchase of privately owned lands in the Appalachian Mountains. Its administration is in the hands of a commission whose active agents are the Forest Service and the Geological Survey, and one of the features of the bill is the clause which provides that the Geological Survey must affirm that the purchase of the lands will favorably affect the navigability of the streams on whose headwaters they lie, before the purchase can be made. Thus the conservation of waters is involved as well as that of the forests and of lands through the prevention of erosion. Those of you who for years advocated such a bill and assisted in its final enactment will agree with me, I believe, in the statement that its passage would not have been possible without the preliminary education of public opinion accomplished by the great pioneer advocates of the Conservation principles.

There is and will continue to be need for revision of the laws under which the administrative officers of the Government work to the end that these officers may administer our public resources more economically, more effectively, with less waste and therefore more thoroughly in the public interest. The enactment of laws does not anticipate the need for their enactment. There must always be widespread recognition of that need before public opinion crystallizes into statute. For, after all, the enactment of a law is nothing more nor less than the recognition on the part of our lawmakers of a public necessity which you and I as citizens force upon their attention. Until new laws can be secured, the task of the administrative officer is to administer with the greatest efficiency possible those laws that do exist. Under the stimulus of an active public opinion an interpretation may be given old laws which will enable them to fit the newer and changed conditions, for no enactment is absolutely rigid in its terms. An example of this adaptation of a law long upon our statute books to the passing of pioneer conditions in the West and the substitution for them of those changed conditions that result from augmented population, is that of the coal land law to which your attention has been called. The statute has not been altered since its passage in 1873, but coal lands are being sold under it now at prices which are based upon real values instead of at the lowest possible price under the law, as was true prior to 1906.

Under the stimulus of the changed character of public opinion, which has resulted from Conservation agitation, all of our public land laws are being carefully scrutinized to determine whether they do not admit of an interpretation and of an administration that is more in consonance with Conservation principles than the interpretation and administration of the past. Among the statutes thus scrutinized is the Carey Act, a law only less vital to the West than the Reclamation Act. In general it provides that public lands may be transferred by the Federal Government to the State in which they lie if that State will enter into a contract for their irrigation, by the terms of which they will eventually be delivered to bona fide homesteaders in tracts of suitable size. Undoubtedly, there have been instances in the past of careless administration of this law. The Federal Government has considered that its responsibility to the settler had ceased when the lands were turned over to the State in trust to him. The State, in turn, has considered that its responsibility ceased when the contract with the irrigating company was signed, and this company has been left free to deal with its actual and prospective settlers in a fashion that was intended too frequently to bring profits to a promoting company rather than water upon arid lands. It has thus happened that settlers, depending upon the State and through the State upon the Federal Government for protection of their interests, have found when the time came to apply for patents to their lands that although they had paid to a company large sums for water supplies, the water was not delivered, the land could not be reclaimed as the law required, and they were therefore unable to secure patent to it; but the irrigating or promoting company to which their funds had gone had disappeared and was inaccessible under the law. The genuine farmer, who at the sacrifice of hard-earned funds and years of labor was intended to be the beneficiary of this law, became instead its victim. This condition is believed to be past. The Federal Government and many of the States are now exhibiting a keen recognition of their responsibilities and of scrutinizing with the utmost care the water supply of each proposed project, the practicability of the engineering features of that project, and the financial standing and responsibility of its backers. A recent interesting example of this changed attitude occurred in one of the Western States, which in the past has administered this law carelessly, but I am glad to record is now exhibiting due care in meeting its responsibilities. In this case, literature issued by the promoters came to the attention of the Department of the Interior. In this literature statements were made to prospective buyers as to the available water supply and as to the acreage to which it would be applied that were known from the departmental records to be highly misleading. The attention of the Governor of the State was called to this condition of affairs by an emphatic letter from the Secretary of the Interior.

The State in turn called upon the promoting company for an explanation. The representatives of the company hastened to Washington for a hearing. As a result of that hearing, the acreage segregated in the project was promptly reduced, the company was forced to agree to cease its sale of water rights to private lands until the rights of the Government lands to which it was inviting settlers were satisfied, and thus the situation so full of menace to prospective settlers was promptly corrected. Other examples of this type of action which represents closer, more careful administration of old laws might be multiplied. Each of them marks a step in the application of the principles for which the Conservation Congress stands.

If the first use of water by man is in the direct sustenance of life and its second is for the production of food supplies through irrigation, perhaps its third most important use is the development of power for all of those manifold purposes tending toward the amelioration of life and the increase of its comforts, for which power may be used. Cities are lighted; street cars are moved; ores are smelted; manufacturing plants are supplied with their motive power; homes are heated; and water is pumped for irrigation by the use of hydro-electric power. No question has been the subject of more bitter controversy than that of the control of this tremendous resource. It has been energetically sought on the one hand by those who seek opportunities for profit and desire that no control be exercised over those opportunities by the power of the State. On the other hand, public opinion, working largely through its State and Federal representatives, has demanded that this resource whose magnitude can be but rudely estimated, and whose future value but guessed at, be so controlled that communities depending upon it shall not be unduly taxed for the purpose of piling up private profits. Here again, both public opinion and Federal officers have repeatedly urged the enactment of new laws which will make possible the exercise of reasonable control in the public interests and at the same time properly safeguard capital which must be invested in order that the resources now wasted may develop and become useful. Bills have been introduced and debated in Congress; conferences have been held with representatives of the public and of capital, but the plans thus far considered have brought no fruition in amended legislation, although some excellent bills are under consideration and it is believed will soon become law. Here again, the task of the administrative officer is to so interpret and apply the laws now upon our statute books, pending the enactment of others more satisfactory, that development may continue and the rights of the public of this generation and the next be at the same time duly safeguarded. Here also there has been progress in the interpretation of law. The responsibility for the administration of the laws for the development of water powers in the national forests lies in the Forest Service where it is admirably exercised in the public interest. The law which provides for the development of powers on the public domain, whether within or without the reserves, is a permissory law, one that authorizes the department having jurisdiction to permit the development of these water powers under general regulations to be fixed by the Secretary. After a thorough study of the situation, the Forest Service on December 28, 1910, issued certain regulations providing for the development of powers under this permissory law, the permit being by the terms of the law itself subject to cancellation at any time and the regulations under it providing for moderate charges upon the developing company. With these regulations in force in the national forests, and no similar procedure provided for on the public lands outside the forests which are under the jurisdiction of the Interior Department, applicants for the privilege of developing water powers which lay in part within and in part without the forest reserves found themselves under two jurisdictions without any provision for uniform procedure. The problem as to the precise amount of control that could be exercised on the Interior Department lands under the act of 1901 has not been solved until recently; but as a result of this final solution, there were approved by Secretary Fisher on the 24th of August, 1912, regulations controlling the issue of permits for power development outside of the national forests that are in substantial accord with those heretofore in force within the forests. These regulations provide for the exercise of the authority of the Secretary in a definite, uniform, and systematic manner that much more fully safeguards the rights of the public than the policy heretofore pursued in relation to public water powers. The situation, therefore, seems to be as well safeguarded as it can be under the present statutes, at least so far as hydro-electric powers on other than navigable streams are concerned, and this end has been accomplished not by new legislation, which we all recognize as badly needed, but by a proper interpretation and acceptance of responsibility under old legislation.

An incidental phase of the effort to administer a law which provides for no definite tenure of lands having power values has been the constantly repeated attempt of interests desiring to acquire valuable water powers to secure them under the irrigation laws, those laws having great advantage from the commercial viewpoint of providing for a grant instead of a revocable permit. Application after application has been filed with the Department of the Interior in which it is stated solemnly that the rights of way are desired for purposes of irrigation, when it is perfectly obvious to the engineering advisers of the Secretary that the power value is the dominant value and that if the waters are used for irrigation at all, it will be merely in order to effect a technical compliance with the law under which they are acquired. Refusal to approve rights of way of this type have been followed by appeals and by emphatic protests on the part of the applicants. These protests take various forms. Among them are attempts to influence public opinion through various congresses similar to this Congress, and other attempts to secure the enactment of special legislation which will grant to the applicant that which he is unable to secure through the administrative officers. In a particularly interesting case of this type recently acted upon by the Department of the Interior, the acting Secretary expressed the present policy of the department in these emphatic terms, which I am sure will appeal to every member of this Congress. He said:

> I consider it the imperative duty of every supervisory officer of the Government upon whom any duty devolves to conserve the paramount interests of the people, to protect these natural power sites from exploitation under any law which successfully invoked would turn them over to private interests charged with a perpetual easement against the United States.

One other type of administrative action in connection with the conservation of water resources has recently been inaugurated which may well be brought to the attention of this Congress. This is a new exercise by the President of the power of withdrawal conferred upon him by the so-called withdrawal act, approved June 25, 1910, and amended August 24, 1912. By this action those lands in arid States upon which small water supplies essential to the control of the adjoining range are situated are withheld from entry. Those of you who are acquainted with the range industry of Wyoming, Utah, Arizona, and New Mexico realize that the use and control of the ranges are exercised not so much through the ownership of the range lands themselves as through the ownership of small tracts which include the springs and other watering places that alone make the ranges accessible and of value. Literal war has been waged between rival stock interests in parts of the West over the control of springs. Large interests have frequently forced their rivals to abandon the range in a particular area by acquiring through the application of scrip or by a real or pretended exercise of homestead rights the lands on which the springs that alone give value to the range are located. Laws have from time to time been considered which will provide properly for the disposition of those remaining parts of the public domain that are chiefly valuable for grazing purposes. It is recognized that the homestead and desert land laws are inappropriate for the acquisition of range lands in that they do not provide for a sufficient acreage to make the stock industry possible. If the time shall come when such a law is placed upon the statute books, and at that time all of the water supplies adjacent to the ranges shall have been acquired by private interests, the Government will be unable to dispose of its range lands even under a favorable law except to those who already control the water supplies which are the key to the situation. Recognizing this important condition and desiring likewise to provide for fair play between rival stock men on the remaining public lands, the President, upon the recommendation of the Secretary of the Interior, has inaugurated the policy of withholding from entry lands upon which these desert watering places exist, and in pursuance of this policy the first desert water hole withdrawal was made in March, 1912.

It will be realized from this brief review that the process of translating the Conservation doctrines into action is well under way. Before and since the First Conservation Congress met, Federal bureaus have advocated practical measures for the proper use of our natural resources, water among them. With the enlightenment of public opinion dating from the organization of the National Commission and the meeting of the Governors the work has been greatly facilitated. It is advancing now not only through the medium of the unorganized effort of individuals, associations, and isolated bureaus and divisions in the public service, but by the organized efforts of an enthusiastic body of supporters. Laws embodying its principles have passed, proposed laws inimical to those principles have been defeated, old laws have been re-examined and reinterpreted to accord more fully with Conservation doctrines in the public interests. Party platforms are no longer complete without a Conservation plank and indeed it may almost be said that a new party has been founded upon the Conservation idea. On the whole the country and this Congress have ample ground for optimism in considering the great advance that has been made.

WILD LIFE PROTECTION.

Report of Standing Committee, Dr. W. T. HORNADAY, New York City, Chairman.

The Committee on Wild Life Protection wishes to call the attention of the Congress to the enormous losses that are being inflicted upon the farming and fruit-growing interests of the United States through the destruction of insect-eating birds. While the main facts of the situation are known to many persons, the mass of the people of the United States are sound asleep on this subject. The 5,000,000 men and boys who are slaughtering our birds are levying tribute on every American pocketbook. An immense number of birds of great economic value are being slaughtered annually, and many of our most useful and valuable bird species are on the toboggan slide toward extermination. The destruction of our insect-eating birds means a great increase in the armies of destructive insects, a great decrease in our agricultural products, and a great loss to consumers and to farmers. The value of the birds destroyed as "game" and for "food" is declared to be not equal to one-thousandth of the value they would save to the national wealth, if permitted to live.

The committee will distribute a campaign circular containing a table of figures showing the annual losses to the people of the United States by insect pests. Those figures were taken from an official report published in the "Yearbook of the Department of Agriculture." The farmers who grow cereal crops lose about $200,000,000 per annum. The fruit-growers lose $27,000,000 per annum. Hay loses $53,000,000; cotton, $60,000,000; and truck crops, $53,000,000.

The committee's circular gives the cost of certain insects per species to the people of the United States. For example, the codling moth and curculio apple pests cost the American people $8,250,000 a year for spraying operations, and $12,000,000 per year in annual shrinkage in the apple crop. The chinch bug wheat pest sometimes costs $20,000,000 per year, and the cotton boll weevil the same amount. The tree insect pests damage trees and timber to a total of $100,000,000 a year.

Your committee contends that the American people *do not realize* that scores of species of the birds that sportsmen and pot-hunters are regularly allowed to shoot for sport are of *immense value* to agriculture. How many men are there out of every thousand who know that at least thirty species of shore birds feed upon noxious insects, and are immensely valuable to our agricultural industries? The gunners who shoot legally are destroying 154 species of birds that legally are classed as game birds, even in the North.

Very few Americans out of every thousand know the *immense value* of our song birds, swallows, woodpeckers, blackbirds, quail, doves and nighthawks in destroying countless millions of noxious insects.

THE LOGICAL CONCLUSION.

In view of the decrease already accomplished in the general volume of the bird life of America, in view of the enormous losses annually inflicted upon the people of this country by the ravages of insects, and in view of the destruction of wild life that now is furiously proceeding throughout all America, the McLean bill, now before Congress, to provide Federal protection for all migratory birds, becomes the most important wild life measure that ever came before the Congress of the United States in any form. In view of the annual losses to the wealth of this country that will continue so long as the McLean bill fails to pass, it is impossible for any one to put forth one good reason, unless it be on purely technical grounds, against that measure. By the inexorable logic of the situation, any man who opposes the enactment of a law for the Federal protection of migratory birds becomes by that opposition an enemy to the public welfare. The bills introduced in Congress by Representatives Weeks and Anthony have dragged long enough. They provided for the protection of migratory *game* birds, only. Now it is time to strengthen their proposition, as Senator McLean has done, by providing also for the protection of all the migratory insectivorous birds.

Unless the people of America wish to shut their eyes to their own interests, and pay out millions of dollars annually in the form of increased cost of living, they should arouse from their lethargy and put up to Congress such a demand for the passage of the McLean bill that it will be enacted into law at the next session of Congress. It is Senate Bill No. 6497, and on the Senate calendar it is No. 606. We can not afford to wait until 1914 or 1915; and Congress has full power to act next winter.

How many people in the North know that the negroes and poor whites of the South annually slaughter millions of valuable insect-eating birds for food? Around Avery Island, Louisiana, during the robin season (in January when the berries are ripe), Mr. E. A. McIlhenny says that during ten days or two weeks, at least 10,000 robins are each day slaughtered for the pot. “Every negro man and boy who can raise a gun is after them!”

There are seven States in which the robin is regularly and legally being killed as game! They are Louisiana, Mississippi, Maryland, North Carolina, Tennessee, Virginia and Florida.

There are five States that expressly permit the killing of blackbirds as “game”: Louisiana, South Carolina, Tennessee, District of Columbia, Pennsylvania.

Cranes are killed and eaten in Colorado, Nevada, Nebraska, North Dakota and Oklahoma.

In twenty-six States doves are regularly killed as game—much to the loss of the farmers.

The bobwhite quail is a great destroyer of the seeds of noxious weeds. In our fauna he has no equal. And yet this fact is ignored. Throughout the North and most of the South that species is mercilessly shot, and as a result it is fast *becoming extinct*. In New York State it will soon be as extinct as the mastodon, unless given a ten-year close season at once. Its value as a plentiful game bird is gone.

The shore birds are *fast* becoming exterminated by sportsmen and pot-hunters who kill them for food, “according to law.” The Eskimo curlew is totally extinct, and other species are fast going over the same road. Nothing in this world will save this group of birds except *a law for the Federal protection of migratory birds*, such as the McLean bill, now before Congress. The way the whole group of shore birds is being exterminated is nothing less than a crime. And yet, at least thirty members of this group are of a great value to all of us, because of the great numbers of crop-destroying insects that they annually consume.

THE DUTY OF THE HOUR.

The *only way* in which all these valuable migratory birds can be saved to us is through the strong arm of the National Government, and a Federal law for the protection of *all* migratory birds! Protection of game birds alone will not answer. Too many other birds are being killed for food, especially in the South.

The Wild Life Protection Committee urges all delegates to take home with them the burden that rests on every good citizen regarding the enactment into law of a satisfactory measure for the preservation of the insect-eating birds. If any opposition should arise on account of the feature of the bill which covers the ducks, geese and swans, and other migratory wild fowl, the committee is quite willing that those birds should be stricken out of the bill entirely, in order that the protection of the crop-saving birds may be secured. It is believed that no sensible person can possibly raise any objection to the protection of the insectivorous birds by the passage of the McLean or Weeks bill, in case the water fowl are left out. It is, however, regarded as extremely necessary that the shore birds should be included because of their immense value to agriculture.

In concluding, the committee urges all delegates to take this matter up with your members of Congress, and urge them to vote for, and work for, whatever bill may finally be agreed upon as best calculated to protect the insectivorous birds, and be free from objections regarding its constitutionality. A number of able lawyers have decided that it will be wholly within the spirit and letter of the Constitution of the United States for the Federal Government to protect all insectivorous birds through a law of Congress.

VITAL RESOURCES OF THE NATION.

Dr. Henry Sturgis Drinker, President of Lehigh University, a Delegate from the State of Pennsylvania, from Lehigh University, and from the American Forestry Association.

What subject is there to which the constant attention of Conservationists, of patriotic men and women, could be better devoted than to the care of the vital resources of the nation—the care of the lives of all our people, not of a selected few, the teaching and the impressing of the lessons of steady life, of sobriety, of continence, and of due rest and recuperation from the wear and tear of our American life. Surely we have good reason to be proud of the intelligence and activity of our people, formed as they are of the intermingling of many peoples, with a resulting product as a nation that is markedly free from in-breeding and its usually unsatisfactory outcome.

I think it was Mr. Lieber, in the course of his gracious and cordial opening address of welcome to the Congress, who referred to our duty to endeavor to alleviate the condition of the sweat-shop and mine workers, but is there not another and equally great duty of which we are habitually more neglectful? What is our duty, the duty of society, to those self-sacrificing, altruistic men, devoted to public service, men such as Dr. Wallace, Mr. White, Mr. Farquhar, who devote themselves to and ably lead great movements like this Congress for the betterment of conditions among our people—men who are not only captains of industry, but generals in the army of public service, and leaders and exemplars in the pursuit of public duty? What should we, as a body, say to them and to others like them (for, thank God, America owns a great army of good men like them), who uphold the good cause of public service? They become in leading these great movements, in a measure, the custodians of the public welfare, but —"Quis custodiet ipsos custodes"? Who shall watch these very guards, and see that they conserve the intelligence, patriotism and energy, that goes out from them to public welfare, that it may not be prematurely exhausted? Surely we should take measures to have them feel how the Nation values them as a public asset, and how they owe it to their country as well as to their homes to heed and to preach to others the wise words of dear old Mark Twain, who (writing from Naples in 1867) sent us these words, pregnant with the lesson of the higher Conservation:

> "We walked up and down one of the most popular streets for some time, enjoying other people's comfort, and wishing we could export some of it to our restless, driving, vitality-consuming marts at home. Just in this one matter lies the main charm of life in Europe—comfort. In America, we hurry—which is well; but when the day's work is done, we go on thinking of losses and gains, we plan for the morrow, we even carry our business cares to bed with us, and toss and worry over them when we ought to be restoring our racked bodies and brains with sleep. We burn up our energies with these excitements, and either die early, or drop into a mean and lean old age, at a time of life which they call a man's prime in Europe. When an acre of ground has produced long and well, we let it lie fallow and rest for a season; we take no man clear across the continent in the same coach he started in—the coach is stabled somewhere on the plains and its heated machinery allowed to cool for a few days; when a razor has seen long service and refuses to hold an edge, the barber lays it aside for a few weeks, and the edge comes back of its own accord. We bestow thoughtful care upon inanimate objects, but none upon ourselves. What a robust people, what a nation of thinkers we might be, if we would only lay ourselves on the shelf occasionally and renew our edges."

As the official call for this Congress stated, we have in previous meetings dealt with four great subjects—our forests, waters, lands, and minerals, but in taking for its theme this year the subject of "Vital Resources," the Congress is studying the very life of the Nation, is seeking to benefit our people not only by the conservation of our material natural resources, but to do good to them by bringing home the duty of life Conservation in our whole Nation; and what greater task can patriotic men and women devote themselves to than this, and what words can epitomize the sentiment underlying this service better than those in Sophocles' "Oedipus," where it is said:

> "Methinks, no work so grand
> Hath man yet compassed, as with all he can
> Of chance or power, to help his fellow-man."

CONSERVATION OF THE SOIL.

Hon. James J. Hill, of St. Paul, Minn.

Just as all industry depends upon the production and increase of the fruits of the earth, so all other forms of Conservation must be held subordinate to the preservation of the productivity of the soil. To preserve and defend the public health, to see that human beings are brought into the world and kept there under favoring conditions, and to lengthen their term of life will but add to the total of human misery unless they are well fed and housed and clothed. For this, as for the material of all their varied activities, they must come back in the last analysis to the soil. Earth is the mother not only of mankind but of all human industry.

In the years during which the necessity of this most imperative form of Conservation has been the subject of my thought and the theme of most of my public utterances, much has been accomplished. The interest of the public is awake. It is not necessary any longer to urge a Conservation movement, but rather to direct the energy already enlisted in its behalf into wise channels. While the farmer is still subject to some unfavorable legislative discrimination, we know that his prosperity must be made a first object before prosperity can visit others. The progress of the farm is put first in many schemes of public improvement where, a few years ago, it would have been mentioned perfunctorily if at all.

Education in agriculture has made much progress. The number of institutions teaching agriculture increased more than sixty per cent. in nineteen months. They had ten per cent. more students in agriculture in 1910 than in 1909, and more than eight times as many students taking the teachers' course in agriculture. Colleges and high schools give place to some form of agricultural instruction; and the necessity of fostering soil Conservation is recognized today as never before.

What we need to do at once belongs rather to the practical than to the theoretical side of Conservation. There is little reason to doubt that the farmer of the future should be a highly intelligent man, commanding from his acres crops that are far beyond those of today in their abundance. But the present generation may and should do far better for itself, in its own time, while it is also preparing the way for the more careful and productive agriculture which should follow.

I use intentionally the words "careful" and "productive" instead of the word "scientific," as applied to soil treatment and crop raising, because they express the simple and easy processes within the reach of men of the present generation as well as the new; because they avoid a misleading implication that attaches to the word "scientific." It is true that the best methods of soil treatment and crop growing are scientific; but they require only that form of popular science which is within the comprehension and use of every farmer.

The essentials of soil Conservation have been known for centuries. They were practiced in Babylonia, just as irrigation was resorted to there on a splendid scale. They have been the property of the Chinese for four thousand years, and maintained there a dense population in spite of croppings so frequent and severe that it would seem impossible for any soil to stand such treatment without exhaustion. The latest bulletin of the best agricultural institution is scarcely more instructive or helpful than a study of the "forty centuries of agriculture" included in the experience of these skilled and laborious people of the Orient.

The soil is a living thing, and must receive the treatment due to all organic and vital beings from which we expect service or tribute. The first requisite is that the individual man learn with what manner of soil he is dealing. There is now an agricultural college or experiment station within the reach of every farmer in the country. Some are and all should be equipped for a scientific analysis of all soils submitted to them. From this the cultivator may learn the first two things indispensable to any intelligent conduct of his industry: First, to what crops his land is best adapted; second, what elements of fertility have been drawn from it so lavishly that they need to be restored. This information having been given by competent authority, every farmer may do all the rest for himself.

There is no secret and no mystery about the processes involved. If farmers will rotate their crops, fertilize plentifully and intelligently, keep live stock to diversify their industry, refresh the land and utilize waste products, and cultivate thoroughly and frequently, the problem of soil Conservation is solved. The earth has been kept as productive for thousands of years as it was when it produced its first crop of cultivated cereals wherever these few and simple conditions have been observed. If seed is carefully selected, after a test for germination, and the practices mentioned are followed, there is no reason why the yield per acre of the principal crops of the United States should not equal those of England, Germany or many other countries which produce twice as much as we do with far inferior natural advantages.

Dr. Knapp, of the Department of Agriculture, said: “It has been found that the best seed bed added 100 per cent. to the average crop on similar lands, with an average preparation; planting the best seed made a gain of 50 per cent.; and shallow, frequent cultivation was equal to another 50 per cent., making a total gain of 200 per cent., or a crop three times the average. With better teams and implements, this crop is made at less cost per acre.” A bulletin of the Bureau of Plant Industry, at Washington, says: “It is possible within a few years to double the average production of corn per acre in the United States, and to accomplish it without any increase in work or expense.” It declares that twice twenty-six bushels, which is about what we now get, is a fair crop where these conditions are observed, three times twenty-six bushels a good crop and four times twenty-six bushels frequently produced. A similar increase in other farm growths is just as possible.

In a high sense this is conservation of the soil, because it shows the way to make one acre do the work of two or three or four. It is conservation of the soil in a still better sense, because the land, when so intelligently and considerately treated, instead of “wearing out,” not only maintains its productive power indefinitely but actually increases in fertility and value. These are facts which all history attests. They are facts which the most recent scientific research supports. The work before the promoters of the Conservation movement today is one not of discovery but of education. It is to assist in bringing home the truth to the minds and embodying it in the daily practice of the present farm population of the United States.

This tremendous task can be accomplished only by local demonstration and the force of practical example. Small model farms should be operated, preferably consisting of a few acres selected from ordinary neighborhood farms and treated intelligently, in every State, county and township. We have made a beginning of this work in the Northwest; and the results, though not yet completely enough ascertained for tabulation until the tale of threshing and marketing is ended, are as amazing as they are encouraging. Some of the States are providing for traveling instructors and supervisors in agriculture, following the policy successfully adopted in the most enlightened countries of Europe, thus raising the level of agricultural practice and educating the millions who are beyond the reach of the institutions where formal instruction is given to the young. It is imperative that we reach the older people, and the large percentage of the children of the farm who never get beyond the district school, if we are in earnest in the work we have undertaken.

To this practical side of soil Conservation this Congress should give its hearty approval. It should urge upon the people of every community the adoption of the demonstration tract and the local instructor, with as much earnestness as it has championed the saving of forests and the reclamation of arid lands. Ten per cent. of the money now expended in formal instruction in the institutions where agriculture is taught, or supposed to be taught, would put every farmer in touch with the man who could and should help him in the treatment of his land as readily and surely as the doctor helps his family when they are sick. It would be more than repaid every year in the value of the crop increase. It would be repaid over again in the healing of sick soils, the renovation of old lands, the preservation undiminished in every acre of our arable area of those elements of fertility without which plant life languishes, and the wilderness and the desert in a few generations sweep away the traces of man’s unworthy occupation. It is well worth the hearty and undivided support of public-spirited men. For without just such Conservation the time will come when our country will be unable to support its own people; the diminishing percentage of its population engaged in tilling the land will still further decline; and it will scarcely be worth while to consider how best human life may be prolonged and made sturdier and wholesomer physically by vital Conservation, because it will lack the sustenance that it can not longer draw in sufficient quantity and quality from nature's withered breasts.

WAR, THE POLICY OF WASTE—PEACE, THE POLICY OF CONSERVATION.

Mrs. Elmer Black, New York City.

In advancing some arguments bearing on that broad assertion permit me at the outset to express my satisfaction that the questions this Congress has set itself to consider have come to be recognized as among the most urgent of all the world's humanitarian problems. For the peace movement and the Conservation movement are as closely interrelated as, in the pacifist view, the interests of the entire human race are mutual and not antagonistic. The advance of your program is the advance of ours; both are essential to the progress of mankind.

I do not suppose any one will cavil at my plea that when we talk of natural resources we must not merely include inanimate things—timber, minerals, lands, oil and waters—but the brain and sinews of the people as well.

An observant traveler in the United States, asked recently what he considered the greatest asset of the American nation, replied: "The American nation itself, with its self-reliance, ingenuity, the blended genius resulting from race fusion, and the boundless belief in its ability to reach any goal it sets out to attain." With that contention in mind, I would at once emphasize the fact that neither the material resources of the world nor these higher resources of human equipment can be utilized or developed to their full complement till the profligate policy of international strife is purged from the activities of mankind.

I venture to assert that no war can be waged today that can be justified ethically or economically. With the bringing together of the civilizations of the world, the development ever closer of the bonds of communication, and the institution of the International Court of Arbitration, the last excuse of the war makers has disappeared. No nation today need go to war if the cause it advocates is just. When the plea of “questions of national honor” is advanced it will usually be found that the case behind the plea is so faulty as to entail risk if presented to the judgment of an impartial tribunal, or that there is the secret reason of a desire for aggression in order that some other nation may be robbed of territory.

But assuming for the sake of argument that this latter case can be justified on the ground of imperial advancement and the “survival of the fittest”—a conclusion I do not in reality concede—I still contend that war is a ghastly blunder, inevitably inflicting such loss to the treasuries alike of victor and vanquished that both are laden with debts so great that generations yet unborn are foredoomed to carry an unnatural charge.

It requires no casuist to demonstrate that such a policy is detrimental to human progress and diametrically opposed to thrifty administration. If we think for a moment of what might be accomplished if the war expenditures of nations were devoted to the proper development of the world’s bountiful stores of wealth, the advancement of health and science and the promotion of communal betterment, the imagination reels at the vista of progress that is opened up.

Let us take a few comparisons. The Panama Canal, uniting two oceans and bringing into closer contact the peoples of East and West, is being constructed at a cost of $400,000,000. Against that accomplishment set down the blood and treasure poured out in reckless waste in the Crimean, South African and the Russo-Japanese wars. On the one hand we have a constructive policy in which the nation’s toil and money is conserved and invested so as to operate at compound interest for the benefit not only of American citizens but also of the whole human race. On the other hand, there is a destructive and prodigal policy that has disappointed in after days even those most closely concerned with the crimson fruits of victory.

Speaking of the Crimean War, Lord Salisbury, the late Premier of England, said in his cynical way, “We put our money on the wrong horse.”

The South African War cost no less than $1,331,655,000 and added no less than $795,880,000 to the national debt of England. The flower of

British manhood perished on the veldt that the Dutchmen of the Transvaal might be forever relegated to the strata of the subjugated. Yet today, a few short years after that deadly struggle, South Africa is united; the Dutch are enjoying self-government, and, in fact, are politically in the ascendant over their nominal rulers.

Russia lost her entire fleet, wrecked her army and set the forces of internal discontent seething once more within her boundaries. Japan, the nominal victor, so poured forth her wealth that even her amazing vitality is shackled by the bonds of financial stringency. Today both are suffering from the gigantic, blundering conflict—and in the end are compelled peacefully to agree to recognize their respective interests in Northern Asia.

England's naval expenditure amounts to nearly $250,000,000 a year, and every ten years great costly Dreadnoughts are thrown on the scrap heap—a total waste. Now England has spent on irrigation in India $150,000,000, and I would ask your attention to the fact that this expenditure has not only brought health and prosperity to hundreds of thousands, reduced the dangers of famine and made the desert blossom as the rose, but there is a profit on the capital invested of six and three-quarters per cent.

Taking that as a specimen of contrasts, one is amazed at the mental spectacle of the immense strides that could be made in the world's prosperity if the expenditure on war and preparations for war was devoted to the Conservation and development of natural resources. The armed peace in Europe in thirty-seven years has cost $150,000,000,000. Yet there are resources waiting to be developed for the benefit of the struggling millions who are crushed beneath the iron heel of Mars; there are reeking human rookeries in the cities of Europe that are a menace to the human race; there are schemes for waterways that would open up wealth practically untapped, to the end that productive machinery might be set in motion for the continual benefit of nations yet to come.

When a Dreadnought fires a single shot from its big guns as much money is dispersed into the air as would pay a workman's wages for three years or secure a clever student's college course for a full twelve months. For every cruiser scrapped in naval frenzy a fully staffed scientific laboratory could be run for years in conflict against man's mortal enemies, the disease bearing bacilli. For years the inventive faculties of the world have been turned to the production of implements of death and destruction. In a saner age of Conservation and peace this concentrated genius will be focused on the preservation of life, the clothing of the desert with verdure, the elimination of space, the improvement of communications, the harnessing of natural forces to the service of man that even today are seen but as through a glass, darkly.

The world is spending every year eight billion dollars on militarism. The expenditure of that ocean of treasure leads nowhere but to the slippery slope of bankruptcy. It creates nothing by which future generations will benefit or of which any but the superficial can be proud. It robs the treasury of the busy bees of commerce and industry and withdraws from active participation in constructive affairs seventeen million men, the strongest and best types, whose brain and muscle should be used for the advancement of their kind. Women, in consequence, as in Germany, have often to undertake work for which nature did not equip them, and so a double wrong is wrought upon the human race.

If the interest only on the money spent on militarism were used on education, 32,000,000 more students would be accommodated at college every year; or the housing problem of every land could be solved as if by a magic wand, to the immeasurable conservation of human health and vigor.

During the South African War the "Investors' Review" of London said: "In one short eighteen months the war party now sitting on our necks has dissipated more money than the working class managed to accumulate out of their wages during the whole sixty-four years of the reign of Queen Victoria."

Chancellor Lloyd-George, as recently as the last budget, said in the House of Commons that the money spent on building Dreadnoughts in England in one year would add a dollar a week to the income of every workingman's family in Europe.

The United States, though not yet in the same parlous plight as the European nations, is heading in that direction, and devotes the enormous proportion of seventy per cent. of its national expenditure to preparations for war. That is to say, every year we waste more than would construct an epoch-making Panama Canal.

Were it possible immediately to reconstruct the scheme of things so that reason ruled, there would be no need to cry aloud for the development of the barren lands of our continent. Waterways would be extended, irrigation works would carry the life-giving fluid to arid areas that need but that to release their dormant fertility, herbage and fruit would spring from regions now productive of only scrub and cacti, and the reforestation of natural timber land would cause the birth of new resources formerly despoiled by the ignorance and greed of man.

But, some may say, surely these shipyards, barracks, Dreadnoughts, and war equipment circulate money and employ millions of men. That, I contend, is a common fallacy a little thought will dissipate. Money being but the tool used in the purchase of labor or the product of labor, is in itself of no account. But the building of a Dreadnought to fire away the product of labor in thin air is to bring all that has gone to prepare for that achievement to nought. The chain of production is broken; henceforth mankind is so much the worse off for that loss of money, labor and its products. But in the true conservation of resources, the labor and its products employed in peaceful and constructive enterprises achieve benefits that flow on continuously in an ever-widening stream, till the entire humanity is made to feel the blending of each man's labor in the commonweal.

Brain, sinew, time, energy and material resources are being thrown into the melting pot of war, yet but for that very war the evolution of the human race would be eons further on.

Every time a war scare sends the exchanges of the world's capitals into panic, deadly injury is done to the commerce of the entire globe. Ruin has not to wait for the actual outbreak of hostilities. The wolf of war has but to bare his fangs in menace to cause a premonitory slump to spread devastation through the ranks of the investors. Yet instead of meeting, as the delegates of this Congress are meeting, to debate the best methods of insuring national and international thrift, in statesmanlike preparation for the future needs of the human race, many powerful minds are spending their time devising nightmares with which to scare humanity into ruinous expenditure.

I venture to predict that in the future it will be found that the chief instrument of Conservation has been established already in the arbitration court at The Hague, and generations yet to come will say of its founders, “They builded better than they knew.”

Meanwhile the omens are favorable for the causes of peace and Conservation. The object lessons are there for all the world to see. I have mentioned the Panama Canal in the Western Hemisphere as an instance of constructive Conservation. The Eastern Hemisphere also has its encouragement. Right in the reputed cradle of the human race some of the world’s most brilliant engineers are executing works that will unchain rivers and cleave through mountains to the end that on Mesopotamia’s broad lands the Garden of Eden shall be re-established.

I rejoice to know that this movement towards Conservation nowhere has attained greater volume than in our own land; for with our advantages it seems to me that we are specially equipped for the ennobling task of removing obstacles from the path along which our race must tread in the accomplishment of its high destiny.

THE CONSERVATION OF NAVIGABLE STREAMS.

Mr. Jacob P. Dunn, Indianapolis.

The objects of the conservation of natural resources divide naturally into two classes. The first relates to the development of lands in private ownership, such as the encouragement of forestation and renewing the fertility of the soil, in which the interest of the State is the indirect one of increasing the supply, or cheapening the cost of products that are of material benefit to the entire community. The second relates to the preservation and utilization of public property, such as forest lands and mineral resources, in which the State has the direct interest of securing special revenues, whereby the burdens of taxation may be reduced, and of promoting the public welfare by furnishing facilities for commerce and industry. To this second class belongs the conservation of navigable streams, and this subject has already been brought prominently before the public by the discussion of proposals for improvement of the navigation of the Mississippi River and its tributaries. But this Mississippi project has a vastly greater significance than the general public has fully considered; for it means that hundreds of streams that are now navigated only in a small way, or not navigated at all, will later be made navigable in a practical and useful way.

Moreover, this subject is of special importance to the great region formerly included in the territory northwest of the Ohio River, including the present States of Ohio, Indiana, Illinois, Michigan and Wisconsin, because the Ordinance of 1787, by which that territory was created, expressly provided that: “The navigable waters leading into the Mississippi and St. Lawrence, and the carrying places between the same, shall be common highways, and forever free, as well to the inhabitants of the said territory as to the citizens of the United States, and those of any other States that may be admitted into the confederacy, without any tax, duty, or impost therefor.”

These words show that the word “navigable” was not then used in its present common acceptation. When we speak of a navigable stream we commonly mean one that can be navigated by steamboats, but there were no steamboats in 1787, and all of the commercial navigation of this region at that time was by means of canoes and the small vessels known as bateaux and pirogues. That this navigation was what was intended is conclusively shown by the reservation of “the carrying places,” i. e., the stretches of watershed between the headwaters of the streams of the two drainage systems over which both the boats and their loads were transported bodily. This meaning has usually been adhered to by the courts (2 Mich. 219; 19 Oregon, 375; 33 W. Virginia 13; 20 Barbour, N. Y., 9; 14 Ky., 521; 87 Wis. 134), and the general rule is that any stream that will carry commerce, even by floating logs, is a navigable stream. (51 Illinois, 266; 42 Wis. 203.)

The United States Government followed out this theory consistently. By the act of Congress of 1796, for the survey and sale of the public lands of this region, it was expressly declared that “all navigable streams within the territory to be disposed of by virtue of this act shall be deemed to be and remain public highways.” As such their beds were always excluded from the lands surveyed and sold. The government surveyors did not include any of the larger streams in their surveys, but “meandered” them, and when the land was sold, it was sold in fractional sections, running to the meander lines. The beds of the streams and the land bordering them was thus reserved as public property, and when the several States were formed and admitted to the Union, the title was transferred to them from the general government. It is of course to be remembered that Congress has ultimate power over navigable streams, but it is well established law that a State has plenary power over navigable streams that are entirely within its borders, at least until Congress acts.

The acceptance of this provision as to navigable streams was made a prerequisite to the admission of Indiana to the Union by the enabling act of 1816. It was formerly accepted by ordinance of the constitutional convention of Indiana in 1816. And yet Indiana stands today in the unique position of being the only State in the Northwest in which the public rights thus established have been nullified, or at least clouded, by an absurd decision of the Supreme Court of the State, made in 1876. (54 Ind. 471.) Inasmuch as this case deals with White River in Marion County, and as this stream at this point furnishes a typical illustration of the whole subject, I will call your attention to it in detail.

Prior to this decision, every department of the government of Indiana fully recognized the binding force of this compact with the United States, and accepted as conclusive the United States surveys in the determination of what streams were navigable under that compact. The bed of White River in Marion County was not included in these surveys, and it was never sold by the United States or by the State. The Legislature of Indiana always recognized this rule, and always applied it to "White River in Marion County." The act of January 17, 1820, declared White River navigable as high as "the Delaware towns," meaning presumably to Muncie, and made it and the other streams named "public highways," making it a penal offense to obstruct "any stream declared navigable by this act," except only that mill-dams might be erected under certain conditions, by persons who had "purchased from the United States the bed of any stream by this act declared navigable." This law was never repealed, but was modified by the act of February 10, 1831, which declared White River navigable as high as Yorktown in Delaware County. This last law is notable as recognizing that a stream need not be navigable at all seasons, for it prohibited any obstruction that would "injure or impede the navigation of any stream, reserved by the ordinance of Congress of 1787 as a public highway, at a stage of water when it would otherwise be navigable."

Indianapolis was located on this stream because it was navigable for the water-craft then in use. The Legislature of 1825, on petition from the people of Indianapolis, made Alexander Ralston a commissioner to survey the stream, and report on the probable expense of keeping it free from obstruction. He made the survey that summer, and reported the distance from Sample's Mills, in Randolph County, to Indianapolis, 130 miles; from here to the forks, 285 miles; from there to the Wabash, 40 miles, and that for this distance of 455 miles the stream could be made navigable for three months in the year by an expenditure of $1,500. He found two falls or rapids, one of eighteen inches, eight miles above Martinsville, and one of nine feet in about one hundred yards, ten miles above the forks.

On this report, the Legislature on January 21, 1826, passed a law to improve the navigation of White River as high as Sample's Mills, in Randolph County, directing that all persons liable for road work living within two miles of the stream, in the counties bordering on the stream, be called out to improve the stream as a highway. This law was made general by the act of May 31, 1852, which empowered county boards to declare streams navigable, and to work them as highways; and this act is still continued in force by the act of April 15, 1905. (Burns' Stats., Sec. 7672.)

The act of January 28, 1828, appropriated $1,000 for improving the navigation of White River as high as Anderson, in Madison County. The act of January 23, 1829, "relative to navigable streams declared highways by the ordinance of Congress of 1787," prohibited any obstruction of any stream or river "which is navigable, and the bed or channel of which has not been surveyed or sold as land by the United States." So the law of 1852 made it a penal offense to obstruct "any navigable stream, the bed or channel whereof may not have been surveyed or sold by the United States." (Rev. Stats. 1852, Vol. 2, p. 432.) This is continued in force, in the same language, by the act of April 15, 1905. (Burns' Stat., Sec. 2650.)

The executive department never questioned the correctness of this rule, and some of the Governors took a great deal of interest in the matter. After the general introduction of steamboat navigation, Governor Noble was ambitious to add that to the ordinary commerce by flatboats and keel boats, and in 1828 he offered a reward of $200 to the first captain who would bring a steamboat to this point, and also to sell his cargo free of charge. In pursuance of this a small steamboat from Cincinnati was actually brought up the river to Indianapolis in 1831. The early courts also recognized the rule that the survey and sale of the bed of a stream was the conclusive test of its navigability, under the law. (3 Blackf. 193.) The State asserted actual ownership of the bed of the stream in this county, and for years maintained an agent at the Washington street crossing to sell sand and gravel from it on the State’s account.

In the face of all this, when the question came before the Supreme Court, in 1876, the court, by Judge Perkins, without any real examination of the law or the facts, said: “The court knows judicially, as a matter of fact, that White River, in Marion County, is neither a navigated nor a navigable stream”; and as to the bed not being surveyed or sold he said: “The idea that the power was given to a surveyor, or his deputy, upon casual observation, to determine the question of the navigability of rivers, and thereby conclude vast public and private rights, is an absurdity.” On this assumption he proceeded to wipe the “vast public rights” out of existence. A little examination would have shown him that the surveys were not irresponsible acts of the surveyors, but official acts in pursuance of law, under the direction of superior officers, and confirmed and ratified not only by those superiors, but by the United States and the State of Indiana. (54 Ind. 471.)

The court abandoned the reasoning of this case two years later, when it held that the Wabash River, in Warren County, was “a navigable stream, the bed of which has neither been surveyed nor sold.” (64 Ind. 162.) But it cited the decision of 1876 as authoritative in another case in 1900 (155 Ind. 477), and this again without any examination of the law or the facts. It is worthy of note that the United States Government has uniformly declined to recognize this decision as law, and as late as 1899 refused to be bound by it. (Indianapolis News, November 7, 1899.)

Fortunately, opportunity has arisen for a reconsideration of this question in a case arising in the Kankakee swamp lands (State vs. Tuesbury Land Co., Starke Circuit Court). In the northern end of Indiana, particularly near the Kankakee River, there was a large amount of swamp land which was not included in the United States surveys nor sold by the United States. This was transferred to the State many years ago, and part of it was reclaimed and sold by the State. In 1891 reclamation was entered on a large scale by removing the ledge of rock at Momence, Illinois, which dammed the Kankakee, and caused most of those swamps. As soon as these lands were drained, adjacent owners set up claims to the thread of the stream as riparian owners, and a judgment was obtained in the Starke Circuit Court upholding such claims. If valid this means that the great expense to which the State has gone in reclaiming the lands is money thrown away.

As soon as he learned of it, Governor Marshall, who is very practical in his statesmanship, directed the Attorney-General to take steps to secure a reversal of the judgment or appeal it, and a new trial has been secured in the case, which is to be heard shortly. The Kankakee is one of the most noted of the streams referred to in the Ordinance of 1787 as “navigable waters,” which are reserved forever as “public highways,” and there should be no riparian rights in it.

There is certainly good reason to expect a reversal of the Indiana decision, if not by our Supreme Court, by the Supreme Court of the United States, for two special reasons: (1) The question of the navigability of a stream is not primarily a judicial question, but one of public policy to be determined by the legislative department, and both Congress and our State Legislature have consistent records for the navigability of these streams. (2) In this case the navigability is a matter of solemn compact between the State and the United States; and as the constitutions of both prohibit any law impairing the obligation of a contract, it is hardly to be assumed that the courts would undertake to annul a contract of this character.

Unquestionably White River, like most of the other streams of Indiana, is not as practically navigable today as it was eighty years ago, and for two very simple reasons. First, at that time the only timber that got into the river was trees on the bank that were thrown in by the banks caving, and these were usually held to the banks by their roots. But after settlement began every freshet carried quantities of logs, rails and boards down the river, to form drifts; and these in turn caused the formation of sand and gravel bars. Second, when the land was cleared and cultivated, the ground washed much more readily than it did before, and much greater quantities of sand and gravel were carried into the river to form bars. These bars constitute the chief obstruction to practical navigation now.

But by a change in recent conditions of life, these bars furnish the means for making the river practically navigable. Within the last two decades there has grown up a special demand for this sand and gravel; and especially has this demand been increased by the call for good roads; for washed gravel is one of the best materials available for road-making, and by these streams, nature has distributed it very widely over the State. This demand has developed the industry of removing sand and gravel from the river beds by means of suction pumps, and since 1897, when it began, this industry has reached proportions that are not generally known to the public. At Indianapolis there have been six steam pumps working for several years. They are mounted on scow boats, fifty to sixty-five feet in length, and twenty to twenty-five feet in width, and by centrifugal suction power, draw up a mixture of sand, gravel and water through eight-inch pipes. The pipe entrance is screened to prevent the entrance of stones over four or five inches in diameter, in order to avoid clogging the pipe.

These six pumps take out 180,000 cubic yards of sand and gravel in a year, at a cost of 20 to 25 cents a cubic yard. The material is separated by passing over screens into two grades of sand and two of gravel, and is sold at a good profit for street improvement, roofing, asphalt mixture, concrete, mortar and locomotive sand. Formerly Lake Michigan sand was shipped here in considerable quantities, but now the demand is fully met by this local industry. The pumps take out the material for a depth of about fifteen feet, and in the course of their work they have made about three miles of Indianapolis river front practically navigable for any kind of river craft. The boats can easily be run to any point on the river and used for removing bars at any place. At present the proprietors of the boats are paying the adjacent landowners for the privilege of taking out material that rightfully belongs to the State, and of which the public ought to have the benefit.

The practical situation is this: Indiana has an almost inexhaustible supply of the best and cheapest road material known, which rightfully belongs to the State. By using this material it will make actually navigable hundreds of miles of waterways that are now of no use in commerce. It is quite common for the unthinking to joke about the absurdity of making small streams navigable, but there is nothing absurd about it. Over half a century ago Indiana constructed 453 miles of canal, at an average cost of $15,000 a mile, which has since been practically abandoned, not as is generally supposed, because of the competition of rail roads, but because it was high line canal, and was built up, in part, instead of being dug out, and without proper precaution for making it water-tight. The State was not alone in its experience. There are in the United States over 1,950 miles of abandoned high-line canal, that cost over $44,000,000. But there are also plenty of low-line canals in practical and profitable operation.

The mistake that was made in Indiana was in not utilizing the natural water-courses. At an expense of less than $15,000 per mile, White River can easily be made navigable for steamboats from Indianapolis to its mouth, where there is actual steamboat navigation now. The fall in the river is only 269 feet in the 285 miles, or less than a foot to the mile. The tested flow of White River at this point is over 1,000 cubic feet per second. With not to exceed half a dozen dams, and the principal bars pumped out and put into roads, the thing is accomplished. Not only is there nothing impracticable about it, but it is as certain to be done, in the not distant future, as it is that the sun will rise tomorrow morning. The advantages of water communication with the great coal and building stone region of the State, as well as direct connection with the Wabash, Ohio and Mississippi Rivers, is too obvious for discussion. With our Supreme Court decisions put on a rational and just basis, there is nothing to prevent a speedy accomplishment of the work.

REPORT FROM THE NATIONAL FERTILIZER ASSOCIATION.

Presented by Mr. John D. Toll, Secretary of the Educational Bureau of the National Fertilizer Association, and Mr. Charles S. Rauh, of Indianapolis, Official Delegates to the Fourth National Conservation Congress.

In the last analysis man must have food. The law of supply and demand becomes operative the minute that a nation is born. Scarcity of food produces abnormal prices. As scarcity increases, the prices become almost prohibitive. The attention of every citizen of this country has been called to the rapid increase in the cost of food products during the last decade. This increase has been due to several causes, among which we may note the following: (1) Increase of population has exceeded the increase in production of foodstuffs; (2) scarcity of new lands to be developed to meet the needs of a growing population; (3) decrease in productivity of some of the formerly productive soils of this country.

Other causes have undoubtedly contributed to this situation, but the one pertinent to the present discussion is that of decreasing soil fertility as it is related to the production of crops. The production of crops depends (1) upon the fertility of the soil; (2) climatic conditions; (3) quality of seed used; and (4) the culture and care given to the crop. Three of these governing factors are under the control of the farmer. Therefore, inasmuch as they are under his control, so also is the supply of food under his control to an equal degree.

According to the latest available statistics, out of 1,755,132,800 acres in the United States, there are 383,891,682 acres of improved land. A considerable amount of this land is being used for the production of crops of various kinds. These crops in 1911 totaled the enormous production of $5,504,000,000.

The plant food consisting of nitrogen, phosphoric acid and potash which these crops take from the soil is not all returned to the land, and, as a result, we have a yearly drain upon the fertility account of our soils.

The loss of all three essential elements of plant food is being partly met by the wise Conservation and use of barn manure.

The loss of nitrogen in this depletion is also being partly met by the growing of legumes which have the power of taking nitrogen from the air and fixing it in the soil where it is available as plant food.

If any one of these elements is lacking, crop production suffers a decrease in quantity and a deterioration in quality, to wit: If phosphoric acid is the lacking element in soil, the production of ear corn may be large in quantity but the ears are soft and immature, consequently are inferior for stock food or for human food. In the same way a scarcity of phosphoric acid decreases the yield of wheat and causes an inferiority in its quality.

The lack of nitrogen and potash has equally disastrous effects on the production of the crop. It is, therefore, not only a supply of plant food that is necessary, but the balancing of it in order that our soil may produce a maximum crop of best quality. The fertilizer industry is an industry engaged entirely in the supply of these plant food elements. It is, therefore, a direct contributor to the maintenance of the human race, in that it deals in the elements of plant food.

Not only since its inception in 1840 has this industry contributed to the supply of plant food, which in the end means human food, but it has noted the wasteful methods of agriculture being practiced, unconsciously frequent, in many parts of this country, and it has put forth continuous efforts along the lines of educating the American farmer to grow larger yields of better crops, to maintain and increase the fertility of the soil, and increase the average yield per acre.

The fertilizer industry has, through its national association, within the past two years established several movements, the great purpose of which is to assist in the dissemination of knowledge of modern methods of agriculture.

Another line of effort wherein the fertilizer industry has been a direct contributor to the conservation of vital resources is in its great manufacturing plants. To produce the phosphoric acid which is supplied in fertilizers, the industry obtains the barren phosphatic rock from Tennessee, Carolinas, Florida, etc., grinds this material and treats it chemically so as to make the phosphoric acid available. By this means, it supplies to the great crop producing States of this country the element of plant food, which, to a large extent, determines the maturity and quality of crop production.

The industry also takes the waste material of the packing houses, such as bone, offal, blood, etc., dries and grinds it and produces an ingredient of fertilizers which was formerly thrown away. This packing house material supplies nitrogen and phosphoric acid.

Furthermore, the garbage of the cities and towns of this country is collected and reduced to the form of fertilizer ingredient, where it formerly was burned or otherwise destroyed.

Not only is all this done, but the former waste products of the cotton and tobacco industry are similarly reduced to a form of plant food to be mixed with other materials and returned to the soil.

Still further, besides gathering up the waste and otherwise barren products of the country, the industry has developed processes whereby the gases from gas and coke manufacture are collected and reduced to sulphate of ammonia in which form it constitutes one of the nitrogenous ingredients in fertilizers. This ammonia sulphate is used as an ingredient in fertilizers supplying nitrogen.

Of late years the industry has gone even further than this, in that a process has been discovered whereby the nitrogen of the air is harnessed, and the product reduced to such forms that it supplies available nitrogen for plant food.

In assembling and preparing these essential elements of plant food, the industry, as we have pointed out, not only prepares material to return to the soil to supply the elements which have been taken out of it, but it actually provides in deficient soils elements in which they may be deficient, and by so doing makes more productive lands which, on account of their balanced plant food, could not produce paying results before being treated.

The State of Georgia in 1911 spent over twenty million dollars for fertilizers, with the result that they raised more and better cotton than was ever raised in this State before. The State of Maine in the same year used 150,000 tons of fertilizers on their potato fields, with the result that their good potato growers produced from two to four hundred bushels of potatoes per acre. The total State productions exceed 25,000,000 bushels.

The fertilizer industry, we believe, occupies a most prominent part in the problem of producing sustenance for future generations and in conserving the vital resources of the world, inasmuch as it is making a close study of and doing a wonderful work in devising means whereby practically all of the former waste materials of this country may be reduced to available plant food and returned to the soil from which they were taken.

Summarizing, the fertilizer industry, as we have pointed out, is an important factor in the maintenance of the human race, of this and other continents, in that (1) it supplies plant food to balance up the plant food in the soil and to make up the deficiencies which have occurred as a result of continuous cropping; (2) assists in educating the farmers to conserve the fertility of their soils by employing scientific methods of farming; (3) it makes use of waste products of other industries which have formerly been destroyed, and returns them to the soil in the form of food for future crops.

The fertilizer industry, therefore, must be recognized as one of the greatest agencies of conservation of vital resources.

DR. W J McGEE: AN APPRECIATION OF HIS SERVICES FOR CONSERVATION.

Mr. W. C. Mendenhall, Washington, D. C.

Dr. W J McGee had mastered and advocated the fundamental principles of Conservation long before the majority of those now most active in the movement had come to appreciate the real meaning of the word. He was one of the founders of this movement, was at all times one of the most stimulating thinkers in its councils, and in him Mr. Pinchot found one of his most loyal supporters and friends. Doctor McGee was the personification of strength and steadfastness in the pioneer period of Conservation, when the meaning of the word was unknown to the multitude, and when the mere suggestion that our natural resources are not inexhaustible but may be depleted to the vanishing point by wasteful use was regarded as a wild heresy. During the preceding sessions of this Congress he was its accepted authority on problems involving that most widespread and universally distributed of our natural resources—water. He has dealt with this resource from the points of view of transportation, of irrigation, and of power, and from the standpoint of biology in which it is recognized as fundamental in all life.

Doctor McGee's mind was of the type of the intellectual pioneer, intensely individual and original. He was masterful in the alignment of facts and stimulating in the recognition and boldness of his expression of the generalizations and far-reaching conclusions to which his marshalled facts pointed. Like most men of brilliant imagination, he was at times impatient of the slow processes of research, or let us say rather that his impatience was with that timidity in reaching conclusions so often displayed by those engaged in research, rather than with the process itself. He believed that the scientist's practical rule of life should be the acceptance, as a basis of action, of the conclusions indicated by such facts as are known, even though those conclusions may not at present be definitely established.

Doctor McGee's death at the Cosmos Club on September 4, 1912, removed from the domain of science and from the forum of public discussion one of its leading personalities. His career embraced an unusually wide range of activities and in each of these he attained distinction. As a geologist he was one of the group assembled by Major Powell during the formative period of the United States Geological Survey, a group which made American geology classic and its leaders world-leaders in their science. In this field McGee's name is associated with the names of

Powell, Dutton, Gilbert, Holmes, Emmons, and Hague. Later, with the establishment of the Bureau of Ethnology, into which he followed Major Powell, he became a pioneer in ethnological research, although retaining continually his interest in geologic and geographic problems. At the time of his death and for a few years prior thereto he was the erosion and hydrologic expert of the Department of Agriculture, his immediate connection with that department being through the Bureau of Soils. His last years are distinguished by a number of papers on the subject of Conservation, in which he was so vitally interested; these articles are broad in their scope, thoroughly original and stimulating in their expression, and point out fearlessly some dangers of present practice and suggest methods of remedy. But a few days before his death he completed the correction of the galley proofs of the last of his papers, faithful to his work and to his duty even while descending into the Valley of the Shadow.

This very brief sketch would not be complete without expressed recognition of the fact that Doctor McGee's attitude in the face of death was in accordance with the best traditions of the science to which his life had been devoted and was as admirable and as deeply stirring and stimulating as any act of his career. For a year or more he had recognized the fact that he was afflicted with cancer and that his days were numbered. He faced this fact calmly and prepared patiently for the inevitable by carefully completing all work on hand and by disposing by will of his body and his brain to his friend and fellow scientist, Dr. Spitska, to be used in the way most likely to be beneficial to humanity. So long as his faculties remained undimmed, he maintained the same keen interest in scientific questions and current affairs that had marked his career at its height. Those of his friends who visited him during the last few weeks of his life heard not a single complaint nor an expression of regret, but found themselves chatting easily with an old and honored friend who gave no indication of the fact that he knew definitely that his career was soon to be stayed by the hand of Death. Thus, rising superior to the weakened, pain-racked body, he met with philosophic calm and sublime courage the final inevitable test. It is not given to man to do more than this.

REPORTS OF STANDING COMMITTEES.

All the standing committees were represented at the Fourth National Conservation Congress and made reports. Members of these committees also addressed the Congress.

Forests: See paper by Prof. Henry S. Graves, Chairman, page 318; also addresses of Mr. E. T. Allen and Major E. G. Griggs, pages 312 and 183, respectively.

Lands: Prof. L. H. Bailey, being unavoidably prevented from attending the Congress, the report of the Lands Committee was presented by Dr. George E. Condra, page 123. See also address of Mr. Charles S. Barrett, page 132.

Waters: Owing to the death of Dr. W J McGee, Chairman, the report of this committee was submitted by Dr. W. C. Mendenhall. See page 335.

Minerals: See address of Dr. Joseph A. Holmes, Chairman, page 200.

Vital Resources: See address of Mr. A. B. Farquhar, page 214.

Food: See address of Dr. Harvey W. Wiley, Chairman, page 75. Also see paper by Mr. F. G. Urner, page 327.

Homes: See address of Mrs. Matthew T. Scott, President-General, Daughters of the American Revolution, page 250. Also report by Mrs. Orville T. Bright, of Chicago, Vice-President of the National Congress of Mothers, page 196.

Child Life: See address and report of the Hon. Ben B. Lindsey, Chairman, page 170.

Education: See report of Dr. C. E. Bessey, Chairman, page 66; see also address of Prof. E. T. Fairchild, page 134.

Civics: See address of Mr. Ralph M. Easley, Chairman, page 272.

National Parks: The report of this committee was submitted in writing, signed by Dr. W J McGee, Chairman, and Col. Malcolm H. Crump. See page 182.

General (including Wild Life): Dr. W. T. Hornaday, Chairman, presented the report of this committee. See page 344. Dr. T. Gilbert Pearson gave an address on “Bird Slaughter and the Cost of Living.” See page 72.

INDEX.

Transcriber’s note

Consolidated the different spelling of cooperation and coöperation with the most used co-operation throughout book

Added missing punctuation where needed

pg 4 Changed Relation to Pulbic to: Public

pg 5 Changed Scott 250-258 to: 250-254

pg 5 Added letter h to Pittsburg in: Pittsburg, Mr. George M. Lehman

pg 10 Changed W J McGee. to: McGee,

pg 11 Changed Okland to: Oakland, Cal.

pg 14 Changed from the State Conservations to: Conservation

pg 16 Added period to: of less than 25,000.

pg 27 Changed take Russion with her to: Russia

pg 28 Changed spelling of: and familiarizing themeslves to: themselves

pg 29 Changed comma to period at: drainage and the like.

pg 31 Changed spelling of: $600,000 to $300,000 annully to: annually

pg 37 Changed something for beyond to: something far beyond

pg 43 Changed which will create waterpower to: water power

pg 45 Changed case of First, Is the river to: is

pg 54 Changed one of reforstation to: reforestation

pg 58 Changed spelling of older communities undistrubed to: undisturbed

pg 74 Changed This we as through to: ask

pg 80 Changed principles of serum phophylaxis to: prophylaxis

pg 85 Changed spelling nature of the phagocytosthe to: phagocytose

pg 88 Changed spelling in any localtiy to: locality

pg 89 Changed spelling the use of stimulii to stimuli (3 places)

pg 98 Changed spelling other official public healh to: health

pg 100 Changed spelling height of absudity to: absurdity

pg 108 Changed spelling already passed the prosphorus to: phosphorus

pg 111 Changed spelling simply to breath to: breathe

pg 124 Changed spelling speculation and over valuation to: overvaluation (other matches in book)

pg 135 Matched spelling of little short of marvellous to: spelled marvelous in other places

pg 135 Changed world-wide movement, by friends to: my friends

pg 135 Changed that not may purpose to: my purpose

pg 137 Changed they have not he to: the

pg 138 Added comma to million boys and girls,

pg 141 Changed low standards or decency to: of decency

pg 144 Changed spelling ethical and spirtual to: spiritual

pg 150 Removed unnecessary quote after: to have adequate recreation.

pg 150 Changed spelling A Japanese physican to: physician

pg 152 Changed two instances of clamy hands, clamy feet to: clammy

pg 152 Changed insomnia, fugative pains to: fugitive pains

pg 154 Changed spelling temperance and sanitied to: sanitized

pg 155 Changed alloy steel because it work to: works

pg 166 Changed mathematics to compete to: compute
pg 166 Changed which tends to drawf to: dwarf
pg 172 Changed waterfalls and vendure to: verdure
pg 176 Changed spelling refuse to perfrom to: perform
pg 179 Changed where these things out to: ought
pg 206 Chart total for Scalded or Burned does not add up 197, should be 297
pg 210 Changed human factor with its attendent to: attendant
pg 214 Changed President White—Whe to: We
pg 217 Changed Similiar to the work to: Similar
pg 235 Changed single quote to double after: partners in the creation?
pg 236 Changed not write an immoral to: immortal
pg 247 Changed deux ex machina to: deus
pg 255 Changed abundance make it posible to: possible
pg 258 Changed I feel that this Congres to: Congress
pg 264 Changed each year and very probaly to: probably
pg 269 Changed in procuring the necesary to: necessary
pg 279 Changed released the rich, wheras to: whereas
pg 280 Changed mind that this resume to: resumé
pg 290 Changed Pittsburg having been to: Pittsburgh
pg 290 Changed In conection with complete to: In connection
pg 293 Changed also for deeping to: deepening
pg 294 Changed now successfully empolyed to: employed
pg 295 Changed derived by preventitive to: preventative
pg 303 Changed in the lower straum to: stratum
pg 304 Removed comma after if we go on and plant
pg 307 Removed period after Office of Public Roads
pg 308 Changed time when these Stats to: States
pg 309 Removed comma after Mr. Chairman, Knoxville
pg 313 Changed effort to secure reforstation. to: reforestation.
pg 316 Added period after: given the necessary publicity
pg 325 Changed In the hands of sopilsmen to: spoilsmen
pg 326 Changed and large quanities to: quantities
pg 333 Changed no criteron of their quality. to no criterion
pg 338 Added period to: governmental activity in this direction
pg 346 Changed North Carolina, Tennsesee to: Tennessee
pg 346 Changed extinct as the mastadon to: mastodon
pg 353 Changed International Court of Aribtration to: Arbitration
pg 358 Added double quote to: always applied it to “White River
pg 363 Changed be large in quanaity to: quantity
pg 364 Changed Not only since its inseption to: inception
pg 365 Removed the word for from: soil in the form for of food
pg 367 Changed he knew definitely that has to: his
pg 369 Added period to: Baumgartner, J.
pg 369 Changed capitalization of In relation to the home, 252 to: in relation
pg 370 Changed Investigation of Pittsburgh Flood Commisson to: Commission

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”

- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.

- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain "Defects," such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the "Right of Replacement or Refund" described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations ($1 to $5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works